MBSniffer v3.2 Technical Code Documentation

Architecture, runtime flows, every callable, regression tests, build automation, and complete source-code screenshots

Documented package

`MBSniffer_v3.2_Package.zip`

Source snapshot SHA-256

`50e3157c8abe93187acd4e4b90dd437fd10cc3154264e8965b60038bc8cde1d2`

Documentation build date

23 October 2026

Verification

64/64 regression tests passed before documentation generation.

This document describes the exact source snapshot above. It does not document proposed future Modbus TCP or Slave-Simulator features that are not present in this package.

The final appendix reproduces every Python and Windows batch source line as code screenshots. The explanatory chapters refer to the same line numbers.

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1. Documented source snapshot

This reference was generated from the current MBSniffer v3.2 package, without modifying the application source.

Python source lines	Documented Python + BAT lines
13,056	13,462

Package structure

```
MBSniffer_v3.2_Package.zip
├── tools/  (documentation and package builders)
├── MBSniffer_v3.2_Technical_Code_Documentation.pdf  (this document)
├── MBSniffer.bat
├── README.md
├── README.en.md
├── README.txt
└── MBSniffer v3.2/
    ├── MBSniffer.py
    ├── mb_config.py
    ├── mb_protocol.py
    ├── mb_capture.py
    ├── mb_diagnostics.py
    ├── mb_view.py
    ├── mb_slave_finder.py
    ├── mb_master.py
    ├── mb_master_ui.py
    ├── mb_i18n.py
    ├── mb_theme.py
    ├── mb_widgets.py
    ├── mb_gui.py
    ├── tests/test_regression.py
    ├── tests/test_master.py
    ├── run_tests.bat
    ├── build_exe.bat
    ├── MBSniffer.ico
    ├── DEVELOPMENT_NOTES.md
    └── CHANGELOG.md
```

Source inventory

Python module	Lines	Primary responsibility
`MBSniffer.py`	29	Application launcher and top-level entry point.
`mb_config.py`	464	Central configuration, constants, user settings validation and filesystem paths.

Python module	Lines	Primary responsibility
`mb_protocol.py`	1007	Protocol engine for passive Modbus RTU parsing and session metrics.
`mb_capture.py`	883	Serial-port management and passive acquisition pipeline.
`mb_diagnostics.py`	286	Presentation-oriented diagnostics derived from captured records.
`mb_view.py`	1853	Traffic view, filtering, sorting, Frame Inspector, Bus Health updates, CSV export and TXT logging.
`mb_slave_finder.py`	1587	Active Modbus RTU Slave discovery and Device Identification.
`mb_master.py`	516	Modbus RTU master engine, frames and data formats.
`mb_master_ui.py`	1150	Master (poll) tab user interface.
`mb_i18n.py`	555	Runtime Portuguese/English translation layer.
`mb_theme.py`	692	Fixed-geometry Light/Dark theme implementation.
`mb_widgets.py`	418	Custom reusable Tkinter widgets.
`mb_gui.py`	2529	Main application composition and global UI behavior.
`test_regression.py`	787	Automated regression suite.
`test_master.py`	300	Master regression tests.

Integrity hashes

Source file	Lines	SHA-256
`MBSniffer v3.2/MBSniffer.py`	29	`c5eae72b053b4d96a112c492ec200b5a046724ff0a0893b3e50f86ae1df564bb`
`MBSniffer v3.2/mb_config.py`	464	`0c4c334410d1117bfbe2f8fc7bc9ecc1fd68663124f87539e590cfb22590b829`
`MBSniffer v3.2/mb_protocol.py`	1007	`8c4c41051e41f0c998692206ed0204d91f057cce5fce726b15ad2cce503c1ef2`
`MBSniffer v3.2/mb_capture.py`	883	`2785386c053f477dd0095b8cb6446264da208b9cd0ed2fa6807047f965e01a0c`
`MBSniffer v3.2/mb_diagnostics.py`	286	`d5edc15bf0259d130fdf9b42ffaf2caca770782101aeed1886f980e0fcf24bda`
`MBSniffer v3.2/mb_view.py`	1853	`1ae5a23ae5cecd50e22a50868bce04ef1c33168b2aef091e0b32384f5983324f`
`MBSniffer v3.2/mb_slave_finder.py`	1587	`54bc1c44f963798a8a2a0912d872306224fb87ffea53489f0a97b7ddb0f90789`
`MBSniffer v3.2/mb_master.py`	516	`5b0aa32c4982020966f0dfc7dbdec0d284dc60a45a05a68e11e80bb5d579f5d7`

Source file	Lines	SHA-256
`MBSniffer v3.2/mb_master_ui.py`	1150	`c898de2286993ea45266f4a3566defe7248a09d41c19050f52cbcc955923bca0`
`MBSniffer v3.2/mb_i18n.py`	555	`92a5d8fbd7c244940c8f6ba7ad44aae325d27b107ff02cfaa511cc388d14aa1c`
`MBSniffer v3.2/mb_theme.py`	692	`88c0c2592f4b6e0666aa42e4d163ae120845b154f7faa3b9f6a007fef7ed8265`
`MBSniffer v3.2/mb_widgets.py`	418	`fb45d18d621a734220afa46d92dfb882228a5cba2702e4201ccf02120b2bff97`
`MBSniffer v3.2/mb_gui.py`	2529	`52f2770e0766f002518b4adbd48180147ad60319194dd21ee8bfce1f57eb65d4`
`MBSniffer v3.2/tests/test_regression.py`	787	`b5a3f7e75c0b385f8cb874fdf3b16c8e867108c59406078d966f333db4dba70c`
`MBSniffer v3.2/tests/test_master.py`	300	`bd82e2e3af578d5b0d6e21c1bf55a1d2abcaba510b629198a5447bfa9089a19c`
`MBSniffer.bat`	210	`3c1a036c6ef094c26390c883d8f59aa5e226da5394100edad2e523410fdf5a1d`
`MBSniffer v3.2/build_exe.bat`	151	`fbe1caf24e1c48fe6a55f959a61ecc0d0f6a50b376f02297bb6c7d642b8b0775`
`MBSniffer v3.2/run_tests.bat`	45	`8066035cd8d62a7aded1f8314b457d438795a8c59319bf1290c47bc63d34372f`

> **Regression status.** The supplied suite was executed directly from this source tree. All 64 tests passed.

2. Version history and changes in this version

Release history of the documented code base. The detailed list below applies to v3.2, the version described by every other chapter of this document.

Version	Date	Summary
v3.0	21 September 2026	Bilingual PT-PT / English interface, FC43/14 Device Identification, regression suite and first Technical Code Documentation.
v3.1	23 October 2026	Broadcast/echo-write pairing fixes, USB split-frame reassembly, in-place timeout marking, Finder attempts and idle-bus check, address-mapped read data, table-driven CRC and centralised translations.
v3.2	23 October 2026	Master (poll) tab with one window per Slave / data area, cyclic reads, writes, 16/32-bit data formats and per-window counters; the release ZIP now also carries this PDF and the documentation tools.

Changes in v3.2

v3.2 adds the first phase of a Modbus RTU master in the style of Modbus Poll: a **Master** tab with several poll windows, one per Slave or data area. The feature list was taken from the public Modbus Poll documentation. Every other chapter of this document is regenerated from the v3.2 source, so the module reference, line numbers, hashes, tests and Appendix A describe v3.2.

Master tab (phase 1)

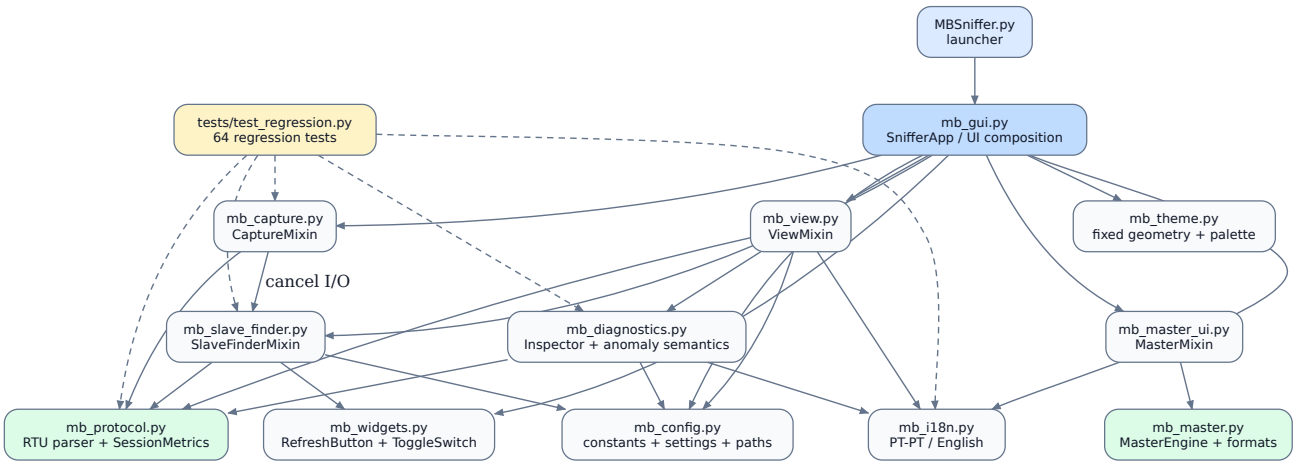
- **New modules:** `mb_master.py` (frames, data formats and `MasterEngine`, no widgets) and `mb_master_ui.py` (`MasterMixin`, the tab and its windows). See chapter "Master (poll) tab".
- **Windows:** Slave ID, FC01–FC04, address, quantity, scan rate, address base 0/1 and data format, with Apply, Pause, Read once, Duplicate, Pop out and Close; up to 32 windows arranged in a responsive grid or popped out to their own top-level window.
- **One bus, one request at a time:** all windows share one serial port; writes and "Read once" are served first, then the window whose scan time is due earliest, with the configured delay between polls (at least t3.5).
- **Data formats:** signed/unsigned 16-bit, hex, binary and signed/unsigned/float 32-bit in byte orders ABCD, CDAB, BADC and DCBA.
- **Writes:** FC05/FC15 from coil windows and FC06/FC16 from holding-register windows, several values separated by ";", broadcast with Slave 0, automatic re-read after an accepted write.
- **Diagnostics:** Tx, error and timeout counters, last response time and last error (Timeout, Exception, invalid response) per window.
- **Safety:** the same confirmation and idle-bus check as Bus Slave Finder; while connected, Sniffer, Simulation and Finder are locked through the global busy flag.
- **Workspace:** `.mbsw` JSON files store the connection, windows, formats and per-address names.

Packaging and documentation

- The release ZIP holds the application files, the README files, this PDF and `tools/`, and nothing related to git or GitHub, which the user manages. `tools/build_package.py` builds it from the working folder.
- Because this PDF travels inside the ZIP, the cover quotes a **Source snapshot SHA-256** (one hash over every documented file) instead of the ZIP hash.
- The regression suite grew from 45 to 64 tests, including an in-memory Modbus slave used to test the Master engine.

3. Architecture

MBSniffer uses one Tkinter root object with four behavior mixins. Protocol parsing and metrics are separated from the GUI, while the active Finder and Master have their own worker logic. This is a pragmatic modular architecture rather than a strict MVC/MVVM framework.



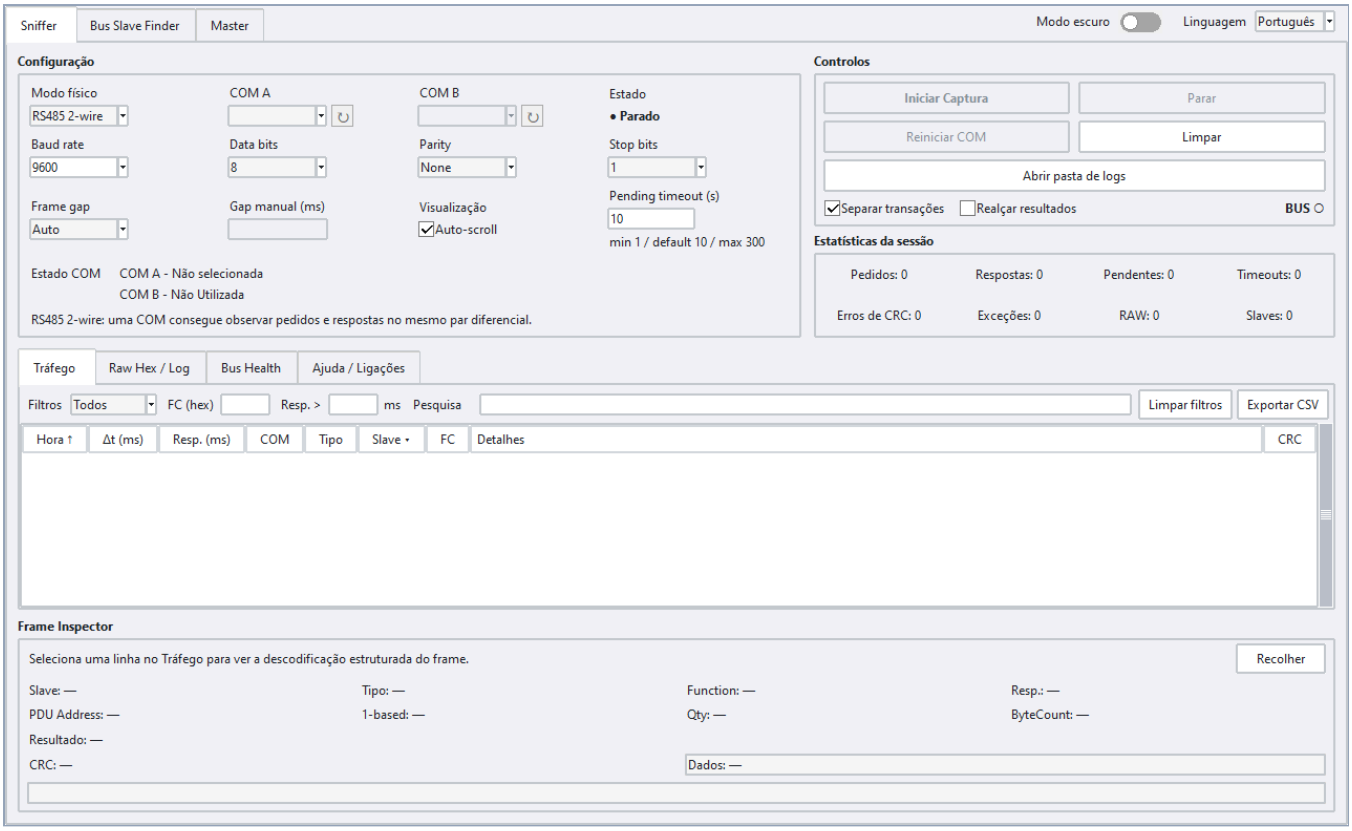
Module dependency and responsibility overview.

Composition of SnifferApp

`SnifferApp(MasterMixin, SlaveFinderMixin, CaptureMixin, ViewMixin, tk.Tk)` composes the application. `mb_gui.py` constructs and coordinates the UI; the four mixins supply the Master tab, active discovery, passive capture, and presentation/logging behavior.

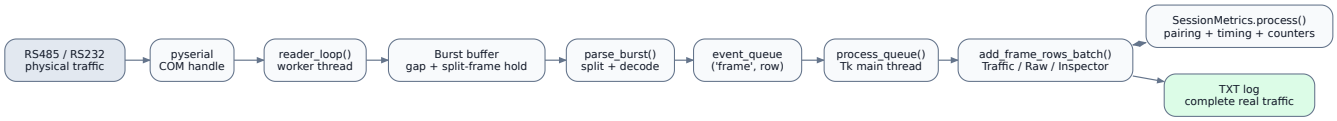
Important architectural boundaries

- **Protocol vs UI:** CRC, frame recognition, parsing and SessionMetrics are in `mb_protocol.py` and have no Tkinter dependency.
- **Workers vs Tk:** serial reader/discovery threads communicate through `event_queue`; Tk widgets are updated from the main thread.
- **Source history vs presentation:** filters and sorting rebuild the view from `_frame_history`; they do not modify the raw session record or statistics.
- **Passive vs active:** the Sniffer only reads. Bus Slave Finder intentionally transmits requests and is protected by a safety confirmation/global busy state.
- **Appearance vs geometry:** Light/Dark changes colors; language changes text; both are designed not to rebuild widget geometry.



Current Sniffer UI generated from the documented source snapshot.

4. Passive capture pipeline



From physical bytes to UI records and disk logging.

1. **Port selection.** `selected_port_plan()` maps RS485/RS232 mode to one or two receive channels.
2. **Serial configuration.** `serial_config()` translates Baud/Data bits/Parity/Stop bits into pyserial constants.
3. **Gap calculation.** Automatic mode uses the shared Modbus t3.5 helper; manual mode validates the user-entered millisecond value.
4. **Reader workers.** One daemon thread per active receive channel reads bytes, records the arrival time of every read and closes a burst after the configured silence. If the burst ends with the start of an unfinished known frame, the reader waits up to `SNIFFER_SPLIT_FRAME_HOLD_MS` for the rest (USB latency-timer splits).
5. **Parsing.** Each burst is split/resynchronised and decoded by `parse_burst()`; each frame keeps the arrival time of the read holding its first byte.
6. **Thread boundary.** Parsed rows are placed in `event_queue`.
7. **Main-thread processing.** `process_queue()` drains events in batches and passes frame batches to `add_frame_rows_batch()`.
8. **Metrics/view/log.** The view runs SessionMetrics, updates Traffic/Raw/Inspector/Bus Health and writes complete real traffic to the lazy TXT log.

> **Threading rule.** The serial threads do not call Tkinter methods. The queue is the explicit synchronization boundary between workers and the GUI thread.

5. Frame model and parser behavior

RTU CRC and frame timing

`crc16_modbus()` implements the Modbus CRC-16 algorithm with a 256-entry lookup table (`_CRC16_TABLE`). `append_crc()` appends low byte then high byte. `modbus_rtu_interframe_gap_seconds()` provides the common t3.5 timing used by Sniffer and Finder.

Recognised frame shapes

`_candidate_prefixes_at()` recognises structural lengths for FC01/02/03/04, FC05/06/08, FC07, FC15/16, FC22, FC23 and standard Exception frames. Unknown Function Codes are deliberately handled conservatively.

Resynchronisation

The parser first tries a recognised, CRC-valid frame at offset zero. If that fails, it searches later offsets but only using recognised frame shapes. This is the performance and false-positive safeguard introduced after the earlier brute-force CRC scan.

- Known-shape prefix before a recovered valid frame + bad CRC → `CRC ERROR`.
- Unstructured bytes before a recovered frame → `RAW/UNSYNC`.
- Known-shape complete remainder + bad CRC → `CRC ERROR`.
- Unsupported-FC complete remainder with valid CRC → `UNKNOWN / OK`.
- Otherwise → `RAW/UNPARSED`.

Bursts split by USB adapters

`burst_may_continue()` re-runs the splitter on the burst and inspects only the final unvalidated segment: if it starts with a plausible Slave ID and a known Function Code but is shorter than every candidate length, more bytes are expected. The reader re-checks only when the buffer length changes, so a long noise burst is not re-parsed on every poll.

Read-response bits

FC01/FC02 responses are decoded least-significant bit first by `read_bits_from_response()`. `read_bits_details()` groups the bits per data byte, shows at most 32 of them in Traffic and trims the padding of the last byte once the matching request quantity is known.

Parser row schema

Field	Meaning
`time`	Display timestamp string.
`channel`	BUS, A→B, B→A or SIM source.
`type`	REQUEST, RESPONSE, EXCEPTION, RAW/UNSYNC, RAW/UNPARSED, UNKNOWN or REQ/RESP (FC05/06/08/22, resolved to REQUEST/RESPONSE by SessionMetrics).
`slave`, `fc`	Decoded address and hexadecimal Function Code when available.
`details`	Human-readable protocol details.
`crc`	OK, ERROR or ?.
`raw`	Original frame bytes rendered as hexadecimal.
`pdu_address`, `one_based`, `qty`	Addressing/quantity metadata when decoded.
`_frame_len`	Number of observed bytes represented by the row.
`_epoch`	Floating timestamp used for timing/pairing: arrival time of the read holding the frame's first byte.

6. Session metrics and Bus Health

`SessionMetrics` is stateful per capture/session. It records counters, pending requests, response samples and diagnostic aggregates before GUI filters are applied.

Request/response pairing

A request key is `(Slave ID, base Function Code)`. Valid REQUEST rows are queued FIFO per key. RESPONSE/EXCEPTION rows remove the oldest matching pending request and compute response time from the two epoch timestamps.

> **Pairing limitation.** Modbus RTU has no transaction identifier. The implementation therefore uses FIFO matching by Slave and Function Code. If a non-standard master pipelines multiple same-Slave/same-FC requests without waiting for responses, pairing cannot be proven uniquely from RTU headers alone.

Echo-shaped functions

FC05, FC06, FC08 and FC22 requests and their normal responses have the same length, so the parser emits `REQ/RESP`. `_resolve_echo_direction()` classifies the frame before accounting: RESPONSE when an older pending request with the same key exists (identical bytes for the exact-echo FCs 05/06/22; opposite channel in RS232 dual RX), otherwise REQUEST.

> **Echo limitation.** On a single RS485 channel, if a master repeats the same write to a Slave that never answers, the repetition is indistinguishable from an echo response and is paired as one.

Broadcast

Requests to Slave 0 are counted as requests and in `broadcast_requests`, but are never pending: by definition they receive no response and must not become timeouts.

Timeouts

Pending requests older than the configured Pending timeout are expired. The timeout is attached to the original REQUEST record through its sequence ID; MBSniffer does not invent a fake timeout frame. The view retags only the affected rows instead of rebuilding the table.

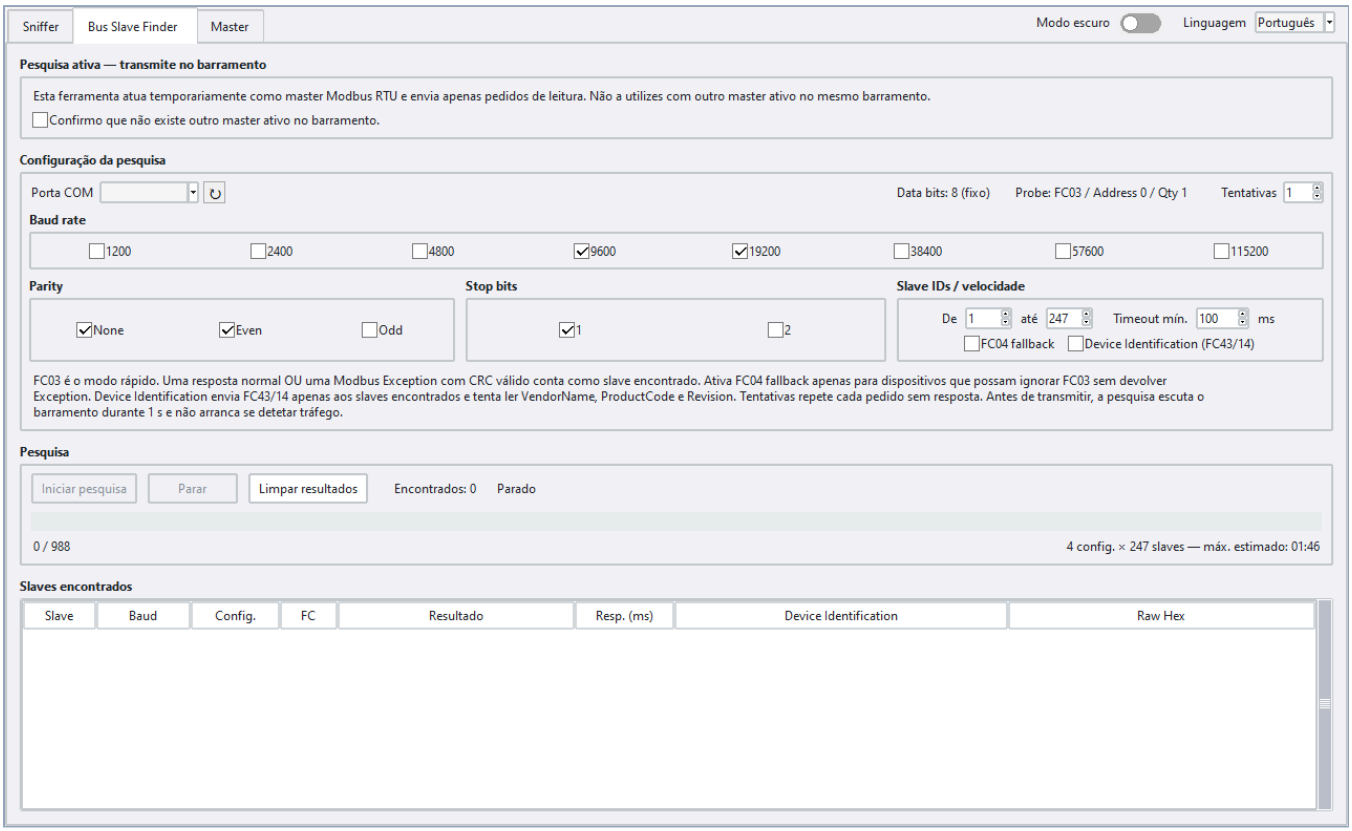
Bus Health calculations

- **Request rate:** requests divided by observed elapsed session time.
- **Response time:** average, minimum, maximum and nearest-rank P95 from valid paired responses.

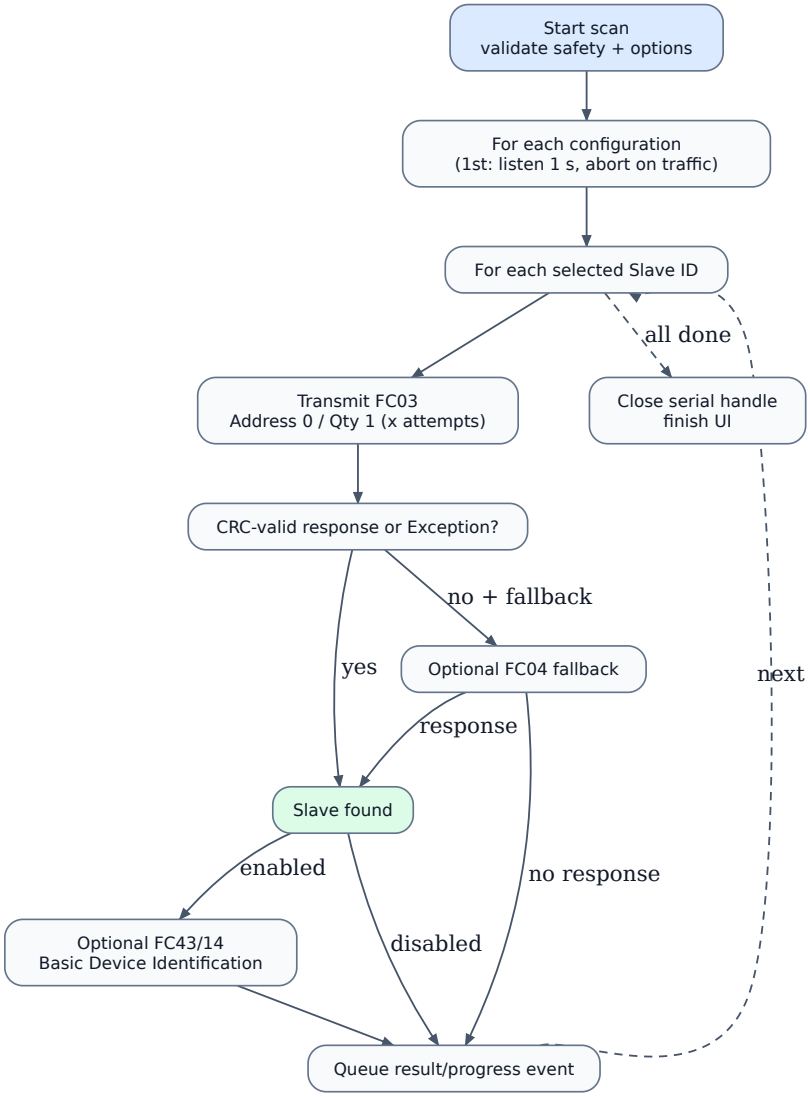
- **Slowest Slave:** highest average response time among sampled Slaves.
- **Unvalidated bytes:** bytes in explicit CRC-error frames plus RAW blocks divided by all observed bytes.
- **Timeout rate:** expired requests / observed non-broadcast requests.
- **Approximate bus utilisation:** observed bytes × configured serial bits per character / (baud × elapsed time).
- **Exceptions / Slave:** counts of Exception responses grouped by Slave ID.

> **Physical-layer boundary.** Bus utilisation and unvalidated-byte percentage are software/capture indicators. The program does not measure analog RS485 voltage, bias, reflections, ringing, common-mode level or rise/fall times.

7. Active Bus Slave Finder



Current active Finder UI generated from the documented source.



Discovery flow for each selected configuration and Slave ID.

Discovery semantics

The standard probe is FC03, Address 0, Quantity 1. A CRC-valid normal response proves the Slave exists. A CRC-valid Modbus Exception also proves the Slave exists, because the addressed device had to receive and interpret the request to generate that Exception.

FC04 fallback is optional and is attempted only after FC03 produces no accepted response. **Attempts** (1–3) repeats each unanswered probe before moving on. Device Identification is also optional and is performed only after a Slave has already been found.

Idle-bus check

Before the first transmission of a scan, `bus_activity_detected()` listens for `DISCOVERY_IDLE_CHECK_SECONDS` on the first configuration. Any byte, even one framed at another baud rate, means something else is transmitting: the worker raises an error naming the port and the first bytes, and nothing is sent. This complements the safety checkbox.

FC43/14 Device Identification

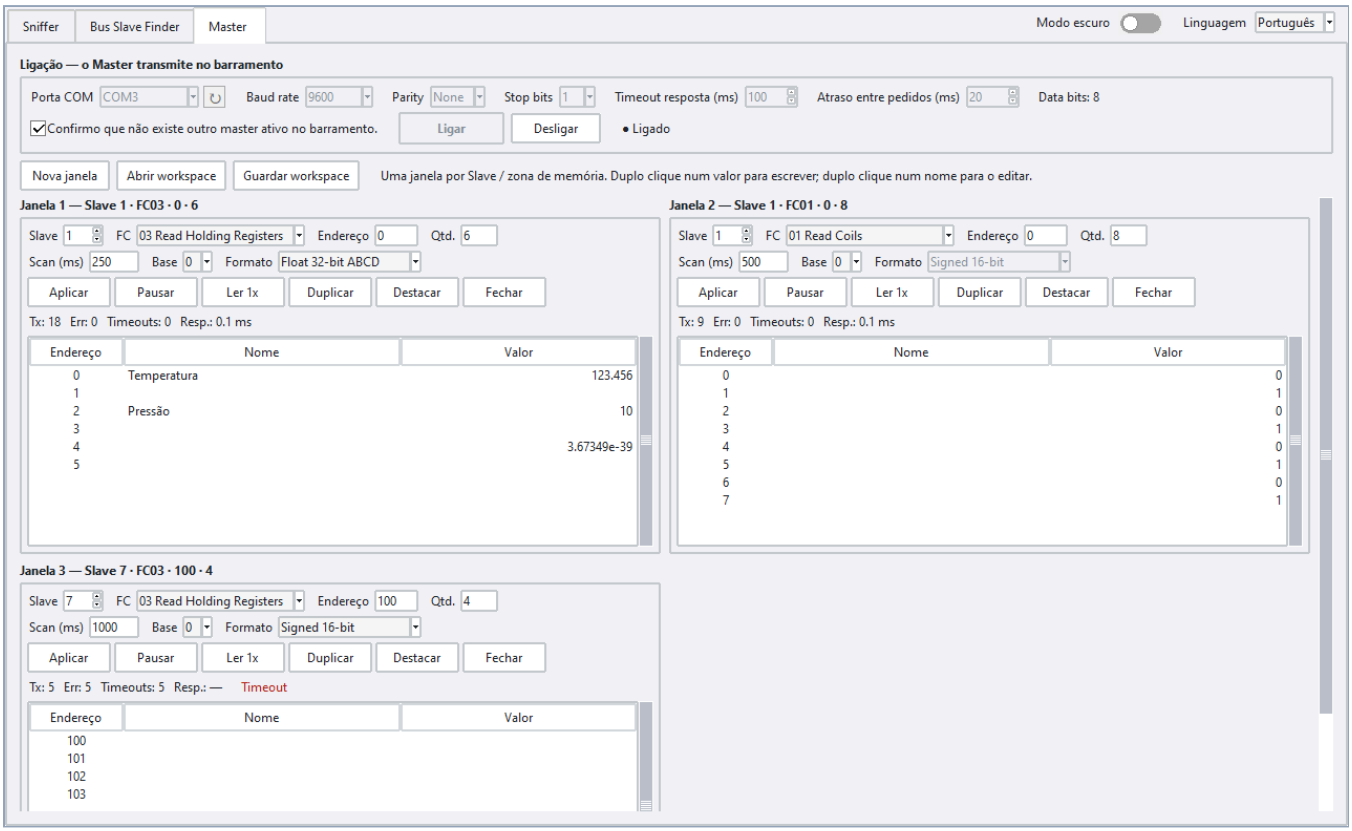
The request uses Function Code 0x2B, MEI type 0x0E and Basic Device Identification read code 0x01. The parser reads variable-length objects, validates CRC and follows `More Follows` pages up to the configured page cap. Display output focuses on object 0x00 VendorName, 0x01 ProductCode and 0x02 MajorMinorRevision.

Driver/error handling

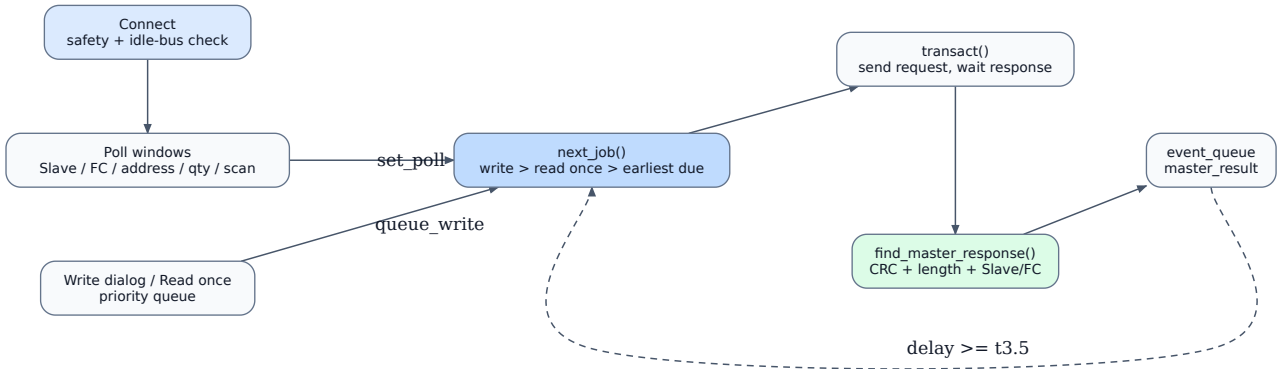
A serial write/read failure is treated as a communication/driver error, not as an ordinary "Slave did not respond" result. Write timeout recovery can cancel/clear driver buffers and retry once. Stop requests use `threading.Event` and best-effort cancel_read/cancel_write support.

> **Operational safety.** This tab actively transmits Modbus RTU requests. It must not be run on a bus that already has another active master unless the system is explicitly designed for multi-master behavior.

8. Master (poll) tab



Master tab connected to the in-memory test slave of tests/test_master.py.



Master engine: one request at a time over one serial port.

The Master tab reads and writes values as a Modbus RTU master, in the style of Modbus Poll. Each **poll window** describes one data area of one Slave; the state of a window lives in a `MasterWindow` object, independent of where its pane is shown.

Scheduling on one bus

Modbus RTU allows a single request on the bus at a time, so one `MasterEngine` thread serves every window over one serial port. Its `next_job()` returns, in this order: a queued write, a queued "Read once", or the enabled window whose next due time is earliest. A window becomes due again `scan_ms` after its previous request started (0 polls continuously). After every transaction the engine leaves the configured

delay between polls on the bus, never less than Modbus t3.5. Response time and the response timeout count from the end of the transmitted request.

Frames and responses

`build_read_request()` and `build_write_request()` build FC01–04 and FC05/06/15/16 frames within the Modbus limits (2000 bits / 125 registers per read, 1968 coils / 123 registers per write). `find_master_response()` scans the received bytes for the CRC-valid frame of the expected length from the addressed Slave, or a 5-byte exception. Read requests echoed by an RS485 adapter are skipped because their length and CRC do not fit. Write responses must echo the request (FC05/06) or its address/quantity (FC15/16). Broadcast writes (Slave 0) are not answered and are reported as sent.

Data formats

Format	Meaning
Signed / Unsigned 16-bit	One register as a two's-complement or unsigned integer.
Hex / Binary 16-bit	One register as `0xHHHH` or two binary bytes.
Signed / Unsigned / Float 32-bit	Two registers. Register n carries bytes A B and register n+1 bytes C D; the order names the value's bytes: ABCD big-endian, CDAB word swap, BADC byte swap, DCBA little-endian. The value is shown on the first register's row.

Writing

Double-clicking a value in an FC01 or FC03 window opens the write dialog. `master_build_write()` validates the Slave (0–247), the address in the window's base and the values, encodes them with the window format and builds an FC05, FC06, FC15 or FC16 request; 32-bit formats require FC16. Several values separated by ";" go to consecutive addresses. An accepted write queues a "Read once" of the window so the table shows the new value.

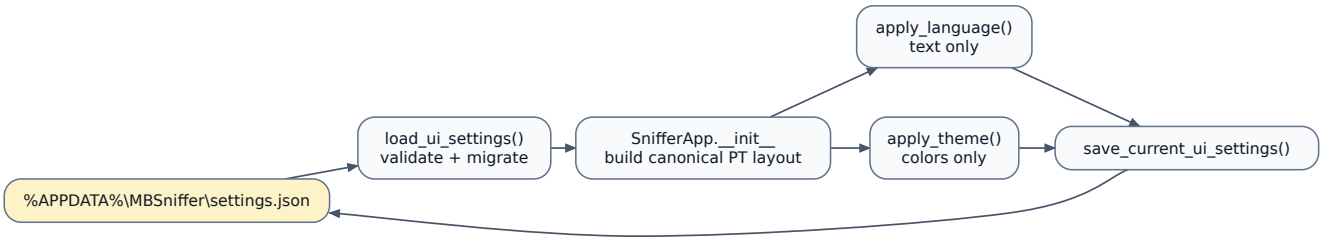
Window life cycle

- **Apply** validates the fields and updates the engine; a changed data area resets the counters. Results of a request sent with the previous definition are ignored.
- **Pop out** rebuilds the pane inside a `tk.Toplevel`; closing it or **Dock** returns it to the grid. The grid uses as many equal columns as the tab width allows.
- Panes created after start-up receive the application's mouse-wheel guard tag: the wheel scrolls the window grid and never changes combobox/spinbox values.
- **Workspace** files (`.mbsw`, JSON) store the connection parameters, every window and its names; invalid entries are skipped with a warning.

Operational safety. Like Bus Slave Finder, the Master transmits. Connecting requires the safety confirmation and an idle-bus check; while it is connected the global busy flag locks Sniffer, Simulation and Finder.

Phase boundary. Logging, scaling, conditional colours, charts, Test Center, FC22/23 and Modbus TCP are planned later phases and are not part of v3.2.

9. UI, language and theme design



Settings, language and theme lifecycle.

Canonical-language strategy

Portuguese (Portugal) is the canonical static UI language. `canonical_ui_text()` can map known English captions back to PT-PT, then `translate_text()` renders either canonical Portuguese or an explicit English translation. Technical English terms without an explicit mapping are left unchanged.

Messages with live values are stored as PT-PT templates in `PT_TO_EN` and rendered with `SnifferApp.trf(template, **values)`. Code never branches on `current_language`; a regression test enforces this.

Runtime language changes

`apply_language()` updates widget text, table headings, context menus, Help content and live status strings. Protocol bytes and parser data are not translated; only presentation fragments such as `Registos` / `Registers` are changed.

Language-independent geometry

The UI is built using the canonical Portuguese captions, then `_freeze_language_layout_geometry()` measures key dimensions and freezes them. This prevents shorter/longer English captions from moving the surrounding rectangles or controls.

Light/Dark invariant

The application uses one ttk `clam` construction. `mb_theme.py` configures geometry once and palette functions only change colors/state maps. Custom Canvas widgets receive palette updates separately.

Mouse-wheel behavior

An application bindtag lets Treeviews/Text/Listboxes/scrollbars consume wheel events normally while stopping wheel propagation elsewhere. This preserves table scrolling without allowing the Notebook to change tabs with the wheel.

10. Persistence, logs and export

User settings

On Windows, preferences are stored in `%APPDATA%\MBSniffer\settings.json`. Loading is defensive and every persisted choice/range is validated before use. Saves use a temporary file and replace operation.

Persisted data includes language, Light/Dark, last main tab, serial format, frame-gap mode/value, Pending timeout, auto-scroll, transaction separators, highlighting and Finder options (including attempts and the minimum timeout, migrated from the old 40 ms default to 100 ms) and the Master connection parameters. Master windows are not part of the settings: they are saved explicitly in workspace files. COM names, window geometry, traffic/results and the Finder safety confirmation are intentionally not persisted.

TXT capture log

Real-capture logs are lazy: the file is created only when traffic actually arrives. The disk log is complete and independent of table filters, sorting and the bounded GUI history.

CSV export

CSV exports the records currently retained in `_frame_history`, not necessarily an arbitrarily long complete session. It includes timing, channel, type, address metadata, CRC, raw bytes, timeout status and anomaly classification.

Bounded GUI history

The UI retains up to approximately `MAX_UI_FRAMES = 20000` records and prunes in chunks after a hysteresis threshold. Raw Text has a similar bounded-line policy. This prevents the GUI from growing indefinitely while keeping the capture log complete.

11. Module-by-module reference

The following chapters provide a complete callable inventory for every Python module. Each function/method is tied to its exact source-line range. The complete source itself appears as screenshots in Appendix A.

MBSniffer.py

Application launcher and top-level entry point. It sets the development simulation switch before importing the GUI, applies the Windows application identity, creates SnifferApp and enters Tkinter's main event loop.

Source lines	SHA-256
29	`c5eae72b053b4d96a112c492ec200b5a046724ff0a0893b3e50f86ae1df564bb`

Imports: `mb_config, mb_config.set_windows_app_user_model_id, mb_gui.SnifferApp`

Module constants / configuration data

Name	Definition
`debug_sim`	`0`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`main` main()	23-25	Main.	`set_windows_app_user_model_id, SnifferApp().mainloop, SnifferApp`

mb_config.py

Central configuration, constants, user settings validation and filesystem paths. It defines versioned application constants, discovery limits, GUI history limits, Modbus exception metadata, persistent settings behavior, application/log directories and PyInstaller resource lookup.

Source lines	SHA-256
464	`0c4c334410d1117bfbe2f8fc7bc9ecc1fd68663124f87539e590cfb22590b829`

Imports: `ctypes, json, os, sys, pathlib.Path`

Module constants / configuration data

Name	Definition
`APP_NAME`	`'MBSniffer'`
`APP_VERSION`	`'3.2'`
`debug_sim`	`0`
`DISCOVERY_BAUD_RATES`	`(1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200)`
`DISCOVERY_PARITIES`	`('None', 'Even', 'Odd')`
`DISCOVERY_STOP_BITS`	`('1', '2')`
`DISCOVERY_DEFAULT_MIN_TIMEOUT_MS`	`100`
`DISCOVERY_LEGACY_DEFAULT_MIN_TIMEOUT_MS`	`40`
`DISCOVERY_MIN_TIMEOUT_MS`	`10`
`DISCOVERY_MAX_TIMEOUT_MS`	`2000`
`DISCOVERY_PROBE_ADDRESS`	`0`
`DISCOVERY_PROBE_QTY`	`1`
`DISCOVERY_WRITE_TIMEOUT_SECONDS`	`1.5`
`DISCOVERY_WRITE_RETRIES`	`1`
`DISCOVERY_RX_BUFFER_LIMIT`	`1024`
`DISCOVERY_MIN_ATTEMPTS`	`1`
`DISCOVERY_MAX_ATTEMPTS`	`3`
`DISCOVERY_DEFAULT_ATTEMPTS`	`1`
`DISCOVERY_IDLE_CHECK_SECONDS`	`1.0`

Name	Definition
UI_QUEUE_BATCH_SIZE	200
UI_QUEUE_IDLE_INTERVAL_MS	50
UI_QUEUE_BUSY_INTERVAL_MS	10
MIN_PENDING_REQUEST_TIMEOUT_SECONDS	1.0
DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS	10.0
MAX_PENDING_REQUEST_TIMEOUT_SECONDS	300.0
PENDING_REQUEST_TIMEOUT_SECONDS	DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS
MASTER_MAX_READ_BITS	2000
MASTER_MAX_READ_REGISTERS	125
MASTER_MAX_WRITE_BITS	1968
MASTER_MAX_WRITE_REGISTERS	123
MASTER_DEFAULT_RESPONSE_TIMEOUT_MS	1000
MASTER_MIN_RESPONSE_TIMEOUT_MS	10
MASTER_MAX_RESPONSE_TIMEOUT_MS	10000
MASTER_DEFAULT_DELAY_MS	20
MASTER_MIN_DELAY_MS	0
MASTER_MAX_DELAY_MS	10000
MASTER_DEFAULT_SCAN_MS	1000
MASTER_MAX_SCAN_MS	3600000
MASTER_MAX_WINDOWS	32
MASTER_WORKSPACE_FORMAT	'MBSniffer Master workspace'
MASTER_WORKSPACE_VERSION	1
SNIFFER_SPLIT_FRAME_HOLD_MS	40.0
SLOW_RESPONSE_THRESHOLD_MS	500.0
MODBUS_EXCEPTION_CODES	{1: ('Illegal Function', 'A função Modbus pedida não é suportada/permitida pelo dispositivo.'), 2: ('Illegal Data Address', 'O endereço ou a gama d…
MAX_UI_FRAMES	20000
MAX_RAW_TEXT_LINES	20000

Name	Definition
`UI_PRUNE_CHUNK_FRAMES`	`2000`
`RAW_TEXT_PRUNE_CHUNK_LINES`	`2000`
`RAW_TRANSACTION_SEPARATOR`	`'—' * 72 + '\n'`
`SIMULATION_FUNCTION_CODES`	`(1, 2, 3, 4)`
`SIMULATION_EXCEPTION_CODES`	`(1, 2, 3, 4, 6)`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`resource_path` resource_path(filename)	126-132	Return a bundled/source resource path for normal Python or PyInstaller.	`Path, Path(__file__).resolve`
`set_windows_app_user_model_id` set_windows_app_user_model_id()	134-146	Give Windows a stable application identity so the taskbar can associate the Tk window with MBSniffer instead of a generic Python process.	`ctypes.windll.shell32.SetCurrentProcessExplicitAppUserModelID`
`get_settings_file` get_settings_file()	150-177	Return the persistent per-user settings file.	`folder.mkdir, os.environ.get, Path.home, Path`
`_validated_choice` _validated_choice(value, allowed, default)	180-181	Validated choice.	—
`_validated_bool` _validated_bool(value, default)	184-185	Validated bool.	`isinstance`
`_validated_number_string` _validated_number_string(value, minimum, maximum, default)	188-195	Validated number string.	`str, float`
`_validated_int_string` _validated_int_string(value, minimum, maximum, default)	198-205	Validated int string.	`str, int`
`load_ui_settings` load_ui_settings()	208-403	Reads settings.json defensively and validates every persisted choice/range.	`get_settings_file, isinstance, data.get, json.loads, sniffer_in.get`

Callable	Lines	Responsibility	Representative direct calls
`save_ui_settings` save_ui_settings(settings=None, **updates)	406-430	Merges updates with validated current settings and uses a temporary file plus replace operation for atomic persistence; settings failures never prevent normal application use.	`get_settings_file, load_ui_settings, isinstance, current.update, path.parent.mkdir`
`get_application_directory` get_application_directory()	433-454	Resolves a user-facing application directory consistently for PyInstaller EXE, packaged BAT/source mode and development mode.	`getattr, (launcher_dir / 'MBSniffer.bat').is_file, Path(__file__).resolve, Path.cwd, Path(sys.executable).resolve`
`get_logs_folder` get_logs_folder()	456-464	Creates and returns the MBSniffer Logs directory beside the launcher/EXE.	`folder.mkdir, get_application_directory`

Detailed behavior of the key paths

`load_ui_settings` lines 208-403

Reads settings.json defensively and validates every persisted choice/range. Invalid/corrupt data falls back safely. It also migrates the older highlight_anomalies key to highlight_results and the old 40 ms Finder timeout default to 100 ms.

`save_ui_settings` lines 406-430

Merges updates with validated current settings and uses a temporary file plus replace operation for atomic persistence; settings failures never prevent normal application use.

`get_application_directory` lines 433-454

Resolves a user-facing application directory consistently for PyInstaller EXE, packaged BAT/source mode and development mode.

`get_logs_folder` lines 456-464

Creates and returns the MBSniffer Logs directory beside the launcher/EXE.

mb_protocol.py

Protocol engine for passive Modbus RTU parsing and session metrics. It implements CRC16, RTU t3.5 timing, frame-shape recognition, resynchronisation, decoding, request/response pairing, timeout accounting, echo-write and broadcast classification, split-frame detection, Bus Health metrics and debug traffic generation.

Source lines
1007

SHA-256
`8c4c41051e41f0c998692206ed0204d91f057cce5fce726b15ad2cce503c1ef2`

Imports: `math, random, time, bisect.bisect_right, collections.defaultdict, deque, datetime.datetime, mb_config.MODBUS_EXCEPTION_CODES, PENDING_REQUEST_TIMEOUT_SECONDS, SIMULATION_FUNCTION_CODES, SIMULATION_EXCEPTION_CODES, SLOW_RESPONSE_THRESHOLD_MS`

Module constants / configuration data

Name	Definition
`_CRC16_TABLE`	`_build_crc16_table()`
`KNOWN_FRAME_FUNCTION_CODES`	`(1, 2, 3, 4, 5, 6, 7, 8, 15, 16, 22, 23)`
`EXACT_ECHO_FUNCTION_CODES`	`(5, 6, 22)`
`BROADCAST_UNIT_ID`	`0`
`READ_BITS_DETAILS_LIMIT`	`32`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`_build_crc16_table` _build_crc16_table()	19-29	Build crc16 table.	`range, tuple, table.append`
`crc16_modbus` crc16_modbus(data)	36-41	Computes the standard Modbus CRC-16 value (initial 0xFFFF, reflected polynomial 0xA001) with one lookup in the precomputed 256-entry _CRC16_TABLE per byte. append_crc() stores the low CRC byte first, matching Modbus RTU wire order.	—
`crc_ok` crc_ok(frame)	43-48	Crc ok.	`crc16_modbus, len`

Callable	Lines	Responsibility	Representative direct calls
`append_crc` append_crc(payload)	50-52	Append crc.	`crc16_modbus, bytes`
`hex_bytes` hex_bytes(data)	54-55	Hex bytes.	`' '.join`
`modbus_rtu_interframe_gap_seconds` modbus_rtu_interframe_gap_seconds(baud, bits_per_char)	57-72	Implements the automatic t3.5 rule used by capture and discovery: 3.5 character times at or below 19200 bit/s, fixed 1.750 ms above 19200 bit/s.	`int, float, ValueError`
`plausible_request_qty` plausible_request_qty(fc, qty)	74-79	Plausible request qty.	—
`_plausible_unit_id` _plausible_unit_id(value)	89-94	Return True for Modbus RTU unit addresses that can appear on the wire.	`int`
`_candidate_prefixes_at` _candidate_prefixes_at(buf, start=0)	97-168	Infers only structurally plausible lengths for supported Modbus frame shapes.	`set, len, int, _plausible_unit_id, out.append`
`candidate_prefixes` candidate_prefixes(buf)	171-173	Compatibility wrapper for callers/tests that inspect offset zero.	`_candidate_prefixes_at`
`_crc_ok_at` _crc_ok_at(buf, start, length)	176-183	CRC check without copying the whole remaining capture buffer.	`crc16_modbus, len, memoryview`
`find_valid_prefix` find_valid_prefix(buf, start=0)	186-191	Find a CRC-valid, structurally recognised Modbus frame at *start*.	`_candidate_prefixes_at, _crc_ok_at, len`
`guess_invalid_whole_frame` guess_invalid_whole_frame(buf)	194-201	Classify a complete known-shape frame whose CRC is invalid.	`_candidate_prefixes_at, len, _plausible_unit_id`
`_unknown_whole_frame_is_valid` _unknown_whole_frame_is_valid(buf)	204-217	Conservative fallback for one unsupported FC occupying the whole burst.	`_candidate_prefixes_at, crc_ok, _plausible_unit_id, len`
`split_capture_buffer` split_capture_buffer(data)	220-269	Walks a burst from left to right.	`bytearray, find_valid_prefix, max, range, guess_invalid_whole_frame`
`burst_may_continue` burst_may_continue(data)	271-302	Reports whether a burst ends with the start of a known frame that has not fully arrived yet, so the reader can wait for bytes delayed by a USB adapter latency timer instead of emitting two RAW rows.	`split_capture_buffer, any, _plausible_unit_id, len, _candidate_prefixes_at`

Callable	Lines	Responsibility	Representative direct calls
`read_bits_from_response` read_bits_from_response(frame, qty=None)	305-322	Return FC01/FC02 coil/input states, least-significant bit first.	`len, range, int`
`read_bits_details` read_bits_details(frame, qty=None)	330-338	Formats FC01/FC02 response bits in address order, grouped per data byte, trimmed to the request quantity when known and capped at 32 visible bits.	`read_bits_from_response, ''.join, ' '.join, len, str`
`decode_frame` decode_frame(frame, kind_hint)	341-449	Extracts Slave, FC, PDU address, one-based address, quantity and selected payload details from a split frame.	`str(kind_hint).startswith, str, len, MODBUS_EXCEPTION_CODES.get, read_bits_details`
`format_clock_time` format_clock_time(epoch)	451-452	Format clock time.	`datetime.fromtimestamp(float(epoch)).strftime, datetime.fromtimestamp, float`
`parse_burst` parse_burst(data, timestamp, channel='', event_time=None, chunk_times=None)	455-498	Transforms split frames into the row dictionary consumed by the rest of the program.	`split_capture_buffer, time.time, float, len, decode_frame`
`class SessionMetrics`	**500-872**	**Tracks pairing, counters and bus-health diagnostics for one session.**	**Class container**
`SessionMetrics.__init__` __init__(self, pending_timeout_seconds=PENDING_REQUEST_TIMEOUT_SECONDS)	503-505	Init.	`float, self.reset`
`SessionMetrics.reset` reset(self)	507-531	Reset.	`set, defaultdict, deque`
`SessionMetrics._key` _key(self, row)	533-543	Key.	`int, row.get`
`SessionMetrics.pending_count` pending_count(self)	545-546	Pending count.	`sum, len, self.pending.values`
`SessionMetrics.pop_expired_requests` pop_expired_requests(self)	548-551	Pop expired requests.	—
`SessionMetrics.expire_pending` expire_pending(self, now=None)	553-591	Expire stale requests and retain their sequence IDs for the GUI.	`float, list, time.time, self.pending.keys, self.pending.pop`
`SessionMetrics.force_timeout` force_timeout(self, slave, fc)	593-617	Debug/simulation helper: expire the oldest pending request for one key.	`self.pending.get, pending_list.pop, dict, self.pending.pop, int`

Callable	Lines	Responsibility	Representative direct calls
`SessionMetrics._resolve_echo_direction` _resolve_echo_direction(self, row)	619-651	Classifies FC05/06/08/22 frames as REQUEST or RESPONSE from the pending queue: same key, identical bytes for exact echoes and the opposite channel in RS232 dual RX.	`self._key, self.pending.get, row.get, candidate.get`
`SessionMetrics._record_response_time` _record_response_time(self, row, response_ms)	653-664	Record response time.	`float, self.response_times_ms.append, self.response_times_by_slave[slave].append, int, row.get`
`SessionMetrics.process` process(self, row)	666-765	This is the session accounting hot path.	`float, self.expire_pending, self.pop_expired_requests, row.get, self._record_response_time`
`SessionMetrics._p95` _p95(values)	768-773	P95.	`sorted, max, min, float, len`
`SessionMetrics.health_summary` health_summary(self, *, baud=9600, data_bits=8, parity='None', stop_bits=1, now=None)	775-872	Builds the Bus Health model: request rate, response average/min/max/P95, slowest Slave, unvalidated-byte rate, timeout rate, active Slave count, approximate bus utilisation and exception distribution.	`list, self._p95, self.response_times_by_slave.items, time.time, max`
`u16_bytes` u16_bytes(value)	874-876	U16 bytes.	`bytes, int`
`corrupt_crc` corrupt_crc(frame)	878-882	Corrupt crc.	`bytearray, bytes, len`
`build_simulation_transaction` build_simulation_transaction(slave, fc, rng)	884-928	Build one syntactically valid Modbus request/response pair.	`int, rng.randint, append_crc, bytes, b''.join`
`build_random_simulation_plan` build_random_simulation_plan(rng=None)	930-952	Build exactly four simulation transactions: FC01, FC02, FC03 and FC04, one each.	`rng.sample, zip, random.Random, range, build_simulation_transaction`
`build_anomaly_simulation_plan` build_anomaly_simulation_plan(rng=None)	956-1006	Creates the deterministic categories required to preview diagnostic highlighting: normal response, slow response, Exception, bad CRC, timeout and RAW bytes.	`rng.sample, build_simulation_transaction, append_crc, corrupt_crc, random.Random`

Detailed behavior of the key paths

`crc16_modbus` lines 36-41

Computes the standard Modbus CRC-16 value (initial 0xFFFF, reflected polynomial 0xA001) with one lookup in the precomputed 256-entry _CRC16_TABLE per byte. append_crc() stores the low CRC byte first, matching Modbus RTU wire order.

`modbus_rtu_interframe_gap_seconds` lines 57-72

Implements the automatic t3.5 rule used by capture and discovery: 3.5 character times at or below 19200 bit/s, fixed 1.750 ms above 19200 bit/s.

`_candidate_prefixes_at` lines 97-168

Infers only structurally plausible lengths for supported Modbus frame shapes. This constrains resynchronisation and avoids the former brute-force CRC scan across arbitrary lengths.

`split_capture_buffer` lines 220-269

Walks a burst from left to right. A valid recognised frame is emitted immediately; otherwise the parser searches later offsets for a recognised CRC-valid frame. Bytes before the recovered frame become CRC ERROR if they exactly match a known frame shape, otherwise RAW/UNSYNC. If nothing can be recovered, the remainder becomes CRC ERROR, UNKNOWN/OK, or RAW/UNPARSED depending on its structure and CRC.

`burst_may_continue` lines 271-302

Reports whether a burst ends with the start of a known frame that has not fully arrived yet, so the reader can wait for bytes delayed by a USB adapter latency timer instead of emitting two RAW rows.

`read_bits_details` lines 330-338

Formats FC01/FC02 response bits in address order, grouped per data byte, trimmed to the request quantity when known and capped at 32 visible bits.

`decode_frame` lines 341-449

Extracts Slave, FC, PDU address, one-based address, quantity and selected payload details from a split frame. It handles Exceptions and common read/write Function Codes without modifying the raw bytes.

`parse_burst` lines 455-498

Transforms split frames into the row dictionary consumed by the rest of the program. It attaches display time, channel, decoded fields, raw hex, frame length and a floating epoch timestamp; with chunk_times each frame gets the arrival time of its own serial read.

`SessionMetrics.process` lines 666-765

This is the session accounting hot path. It expires old pending requests first, resolves REQ/RESP frames, updates byte/counter statistics, stores valid non-broadcast requests by (Slave, FC), FIFO-pairs responses/exceptions to pending requests, calculates response time and returns timing/pairing information to the view.

`SessionMetrics._resolve_echo_direction` lines 619-651

Classifies FC05/06/08/22 frames as REQUEST or RESPONSE from the pending queue: same key, identical bytes for exact echoes and the opposite channel in RS232 dual RX. Broadcast frames are always requests.

`SessionMetrics.health_summary` lines 775-872

Builds the Bus Health model: request rate, response average/min/max/P95, slowest Slave, unvalidated-byte rate, timeout rate, active Slave count, approximate bus utilisation and exception distribution.

`build_anomaly_simulation_plan` lines 956-1006

Creates the deterministic categories required to preview diagnostic highlighting: normal response, slow response, Exception, bad CRC, timeout and RAW bytes.

> **Design note.** Important design boundary: this module contains no Tkinter code and can be regression-tested independently. That separation is valuable if the project is later rebuilt for a thesis or extended with additional front ends.

mb_capture.py

Serial-port management and passive acquisition pipeline. It opens selected COM ports, starts reader threads, converts inter-frame silence into capture bursts, parses those bursts, queues GUI events, controls simulation, handles COM restart and performs orderly shutdown.

Source lines	SHA-256
883	`2785386c053f477dd0095b8cb6446264da208b9cd0ed2fa6807047f965e01a0c`

Imports: `queue, random, threading, time, traceback, datetime.datetime, tkinter, tkinter.messagebox, mb_config.APP_NAME, APP_VERSION, MIN_PENDING_REQUEST_TIMEOUT_SECONDS, MAX_PENDING_REQUEST_TIMEOUT_SECONDS, SNIFFER_SPLIT_FRAME_HOLD_MS, UI_QUEUE_BATCH_SIZE, UI_QUEUE_IDLE_INTERVAL_MS, UI_QUEUE_BUSY_INTERVAL_MS, debug_sim, mb_protocol.parse_burst, burst_may_continue, build_anomaly_simulation_plan, modbus_rtu_interframe_gap_seconds, mb_slave_finder.cancel_serial_io`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class CaptureMixin`	**33-882**	**Methods for serial capture, COM management and simulation.**	**Class container**
`CaptureMixin.set_sniffer_refresh_state` set_sniffer_refresh_state(self, state)	36-41	Apply the same state to both Sniffer COM refresh buttons.	`getattr, button.configure`
`CaptureMixin.update_mode_ui` update_mode_ui(self)	43-63	Update mode ui.	`self.mode_var.get, self.port_b_combo.configure, self.update_com_status, self.mode_help_var.set, self.tr`
`CaptureMixin.update_gap_state` update_gap_state(self)	65-68	Update gap state.	`self.gap_entry.configure, self.gap_mode_var.get`
`CaptureMixin.get_serial_modules` get_serial_modules(self)	70-76	Get serial modules.	—
`CaptureMixin.refresh_ports` refresh_ports(self)	78-115	Refresh ports.	`self.get_serial_modules, hasattr, self.update_com_status, self.port_a_var.set, self.port_b_var.set`

Callable	Lines	Responsibility	Representative direct calls
`CaptureMixin.detected_ports` detected_ports(self)	117-125	Return the set of COM device names currently reported by Windows/pyserial.	`self.get_serial_modules, set, list_ports.comports`
`CaptureMixin.port_status_text` port_status_text(self, label, port, enabled=True)	127-143	Port status text.	`self.serial_by_port.get, self.detected_ports, self.tr`
`CaptureMixin.update_com_status` update_com_status(self)	145-164	Update COM status without opening/probing a port.	`self.port_a_status_var.set, self.port_b_status_var.set, hasattr, self.mode_var.get, self.port_status_text`
`CaptureMixin.poll_com_status` poll_com_status(self)	166-178	Poll com status.	`self.update_com_status, hasattr, self.update_stats_labels, self.after`
`CaptureMixin.selected_port_plan` selected_port_plan(self)	180-198	Translates physical mode into one or two receive channels.	`self.mode_var.get, self.port_a_var.get().strip, self.port_b_var.get().strip, ValueError, plan.append`
`CaptureMixin.open_selected_ports` open_selected_ports(self, serial, config)	200-213	Open the selected capture ports and return [(channel, serial_obj), ...].	`self.selected_port_plan, opened.append, self.open_serial_port, ser_obj.close`
`CaptureMixin.install_opened_ports` install_opened_ports(self, opened)	215-220	Install opened ports.	`self.update_com_status, getattr`
`CaptureMixin.start_reader_threads` start_reader_threads(self, opened, gap_ms)	222-232	Start reader threads.	`threading.Thread, self.reader_threads.append, t.start`
`CaptureMixin.restart_com_port` restart_com_port(self)	234-330	Close and reopen the selected COM port(s).	`self.get_serial_modules, self.restart_btn.configure, self.status_var.set, self.stop_event.set, list`
`CaptureMixin.set_simulation_button_state` set_simulation_button_state(self, state)	332-335	Change Simulation button state only when the optional button exists.	`self.sim_btn.configure`
`CaptureMixin.selected_pending_timeout_seconds` selected_pending_timeout_seconds(self)	337-356	Selected pending timeout seconds.	`float, ValueError, self.pending_timeout_var.get().strip().replace, self.trf, self.tr`
`CaptureMixin.calculated_gap_ms` calculated_gap_ms(self)	358-370	Returns the manual gap after validation, or computes the automatic RTU t3.5 value from baud/data/parity/stop settings using the shared protocol timing helper.	`int, float, self.gap_mode_var.get, self.baud_var.get, self.data_var.get`

Callable	Lines	Responsibility	Representative direct calls
`CaptureMixin.serial_config` serial_config(self, serial)	372-394	Serial config.	`int, self.baud_var.get, self.parity_var.get, self.stop_var.get, self.data_var.get`
`CaptureMixin.open_serial_port` open_serial_port(self, serial, port, config)	396-428	Open serial port.	`serial.Serial, ser.open, ser.reset_input_buffer`
`CaptureMixin.start_capture` start_capture(self)	430-488	Validates state and serial settings, opens the selected port(s), resets the current session, prepares lazy logging, starts reader workers and updates the GUI to capturing state.	`self.get_serial_modules, self.mode_var.get, self.port_a_var.get().strip, self.port_b_var.get().strip, self.install_opened_ports`
`CaptureMixin.reader_loop` reader_loop(self, ser, gap_ms, channel, generation)	490-567	Runs in a daemon thread per capture channel.	`max, min, bytearray, parse_burst, self.event_queue.put`
`CaptureMixin.stop_capture` stop_capture(self)	569-572	Stop capture.	`self.stop_event.set, self.status_var.set, self.tr`
`CaptureMixin.run_simulation` run_simulation(self)	574-675	Runs the internal diagnostic scenario without opening serial hardware.	`self.reset_session_metrics, self.close_log, self.status_var.set, self.start_btn.configure, self.restart_btn.configure`
`CaptureMixin.process_queue` process_queue(self)	677-822	Main-thread bridge between workers and Tkinter.	`range, flush_frames, self.metrics.expire_pending, time.time, self.metrics.pop_expired_requests`
`CaptureMixin.on_close` on_close(self)	824-882	Performs defensive shutdown: saves settings, signals all workers, cancels serial I/O, joins threads with timeouts, closes serial handles/logs and destroys the Tk root.	`self.stop_event.set, self.discovery_stop_event.set, cancel_serial_io, getattr, callable`

Detailed behavior of the key paths

`selected_port_plan` lines 180-198

Translates physical mode into one or two receive channels. RS485 and single-RX RS232 use COM A; dual-RX RS232 requires distinct COM A and COM B and labels their directions A→B/B→A.

`calculated_gap_ms` lines 358-370

Returns the manual gap after validation, or computes the automatic RTU t3.5 value from baud/data/parity/stop settings using the shared protocol timing helper.

`start_capture` lines 430-488

Validates state and serial settings, opens the selected port(s), resets the current session, prepares lazy logging, starts reader workers and updates the GUI to capturing state.

`reader_loop` lines 490-567

Runs in a daemon thread per capture channel. It accumulates serial bytes and their read times into a burst, considers the burst complete after the configured inter-frame silence (or after the split-frame hold when the burst ends mid-frame), then calls parse_burst() and posts frame events to the GUI queue.

`run_simulation` lines 574-675

Runs the internal diagnostic scenario without opening serial hardware. It feeds the same parser/event path used by real capture, including forced timeout handling.

`process_queue` lines 677-822

Main-thread bridge between workers and Tkinter. It drains events in bounded batches, batches frame insertions for performance, handles capture/discovery status events, expires requests and reschedules itself with a faster cadence when backlog exists.

`on_close` lines 824-882

Performs defensive shutdown: saves settings, signals all workers, cancels serial I/O, joins threads with timeouts, closes serial handles/logs and destroys the Tk root.

> **Design note.** Tkinter widgets are never updated directly by reader threads. Workers communicate through event_queue; process_queue() runs on the Tk main thread.

mb_diagnostics.py

Presentation-oriented diagnostics derived from captured records. It maps Function Codes to names, recalculates/display CRC fields, extracts payloads, classifies anomalies/results and builds the structured Frame Inspector text.

Source lines	SHA-256
286	`d5edc15bf0259d130fdf9b42ffaf2caca770782101aeed1886f980e0fcf24bda`

Imports: `mb_config.MODBUS_EXCEPTION_CODES, SLOW_RESPONSE_THRESHOLD_MS, mb_protocol.crc16_modbus, read_bits_from_response, mb_i18n.LANGUAGE_PORTUGUESE, translate_details, translate_text`

Module constants / configuration data

Name	Definition
`FUNCTION_NAMES`	`{1: 'Read Coils', 2: 'Read Discrete Inputs', 3: 'Read Holding Registers', 4: 'Read Input Registers', 5: 'Write Single Coil', 6: 'Write Single Regis…`
`RESULT_LABELS`	`{'success': ('✓ Sucesso', 'Verde'), 'slow': ('⚠ Resposta lenta', 'Amarelo'), 'exception': ('Exception', 'Vermelho'), 'crc': ('× CRC Error', 'Vermel…`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`raw_hex_to_bytes` raw_hex_to_bytes(raw)	26-30	Raw hex to bytes.	`bytes.fromhex, str`
`fc_number` fc_number(record)	33-41	Fc number.	`str(record.get('fc', '') or '').strip, int, record.get, str`
`function_label` function_label(record, language=LANGUAGE_PORTUGUESE)	44-49	Function label.	`fc_number, FUNCTION_NAMES.get, translate_text`
`crc_fields` crc_fields(record)	52-62	Crc fields.	`str(record.get('type', '')).startswith, raw_hex_to_bytes, crc16_modbus, record.get, len`

Callable	Lines	Responsibility	Representative direct calls
`response_payload` response_payload(record, language=LANGUAGE_PORTUGUESE)	65-127	Return (byte_count, data_hex, decoded_data) where available.	`raw_hex_to_bytes, str, record.get, len, MODBUS_EXCEPTION_CODES.get`
`anomaly_kind` anomaly_kind(record, threshold_ms=SLOW_RESPONSE_THRESHOLD_MS)	130-153	Classifies a record with explicit priority: bad CRC, timeout, Exception, RAW, then slow response.	`bool, str(record.get('type', '')).startswith, str, record.get, float`
`result_info` result_info(record, successful_request=False, language=LANGUAGE_PORTUGUESE)	168-207	Turns the anomaly/pairing state into a semantic result key, user-facing label and color name.	`anomaly_kind, str, record.get, translate_text`
`inspector_values` inspector_values(record, successful_request=False, language=LANGUAGE_PORTUGUESE)	210-264	Builds every Frame Inspector field, including recalculated CRC received/calculated values, decoded payload, timeout suffix and translated result text.	`crc_fields, response_payload, str, result_info, record.get`
`decoded_record_text` decoded_record_text(record, successful_request=False, language=LANGUAGE_PORTUGUESE)	267-286	Serialises the same Frame Inspector information into newline-separated text for the copy-decoded-frame context command.	`inspector_values, '\n'.join`

Detailed behavior of the key paths

`anomaly_kind` lines 130-153

Classifies a record with explicit priority: bad CRC, timeout, Exception, RAW, then slow response. The priority prevents conflicting colors/results.

`result_info` lines 168-207

Turns the anomaly/pairing state into a semantic result key, user-facing label and color name. Successful completed pairs are distinguished from pending requests.

`inspector_values` lines 210-264

Builds every Frame Inspector field, including recalculated CRC received/calculated values, decoded payload, timeout suffix and translated result text.

`decoded_record_text` lines 267-286

Serialises the same Frame Inspector information into newline-separated text for the copy-decoded-frame context command.

mb_view.py

Traffic view, filtering, sorting, Frame Inspector, Bus Health updates, CSV export and TXT logging. It owns the bounded in-memory frame history, converts parser rows into richer view records, pairs display state with SessionMetrics, maintains table/raw views and keeps the real capture log independent from GUI filtering.

Source lines
1853

SHA-256
`1ae5a23ae5cecd50e22a50868bce04ef1c33168b2aef091e0b32384f5983324f`

Imports: `csv, math, os, subprocess, sys, time, datetime.datetime, tkinter, tkinter.filedialog, messagebox, mb_config.MAX_UI_FRAMES, MAX_RAW_TEXT_LINES, UI_PRUNE_CHUNK_FRAMES, RAW_TEXT_PRUNE_CHUNK_LINES, SLOW_RESPONSE_THRESHOLD_MS, get_application_directory, get_logs_folder, mb_diagnostics.anomaly_kind, decoded_record_text, inspector_values, mb_i18n.canonical_ui_text, translate_details, mb_protocol.read_bits_details`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class ViewMixin`	**32-1852**	**Methods for rendering, filtering, statistics and logging.**	**Class container**
`ViewMixin.reset_session_metrics` reset_session_metrics(self)	35-37	Reset session metrics.	`self.metrics.reset, self.update_stats_labels`
`ViewMixin.update_stats_labels` update_stats_labels(self)	39-104	Converts SessionMetrics into both compact session counters and Bus Health values.	`self.stat_requests_var.set, self.stat_responses_var.set, self.stat_pending_var.set, self.stat_timeouts_var.set, self.stat_crc_var.set`
`ViewMixin.pulse_bus_activity` pulse_bus_activity(self)	109-128	Pulse bus activity.	`getattr, var.set, self.after, self.after_cancel`
`ViewMixin._clear_bus_activity_pulse` _clear_bus_activity_pulse(self)	130-134	Clear bus activity pulse.	`getattr, var.set`
`ViewMixin.schedule_advanced_filter_refresh` schedule_advanced_filter_refresh(self, *_args)	136-149	Schedules advanced filter refresh.	`getattr, self.after, self.after_cancel`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.clear_advanced_filters` clear_advanced_filters(self)	151-171	Clear advanced filters.	`getattr, self.advanced_filter_type_var.set, self.advanced_filter_fc_var.set, self.advanced_filter_response_ms_var.set, self.advanced_filter…`
`ViewMixin.normalized_fc_filter` normalized_fc_filter(value)	174-186	Normalized fc filter.	`str(value or '').strip().upper, text.startswith, int, str(value or '').strip, str`
`ViewMixin.record_passes_advanced_filters` record_passes_advanced_filters(self, record)	188-249	Implements Type, FC, response-time and free-text filtering.	`getattr, canonical_ui_text, str, self.normalized_fc_filter, self.advanced_filter_response_ms_var.get().strip`
`ViewMixin.configure_anomaly_tags` configure_anomaly_tags(self, dark=None)	251-282	Configure anomaly tags.	`palette.items, hasattr, bool, self.tree.tag_configure, getattr(self, 'dark_mode_var', tk.BooleanVar(value=False)).get`
`ViewMixin.anomaly_tag_for_record` anomaly_tag_for_record(self, record)	284-321	Maps the semantic result of a record to the Treeview tag used for colour highlighting, including completed normal REQUEST/RESPONSE pairs.	`anomaly_kind, str, getattr, self.highlight_anomalies_var.get, record.get`
`ViewMixin.on_highlight_anomalies_changed` on_highlight_anomalies_changed(self)	323-330	Event handler for highlight anomalies changed.	`self.configure_anomaly_tags, self.configure_inspector_result_styles, self.render_filtered_view, self.update_frame_inspector, getattr`
`ViewMixin.mark_timed_out_records` mark_timed_out_records(self, expired, rerender=True)	332-358	Finds original REQUEST records by their sequence IDs and marks them timed_out rather than inventing artificial timeout frames.	`len, item.get, changed_records.append, canonical_ui_text, self.render_filtered_view`
`ViewMixin.refresh_record_tags` refresh_record_tags(self, records)	360-372	Re-applies the result tag to the Treeview rows of the given records and refreshes the Frame Inspector when one of them is selected.	`self.selected_record, self.anomaly_tag_for_record, self._tree_iids_by_seq.get, any, self.update_frame_inspector`
`ViewMixin.selected_record` selected_record(self)	374-378	Selected record.	`self.tree.selection, self._record_by_iid.get`
`ViewMixin.update_frame_inspector` update_frame_inspector(self, record=None)	380-409	Builds the structured inspector values for the selected traffic record and applies result styling while preserving the raw frame.	`bool, inspector_values, self.inspector_vars.items, self.selected_raw_var.set, getattr`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.toggle_frame_inspector` toggle_frame_inspector(self)	411-423	Toggle frame inspector.	`getattr, callable, self.inspector_body.grid, self.inspector_toggle_btn.configure, self.inspector_body.grid_remove`
`ViewMixin.copy_to_clipboard` copy_to_clipboard(self, text)	425-431	Copy to clipboard.	`self.clipboard_clear, self.clipboard_append, self.update_idletasks, str`
`ViewMixin.copy_selected_raw` copy_selected_raw(self)	433-436	Copy selected raw.	`self.selected_record, self.copy_to_clipboard, record.get`
`ViewMixin.copy_selected_decoded` copy_selected_decoded(self)	438-451	Copy selected decoded.	`self.selected_record, bool, self.copy_to_clipboard, decoded_record_text, record.get`
`ViewMixin.context_filter_selected_slave` context_filter_selected_slave(self)	453-466	Context filter selected slave.	`self.selected_record, str, self.slave_filter_vars.items, self.update_slave_heading, self.render_filtered_view`
`ViewMixin.context_filter_selected_fc` context_filter_selected_fc(self)	468-482	Context filter selected fc.	`self.selected_record, str, self.advanced_filter_fc_var.set, self.render_filtered_view, record.get`
`ViewMixin.context_show_selected_transaction` context_show_selected_transaction(self)	484-513	Context show selected transaction.	`self.selected_record, record.get, self.slave_filter_vars.values, self.update_slave_heading, self.advanced_filter_type_var.set`
`ViewMixin.clear_transaction_filter` clear_transaction_filter(self)	515-517	Clear transaction filter.	`self.render_filtered_view`
`ViewMixin.show_traffic_context_menu` show_traffic_context_menu(self, event)	519-534	Show traffic context menu.	`self.tree.identify_row, self.tree.selection_set, self.tree.focus, self.on_row_select, self.traffic_context_menu.tk_popup`
`ViewMixin.export_session_csv` export_session_csv(self, path=None)	536-638	Exports the frames retained by the GUI history, including timing, decoded addressing, raw data and anomaly/timeout fields.	`list, datetime.now().strftime, filedialog.asksaveasfilename, messagebox.showinfo, open`
`ViewMixin.tree_record_column_values` tree_record_column_values(self, record)	640-651	Tree record column values.	`record.get, translate_details`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.measure_tree_column_width` measure_tree_column_width(self, column, records, heading_text=None)	653-671	Measure the pixel width required by a non-wrapped Treeview column.	`self.tree_heading_font.measure, self.tree_headings.get, str, self.tree_record_column_values(record).get, max`
`ViewMixin.available_details_width` available_details_width(self)	673-691	Return the width that can be assigned to Details without intentionally forcing horizontal scrolling in the current Treeview viewport.	`int, max, self.tree.winfo_width, self.tree.column`
`ViewMixin.wrap_text_to_pixels` wrap_text_to_pixels(self, value, max_px)	693-743	Word-wrap text using the actual Tk font width.	`str, max, '\n'.join, int, text.splitlines`
`ViewMixin.details_horizontal_rule` details_horizontal_rule(self)	745-757	Return a rule that fills the Details column as far as the Treeview cell permits.	`max, int, self.tree_cell_font.measure, self.tree.column, math.ceil`
`ViewMixin.completed_pairs_for_records` completed_pairs_for_records(self, records)	759-799	Return complete visible REQUEST/RESPONSE pairs in current display order.	`pairs.sort, record.get, response.get, seq_to_index.get, min`
`ViewMixin.separator_after_sequences` separator_after_sequences(self, records)	801-818	Return logical frame seq ids that need a separator AFTER them.	`self.completed_pairs_for_records, self.separate_transactions_var.get, set, len`
`ViewMixin.tree_separator_values` tree_separator_values(self)	820-825	Tree separator values.	`self.details_horizontal_rule`
`ViewMixin.insert_tree_separator_after_seq` insert_tree_separator_after_seq(self, seq)	827-845	Insert one Details-width separator after the visual rows of seq.	`self._tree_iids_by_seq.get, self.tree.insert, self.tree.index, self.tree_separator_values`
`ViewMixin.sync_tree_transaction_separators` sync_tree_transaction_separators(self, records=None)	847-867	Synchronize standalone transaction separator rows without rebuilding normal traffic rows.	`self.separator_after_sequences, set, self.records_for_display, self._tree_separator_iid_by_seq.pop, self._raw_by_iid.pop`
`ViewMixin.detail_lines_for_record` detail_lines_for_record(self, record)	869-891	Return wrapped Details as separate visual Treeview rows.	`max, self.wrap_text_to_pixels, wrapped.splitlines, translate_details, int`
`ViewMixin.set_tree_rowheight_for_records` set_tree_rowheight_for_records(self, records)	893-898	Treeview row height stays fixed; wrapping uses continuation rows.	`self.tree_style.configure`
`ViewMixin.update_tree_rowheight_for_record` update_tree_rowheight_for_record(self, record)	900-902	No-op by design: one logical frame may use several compact rows.	—

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.autofit_tree_columns` autofit_tree_columns(self, records=None)	904-947	Non-Details columns fit the longest visible value/title.	`self.measure_tree_column_width, max, self.tree.column, self.set_tree_rowheight_for_records, self.records_for_display`
`ViewMixin.expand_tree_columns_for_record` expand_tree_columns_for_record(self, record)	949-999	Incrementally enlarge only the compact non-Details columns.	`self.tree_record_column_values, self.tree_heading_font.measure, self.tree_cell_font.measure, int, self.measure_tree_column_width`
`ViewMixin.refresh_tree_heading_width` refresh_tree_heading_width(self, column)	1001-1017	Dynamic Hora/Slave headings remain compact.	`self.records_for_display, self.measure_tree_column_width, self.tree.column`
`ViewMixin.schedule_tree_reflow` schedule_tree_reflow(self, _event=None)	1019-1033	Debounce window/Treeview resizing and then re-wrap Details to the new available width.	`self.after, self.after_cancel`
`ViewMixin.reflow_tree_after_resize` reflow_tree_after_resize(self)	1035-1039	Reflow tree after resize.	`self.render_filtered_view, hasattr`
`ViewMixin._slave_sort_key` _slave_sort_key(self, slave)	1041-1045	Slave sort key.	`int, str`
`ViewMixin.rebuild_slave_filter_menu` rebuild_slave_filter_menu(self)	1047-1074	Rebuild slave filter menu.	`self.slave_filter_menu.delete, self.slave_filter_menu.add_command, self.update_slave_heading, self.slave_filter_menu.add_separator, sorted`
`ViewMixin.update_slave_heading` update_slave_heading(self)	1076-1093	Update slave heading.	`self.tree.heading, self.refresh_tree_heading_width, self.tr, self.slave_filter_vars.items, var.get`
`ViewMixin.show_slave_filter_menu` show_slave_filter_menu(self)	1095-1108	Open the slave filter menu at the position where the Slave heading was clicked.	`self.winfo_pointerx, self.slave_filter_menu.tk_popup, self.winfo_pointery, self.slave_filter_menu.grab_release`
`ViewMixin.register_slave_for_filter` register_slave_for_filter(self, slave)	1110-1121	Register slave for filter.	`str(slave or '').strip, tk.BooleanVar, self.rebuild_slave_filter_menu, str`
`ViewMixin.select_all_slaves` select_all_slaves(self)	1123-1128	Select all slaves.	`self.slave_filter_vars.values, self.update_slave_heading, self.render_filtered_view, var.set`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.select_no_slaves` select_no_slaves(self)	1130-1135	Select no slaves.	`self.slave_filter_vars.values, self.update_slave_heading, self.render_filtered_view, var.set`
`ViewMixin.on_slave_filter_changed` on_slave_filter_changed(self)	1137-1147	Event handler for slave filter changed.	`self.update_slave_heading, self.render_filtered_view, var.get, bool, all`
`ViewMixin.selected_slave_ids` selected_slave_ids(self)	1149-1153	Selected slave ids.	`self.slave_filter_vars.items, var.get`
`ViewMixin.record_passes_slave_filter` record_passes_slave_filter(self, record)	1155-1165	Record passes slave filter.	`self.selected_slave_ids, str, record.get`
`ViewMixin.time_order_descending` time_order_descending(self)	1167-1168	Time order descending.	`canonical_ui_text(self.time_order_var.get()).startswith, canonical_ui_text, self.time_order_var.get`
`ViewMixin.on_time_order_changed` on_time_order_changed(self)	1170-1179	Event handler for time order changed.	`self.tree.heading, self.render_filtered_view, self.time_order_descending, self.tr`
`ViewMixin.toggle_time_order` toggle_time_order(self)	1181-1186	Toggle time order.	`self.time_order_descending, self.on_time_order_changed, self.time_order_var.set, self.tr`
`ViewMixin.records_for_display` records_for_display(self)	1188-1198	Applies current filtering and time-order state to the bounded source-of-truth frame history.	`self.time_order_descending, records.reverse, self.record_passes_slave_filter, self.record_passes_advanced_filters`
`ViewMixin.raw_line_for_record` raw_line_for_record(self, record)	1200-1223	Raw line for record.	`record.get, translate_details`
`ViewMixin.raw_separator_width_chars` raw_separator_width_chars(self, raw_lines=None)	1225-1245	Width of one Raw Hex separator in visible character cells.	`max, self.raw_text_font.measure, int, self.raw_text.winfo_width`
`ViewMixin.raw_transaction_separator` raw_transaction_separator(self, width_chars=None)	1247-1250	Raw transaction separator.	`max, int`
`ViewMixin.schedule_raw_separator_reflow` schedule_raw_separator_reflow(self, _event=None)	1252-1262	Schedules raw separator reflow.	`self.after, self.after_cancel`
`ViewMixin.reflow_raw_separator_after_resize` reflow_raw_separator_after_resize(self)	1264-1267	Reflow raw separator after resize.	`self.separate_transactions_var.get, self.render_raw_text_only`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.insert_tree_record` insert_tree_record(self, record, index='end', allow_separator=True)	1269-1337	Insert tree record.	`self.detail_lines_for_record, visual_rows.append, self.anomaly_tag_for_record, list, self.tree.insert`
`ViewMixin.render_raw_text_only` render_raw_text_only(self)	1339-1392	Render raw text only.	`self.records_for_display, self.separator_after_sequences, list, self.raw_separator_width_chars, self.raw_transaction_separator`
`ViewMixin.render_filtered_view` render_filtered_view(self)	1394-1454	Render filtered view.	`self.selected_record, self.tree.get_children, self._tree_history.clear, self._tree_iids_by_seq.clear, self._tree_separator_iid_by_seq.clear`
`ViewMixin.add_frame_row` add_frame_row(self, row)	1456-1458	Add frame row.	`self.add_frame_rows_batch`
`ViewMixin.add_frame_rows_batch` add_frame_rows_batch(self, rows)	1460-1658	Central view ingestion path.	`self.pulse_bus_activity, ''.join, self.update_stats_labels, self.prune_ui_history, self.metrics.process`
`ViewMixin.prune_ui_history` prune_ui_history(self)	1660-1683	Keeps memory/UI work bounded by removing old records in chunks after a hysteresis threshold.	`self._successful_request_seqs.intersection_update, self.render_filtered_view, len, self._frame_history.popleft, record.get`
`ViewMixin.on_row_select` on_row_select(self, _event=None)	1685-1687	Event handler for row select.	`self.selected_record, self.update_frame_inspector`
`ViewMixin.finish_capture_ui` finish_capture_ui(self)	1689-1702	Finish capture ui.	`self.status_var.set, self.start_btn.configure, self.stop_btn.configure, self.restart_btn.configure, self.set_simulation_button_state`
`ViewMixin.app_directory` app_directory(self)	1704-1705	App directory.	`get_application_directory`
`ViewMixin.prepare_log_session` prepare_log_session(self, prefix, started_at, header)	1707-1720	Prepare a real-capture log session without creating a file.	`self.close_log, started_at.strftime`
`ViewMixin.ensure_log_started` ensure_log_started(self)	1722-1761	Creates the TXT capture log lazily only when real traffic actually arrives, preventing empty logs from start/stop operations with no data.	`target.open, get_logs_folder, self.log_file.write, messagebox.showerror, ''.join`

Callable	Lines	Responsibility	Representative direct calls
`ViewMixin.queue_log_note` queue_log_note(self, text)	1763-1771	Write a capture note to an already-open log, or keep it in memory until the first traffic frame creates the log.	`self.log_line, self.pending_log_notes.append`
`ViewMixin.log_line` log_line(self, text)	1773-1778	Log line.	`self.log_file.write`
`ViewMixin.close_log` close_log(self)	1780-1795	Close log.	`self.log_file.write, self.log_file.close, datetime.now().isoformat, datetime.now`
`ViewMixin.open_log_folder` open_log_folder(self)	1797-1807	Open log folder.	`get_logs_folder, os.startfile, messagebox.showerror, str, subprocess.Popen`
`ViewMixin.clear_view` clear_view(self)	1809-1852	Clear view.	`self.tree.get_children, self.raw_text.delete, self.selected_raw_var.set, self._tree_history.clear, self._tree_iids_by_seq.clear`

Detailed behavior of the key paths

`update_stats_labels` lines 39-104

Converts SessionMetrics into both compact session counters and Bus Health values. It recalculates health using the currently configured serial parameters so approximate bus utilisation reflects the selected format.

`record_passes_advanced_filters` lines 188-249

Implements Type, FC, response-time and free-text filtering. Filtering is display-only; it does not alter SessionMetrics or the on-disk capture log.

`anomaly_tag_for_record` lines 284-321

Maps the semantic result of a record to the Treeview tag used for colour highlighting, including completed normal REQUEST/RESPONSE pairs.

`mark_timed_out_records` lines 332-358

Finds original REQUEST records by their sequence IDs and marks them timed_out rather than inventing artificial timeout frames. Visible rows are retagged in place; the table is rebuilt only while the Timeouts filter is active.

`refresh_record_tags` **lines 360-372**

Re-applies the result tag to the Treeview rows of the given records and refreshes the Frame Inspector when one of them is selected.

`update_frame_inspector` **lines 380-409**

Builds the structured inspector values for the selected traffic record and applies result styling while preserving the raw frame.

`export_session_csv` **lines 536-638**

Exports the frames retained by the GUI history, including timing, decoded addressing, raw data and anomaly/timeout fields. This differs intentionally from the complete TXT capture log.

`records_for_display` **lines 1188-1198**

Applies current filtering and time-order state to the bounded source-of-truth frame history.

`add_frame_rows_batch` **lines 1460-1658**

Central view ingestion path. It assigns stable sequence IDs, runs SessionMetrics, enriches parser rows into view records (including the matched request address/quantity for read responses), tracks successful pairs, writes complete real traffic to disk, updates visible table/raw views, prunes history and auto-scrolls.

`prune_ui_history` **lines 1660-1683**

Keeps memory/UI work bounded by removing old records in chunks after a hysteresis threshold. The disk log is independent and remains complete.

`ensure_log_started` **lines 1722-1761**

Creates the TXT capture log lazily only when real traffic actually arrives, preventing empty logs from start/stop operations with no data.

> **Design note.** There are deliberately two histories: the bounded GUI history and the complete TXT log. Sorting/filtering only changes presentation; statistics are derived before filtering.

mb_slave_finder.py

Active Modbus RTU Slave discovery and Device Identification. It checks that the bus is idle, builds safe read-only FC03/FC04 probes with configurable attempts, calculates adaptive timeouts, performs robust serial writes/reads, accepts normal or CRC-valid Exception responses as proof of a Slave, optionally reads FC43/14 Basic Device Identification and drives the Finder UI through queued events.

Source lines	SHA-256
1587	`54bc1c44f963798a8a2a0912d872306224fb87ffea53489f0a97b7ddb0f90789`

Imports: `threading, time, tkinter, tkinter.ttk, messagebox, mb_config.DISCOVERY_BAUD_RATES, DISCOVERY_PARITIES, DISCOVERY_STOP_BITS, DISCOVERY_DEFAULT_MIN_TIMEOUT_MS, DISCOVERY_MIN_TIMEOUT_MS, DISCOVERY_MAX_TIMEOUT_MS, DISCOVERY_PROBE_ADDRESS, DISCOVERY_PROBE_QTY, DISCOVERY_WRITE_TIMEOUT_SECONDS, DISCOVERY_WRITE_RETRIES, DISCOVERY_RX_BUFFER_LIMIT, DISCOVERY_MIN_ATTEMPTS, DISCOVERY_MAX_ATTEMPTS, DISCOVERY_DEFAULT_ATTEMPTS, DISCOVERY_IDLE_CHECK_SECONDS, MODBUS_EXCEPTION_CODES, mb_protocol.append_crc, crc_ok, hex_bytes, modbus_rtu_interframe_gap_seconds, mb_widgets.RefreshButton`

Module constants / configuration data

Name	Definition
`DEVICE_ID_FC`	`43`
`DEVICE_ID_MEI_TYPE`	`14`
`DEVICE_ID_READ_BASIC`	`1`
`DEVICE_ID_MORE_FOLLOWS`	`255`
`DEVICE_ID_BASIC_OBJECTS`	`(0, 1, 2)`
`DEVICE_ID_MAX_PAGES`	`8`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`build_discovery_probe` build_discovery_probe(slave, fc)	35-48	Constructs a read-only FC03/FC04 request for Address 0, Quantity 1 and appends the RTU CRC.	`bytes, append_crc, ValueError, int`

Callable	Lines	Responsibility	Representative direct calls
`build_device_identification_probe` build_device_identification_probe(slave, object_id=0)	57-71	Builds FC43 / MEI type 14 / Basic Device Identification requests, beginning at a selectable Object ID.	`int, append_crc, ValueError, bytes`
`_decode_device_id_value` _decode_device_id_value(value)	73-75	Decode a Modbus Device Identification ASCII object for display.	`bytes(value).decode('ascii', errors='replace').strip, bytes(value).decode, bytes`
`find_device_identification_response` find_device_identification_response(buffer, slave)	77-158	Parses FC43/14 responses and Exceptions from a receive buffer, validates CRC, decodes variable-length ASCII objects and exposes More Follows / Next Object ID for paged identification.	`bytes, int, range, max, len`
`read_device_identification_response` read_device_identification_response(ser, slave, timeout_s, stop_event)	160-182	Read until a valid FC43/14 response/exception is received or timed out.	`bytearray, find_device_identification_response, time.perf_counter, max, stop_event.is_set`
`device_identification_timeout_seconds` device_identification_timeout_seconds(baud, parity, stopbits, minimum_timeout_ms=DISCOVERY_DEFAULT_MIN_TIMEOUT_MS)	184-201	Conservative timeout for one FC43/14 response, up to one RTU frame.	`max, discovery_bits_per_char, int, float, modbus_rtu_interframe_gap_seconds`
`device_identification_request_transmit_seconds` device_identification_request_transmit_seconds(baud, parity, stopbits)	203-207	Estimated on-wire duration of the 7-byte FC43/14 request.	`float, discovery_bits_per_char, max, int`
`format_device_identification` format_device_identification(result)	209-224	Return compact Device Identification text for the Finder result table.	`any, result.get, int, objects.get, ' | '.join`
`read_basic_device_identification` read_basic_device_identification(ser, slave, baud, parity, stopbits, minimum_timeout_ms, stop_event, serial_module)	226-301	Repeats FC43/14 requests while More Follows is set, merging objects until VendorName, ProductCode and MajorMinorRevision are available or the paging limit/response terminates.	`range, stop_event.is_set, build_device_identification_probe, write_discovery_request, read_device_identification_response`
`discovery_bits_per_char` discovery_bits_per_char(parity, stopbits)	303-307	Return serial bits/character: start + 8 data + parity + stop.	`float`
`discovery_response_timeout_seconds` discovery_response_timeout_seconds(baud, parity, stopbits, minimum_timeout_ms=DISCOVERY_DEFAULT_MIN_TIMEOUT_MS)	309-330	Calculates a response deadline from serial character time, the expected 7-byte FC03/FC04 response and an inter-frame/processing margin, never below the user-configured minimum.	`max, discovery_bits_per_char, int, float, modbus_rtu_interframe_gap_seconds`

Callable	Lines	Responsibility	Representative direct calls
`discovery_request_transmit_seconds` discovery_request_transmit_seconds(baud, parity, stopbits)	332-334	Estimated on-wire duration of the fixed 8-byte discovery request.	`float, discovery_bits_per_char, max, int`
`find_discovery_response` find_discovery_response(buffer, slave, fc)	336-379	Scans received bytes for the requested Slave and Function Code.	`bytes, int, range, max, len`
`read_discovery_response` read_discovery_response(ser, slave, fc, timeout_s, stop_event)	381-407	Read non-blocking serial bytes until a valid probe response or timeout.	`bytearray, find_discovery_response, time.perf_counter, max, stop_event.is_set`
`bus_activity_detected` bus_activity_detected(ser, listen_s, stop_event)	409-427	Listens passively for the idle-check period and returns any bytes received; a non-empty result means another device is transmitting and the scan must not start.	`bytearray, bytes, time.perf_counter, max, stop_event.is_set`
`cancel_serial_io` cancel_serial_io(ser)	430-440	Best-effort cancellation of pending serial I/O.	`getattr, callable, method`
`recover_discovery_write` recover_discovery_write(ser)	442-457	Best-effort cleanup after a discovery write timeout.	`getattr, callable, ser.reset_output_buffer, ser.reset_input_buffer, cancel`
`write_discovery_request` write_discovery_request(ser, request, baud, parity, stopbits, stop_event, serial_module)	459-523	Transmits one probe without using ser.flush(), retries one write timeout after driver/buffer recovery, waits for estimated on-wire transmission time and raises actual driver errors instead of disguising them as no-response Slaves.	`bytes, getattr, discovery_bits_per_char, max, range`
`format_duration` format_duration(seconds)	525-531	Format duration.	`max, divmod, int, round, float`
`class SlaveFinderMixin`	**534-1586**	**Methods that implement the active Bus Slave Finder tab.**	**Class container**
`SlaveFinderMixin.build_discovery_tab` build_discovery_tab(self, page)	537-912	Build the full-window active Modbus RTU slave-discovery workspace.	`getattr(self, 'ui_settings', {}).get, set, page.columnconfigure, page.rowconfigure, ttk.LabelFrame`
`SlaveFinderMixin.fit_discovery_columns_to_viewport` fit_discovery_columns_to_viewport(self, _event=None)	914-943	Fit Slave Finder columns to the visible Treeview width.	`int, max, sum, len, enumerate`
`SlaveFinderMixin.on_discovery_options_changed` on_discovery_options_changed(self)	945-948	Event handler for discovery options changed.	`hasattr, self.update_discovery_estimate, self.update_discovery_start_state`

Callable	Lines	Responsibility	Representative direct calls
`SlaveFinderMixin.selected_discovery_configs` selected_discovery_configs(self)	950-959	Selected discovery configs.	`self.discovery_baud_vars[baud].get, self.discovery_parity_vars[p].get, self.discovery_stop_vars[s].get`
`SlaveFinderMixin.selected_discovery_slaves` selected_discovery_slaves(self)	961-968	Selected discovery slaves.	`int, list, self.discovery_slave_start_var.get, self.discovery_slave_end_var.get, ValueError`
`SlaveFinderMixin.selected_discovery_min_timeout_ms` selected_discovery_min_timeout_ms(self)	970-980	Selected discovery min timeout ms.	`float, self.discovery_min_timeout_var.get, ValueError, self.trf`
`SlaveFinderMixin.selected_discovery_attempts` selected_discovery_attempts(self)	982-992	Selected discovery attempts.	`int, self.discovery_attempts_var.get, ValueError, self.trf`
`SlaveFinderMixin.discovery_config_label` discovery_config_label(self, parity, stopbits)	994-996	Discovery config label.	—
`SlaveFinderMixin.update_discovery_estimate` update_discovery_estimate(self)	998-1055	Update discovery estimate.	`bool, self.discovery_progress_text_var.set, self.discovery_estimate_var.set, self.selected_discovery_configs, self.selected_discovery_slaves`
`SlaveFinderMixin.update_discovery_start_state` update_discovery_start_state(self)	1057-1073	Update discovery start state.	`self.discovery_start_btn.configure, hasattr, bool, self.selected_discovery_slaves, self.selected_discovery_min_timeout_ms`
`SlaveFinderMixin.clear_discovery_results` clear_discovery_results(self)	1075-1084	Clear discovery results.	`self.discovery_tree.get_children, self.discovery_found_var.set, self.discovery_progress_var.set, self.discovery_status_var.set, self.update…`
`SlaveFinderMixin.open_discovery_serial_port` open_discovery_serial_port(self, serial, port, baud, parity, stopbits)	1086-1124	Open discovery serial port.	`serial.Serial, int, ser.open, ser.reset_input_buffer, ser.reset_output_buffer`
`SlaveFinderMixin.start_discovery_scan` start_discovery_scan(self)	1126-1217	Validates the safety confirmation, COM selection, Slave range, attempts, selected serial configurations and global busy state.	`self.get_serial_modules, bool, self.clear_discovery_results, self.discovery_stop_event.clear, self.discovery_progress_var.set`

Callable	Lines	Responsibility	Representative direct calls
`SlaveFinderMixin.stop_discovery_scan` stop_discovery_scan(self)	1219-1226	Stop discovery scan.	`self.discovery_status_var.set, self.discovery_stop_event.set, self.discovery_stop_btn.configure, cancel_serial_io, self.tr`
`SlaveFinderMixin.discovery_worker` discovery_worker(self, serial, port, configs, slaves, minimum_timeout_ms, fallback_fc04, identify_devices, attempts=DISCOVERY_DEFAULT_ATTEMPTS)	1228-1507	Iterates selected serial configurations and Slave IDs.	`time.perf_counter, str, self.event_queue.put, len, self.discovery_stop_event.is_set`
`SlaveFinderMixin.apply_discovery_progress` apply_discovery_progress(self, payload)	1509-1524	Apply discovery progress.	`int, max, self.discovery_progress_var.set, self.discovery_progress_text_var.set, self.discovery_found_var.set`
`SlaveFinderMixin.add_discovery_result` add_discovery_result(self, payload)	1526-1543	Add discovery result.	`self.discovery_found_var.set, self.discovery_tree.insert, self.trf, payload.get`
`SlaveFinderMixin.finish_discovery_ui` finish_discovery_ui(self, payload)	1545-1586	Returns both Finder and Sniffer controls to idle state and presents completed/stopped/error status without leaving the application locked in a busy state.	`self.discovery_stop_event.clear, self.discovery_stop_btn.configure, self.discovery_clear_btn.configure, self.discovery_refresh_btn.configur…`

Detailed behavior of the key paths

`build_discovery_probe` lines 35-48

Constructs a read-only FC03/FC04 request for Address 0, Quantity 1 and appends the RTU CRC. Input validation restricts Slave IDs to 1..247 and FC to 03/04.

`find_discovery_response` lines 336-379

Scans received bytes for the requested Slave and Function Code. A CRC-valid normal register response or CRC-valid Modbus Exception is accepted; echoes and unrelated traffic are ignored.

`discovery_response_timeout_seconds` lines 309-330

Calculates a response deadline from serial character time, the expected 7-byte FC03/FC04 response and an inter-frame/processing margin, never below the user-configured minimum.

`bus_activity_detected` **lines 409-427**

Listens passively for the idle-check period and returns any bytes received; a non-empty result means another device is transmitting and the scan must not start.

`write_discovery_request` **lines 459-523**

Transmits one probe without using ser.flush(), retries one write timeout after driver/buffer recovery, waits for estimated on-wire transmission time and raises actual driver errors instead of disguising them as no-response Slaves.

`build_device_identification_probe` **lines 57-71**

Builds FC43 / MEI type 14 / Basic Device Identification requests, beginning at a selectable Object ID.

`find_device_identification_response` **lines 77-158**

Parses FC43/14 responses and Exceptions from a receive buffer, validates CRC, decodes variable-length ASCII objects and exposes More Follows / Next Object ID for paged identification.

`read_basic_device_identification` **lines 226-301**

Repeats FC43/14 requests while More Follows is set, merging objects until VendorName, ProductCode and MajorMinorRevision are available or the paging limit/response terminates.

`start_discovery_scan` **lines 1126-1217**

Validates the safety confirmation, COM selection, Slave range, attempts, selected serial configurations and global busy state. It disables conflicting controls and launches the discovery worker.

`discovery_worker` **lines 1228-1507**

Iterates selected serial configurations and Slave IDs. It opens one configuration at a time, checks once that the bus is idle, probes FC03 then optional FC04 with the selected attempts, optionally performs Device Identification for found Slaves, publishes progress/results, and always closes the active serial handle.

`finish_discovery_ui` **lines 1545-1586**

Returns both Finder and Sniffer controls to idle state and presents completed/stopped/error status without leaving the application locked in a busy state.

> **Design note.** The Finder is active traffic generation, unlike the passive Sniffer. The UI requires an explicit confirmation that another master is not active and the global busy lock prevents simultaneous capture/scan inside this application.

mb_master.py

Modbus RTU master engine, frames and data formats. It builds FC01-04 read and FC05/06/15/16 write requests, matches responses and exceptions, converts registers to and from 16/32-bit formats in four byte orders and runs MasterEngine, which serves every poll window over one serial port one request at a time. It contains no widgets.

Source lines	SHA-256
516	`5b0aa32c4982020966f0dfc7dbdec0d284dc60a45a05a68e11e80bb5d579f5d7`

Imports: `math, struct, threading, time, collections.deque, dataclasses.dataclass, mb_config.MASTER_MAX_READ_BITS, MASTER_MAX_READ_REGISTERS, MASTER_MAX_WRITE_BITS, MASTER_MAX_WRITE_REGISTERS, mb_protocol.append_crc, crc_ok, hex_bytes, modbus_rtu_interframe_gap_seconds, mb_slave_finder.bus_activity_detected, cancel_serial_io, discovery_bits_per_char, write_discovery_request`

Module constants / configuration data

Name	Definition
`READ_FUNCTION_CODES`	`(1, 2, 3, 4)`
`BIT_FUNCTION_CODES`	`(1, 2)`
`WRITABLE_READ_FCS`	`{1: (5, 15), 3: (6, 16)}`
`BYTE_ORDERS`	`('ABCD', 'CDAB', 'BADC', 'DCBA')`
`REGISTER_FORMATS`	`(('int16', 'Signed 16-bit', 1), ('uint16', 'Unsigned 16-bit', 1), ('hex16', 'Hex 16-bit', 1), ('bin16', 'Binary 16-bit', 1)) + tuple(((f'{kind}_{or…`
`FORMAT_LABELS`	`{key: label for key, label, _size in REGISTER_FORMATS}`
`FORMAT_KEYS_BY_LABEL`	`{label: key for key, label, _size in REGISTER_FORMATS}`
`FORMAT_SIZES`	`{key: size for key, _label, size in REGISTER_FORMATS}`
`DEFAULT_FORMAT`	`'int16'`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`_value_bytes_from_registers` _value_bytes_from_registers(high, low, order)	65-67	Value bytes from registers.	`bytes, 'ABCD'.index`
`_registers_from_value_bytes` _registers_from_value_bytes(value, order)	70-74	Registers from value bytes.	`enumerate, 'ABCD'.index`
`format_float` format_float(value)	77-82	Format float.	`math.isnan, math.isinf`
`format_register_values` format_register_values(registers, fmt)	85-119	Turns raw registers into display strings for the selected format; 32-bit values are shown on the first register of each pair and an unpaired last register shows a dash.	`fmt.split, range, len, _value_bytes_from_registers, out.extend`
`_parse_int` _parse_int(text)	122-129	Parse int.	`str(text).strip().replace(' ', '').replace, text.lower, lowered.startswith, int, str(text).strip().replace`
`encode_register_value` encode_register_value(text, fmt)	132-177	Converts one user-entered value into the register(s) it occupies in the selected format and byte order, rejecting values that do not fit.	`str(text).strip, ValueError, fmt.split, _registers_from_value_bytes, str`
`encode_coil_value` encode_coil_value(text)	180-186	Encode coil value.	`str(text).strip().lower, ValueError, str(text).strip, str`
`_u16` _u16(value)	193-194	U16.	`bytes`
`read_limit` read_limit(fc)	197-198	Read limit.	—
`validate_read_definition` validate_read_definition(slave, fc, address, qty)	201-212	Raise ValueError (PT canonical message) for an invalid poll definition.	`ValueError, int, read_limit`
`build_read_request` build_read_request(slave, fc, address, qty)	215-217	Build read request.	`validate_read_definition, append_crc, _u16, bytes, int`
`build_write_request` build_write_request(slave, fc, address, values)	220-251	Builds FC05/06/15/16 requests within the Modbus write limits; Slave 0 is accepted for broadcast.	`append_crc, int, ValueError, len, _u16`
`expected_response_length` expected_response_length(request)	254-262	Expected response length.	—

Callable	Lines	Responsibility	Representative direct calls
`find_master_response` find_master_response(buffer, request)	269-317	Finds the CRC-valid response of the expected length from the addressed Slave, or a Modbus exception, and decodes read values.	`bytes, expected_response_length, range, len, crc_ok`
`class PollDefinition`	**325-335**	**PollDefinition.**	**Class container**
`PollDefinition.request` request(self)	334-335	Request.	`build_read_request`
`class MasterEngine`	**338-516**	**Serve all poll windows over one serial port, one transaction at a time.**	**Class container**
`MasterEngine.__init__` __init__(self, ser, event_queue, *, baud, parity, stopbits, response_timeout_ms, delay_ms, serial_module=None, idle_check_seconds=0.0)	348-370	Init.	`int, max, modbus_rtu_interframe_gap_seconds, float, threading.Lock`
`MasterEngine.set_poll` set_poll(self, definition)	373-376	Set poll.	`self._wake.set`
`MasterEngine.remove_poll` remove_poll(self, window_id)	378-381	Remove poll.	`self._polls.pop, deque`
`MasterEngine.read_once` read_once(self, window_id)	383-387	Read once.	`self._wake.set, self._once.append`
`MasterEngine.queue_write` queue_write(self, window_id, request, description='')	389-392	Queue write.	`self._wake.set, self._writes.append, bytes`
`MasterEngine.start` start(self)	394-396	Start.	`threading.Thread, self.thread.start`
`MasterEngine.stop` stop(self, join_timeout=1.0)	398-404	Stop.	`self.stop_event.set, self._wake.set, cancel_serial_io, thread.is_alive, thread.join`
`MasterEngine.next_job` next_job(self, now=None)	407-430	Chooses the next transaction: queued writes first, then "Read once" requests, then the enabled window whose next due time is earliest.	`time.perf_counter, min, self._once.popleft, self._polls.get, self._writes.popleft`
`MasterEngine.run` run(self)	433-470	Worker loop: optional idle-bus check, then one transaction at a time with the configured delay between polls, posting master_ready, master_result and master_stopped events.	`self.event_queue.put, bus_activity_detected, self.stop_event.is_set, self.next_job, self.stop_event.wait`

Callable	Lines	Responsibility	Representative direct calls
`MasterEngine.transact` transact(self, request)	472-516	Sends one request and waits for its response until the response timeout, classifying the result as ok, exception, invalid, timeout or broadcast.	`write_discovery_request, time.perf_counter, bytearray, self.stop_event.is_set, hex_bytes`

Detailed behavior of the key paths

`format_register_values` lines 85-119

Turns raw registers into display strings for the selected format; 32-bit values are shown on the first register of each pair and an unpaired last register shows a dash.

`encode_register_value` lines 132-177

Converts one user-entered value into the register(s) it occupies in the selected format and byte order, rejecting values that do not fit.

`build_write_request` lines 220-251

Builds FC05/06/15/16 requests within the Modbus write limits; Slave 0 is accepted for broadcast.

`find_master_response` lines 269-317

Finds the CRC-valid response of the expected length from the addressed Slave, or a Modbus exception, and decodes read values. Adapter echoes of read requests are skipped.

`MasterEngine.next_job` lines 407-430

Chooses the next transaction: queued writes first, then "Read once" requests, then the enabled window whose next due time is earliest.

`MasterEngine.run` lines 433-470

Worker loop: optional idle-bus check, then one transaction at a time with the configured delay between polls, posting master_ready, master_result and master_stopped events.

`MasterEngine.transact` lines 472-516

Sends one request and waits for its response until the response timeout, classifying the result as ok, exception, invalid, timeout or broadcast.

> **Design note.** The engine owns the serial port while the Master is connected. Widgets only change definitions through set_poll(), read_once() and queue_write(); results come back through event_queue.

mb_master_ui.py

Master (poll) tab user interface. MasterMixin builds the connection bar and the grid of poll windows, validates window definitions, renders values and counters, opens the write and name dialogs, pops windows out to their own top-level windows and saves or opens workspace files.

Source lines	SHA-256
1150	`c898de2286993ea45266f4a3566defe7248a09d41c19050f52cbcc955923bca0`

Imports: `json, time, tkinter, tkinter.filedialog, messagebox, ttk, mb_config.DISCOVERY_BAUD_RATES, DISCOVERY_IDLE_CHECK_SECONDS, MASTER_DEFAULT_DELAY_MS, MASTER_DEFAULT_RESPONSE_TIMEOUT_MS, MASTER_DEFAULT_SCAN_MS, MASTER_MAX_DELAY_MS, MASTER_MAX_RESPONSE_TIMEOUT_MS, MASTER_MAX_SCAN_MS, MASTER_MAX_WINDOWS, MASTER_MIN_DELAY_MS, MASTER_MIN_RESPONSE_TIMEOUT_MS, MASTER_WORKSPACE_FORMAT, MASTER_WORKSPACE_VERSION, MODBUS_EXCEPTION_CODES, mb_diagnostics.FUNCTION_NAMES, mb_master.BIT_FUNCTION_CODES, DEFAULT_FORMAT, FORMAT_KEYS_BY_LABEL, FORMAT_LABELS, FORMAT_SIZES, READ_FUNCTION_CODES, WRITABLE_READ_FCS, MasterEngine, PollDefinition, build_write_request, encode_coil_value, encode_register_value, format_register_values, validate_read_definition, mb_theme.theme_colors, mb_widgets.RefreshButton`

Module constants / configuration data

Name	Definition
`MASTER_PARITIES`	`('None', 'Even', 'Odd')`
`MASTER_STOP_BITS`	`('1', '2')`
`MASTER_PANE_MIN_WIDTH`	`600`
`MASTER_PANE_ROWS`	`10`
`MASTER_OK_MESSAGE_SECONDS`	`3.0`
`WHEEL_GUARD_TAG`	`'MBSniffer.MouseWheelGuard'`
`MASTER_WHEEL_TAG`	`'MBSniffer.MasterWheel'`
`FC_CHOICES`	`tuple((fc_choice_label(fc) for fc in READ_FUNCTION_CODES))`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`fc_choice_label` fc_choice_label(fc)	59-60	Fc choice label.	—

Callable	Lines	Responsibility	Representative direct calls
`parse_int_field` parse_int_field(text, minimum, maximum, message)	66-73	Parse int field.	`int, ValueError, str(text).strip, str`
`class MasterWindow`	**76-140**	**State of one poll window (independent of where its pane is shown).**	**Class container**
`MasterWindow.__init__` __init__(self, window_id, number, slave=1, fc=3, address=0, qty=10, scan_ms=MASTER_DEFAULT_SCAN_MS, base=0, fmt=DEFAULT_FORMAT, names=None, enabled=True)	79-94	Init.	`dict, self.reset_counters`
`MasterWindow.reset_counters` reset_counters(self)	96-104	Reset counters.	—
`MasterWindow.definition` definition(self)	106-108	Definition.	`PollDefinition`
`MasterWindow.to_json` to_json(self)	110-116	To json.	`str, sorted, self.names.items`
`MasterWindow.validated_kwargs` validated_kwargs(data)	119-140	Validate one workspace window entry; raise ValueError if invalid.	`validate_read_definition, int, data.get, isinstance, ValueError`
`class MasterMixin`	**143-1146**	**Master (poll) tab GUI.**	**Class container**
`MasterMixin.build_master_tab` build_master_tab(self, page)	147-282	Builds the connection bar, toolbar and scrollable grid of poll windows, restoring the saved connection parameters.	`getattr(self, 'ui_settings', {}).get, page.columnconfigure, page.rowconfigure, ttk.LabelFrame, connection.grid`
`MasterMixin._master_install_wheel_tags` _master_install_wheel_tags(self, root_widget)	285-299	Give new widgets the global wheel guard; plain panels scroll the grid.	`stack.pop, stack.extend, tuple, isinstance, widget.winfo_children`
`MasterMixin._master_wheel` _master_wheel(self, event)	301-318	Master wheel.	`getattr, canvas.yview, canvas.yview_scroll`
`MasterMixin._master_window_title` _master_window_title(self, window)	320-326	Master window title.	`self.trf`
`MasterMixin._master_counters_text` _master_counters_text(self, window)	328-331	Master counters text.	—
`MasterMixin._master_set_message` _master_set_message(self, window, template, style='normal', **values)	333-336	Master set message.	`time.monotonic, self._master_render_status`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin._master_render_status` _master_render_status(self, window)	338-359	Master render status.	`self._master_panes.get, self.trf, pane['counters_var'].set, pane['message_var'].set, pane['message_label'].configure`
`MasterMixin._master_build_pane` _master_build_pane(self, window, parent)	362-475	Master build pane.	`ttk.LabelFrame, frame.columnconfigure, frame.rowconfigure, ttk.Frame, definition.grid`
`MasterMixin._master_update_format_state` _master_update_format_state(self, window)	477-481	Master update format state.	`self._master_panes.get, pane['format_combo'].configure`
`MasterMixin._master_fill_rows` _master_fill_rows(self, window)	483-493	Master fill rows.	`self._master_panes.get, tree.delete, range, self._master_render_values, tree.insert`
`MasterMixin._master_render_values` _master_render_values(self, window)	495-508	Master render values.	`self._master_panes.get, enumerate, format_register_values, str, tree.set`
`MasterMixin._master_schedule_reflow` _master_schedule_reflow(self, _event=None)	510-516	Master schedule reflow.	`self.after, self.after_cancel`
`MasterMixin._master_reflow` _master_reflow(self)	518-538	Arrange docked panes in as many equal columns as the width allows.	`max, canvas.itemconfigure, range, enumerate, canvas.winfo_width`
`MasterMixin.master_add_window` master_add_window(self, kwargs=None)	541-558	Master add window.	`next, MasterWindow, self._master_build_pane, self._master_reflow, len`
`MasterMixin.master_duplicate_window` master_duplicate_window(self, window_id)	560-564	Master duplicate window.	`self.master_windows.get, MasterWindow.validated_kwargs, self.master_add_window, window.to_json`
`MasterMixin.master_close_window` master_close_window(self, window_id)	566-579	Master close window.	`self.master_windows.pop, self._master_panes.pop, self._master_reflow, self.master_engine.remove_poll, pane.get`
`MasterMixin.master_apply_window` master_apply_window(self, window_id)	581-614	Validates the fields of one window, updates its model, rebuilds its rows and hands the new definition to the running engine.	`self.master_windows.get, self._master_panes.get, FORMAT_KEYS_BY_LABEL.get, self._master_fill_rows, self._master_update_format_state`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin.master_toggle_pause` master_toggle_pause(self, window_id)	616-625	Master toggle pause.	`self.master_windows.get, self._master_render_status, self.master_engine.set_poll, window.definition`
`MasterMixin.master_read_once` master_read_once(self, window_id)	627-631	Master read once.	`self.master_engine.read_once, messagebox.showinfo, self.tr`
`MasterMixin.master_toggle_popout` master_toggle_popout(self, window_id)	633-661	Moves a window between the grid and its own top-level window by rebuilding its pane from the model.	`self.master_windows.get, self._master_panes.pop, pane.get, self._master_render_status, self._master_reflow`
`MasterMixin._master_on_double_click` _master_on_double_click(self, window_id, event)	664-677	Master on double click.	`self._master_panes.get, tree.identify_row, tree.selection_set, tree.identify_column, self._master_edit_name`
`MasterMixin._master_dialog` _master_dialog(self, title)	679-687	Master dialog.	`tk.Toplevel, dialog.title, dialog.transient, dialog.resizable, self._master_apply_toplevel_palette`
`MasterMixin._master_edit_name` _master_edit_name(self, window_id, row)	689-715	Master edit name.	`self.master_windows.get, self._master_dialog, ttk.Label(body, text=self.trf('Nome do endereço {address}', address=pdu + window.base)).grid,…`
`MasterMixin._master_edit_value` _master_edit_value(self, window_id)	717-800	Master edit value.	`self.master_windows.get, self._master_panes.get, pane['tree'].selection, int, WRITABLE_READ_FCS.get`
`MasterMixin.master_build_write` master_build_write(self, window, slave_text, address_text, values_text, fc)	802-821	Validates the write dialog and encodes the values with the window format into an FC05/06/15/16 request.	`parse_int_field, build_write_request, ValueError, str(values_text).split, p.strip`
`MasterMixin._master_connection_settings` _master_connection_settings(self)	824-839	Master connection settings.	`parse_int_field, self.master_timeout_var.get, self.master_delay_var.get, self.master_port_var.get().strip, int`
`MasterMixin.master_connect` master_connect(self)	841-899	Checks the busy flag, safety confirmation and settings, opens the port, starts MasterEngine with the idle-bus check and locks the other active features.	`self.get_serial_modules, list, MasterEngine, self.master_windows.values, self._master_lock_other_features`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin.master_disconnect` master_disconnect(self)	901-906	Master disconnect.	`self.master_status_var.set, self.master_disconnect_btn.configure, self.master_engine.stop, self.tr`
`MasterMixin._master_lock_other_features` _master_lock_other_features(self, locked)	908-923	Master lock other features.	`self.master_connect_btn.configure, self.master_disconnect_btn.configure, self.master_refresh_btn.configure, self.start_btn.configure, self.…`
`MasterMixin._master_finish_session` _master_finish_session(self, payload)	925-953	Master finish session.	`self._master_lock_other_features, self.master_status_var.set, (payload or {}).get, engine.stop, self.tr`
`MasterMixin.handle_master_event` handle_master_event(self, event, payload)	956-962	Handle master event.	`self.master_status_var.set, self.tr, self._master_finish_session, self._master_apply_result`
`MasterMixin._master_apply_result` _master_apply_result(self, result)	964-1009	Updates counters, values and the status line of the window that issued a transaction, ignoring results of an outdated definition.	`self.master_windows.get, result.get, self._master_set_message, self._master_render_values, self.master_engine.read_once`
`MasterMixin.master_workspace_data` master_workspace_data(self)	1012-1026	Master workspace data.	`self.master_port_var.get, self.master_baud_var.get, self.master_parity_var.get, self.master_stop_var.get, self.master_timeout_var.get`
`MasterMixin.master_save_workspace` master_save_workspace(self, path=None)	1028-1042	Master save workspace.	`filedialog.asksaveasfilename, open, json.dump, messagebox.showerror, self.tr`
`MasterMixin.master_open_workspace` master_open_workspace(self, path=None)	1044-1091	Loads a .mbsw workspace, validating every window and skipping invalid entries with a warning.	`list, data.get, filedialog.askopenfilename, self.master_close_window, isinstance`
`MasterMixin.master_apply_language` master_apply_language(self)	1094-1110	Master apply language.	`self.master_windows.values, hasattr, self._master_panes.get, self._master_render_status, self.master_status_var.set`
`MasterMixin._master_apply_toplevel_palette` _master_apply_toplevel_palette(self, top)	1112-1117	Master apply toplevel palette.	`bool, getattr(self, 'dark_mode_var', tk.BooleanVar(value=False)).get, top.configure, getattr, tk.BooleanVar`

Callable	Lines	Responsibility	Representative direct calls
`MasterMixin.master_apply_palette` master_apply_palette(self, dark)	1119-1133	Master apply palette.	`theme_colors, hasattr, getattr(self, '_master_panes', {}).values, getattr, self.master_canvas.configure`
`MasterMixin.master_shutdown` master_shutdown(self)	1135-1146	Master shutdown.	`getattr, engine.stop, ser.close`
`fc_choice_label_write` fc_choice_label_write(fc)	1149-1150	Fc choice label write.	—

Detailed behavior of the key paths

`build_master_tab` lines 147-282

Builds the connection bar, toolbar and scrollable grid of poll windows, restoring the saved connection parameters.

`master_apply_window` lines 581-614

Validates the fields of one window, updates its model, rebuilds its rows and hands the new definition to the running engine.

`master_build_write` lines 802-821

Validates the write dialog and encodes the values with the window format into an FC05/06/15/16 request.

`master_connect` lines 841-899

Checks the busy flag, safety confirmation and settings, opens the port, starts MasterEngine with the idle-bus check and locks the other active features.

`_master_apply_result` lines 964-1009

Updates counters, values and the status line of the window that issued a transaction, ignoring results of an outdated definition.

`master_toggle_popout` lines 633-661

Moves a window between the grid and its own top-level window by rebuilding its pane from the model.

`master_open_workspace` lines 1044-1091

Loads a .mbsw workspace, validating every window and skipping invalid entries with a warning.

Design note. Window state lives in MasterWindow objects, so a pane can be destroyed and rebuilt (pop out / dock, language switch) without losing data.

mb_i18n.py

Runtime Portuguese/English translation layer. Portuguese (Portugal) is the canonical UI text. Static translations, runtime-status patterns, protocol-detail fragments and the complete English Help content are kept separate from protocol logic so switching language affects presentation only.

Source lines
555

SHA-256
`92a5d8fbd7c244940c8f6ba7ad44aae325d27b107ff02cfaa511cc388d14aa1c`

Imports: `mb_config.DISCOVERY_IDLE_CHECK_SECONDS, SNIFFER_SPLIT_FRAME_HOLD_MS`

Module constants / configuration data

Name	Definition
`LANGUAGE_PORTUGUESE`	`'Português'`
`LANGUAGE_ENGLISH`	`'English'`
`LANGUAGE_CHOICES`	`(LANGUAGE_PORTUGUESE, LANGUAGE_ENGLISH)`
`PT_TO_EN`	`{'Linguagem': 'Language', 'Modo escuro': 'Dark mode', 'Configuração': 'Configuration', 'Modo físico': 'Physical mode', 'RS232 dual RX: COM A escuta…`
`EN_TO_PT`	`{english: portuguese for portuguese, english in PT_TO_EN.items()}`
`EXCEPTION_DESCRIPTION_EN`	`{1: 'The requested Modbus function is not supported or permitted by the device.', 2: 'The requested address or address range is not valid for this …`
`DETAIL_FRAGMENTS_PT_TO_EN`	`(('Registos (PDU Address=valor):', 'Registers (PDU Address=value):'), ('Bits (PDU Address=valor):', 'Bits (PDU Address=value):'), ('Registos=', 'Re…`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`normalize_language` normalize_language(value)	278-279	Normalize language.	—

Callable	Lines	Responsibility	Representative direct calls
`canonical_ui_text` canonical_ui_text(text)	282-285	Normalises a known English caption back to its canonical PT-PT form.	`str, EN_TO_PT.get`
`translate_text` translate_text(text, language)	288-293	Returns the English translation for known canonical PT-PT UI strings when English is selected; unknown/technical terms are returned unchanged.	`canonical_ui_text, normalize_language, PT_TO_EN.get`
`translate_details` translate_details(text, language)	304-313	Translates selected human-readable fragments embedded inside protocol detail strings (DETAIL_FRAGMENTS_PT_TO_EN) while preserving numbers, addresses, register values and other protocol data.	`str, normalize_language, value.replace`
`exception_description` exception_description(code, portuguese_text, language)	316-319	Exception description.	`normalize_language, EXCEPTION_DESCRIPTION_EN.get`
`english_help_blocks` english_help_blocks(max_ui_frames_text, max_raw_lines_text, min_pending, max_pending, debug_sim=False)	323-526	Returns the complete English Help / Connections document as structured blocks so Help switches language together with the rest of the UI.	`blocks.extend`
`translate_runtime_text` translate_runtime_text(text, language)	529-555	Uses regular expressions to translate live status strings that contain counters, elapsed time, selected configuration or estimated duration.	`str, translate_text, normalize_language, re.match, re.sub`

Detailed behavior of the key paths

`canonical_ui_text` lines 282-285

Normalises a known English caption back to its canonical PT-PT form. This makes language switching reversible instead of translating an already translated string repeatedly.

`translate_text` lines 288-293

Returns the English translation for known canonical PT-PT UI strings when English is selected; unknown/technical terms are returned unchanged.

`translate_details` **lines 304-313**

Translates selected human-readable fragments embedded inside protocol detail strings (DETAIL_FRAGMENTS_PT_TO_EN) while preserving numbers, addresses, register values and other protocol data.

`english_help_blocks` **lines 323-526**

Returns the complete English Help / Connections document as structured blocks so Help switches language together with the rest of the UI.

`translate_runtime_text` **lines 529-555**

Uses regular expressions to translate live status strings that contain counters, elapsed time, selected configuration or estimated duration.

> **Design note.** The translation design treats PT-PT as canonical and leaves technical English terms unchanged unless a translation is explicitly defined.

mb_theme.py

Fixed-geometry Light/Dark theme implementation. It configures a single ttk clam layout once, installs stable checkbox geometry and changes only palette values at runtime. This is deliberately designed so theme switching does not move or resize controls.

Source lines	SHA-256
692	`88c0c2592f4b6e0666aa42e4d163ae120845b154f7faa3b9f6a007fef7ed8265`

Imports: `tkinter, tkinter.ttk`

Module constants / configuration data

Name	Definition
`BOOTSTRAP_ACTIVE`	`False`
`THEME_ENGINE`	`'clam'`
`LIGHT`	`{'root': '#f3f4f6', 'panel': '#f3f4f6', 'panel_alt': '#ffffff', 'input': '#ffffff', 'input_readonly': '#f5f6f7', 'text': '#1f2328', 'muted': '#5963…`
`DARK`	`{'root': '#171a1f', 'panel': '#171a1f', 'panel_alt': '#1f242a', 'input': '#20262c', 'input_readonly': '#252c33', 'text': '#e8eaed', 'muted': '#aab2…`
`_CHECK_IMAGES`	`None`
`_CHECK_ELEMENT`	`'MBS.Check.indicator'`
`_FIXED_GEOMETRY_READY`	`False`
`_INTERP_ID`	`None`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`theme_colors` theme_colors(root=None, dark=False)	70-71	Theme colors.	`dict`
`_ensure_engine` _ensure_engine(style)	74-95	Select/configure the fixed engine once per Tk interpreter.	`id, _configure_fixed_geometry, style.theme_use, style.theme_names`

Callable	Lines	Responsibility	Representative direct calls
`_configure_fixed_geometry` _configure_fixed_geometry(style)	98-296	Defines ttk element layouts, paddings, borders, row heights and widget metrics once.	`style.configure, style.map, _install_fixed_checkbutton_element, style.layout`
`_paint_checkbox_image` _paint_checkbox_image(image, *, fill, border, check=None)	299-314	Repaint one fixed-size checkbox image without changing its dimensions.	`image.put`
`_install_fixed_checkbutton_element` _install_fixed_checkbutton_element(style)	317-367	Creates the custom checkbox indicator element/images used to keep checkbox dimensions stable across palette changes.	`style.layout, repr, tk.PhotoImage, style.element_names, style.element_create`
`_repaint_checkboxes` _repaint_checkboxes(colors, dark)	370-408	Repaint checkboxes.	`_paint_checkbox_image`
`_apply_palette` _apply_palette(root, style, colors, dark)	411-666	Changes ttk colors/state maps and classic Tk colors without changing layout definitions, supporting the invariant that theme switching changes appearance but not geometry.	`style.configure, style.map, _repaint_checkboxes, root.configure`
`apply_theme` apply_theme(root, dark=False, style=None)	669-677	Change palette only.	`_ensure_engine, _apply_palette, ttk.Style, bool`
`apply_light_theme` apply_light_theme(root, style=None)	680-681	Apply light theme.	`apply_theme`
`apply_dark_theme` apply_dark_theme(root, style=None)	684-685	Apply dark theme.	`apply_theme`
`apply_modern_theme` apply_modern_theme(root, dark=False)	688-692	Public startup helper that selects the fixed clam engine and applies either the Light or Dark palette.	`ttk.Style, _ensure_engine, _apply_palette, bool`

Detailed behavior of the key paths

`_configure_fixed_geometry` lines 98-296

Defines ttk element layouts, paddings, borders, row heights and widget metrics once. These values are intentionally independent of Light/Dark selection.

`_install_fixed_checkbutton_element` lines 317-367

Creates the custom checkbox indicator element/images used to keep checkbox dimensions stable across palette changes.

`_apply_palette` **lines 411-666**

Changes ttk colors/state maps and classic Tk colors without changing layout definitions, supporting the invariant that theme switching changes appearance but not geometry.

`apply_modern_theme` **lines 688-692**

Public startup helper that selects the fixed clam engine and applies either the Light or Dark palette.

Design note. The theme module is geometry-sensitive: changing colors must not rebuild ttk layouts. This is why theme engine selection and fixed geometry are separated from palette application.

mb_widgets.py

Custom reusable Tkinter widgets. RefreshButton draws a stable square refresh control on a Canvas; ToggleSwitch draws the Light/Dark switch. Both implement their own state, hover/press behavior and palette updates.

Source lines	SHA-256
418	`fb45d18d621a734220afa46d92dfb882228a5cba2702e4201ccf02120b2bff97`

Imports: `tkinter`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class RefreshButton`	**7-250**	**Canvas-based square button that draws the ↻ glyph and recentres it from its actual rendered bounding box.**	**Class container**
`RefreshButton.__init__` __init__(self, master, command, size=26, cursor='hand2', **kwargs)	20-53	Init.	`super().__init__, self.bind, self._redraw, self._schedule_final_redraw, super`
`RefreshButton._system_color` _system_color(widget, name, fallback)	56-61	System color.	`widget.winfo_rgb`
`RefreshButton._schedule_final_redraw` _schedule_final_redraw(self)	63-77	Redraw after Tk has settled requested widget dimensions.	`self.after, self.after_cancel`
`RefreshButton._final_redraw` _final_redraw(self)	79-84	Final redraw.	`self._redraw`
`RefreshButton.configure` configure(self, cnf=None, **kwargs)	86-97	Configure.	`kwargs.pop, super().configure, str, self._redraw, self._schedule_final_redraw`
`RefreshButton.cget` cget(self, key)	101-104	Cget.	`super().cget, super`
`RefreshButton.invoke` invoke(self)	106-109	Invoke.	`callable, self._command`
`RefreshButton.set_dark_mode` set_dark_mode(self, enabled)	111-114	Set dark mode.	`bool, self._redraw, self._schedule_final_redraw`
`RefreshButton._palette` _palette(self)	116-146	Palette.	`self._system_color`

Callable	Lines	Responsibility	Representative direct calls
`RefreshButton._center_item_bbox` _center_item_bbox(self, item, width, height)	148-181	Center the visible Tk bbox of one Canvas item on the square.	`range, self.bbox, self.move, round, abs`
`RefreshButton._redraw` _redraw(self, _event=None)	183-220	Redraw.	`self._palette, self.delete, self.create_rectangle, self.create_text, self._center_item_bbox`
`RefreshButton._on_enter` _on_enter(self, _event)	222-225	Event handler for enter.	`self._redraw`
`RefreshButton._on_leave` _on_leave(self, _event)	227-230	Event handler for leave.	`self._redraw`
`RefreshButton._on_press` _on_press(self, _event)	232-235	Event handler for press.	`self._redraw`
`RefreshButton._on_release` _on_release(self, event)	237-250	Event handler for release.	`self._redraw, callable, self._command, self.winfo_width, self.winfo_height`
`class ToggleSwitch`	**254-418**	**Canvas-based boolean switch bound to a Tk variable.**	**Class container**
`ToggleSwitch.__init__` __init__(self, master, variable, command=None, width=42, height=22, cursor='hand2', **kwargs)	257-292	Init.	`super().__init__, self.bind, self.variable.trace_add, self._redraw, self._schedule_final_redraw`
`ToggleSwitch._schedule_final_redraw` _schedule_final_redraw(self)	294-300	Schedules final redraw.	`self.after, self.after_cancel`
`ToggleSwitch._final_redraw` _final_redraw(self)	302-307	Final redraw.	`self._redraw`
`ToggleSwitch.set_dark_mode` set_dark_mode(self, enabled)	309-312	Set dark mode.	`bool, self._redraw, self._schedule_final_redraw`
`ToggleSwitch._parent_bg` _parent_bg(self)	314-321	Parent bg.	`self.winfo_rgb`
`ToggleSwitch._press` _press(self, _event=None)	323-326	Press.	—
`ToggleSwitch._toggle` _toggle(self, _event=None)	328-332	Toggle.	`self.variable.set, callable, self.command, bool, self.variable.get`
`ToggleSwitch._enter` _enter(self, _event=None)	334-336	Enter.	`self._redraw`
`ToggleSwitch._leave` _leave(self, _event=None)	338-340	Leave.	`self._redraw`

Callable	Lines	Responsibility	Representative direct calls
`ToggleSwitch._redraw` _redraw(self, _event=None)	342-418	Redraw.	`bool, self.configure, self.delete, self.create_rectangle, self.create_oval`

Detailed behavior of the key paths

`RefreshButton` lines 7-250

Canvas-based square button that draws the ↻ glyph and recentres it from its actual rendered bounding box. It supports normal/hover/pressed/disabled states and palette changes.

`ToggleSwitch` lines 254-418

Canvas-based boolean switch bound to a Tk variable. It toggles the variable, invokes the callback and redraws track/knob colors while preserving fixed dimensions.

mb_gui.py

Main application composition and global UI behavior. SnifferApp combines MasterMixin, SlaveFinderMixin, CaptureMixin and ViewMixin with tk.Tk. It creates all widgets, initializes state, freezes language-independent geometry, performs runtime translation, manages theme/language persistence, prevents mouse-wheel tab switching and keeps Bus Health content responsive.

Source lines	SHA-256
2529	`52f2770e0766f002518b4adbd48180147ad60319194dd21ee8bfce1f57eb65d4`

Imports: `queue, threading, collections.deque, tkinter, tkinter.ttk, tkinter.font, mb_config.APP_NAME, APP_VERSION, DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS, DISCOVERY_IDLE_CHECK_SECONDS, MAX_PENDING_REQUEST_TIMEOUT_SECONDS, MAX_RAW_TEXT_LINES, MAX_UI_FRAMES, MIN_PENDING_REQUEST_TIMEOUT_SECONDS, SLOW_RESPONSE_THRESHOLD_MS, SNIFFER_SPLIT_FRAME_HOLD_MS, debug_sim, load_ui_settings, resource_path, save_ui_settings, mb_protocol.SessionMetrics, mb_slave_finder.SlaveFinderMixin, mb_master_ui.MasterMixin, mb_capture.CaptureMixin, mb_view.ViewMixin, mb_widgets.RefreshButton, ToggleSwitch, mb_theme.apply_modern_theme, apply_theme, theme_colors, mb_i18n.LANGUAGE_CHOICES, LANGUAGE_ENGLISH, LANGUAGE_PORTUGUESE, canonical_ui_text, english_help_blocks, normalize_language, translate_text, translate_runtime_text`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class SnifferApp`	**47-2528**	**SnifferApp.**	**Class container**
`SnifferApp.__init__` __init__(self)	48-153	Creates the root window, loads persisted preferences before widget construction, initializes capture/discovery/log/history state, constructs the UI, freezes language geometry, applies language, installs event guards/persistence and schedules queue/status polling.	`super().__init__, self.title, load_ui_settings, normalize_language, bool`
`SnifferApp._freeze_language_layout_geometry` _freeze_language_layout_geometry(self)	156-216	Measures the canonical Portuguese layout and freezes key container dimensions/minimums so changing to English changes text only, not panel/control geometry.	`getattr, self.update_idletasks, enumerate, max, overlay.place_configure`

Callable	Lines	Responsibility	Representative direct calls
`SnifferApp._guard_mouse_wheel` _guard_mouse_wheel(self, event)	218-234	Allows normal wheel scrolling only on truly scrollable widgets and consumes wheel events elsewhere so a ttk Notebook cannot change tabs from mouse-wheel input.	`getattr, isinstance`
`SnifferApp._install_mouse_wheel_guard` _install_mouse_wheel_guard(self)	236-270	Prepend one application bindtag to every existing widget.	`stack.pop, self.bind_class, tuple, stack.extend, widget.bindtags`
`SnifferApp._apply_classic_widget_palette` _apply_classic_widget_palette(self, dark)	272-325	Apply palette to the small set of classic Tk widgets.	`theme_colors, hasattr, self.raw_text.configure, self.help_text.configure, getattr`
`SnifferApp.configure_inspector_result_styles` configure_inspector_result_styles(self)	327-373	Match the Inspector Result foreground to the Traffic result palette.	`theme_colors, self.ui_style.configure, hasattr, bool, tag_map.items`
`SnifferApp.tr` tr(self, text)	375-377	Translate one canonical UI string to the currently selected language.	`translate_text`
`SnifferApp.trf` trf(self, text, **values)	379-381	Translate a format template and substitute values afterwards.	`self.tr(text).format, self.tr`
`SnifferApp._capture_portuguese_help` _capture_portuguese_help(self)	383-401	Store the original PT-PT Help text and tag ranges exactly as built.	`hasattr, self.help_text.get, self.help_text.tag_names, list, self.help_text.tag_ranges`
`SnifferApp._render_help_language` _render_help_language(self)	403-431	Render Help in English or restore the original PT-PT text byte-for-byte.	`hasattr, self.help_text.configure, self.help_text.delete, self.help_text.see, f'{MAX_UI_FRAMES:_}'.replace`
`SnifferApp._translate_widget_tree` _translate_widget_tree(self)	433-465	Translate static widget captions without changing widget geometry.	`stack.pop, isinstance, stack.extend, widget.keys, widget.winfo_children`
`SnifferApp._translate_traffic_context_menu` _translate_traffic_context_menu(self)	467-483	Translate traffic context menu.	`getattr, menu.entryconfigure, self.tr`
`SnifferApp._apply_table_language` _apply_table_language(self)	485-537	Refresh headings and filter values that are not normal widget captions.	`hasattr, canonical_ui_text, self.advanced_filter_type_combo.configure, self.advanced_filter_type_var.set, self.time_order_var.set`

Callable	Lines	Responsibility	Representative direct calls
`SnifferApp.apply_language` apply_language(self, save=True)	539-600	Performs runtime translation of static widgets, tables, context menus, help content and live status strings, then re-applies saved/fixed geometry and optionally persists the selected language.	`getattr, self._translate_widget_tree, self._apply_table_language, self._translate_traffic_context_menu, self._render_help_language`
`SnifferApp.on_language_changed` on_language_changed(self, _event=None)	602-603	Event handler for language changed.	`self.apply_language`
`SnifferApp.toggle_dark_mode` toggle_dark_mode(self)	605-622	Applies the alternate palette using the already-created ttk geometry and updates custom Canvas widgets; the control structure itself is not rebuilt.	`bool, apply_theme, self._apply_classic_widget_palette, self.save_current_ui_settings, self.dark_mode_var.get`
`SnifferApp._restore_saved_main_tab` _restore_saved_main_tab(self)	624-630	Restore saved main tab.	`self.ui_settings.get, self.main_notebook.select, pages.get`
`SnifferApp._current_main_tab_name` _current_main_tab_name(self)	632-641	Current main tab name.	`self.main_notebook.select, str`
`SnifferApp._schedule_ui_settings_save` _schedule_ui_settings_save(self, *_args)	643-660	Schedules ui settings save.	`getattr, self.after, self.after_cancel`
`SnifferApp.save_current_ui_settings` save_current_ui_settings(self)	662-736	Collects validated user-facing preferences from both Sniffer and Finder and writes them to the shared settings file.	`getattr, save_ui_settings, self.after_cancel, int, bool`
`SnifferApp._install_ui_settings_persistence` _install_ui_settings_persistence(self)	738-782	Debounced persistence for all roaming-safe user preferences.	`self.main_notebook.bind, variable.trace_add, mapping.values, self._schedule_ui_settings_save`
`SnifferApp._schedule_bus_health_layout_sync` _schedule_bus_health_layout_sync(self, _event=None)	784-798	Debounce Bus Health reflow after notebook/window/Inspector resizing.	`getattr, self.after_idle, self.after_cancel`
`SnifferApp._initialize_bus_health_layout` _initialize_bus_health_layout(self)	800-822	Measure the natural full Bus Health height before adaptive reflow.	`getattr, self._sync_bus_health_layout, self.update_idletasks, card.winfo_reqheight, max`
`SnifferApp._apply_bus_health_layout` _apply_bus_health_layout(self, compact)	824-967	Re-grids Bus Health labels/values into equal-height rows.	`getattr, bool, range, label_widget.grid_forget, value_widget.grid_forget`

Callable	Lines	Responsibility	Representative direct calls
`SnifferApp._sync_bus_health_layout` _sync_bus_health_layout(self)	969-1022	Chooses full versus compact Bus Health arrangement from real available height with hysteresis, then reapplies wrapping/column behavior on width changes.	`getattr, health_frame.winfo_height, max, bool, health_frame.winfo_exists`
`SnifferApp._build_ui` _build_ui(self)	1024-2528	Constructs all top-level tabs and controls.	`self.ui_settings.get, self.columnconfigure, self.rowconfigure, ttk.Notebook, self.main_notebook.grid`

Detailed behavior of the key paths

`SnifferApp.__init__` lines 48-153

Creates the root window, loads persisted preferences before widget construction, initializes capture/discovery/log/history state, constructs the UI, freezes language geometry, applies language, installs event guards/persistence and schedules queue/status polling.

`_freeze_language_layout_geometry` lines 156-216

Measures the canonical Portuguese layout and freezes key container dimensions/minimums so changing to English changes text only, not panel/control geometry.

`_guard_mouse_wheel` lines 218-234

Allows normal wheel scrolling only on truly scrollable widgets and consumes wheel events elsewhere so a ttk Notebook cannot change tabs from mouse-wheel input.

`apply_language` lines 539-600

Performs runtime translation of static widgets, tables, context menus, help content and live status strings, then re-applies saved/fixed geometry and optionally persists the selected language.

`toggle_dark_mode` lines 605-622

Applies the alternate palette using the already-created ttk geometry and updates custom Canvas widgets; the control structure itself is not rebuilt.

`save_current_ui_settings` lines 662-736

Collects validated user-facing preferences from both Sniffer and Finder and writes them to the shared settings file. COM port names, window geometry and safety confirmation are intentionally excluded.

`_apply_bus_health_layout` **lines 824-967**

Re-grids Bus Health labels/values into equal-height rows. Full mode uses one metric pair per row; compact mode uses two pairs per row and controlled wrapping to keep content inside each rectangle.

`_sync_bus_health_layout` **lines 969-1022**

Chooses full versus compact Bus Health arrangement from real available height with hysteresis, then reapplies wrapping/column behavior on width changes.

`_build_ui` **lines 1024-2528**

Constructs all top-level tabs and controls. It is intentionally a large declarative layout method: global language/theme overlay, Sniffer configuration/actions/statistics, Traffic/Raw/Bus Health/Help notebook, filters/context menu, Frame Inspector and the Bus Slave Finder page.

> **Design note.** SnifferApp uses mixins to keep protocol capture, Finder, Master and view behavior in separate files while sharing one Tk root and common state.

test_regression.py

Automated regression suite. The tests protect RTU timing, parser/resynchronisation behavior, pairing/timeouts, broadcast and echo-write classification, split-frame reassembly, in-place timeout marking, Slave Finder attempts, idle-bus check and FC43/14 logic, plus reversible Portuguese/English translation. Fake serial classes keep the tests deterministic and hardware-independent.

Source lines	SHA-256
787	`b5a3f7e75c0b385f8cb874fdf3b16c8e867108c59406078d966f333db4dba70c`

Imports: `json, queue, random, tempfile, threading, time, unittest, collections.deque, pathlib.Path, mb_config, mb_slave_finder, mb_capture.CaptureMixin, mb_protocol.SessionMetrics, append_crc, burst_may_continue, crc16_modbus, crc_ok, modbus_rtu_interframe_gap_seconds, parse_burst, read_bits_details, split_capture_buffer, mb_slave_finder.SlaveFinderMixin, mb_view.ViewMixin, mb_diagnostics.inspector_values, mb_i18n.LANGUAGE_ENGLISH, LANGUAGE_PORTUGUESE, english_help_blocks, normalize_language, translate_details, translate_runtime_text, translate_text, mb_slave_finder.build_device_identification_probe, build_discovery_probe, device_identification_timeout_seconds, find_device_identification_response, find_discovery_response, format_device_identification, read_basic_device_identification`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`with_bad_crc` with_bad_crc(frame)	49-52	With bad crc.	`bytearray, bytes`
`device_id_response` device_id_response(slave=1, objects=None, *, more_follows=False, next_object_id=0, conformity=1)	55-83	Device id response.	`bytearray, objects.items, append_crc, bytes, payload.extend`
`class FakeSerial`	**86-116**	**Minimal non-blocking serial fake for active Device ID regression tests.**	**Class container**
`FakeSerial.__init__` __init__(self, response_by_object)	89-92	Init.	`dict, bytearray`
`FakeSerial.in_waiting` in_waiting(self)	95-96	In waiting.	`len`
`FakeSerial.reset_input_buffer` reset_input_buffer(self)	98-99	Reset input buffer.	`self.rx.clear`

Callable	Lines	Responsibility	Representative direct calls
`FakeSerial.reset_output_buffer` reset_output_buffer(self)	101-102	Reset output buffer.	—
`FakeSerial.write` write(self, data)	104-110	Write.	`bytes, self.writes.append, self.response_by_object.get, self.rx.extend, len`
`FakeSerial.read` read(self, size)	112-116	Read.	`min, bytes, int, len`
`class FakeSerialModule`	**119-124**	**FakeSerialModule.**	**Class container**
`class StaticVar`	**127-132**	**StaticVar.**	**Class container**
`StaticVar.__init__` __init__(self, value)	128-129	Init.	—
`StaticVar.get` get(self)	131-132	Get.	—
`class DummyCapture`	**135-144**	**DummyCapture.**	**Class container**
`DummyCapture.__init__` __init__(self, *, mode='Auto', baud='9600', data='8', parity='None', stop='1', manual='2.0')	138-144	Init.	`StaticVar`
`class TimingRegressionTests`	**147-179**	**TimingRegressionTests.**	**Class container**
`TimingRegressionTests.test_t35_uses_character_time_at_or_below_19200` test_t35_uses_character_time_at_or_below_19200(self)	148-158	Regression test: t35 uses character time at or below 19200.	`self.assertAlmostEqual, modbus_rtu_interframe_gap_seconds`
`TimingRegressionTests.test_t35_is_fixed_above_19200` test_t35_is_fixed_above_19200(self)	160-167	Regression test: t35 is fixed above 19200.	`self.subTest, self.assertAlmostEqual, modbus_rtu_interframe_gap_seconds`
`TimingRegressionTests.test_sniffer_auto_gap_uses_fixed_1750_us_above_19200` test_sniffer_auto_gap_uses_fixed_1750_us_above_19200(self)	169-171	Regression test: sniffer auto gap uses fixed 1750 us above 19200.	`DummyCapture, self.assertAlmostEqual, capture.calculated_gap_ms`
`TimingRegressionTests.test_sniffer_manual_gap_is_unchanged` test_sniffer_manual_gap_is_unchanged(self)	173-175	Regression test: sniffer manual gap is unchanged.	`DummyCapture, self.assertAlmostEqual, capture.calculated_gap_ms`
`TimingRegressionTests.test_device_id_timeout_covers_a_full_rtu_frame` test_device_id_timeout_covers_a_full_rtu_frame(self)	177-179	Regression test: device id timeout covers a full rtu frame.	`device_identification_timeout_seconds, self.assertGreaterEqual`
`class ParserRegressionTests`	**182-284**	**ParserRegressionTests.**	**Class container**

Callable	Lines	Responsibility	Representative direct calls
`ParserRegressionTests.test_crc_known_request` test_crc_known_request(self)	183-186	Regression test: crc known request.	`build_discovery_probe, self.assertTrue, self.assertEqual, crc_ok, bytes.fromhex`
`ParserRegressionTests.test_invalid_frame_before_valid_frame_stays_crc_error` test_invalid_frame_before_valid_frame_stays_crc_error(self)	188-193	Regression test: invalid frame before valid frame stays crc error.	`with_bad_crc, build_discovery_probe, split_capture_buffer, self.assertEqual`
`ParserRegressionTests.test_long_unsynchronised_noise_recovers_next_valid_frame` test_long_unsynchronised_noise_recovers_next_valid_frame(self)	195-201	Regression test: long unsynchronised noise recovers next valid frame.	`build_discovery_probe, split_capture_buffer, self.assertEqual, self.assertTrue, bytes`
`ParserRegressionTests.test_supported_function_code_shapes` test_supported_function_code_shapes(self)	203-235	Regression test: supported function code shapes.	`frames.items, append_crc, bytes.fromhex, self.subTest, split_capture_buffer`
`ParserRegressionTests.test_random_noise_never_crashes_and_preserves_all_bytes` test_random_noise_never_crashes_and_preserves_all_bytes(self)	237-244	Regression test: random noise never crashes and preserves all bytes.	`random.Random, range, rng.randint, bytes, split_capture_buffer`
`ParserRegressionTests.test_request_response_pairing_and_response_time` test_request_response_pairing_and_response_time(self)	246-269	Regression test: request response pairing and response time.	`SessionMetrics, metrics.process, self.assertEqual, self.assertAlmostEqual, parse_burst`
`ParserRegressionTests.test_timeout_regression` test_timeout_regression(self)	271-284	Regression test: timeout regression.	`SessionMetrics, metrics.process, self.assertEqual, metrics.pop_expired_requests, parse_burst`
`class SlaveFinderRegressionTests`	**287-369**	**SlaveFinderRegressionTests.**	**Class container**
`SlaveFinderRegressionTests.test_normal_discovery_response` test_normal_discovery_response(self)	288-292	Regression test: normal discovery response.	`append_crc, find_discovery_response, self.assertIsNotNone, self.assertEqual, bytes.fromhex`
`SlaveFinderRegressionTests.test_discovery_exception_counts_as_response` test_discovery_exception_counts_as_response(self)	294-299	Regression test: discovery exception counts as response.	`append_crc, find_discovery_response, self.assertIsNotNone, self.assertEqual, bytes.fromhex`
`SlaveFinderRegressionTests.test_device_identification_request` test_device_identification_request(self)	301-305	Regression test: device identification request.	`build_device_identification_probe, self.assertEqual, self.assertTrue, bytes.fromhex, len`

Callable	Lines	Responsibility	Representative direct calls
`SlaveFinderRegressionTests.test_device_identification_response_parser` test_device_identification_response_parser(self)	307-318	Regression test: device identification response parser.	`device_id_response, find_device_identification_response, self.assertIsNotNone, self.assertEqual, format_device_identification`
`SlaveFinderRegressionTests.test_device_identification_parser_ignores_request_echo` test_device_identification_parser_ignores_request_echo(self)	320-325	Regression test: device identification parser ignores request echo.	`build_device_identification_probe, device_id_response, find_device_identification_response, self.assertIsNotNone, self.assertEqual`
`SlaveFinderRegressionTests.test_device_identification_exception` test_device_identification_exception(self)	327-333	Regression test: device identification exception.	`append_crc, find_device_identification_response, self.assertIsNotNone, self.assertEqual, self.assertIn`
`SlaveFinderRegressionTests.test_device_identification_rejects_bad_crc` test_device_identification_rejects_bad_crc(self)	335-337	Regression test: device identification rejects bad crc.	`with_bad_crc, self.assertIsNone, device_id_response, find_device_identification_response`
`SlaveFinderRegressionTests.test_device_identification_follows_more_follows` test_device_identification_follows_more_follows(self)	339-369	Regression test: device identification follows more follows.	`device_id_response, FakeSerial, read_basic_device_identification, self.assertIsNotNone, self.assertEqual`
`class InternationalizationRegressionTests`	**372-422**	**InternationalizationRegressionTests.**	**Class container**
`InternationalizationRegressionTests.test_language_normalization_defaults_to_portuguese` test_language_normalization_defaults_to_portuguese(self)	373-375	Regression test: language normalization defaults to portuguese.	`self.assertEqual, normalize_language`
`InternationalizationRegressionTests.test_static_ui_translation_is_reversible` test_static_ui_translation_is_reversible(self)	377-383	Regression test: static ui translation is reversible.	`self.assertEqual, translate_text`
`InternationalizationRegressionTests.test_dynamic_status_translation_is_reversible` test_dynamic_status_translation_is_reversible(self)	385-389	Regression test: dynamic status translation is reversible.	`self.assertEqual, translate_runtime_text`

Callable	Lines	Responsibility	Representative direct calls
`InternationalizationRegressionTests.test_register_details_follow_language` test_register_details_follow_language(self)	391-399	Regression test: register details follow language.	`self.assertEqual, translate_details`
`InternationalizationRegressionTests.test_inspector_english_labels_and_crc` test_inspector_english_labels_and_crc(self)	401-411	Regression test: inspector english labels and crc.	`append_crc, row.update, inspector_values, self.assertTrue, self.assertIn`
`InternationalizationRegressionTests.test_english_help_contains_no_portuguese_diacritics` test_english_help_contains_no_portuguese_diacritics(self)	413-422	Regression test: english help contains no portuguese diacritics.	`''.join, self.assertIn, self.assertNotIn, text.casefold, english_help_blocks`
`bitwise_crc16` bitwise_crc16(data)	425-432	Reference CRC used up to v3.0, kept to verify the table-driven CRC.	`range`
`process_frames` process_frames(metrics, frames)	435-442	Run (bytes, channel, epoch) frames through SessionMetrics.	`enumerate, parse_burst, out.append, metrics.process`
`class ProtocolV31RegressionTests`	**445-538**	**ProtocolV31RegressionTests.**	**Class container**
`ProtocolV31RegressionTests.test_crc_table_matches_bitwise_reference` test_crc_table_matches_bitwise_reference(self)	446-450	Regression test: crc table matches bitwise reference.	`random.Random, bytes, self.assertEqual, crc16_modbus, bitwise_crc16`
`ProtocolV31RegressionTests.test_broadcast_request_never_times_out` test_broadcast_request_never_times_out(self)	452-461	Regression test: broadcast request never times out.	`append_crc, SessionMetrics, process_frames, self.assertEqual, metrics.expire_pending`
`ProtocolV31RegressionTests.test_fc06_echo_pair_is_request_then_response` test_fc06_echo_pair_is_request_then_response(self)	463-473	Regression test: fc06 echo pair is request then response.	`append_crc, SessionMetrics, process_frames, self.assertEqual, self.assertAlmostEqual`
`ProtocolV31RegressionTests.test_unanswered_fc05_write_times_out` test_unanswered_fc05_write_times_out(self)	475-480	Regression test: unanswered fc05 write times out.	`append_crc, SessionMetrics, process_frames, self.assertEqual, bytes.fromhex`
`ProtocolV31RegressionTests.test_different_fc06_writes_are_both_requests` test_different_fc06_writes_are_both_requests(self)	482-491	Regression test: different fc06 writes are both requests.	`append_crc, SessionMetrics, process_frames, self.assertEqual, bytes.fromhex`

Callable	Lines	Responsibility	Representative direct calls
`ProtocolV31RegressionTests.test_dual_rx_echo_must_come_from_opposite_channel` test_dual_rx_echo_must_come_from_opposite_channel(self)	493-504	Regression test: dual rx echo must come from opposite channel.	`append_crc, SessionMetrics, process_frames, self.assertEqual, bytes.fromhex`
`ProtocolV31RegressionTests.test_split_frame_is_held_until_complete` test_split_frame_is_held_until_complete(self)	506-511	Regression test: split frame is held until complete.	`append_crc, self.assertTrue, self.assertFalse, burst_may_continue, bytes.fromhex`
`ProtocolV31RegressionTests.test_merged_frames_keep_their_chunk_arrival_times` test_merged_frames_keep_their_chunk_arrival_times(self)	513-527	Regression test: merged frames keep their chunk arrival times.	`append_crc, parse_burst, self.assertEqual, self.assertAlmostEqual, SessionMetrics`
`ProtocolV31RegressionTests.test_read_bits_details_use_request_qty` test_read_bits_details_use_request_qty(self)	529-538	Regression test: read bits details use request qty.	`append_crc, self.assertEqual, bytes.fromhex, read_bits_details`
`class _ChunkedSerial`	**541-568**	**Delivers byte chunks at fixed delays, like a USB latency timer.**	**Class container**
`_ChunkedSerial.__init__` __init__(self, chunks, stop_event)	544-547	Init.	`time.perf_counter, list`
`_ChunkedSerial._due` _due(self)	549-555	Due.	`time.perf_counter, self.stop_event.set`
`_ChunkedSerial.in_waiting` in_waiting(self)	558-559	In waiting.	`len, self._due`
`_ChunkedSerial.read` read(self, size)	561-565	Read.	`self._due, self.chunks.pop`
`_ChunkedSerial.close` close(self)	567-568	Close.	—
`class _ReaderStub`	**571-574**	**_ReaderStub.**	**Class container**
`_ReaderStub.__init__` __init__(self)	572-574	Init.	`threading.Event, queue.Queue`
`class ReaderV31RegressionTests`	**577-591**	**ReaderV31RegressionTests.**	**Class container**
`ReaderV31RegressionTests.test_frame_split_by_usb_latency_is_reassembled` test_frame_split_by_usb_latency_is_reassembled(self)	578-591	Regression test: frame split by usb latency is reassembled.	`append_crc, _ReaderStub, _ChunkedSerial, reader.reader_loop, self.assertEqual`
`class _FakeVar`	**594-599**	**_FakeVar.**	**Class container**

Callable	Lines	Responsibility	Representative direct calls
`_FakeVar.__init__` __init__(self, value)	595-596	Init.	—
`_FakeVar.get` get(self)	598-599	Get.	—
`class _FakeTree`	**602-607**	**_FakeTree.**	**Class container**
`_FakeTree.__init__` __init__(self)	603-604	Init.	—
`_FakeTree.item` item(self, iid, tags=())	606-607	Item.	—
`class _TimeoutView`	**610-627**	**_TimeoutView.**	**Class container**
`_TimeoutView.__init__` __init__(self, type_filter)	611-621	Init.	`_FakeTree, _FakeVar, set, deque`
`_TimeoutView.render_filtered_view` render_filtered_view(self)	623-624	Render filtered view.	—
`_TimeoutView.selected_record` selected_record(self)	626-627	Selected record.	—
`class ViewV31RegressionTests`	**630-657**	**ViewV31RegressionTests.**	**Class container**
`ViewV31RegressionTests.test_timeout_retags_rows_without_full_rebuild` test_timeout_retags_rows_without_full_rebuild(self)	631-636	Regression test: timeout retags rows without full rebuild.	`_TimeoutView, self.assertEqual, view.mark_timed_out_records`
`ViewV31RegressionTests.test_timeouts_filter_still_rebuilds_view` test_timeouts_filter_still_rebuilds_view(self)	638-641	Regression test: timeouts filter still rebuilds view.	`_TimeoutView, view.mark_timed_out_records, self.assertEqual`
`ViewV31RegressionTests.test_inspector_shows_address_of_each_register` test_inspector_shows_address_of_each_register(self)	643-650	Regression test: inspector shows address of each register.	`append_crc, inspector_values, self.assertIn, bytes.fromhex, parse_burst`
`ViewV31RegressionTests.test_inspector_shows_address_of_each_bit` test_inspector_shows_address_of_each_bit(self)	652-657	Regression test: inspector shows address of each bit.	`append_crc, row.update, inspector_values, self.assertIn, bytes.fromhex`
`class _FakeFinderSerial`	**660-685**	**_FakeFinderSerial.**	**Class container**
`_FakeFinderSerial.__init__` __init__(self, incoming=b'')	661-663	Init.	`bytearray`

Callable	Lines	Responsibility	Representative direct calls
`_FakeFinderSerial.in_waiting` in_waiting(self)	666-667	In waiting.	`len`
`_FakeFinderSerial.read` read(self, size)	669-672	Read.	`bytes`
`_FakeFinderSerial.write` write(self, data)	674-676	Write.	`self.writes.append, len, bytes`
`_FakeFinderSerial.reset_input_buffer` reset_input_buffer(self)	678-679	Reset input buffer.	—
`_FakeFinderSerial.reset_output_buffer` reset_output_buffer(self)	681-682	Reset output buffer.	—
`_FakeFinderSerial.close` close(self)	684-685	Close.	—
`class _FinderStub`	**688-714**	**_FinderStub.**	**Class container**
`_FinderStub.__init__` __init__(self, fake_serial)	691-695	Init.	`queue.Queue, threading.Event`
`_FinderStub.tr` tr(self, text)	697-698	Tr.	`translate_text`
`_FinderStub.trf` trf(self, text, **values)	700-701	Trf.	`self.tr(text).format, self.tr`
`_FinderStub.open_discovery_serial_port` open_discovery_serial_port(self, serial, port, baud, parity, stopbits)	703-704	Open discovery serial port.	—
`_FinderStub.run_worker` run_worker(self, slaves, attempts)	706-714	Run worker.	`self.discovery_worker, self.event_queue.empty, events.append, self.event_queue.get`
`class FinderV31RegressionTests`	**717-763**	**FinderV31RegressionTests.**	**Class container**
`FinderV31RegressionTests.setUp` setUp(self)	718-720	SetUp.	—
`FinderV31RegressionTests.tearDown` tearDown(self)	722-723	TearDown.	—
`FinderV31RegressionTests.test_each_silent_slave_gets_all_attempts` test_each_silent_slave_gets_all_attempts(self)	725-730	Regression test: each silent slave gets all attempts.	`_FinderStub, stub.run_worker, self.assertIsNone, self.assertEqual, _FakeFinderSerial`

Callable	Lines	Responsibility	Representative direct calls
`FinderV31RegressionTests.test_traffic_before_scan_blocks_transmission` test_traffic_before_scan_blocks_transmission(self)	732-737	Regression test: traffic before scan blocks transmission.	`_FinderStub, stub.run_worker, self.assertIn, self.assertEqual, _FakeFinderSerial`
`FinderV31RegressionTests.test_legacy_default_timeout_migrates` test_legacy_default_timeout_migrates(self)	739-763	Regression test: legacy default timeout migrates.	`tempfile.TemporaryDirectory, Path, path.write_text, self.assertEqual, json.dumps`
`class SourceHygieneV31Tests`	**766-783**	**SourceHygieneV31Tests.**	**Class container**
`SourceHygieneV31Tests.test_no_inline_language_branches` test_no_inline_language_branches(self)	767-773	Regression test: no inline language branches.	`package.glob, source.read_text, self.assertNotIn, Path(__file__).resolve, Path`
`SourceHygieneV31Tests.test_message_templates_translate` test_message_templates_translate(self)	775-783	Regression test: message templates translate.	`self.assertEqual, translate_text('Encontrados: {count}', LANGUAGE_ENGLISH).format, translate_text`

test_master.py

Master regression tests. An in-memory Modbus slave behind a fake serial port answers reads and writes, so the tests cover data formats, frames from the Modbus specification, response matching, scheduling, writes, broadcast, timeouts, the idle-bus check and workspace validation without hardware.

Source lines
300

SHA-256
`bd82e2e3af578d5b0d6e21c1bf55a1d2abcaba510b629198a5447bfa9089a19c`

Imports: `queue, struct, time, unittest, mb_master.BYTE_ORDERS, MasterEngine, PollDefinition, build_read_request, build_write_request, encode_coil_value, encode_register_value, find_master_response, format_register_values, mb_master_ui.MasterMixin, MasterWindow, mb_protocol.append_crc, crc_ok`

Complete callable reference

The table below includes every top-level function, class and class method in this file. Line numbers refer to the exact v3.2 source snapshot documented here.

Callable	Lines	Responsibility	Representative direct calls
`class FakeSlaveSerial`	**21-103**	**In-memory Modbus RTU slave behind a fake serial port.**	**Class container**
`FakeSlaveSerial.__init__` __init__(self, slave_id=1, *, echo=False, bus_traffic=b'', exception_for=None)	29-36	Init.	`bytearray, range`
`FakeSlaveSerial.in_waiting` in_waiting(self)	39-40	In waiting.	`len`
`FakeSlaveSerial.read` read(self, size)	42-45	Read.	`bytes`
`FakeSlaveSerial.reset_input_buffer` reset_input_buffer(self)	47-48	Reset input buffer.	`self.rx.clear`
`FakeSlaveSerial.reset_output_buffer` reset_output_buffer(self)	50-51	Reset output buffer.	—
`FakeSlaveSerial.cancel_read` cancel_read(self)	53-54	Cancel read.	—
`FakeSlaveSerial.cancel_write` cancel_write(self)	56-57	Cancel write.	—
`FakeSlaveSerial.close` close(self)	59-60	Close.	—

Callable	Lines	Responsibility	Representative direct calls
`FakeSlaveSerial.write` write(self, data)	62-71	Write.	`bytes, self.requests.append, len, self.rx.extend, crc_ok`
`FakeSlaveSerial._answer` _answer(self, req)	73-103	Answer.	`append_crc, b''.join, bytearray, range, bytes`
`drain` drain(event_queue)	106-110	Drain.	`event_queue.empty, events.append, event_queue.get`
`make_engine` make_engine(serial, **overrides)	113-117	Make engine.	`dict, options.update, MasterEngine, queue.Queue`
`class MasterFormatTests`	**120-156**	**MasterFormatTests.**	**Class container**
`MasterFormatTests.test_32_bit_formats_round_trip_in_every_byte_order` test_32_bit_formats_round_trip_in_every_byte_order(self)	121-128	Regression test: 32 bit formats round trip in every byte order.	`encode_register_value, self.assertEqual, len, format_register_values`
`MasterFormatTests.test_float_byte_orders_match_wire_layout` test_float_byte_orders_match_wire_layout(self)	130-136	Regression test: float byte orders match wire layout.	`struct.pack, self.assertEqual, encode_register_value, bytes.fromhex`
`MasterFormatTests.test_16_bit_formats` test_16_bit_formats(self)	138-145	Regression test: 16 bit formats.	`self.assertEqual, format_register_values, encode_register_value`
`MasterFormatTests.test_values_outside_the_format_are_rejected` test_values_outside_the_format_are_rejected(self)	147-153	Regression test: values outside the format are rejected.	`self.assertRaises, encode_coil_value, encode_register_value`
`MasterFormatTests.test_unpaired_register_in_32_bit_format` test_unpaired_register_in_32_bit_format(self)	155-156	Regression test: unpaired register in 32 bit format.	`self.assertEqual, format_register_values`
`class MasterFrameTests`	**159-204**	**MasterFrameTests.**	**Class container**
`MasterFrameTests.test_write_frames_match_modbus_specification_examples` test_write_frames_match_modbus_specification_examples(self)	160-170	Regression test: write frames match modbus specification examples.	`self.assertEqual, bytes.fromhex, build_write_request`
`MasterFrameTests.test_invalid_definitions_are_rejected` test_invalid_definitions_are_rejected(self)	172-178	Regression test: invalid definitions are rejected.	`self.assertRaises, build_write_request, build_read_request`
`MasterFrameTests.test_read_response_found_after_adapter_echo` test_read_response_found_after_adapter_echo(self)	180-185	Regression test: read response found after adapter echo.	`build_read_request, append_crc, find_master_response, self.assertEqual, bytes.fromhex`

Callable	Lines	Responsibility	Representative direct calls
`MasterFrameTests.test_coil_response_is_trimmed_to_quantity` test_coil_response_is_trimmed_to_quantity(self)	187-190	Regression test: coil response is trimmed to quantity.	`build_read_request, find_master_response, self.assertEqual, append_crc, bytes.fromhex`
`MasterFrameTests.test_exception_and_bad_crc` test_exception_and_bad_crc(self)	192-198	Regression test: exception and bad crc.	`build_read_request, append_crc, self.assertEqual, bytearray, self.assertIsNone`
`MasterFrameTests.test_write_response_must_match_request` test_write_response_must_match_request(self)	200-204	Regression test: write response must match request.	`build_write_request, self.assertIsNotNone, append_crc, self.assertIsNone, find_master_response`
`class MasterEngineTests`	**207-276**	**MasterEngineTests.**	**Class container**
`MasterEngineTests.run_engine` run_engine(self, engine, seconds)	208-212	Run engine.	`engine.start, time.sleep, engine.stop, drain`
`MasterEngineTests.test_scheduler_serves_writes_then_read_once_then_due_polls` test_scheduler_serves_writes_then_read_once_then_due_polls(self)	214-226	Regression test: scheduler serves writes then read once then due polls.	`make_engine, engine.set_poll, engine.read_once, engine.queue_write, self.assertEqual`
`MasterEngineTests.test_fast_and_slow_windows_share_one_bus` test_fast_and_slow_windows_share_one_bus(self)	228-242	Regression test: fast and slow windows share one bus.	`FakeSlaveSerial, make_engine, engine.set_poll, self.run_engine, self.assertGreater`
`MasterEngineTests.test_write_is_executed_and_reported` test_write_is_executed_and_reported(self)	244-250	Regression test: write is executed and reported.	`FakeSlaveSerial, make_engine, engine.queue_write, self.assertEqual, build_write_request`
`MasterEngineTests.test_broadcast_write_does_not_wait_for_a_response` test_broadcast_write_does_not_wait_for_a_response(self)	252-260	Regression test: broadcast write does not wait for a response.	`FakeSlaveSerial, make_engine, engine.queue_write, time.perf_counter, engine.transact`
`MasterEngineTests.test_timeout_exception_and_invalid_results` test_timeout_exception_and_invalid_results(self)	262-267	Regression test: timeout exception and invalid results.	`make_engine, self.assertEqual, engine.transact, FakeSlaveSerial, build_read_request`
`MasterEngineTests.test_traffic_on_the_bus_blocks_the_master` test_traffic_on_the_bus_blocks_the_master(self)	269-276	Regression test: traffic on the bus blocks the master.	`FakeSlaveSerial, make_engine, engine.set_poll, self.run_engine, self.assertEqual`
`class MasterWindowTests`	**279-296**	**MasterWindowTests.**	**Class container**

Callable	Lines	Responsibility	Representative direct calls
`MasterWindowTests.test_workspace_entry_round_trip_and_validation` test_workspace_entry_round_trip_and_validation(self)	280-288	Regression test: workspace entry round trip and validation.	`MasterWindow, MasterWindow.validated_kwargs, self.assertEqual, window.to_json, self.assertRaises`
`MasterWindowTests.test_write_dialog_builds_fc16_for_float_values` test_write_dialog_builds_fc16_for_float_values(self)	290-296	Regression test: write dialog builds fc16 for float values.	`MasterWindow, MasterMixin.master_build_write, self.assertEqual, bytes.fromhex, self.assertRaises`

12. Regression-test suite

The documented package was executed with `python -m unittest discover -s tests -p "test_*.py" -v`. All 64 tests passed.

> Verification result: **64 tests run - 64 passed - 0 failures/errors**.

TimingRegressionTests

Protects RTU timing and automatic/manual gap calculations, including the fixed 1.750 ms t3.5 rule above 19200 bit/s and FC43 response timing.

Test	Lines	What it protects
`test_t35_uses_character_time_at_or_below_19200`	148-158	Regression test: t35 uses character time at or below 19200.
`test_t35_is_fixed_above_19200`	160-167	Regression test: t35 is fixed above 19200.
`test_sniffer_auto_gap_uses_fixed_1750_us_above_19200`	169-171	Regression test: sniffer auto gap uses fixed 1750 us above 19200.
`test_sniffer_manual_gap_is_unchanged`	173-175	Regression test: sniffer manual gap is unchanged.
`test_device_id_timeout_covers_a_full_rtu_frame`	177-179	Regression test: device id timeout covers a full rtu frame.

ParserRegressionTests

Protects CRC, resynchronisation, supported frame shapes, random-noise robustness, request/response pairing and timeout behavior.

Test	Lines	What it protects
`test_crc_known_request`	183-186	Regression test: crc known request.
`test_invalid_frame_before_valid_frame_stays_crc_error`	188-193	Regression test: invalid frame before valid frame stays crc error.
`test_long_unsynchronised_noise_recovers_next_valid_frame`	195-201	Regression test: long unsynchronised noise recovers next valid frame.
`test_supported_function_code_shapes`	203-235	Regression test: supported function code shapes.

Test	Lines	What it protects
`test_random_noise_never_crashes_and_preserves_all_bytes`	237-244	Regression test: random noise never crashes and preserves all bytes.
`test_request_response_pairing_and_response_time`	246-269	Regression test: request response pairing and response time.
`test_timeout_regression`	271-284	Regression test: timeout regression.

SlaveFinderRegressionTests

Protects FC03/FC04 discovery and FC43/14 Device Identification parsing, CRC validation, Exceptions, echo rejection and More Follows paging.

Test	Lines	What it protects
`test_normal_discovery_response`	288-292	Regression test: normal discovery response.
`test_discovery_exception_counts_as_response`	294-299	Regression test: discovery exception counts as response.
`test_device_identification_request`	301-305	Regression test: device identification request.
`test_device_identification_response_parser`	307-318	Regression test: device identification response parser.
`test_device_identification_parser_ignores_request_echo`	320-325	Regression test: device identification parser ignores request echo.
`test_device_identification_exception`	327-333	Regression test: device identification exception.
`test_device_identification_rejects_bad_crc`	335-337	Regression test: device identification rejects bad crc.
`test_device_identification_follows_more_follows`	339-369	Regression test: device identification follows more follows.

InternationalizationRegressionTests

Protects language normalization, reversible static/runtime translation, translated register details, Frame Inspector English text and English Help content.

Test	Lines	What it protects
`test_language_normalization_defaults_to_portuguese`	373-375	Regression test: language normalization defaults to portuguese.

Test	Lines	What it protects
`test_static_ui_translation_is_reversible`	377-383	Regression test: static ui translation is reversible.
`test_dynamic_status_translation_is_reversible`	385-389	Regression test: dynamic status translation is reversible.
`test_register_details_follow_language`	391-399	Regression test: register details follow language.
`test_inspector_english_labels_and_crc`	401-411	Regression test: inspector english labels and crc.
`test_english_help_contains_no_portuguese_diacritics`	413-422	Regression test: english help contains no portuguese diacritics.

ProtocolV31RegressionTests

Protects the v3.1 protocol fixes: table CRC against the bit-wise reference, broadcast accounting, FC05/06 echo pairing, dual-RX direction, split-burst detection, per-read timestamps and FC01/02 bit formatting.

Test	Lines	What it protects
`test_crc_table_matches_bitwise_reference`	446-450	Regression test: crc table matches bitwise reference.
`test_broadcast_request_never_times_out`	452-461	Regression test: broadcast request never times out.
`test_fc06_echo_pair_is_request_then_response`	463-473	Regression test: fc06 echo pair is request then response.
`test_unanswered_fc05_write_times_out`	475-480	Regression test: unanswered fc05 write times out.
`test_different_fc06_writes_are_both_requests`	482-491	Regression test: different fc06 writes are both requests.
`test_dual_rx_echo_must_come_from_opposite_channel`	493-504	Regression test: dual rx echo must come from opposite channel.
`test_split_frame_is_held_until_complete`	506-511	Regression test: split frame is held until complete.
`test_merged_frames_keep_their_chunk_arrival_times`	513-527	Regression test: merged frames keep their chunk arrival times.
`test_read_bits_details_use_request_qty`	529-538	Regression test: read bits details use request qty.

ReaderV31RegressionTests

Feeds a frame split by a simulated 16 ms USB latency into the real reader_loop() and requires one CRC-valid RESPONSE.

Test	Lines	What it protects
`test_frame_split_by_usb_latency_is_reassembled`	578-591	Regression test: frame split by usb latency is reassembled.

ViewV31RegressionTests

Protects in-place timeout retagging, the Timeouts-filter rebuild and the address-mapped register/bit data in the Frame Inspector.

Test	Lines	What it protects
`test_timeout_retags_rows_without_full_rebuild`	631-636	Regression test: timeout retags rows without full rebuild.
`test_timeouts_filter_still_rebuilds_view`	638-641	Regression test: timeouts filter still rebuilds view.
`test_inspector_shows_address_of_each_register`	643-650	Regression test: inspector shows address of each register.
`test_inspector_shows_address_of_each_bit`	652-657	Regression test: inspector shows address of each bit.

FinderV31RegressionTests

Runs the real discovery worker against fake serial ports to check attempts, the idle-bus abort and the 40 to 100 ms settings migration.

Test	Lines	What it protects
`test_each_silent_slave_gets_all_attempts`	725-730	Regression test: each silent slave gets all attempts.
`test_traffic_before_scan_blocks_transmission`	732-737	Regression test: traffic before scan blocks transmission.
`test_legacy_default_timeout_migrates`	739-763	Regression test: legacy default timeout migrates.

SourceHygieneV31Tests

Rejects inline language branches in the package and checks that message templates translate.

Test	Lines	What it protects
`test_no_inline_language_branches`	767-773	Regression test: no inline language branches.
`test_message_templates_translate`	775-783	Regression test: message templates translate.

MasterFormatTests

Protects the 16/32-bit data formats, the four byte orders and the rejection of values that do not fit.

Test	Lines	What it protects
`test_32_bit_formats_round_trip_in_every_byte_order`	121-128	Regression test: 32 bit formats round trip in every byte order.
`test_float_byte_orders_match_wire_layout`	130-136	Regression test: float byte orders match wire layout.
`test_16_bit_formats`	138-145	Regression test: 16 bit formats.
`test_values_outside_the_format_are_rejected`	147-153	Regression test: values outside the format are rejected.
`test_unpaired_register_in_32_bit_format`	155-156	Regression test: unpaired register in 32 bit format.

MasterFrameTests

Checks write frames against the Modbus specification examples, definition limits, adapter echoes, exceptions, bad CRC and write response matching.

Test	Lines	What it protects
`test_write_frames_match_modbus_specification_examples`	160-170	Regression test: write frames match modbus specification examples.
`test_invalid_definitions_are_rejected`	172-178	Regression test: invalid definitions are rejected.
`test_read_response_found_after_adapter_echo`	180-185	Regression test: read response found after adapter echo.
`test_coil_response_is_trimmed_to_quantity`	187-190	Regression test: coil response is trimmed to quantity.
`test_exception_and_bad_crc`	192-198	Regression test: exception and bad crc.
`test_write_response_must_match_request`	200-204	Regression test: write response must match request.

MasterEngineTests

Runs MasterEngine against an in-memory slave: scheduling order, fast and slow windows on one bus, writes, broadcast, timeouts, exceptions and the idle-bus abort.

Test	Lines	What it protects
`test_scheduler_serves_writes_then_read_once_then_due_polls`	214-226	Regression test: scheduler serves writes then read once then due polls.

Test	Lines	What it protects
`test_fast_and_slow_windows_share_one_bus`	228-242	Regression test: fast and slow windows share one bus.
`test_write_is_executed_and_reported`	244-250	Regression test: write is executed and reported.
`test_broadcast_write_does_not_wait_for_a_response`	252-260	Regression test: broadcast write does not wait for a response.
`test_timeout_exception_and_invalid_results`	262-267	Regression test: timeout exception and invalid results.
`test_traffic_on_the_bus_blocks_the_master`	269-276	Regression test: traffic on the bus blocks the master.

MasterWindowTests

Protects workspace entry validation and the FC16 float encoding of the write dialog.

Test	Lines	What it protects
`test_workspace_entry_round_trip_and_validation`	280-288	Regression test: workspace entry round trip and validation.
`test_write_dialog_builds_fc16_for_float_values`	290-296	Regression test: write dialog builds fc16 for float values.

13. Build and launch automation

The package contains three Windows batch files. Their complete source is reproduced in Appendix A.

MBSniffer.bat

Windows batch automation included in the package.

Lines	SHA-256
210	`3c1a036c6ef094c26390c883d8f59aa5e226da5394100edad2e523410fdf5a1d`

- Sets the expected versioned source folder and verifies that MBSniffer.py exists.
- Detects Python 3 through py -3, python on PATH, or standard Python 3.13 install locations.
- If Python is absent and WinGet exists, attempts a silent per-user Python 3.13 installation.
- Verifies tkinter; if missing, attempts a Python repair/reinstall.
- Verifies pyserial and installs it with pip/ensurepip when required.
- Prefers pythonw.exe so the GUI opens without a console window, with python as fallback.

MBSniffer v3.2/build_exe.bat

Windows batch automation included in the package.

Lines	SHA-256
151	`fbe1caf24e1c48fe6a55f959a61ecc0d0f6a50b376f02297bb6c7d642b8b0775`

- Normalizes application/distribution paths and finds Python 3.
- Installs/upgrades pyserial and PyInstaller.
- Runs the complete regression suite before building; any failure cancels the build.
- Cleans prior build/spec/output artifacts.
- Creates a one-file, windowed MBSniffer.exe with the icon embedded when available.

MBSniffer v3.2/run_tests.bat

Windows batch automation included in the package.

Lines	SHA-256
45	`8066035cd8d62a7aded1f8314b457d438795a8c59319bf1290c47bc63d34372f`

- Finds a Python 3 interpreter.
- Runs unittest discovery for tests/test_*.py in verbose mode.
- Returns a non-zero exit code and explicit warning when tests fail, so a release should not proceed.

14. Known boundaries and extension points

What the current v3.2 does not implement

- No Modbus TCP transport.
- Master tab is phase 1 only: no data logging, scaling, conditional colours, charts, Test Center, FC22/23 or OLE-style automation.
- No RTU/TCP Slave Simulator.
- No passive Ethernet packet capture.
- No analog RS485 electrical measurement.

Current parser scope

Known frame-shape recognition covers the Function Codes encoded in `_candidate_prefixes_at()`. An unsupported FC can only be accepted conservatively as `UNKNOWN / OK` when it occupies the complete remaining burst and has a valid CRC. This intentionally favors resynchronisation reliability over aggressive guessing.

Natural refactoring seams

- **Transport abstraction:** `mb_protocol.py` is already mostly transport-independent above RTU framing, making it the natural starting point for a future shared Modbus PDU core.
- **Active client engine:** the Finder already contains robust serial request/write/read/timeout primitives that could inform a future general Master engine, but the thesis-quality design should separate those primitives from the Finder UI.
- **Server simulator:** a future Slave Simulator should reuse a common FC encode/decode layer rather than placing protocol behavior directly inside a tab.
- **Testing:** the existing hardware-independent regression suite is a strong base; future transport/client/server work should extend the same pattern.

> **Documentation intent.** These are architectural extension points inferred from the current code structure, not features claimed to exist in v3.2.

Appendix A - Complete source-code screenshots

Every Python and batch source file in the package is reproduced below as syntax-highlighted screenshots with original line numbers. Very long source lines are visually wrapped inside the screenshot for readability; this does not alter the source file. The SHA-256 values in the module chapters identify the exact files represented.

MBSniffer v3.2/MBSniffer.py

29 source lines.

MBSniffer v3.2/MBSniffer.py — lines 1-29

```
#!/usr/bin/env python3
# MBSniffer v3.2
"""MBSniffer launcher."""

# Simulação:
#   0 = simulação inativa (o botão não aparece no GUI)
#   1 = simulação ativa  (o botão aparece no GUI)
#
# Mantido neste ficheiro para preservar o mesmo ponto de configuração usado
# antes da modularização.
debug_sim = 0

import mb_config

# Apply the launcher switch before importing GUI/capture modules, which read
# the value during import.
mb_config.debug_sim = debug_sim

from mb_config import set_windows_app_user_model_id
from mb_gui import SnifferApp


def main():
    set_windows_app_user_model_id()
    SnifferApp().mainloop()


if __name__ == "__main__":
    main()
```

MBSniffer v3.2/mb_config.py

464 source lines.

MBSniffer v3.2/mb_config.py — lines 1-40

```python
#!/usr/bin/env python3
"""MBSniffer application configuration, constants and filesystem helpers."""

import ctypes
import json
import os
import sys
from pathlib import Path

APP_NAME = "MBSniffer"
APP_VERSION = "3.2"

# Simulação:
#   0 = simulação inativa (o botão não aparece no GUI)
#   1 = simulação ativa  (o botão aparece no GUI)
debug_sim = 0

# Active Modbus RTU slave discovery.
DISCOVERY_BAUD_RATES = (1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200)
DISCOVERY_PARITIES = ("None", "Even", "Odd")
DISCOVERY_STOP_BITS = ("1", "2")
DISCOVERY_DEFAULT_MIN_TIMEOUT_MS = 100
# v3.0 default; settings saved by v3.0 with this value migrate to the new default.
DISCOVERY_LEGACY_DEFAULT_MIN_TIMEOUT_MS = 40
DISCOVERY_MIN_TIMEOUT_MS = 10
DISCOVERY_MAX_TIMEOUT_MS = 2000
DISCOVERY_PROBE_ADDRESS = 0
DISCOVERY_PROBE_QTY = 1
DISCOVERY_WRITE_TIMEOUT_SECONDS = 1.5
DISCOVERY_WRITE_RETRIES = 1
DISCOVERY_RX_BUFFER_LIMIT = 1024
# Total probe attempts per FC when a Slave does not answer.
DISCOVERY_MIN_ATTEMPTS = 1
DISCOVERY_MAX_ATTEMPTS = 3
DISCOVERY_DEFAULT_ATTEMPTS = 1
# Passive listen before the first transmission; any byte aborts the scan.
DISCOVERY_IDLE_CHECK_SECONDS = 1.0

# GUI queue processing limits.
UI_QUEUE_BATCH_SIZE = 200
```

MBSniffer v3.2/mb_config.py — lines 41-80

```
41 UI_QUEUE_IDLE_INTERVAL_MS = 50
42 UI_QUEUE_BUSY_INTERVAL_MS = 10
43
44 # Pending-request timeout.
45 MIN_PENDING_REQUEST_TIMEOUT_SECONDS = 1.0
46 DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS = 10.0
47 MAX_PENDING_REQUEST_TIMEOUT_SECONDS = 300.0
48 PENDING_REQUEST_TIMEOUT_SECONDS = DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS
49
50 # Master (poll) tab. Modbus limits per request; RTU response timeout and the
51 # silent gap between two requests are user settings.
52 MASTER_MAX_READ_BITS = 2000
53 MASTER_MAX_READ_REGISTERS = 125
54 MASTER_MAX_WRITE_BITS = 1968
55 MASTER_MAX_WRITE_REGISTERS = 123
56 MASTER_DEFAULT_RESPONSE_TIMEOUT_MS = 1000
57 MASTER_MIN_RESPONSE_TIMEOUT_MS = 10
58 MASTER_MAX_RESPONSE_TIMEOUT_MS = 10000
59 MASTER_DEFAULT_DELAY_MS = 20
60 MASTER_MIN_DELAY_MS = 0
61 MASTER_MAX_DELAY_MS = 10000
62 MASTER_DEFAULT_SCAN_MS = 1000
63 MASTER_MAX_SCAN_MS = 3600000
64 MASTER_MAX_WINDOWS = 32
65 MASTER_WORKSPACE_FORMAT = "MBSniffer Master workspace"
66 MASTER_WORKSPACE_VERSION = 1
67
68 # Sniffer burst assembly: extra wait when a burst ends with the start of a
69 # recognised frame. USB-serial adapters deliver bytes in latency-timer
70 # chunks (FTDI default 16 ms), which can split one RTU frame in two.
71 SNIFFER_SPLIT_FRAME_HOLD_MS = 40.0
72
73 # Diagnostic thresholds.
74 SLOW_RESPONSE_THRESHOLD_MS = 500.0
75
76 # Standard Modbus exception response codes.
77 MODBUS_EXCEPTION_CODES = {
78     0x01: (
79         "Illegal Function",
80         "A função Modbus pedida não é suportada/permitida pelo dispositivo."
```

MBSniffer v3.2/mb_config.py — lines 81-120

```
    ),
    0x02: (
        "Illegal Data Address",
        "O endereço ou a gama de endereços pedida não é válida neste dispositivo."
    ),
    0x03: (
        "Illegal Data Value",
        "Um valor, quantidade ou campo do pedido é inválido para esta função."
    ),
    0x04: (
        "Server Device Failure",
        "O dispositivo encontrou um erro interno ao tentar executar o pedido."
    ),
    0x05: (
        "Acknowledge",
        "O pedido foi aceite, mas necessita de mais tempo para ser processado."
    ),
    0x06: (
        "Server Device Busy",
        "O dispositivo está ocupado e não consegue processar o pedido neste momento."
    ),
    0x08: (
        "Memory Parity Error",
        "Foi detetado um erro de paridade ao aceder à memória do dispositivo."
    ),
    0x0A: (
        "Gateway Path Unavailable",
        "O gateway não conseguiu estabelecer o caminho para o dispositivo de destino."
    ),
    0x0B: (
        "Gateway Target Device Failed to Respond",
        "O gateway não recebeu resposta do dispositivo de destino."
    ),
}

# Bounded GUI history. The disk log remains complete.
MAX_UI_FRAMES = 20000
MAX_RAW_TEXT_LINES = 20000
UI_PRUNE_CHUNK_FRAMES = 2000
RAW_TEXT_PRUNE_CHUNK_LINES = 2000
```

MBSniffer v3.2/mb_config.py — lines 121-160

```
121 RAW_TRANSACTION_SEPARATOR = "—" * 72 + "\n"
122
123 SIMULATION_FUNCTION_CODES = (1, 2, 3, 4)
124 SIMULATION_EXCEPTION_CODES = (1, 2, 3, 4, 6)
125
126 def resource_path(filename):
127     """Return a bundled/source resource path for normal Python or PyInstaller."""
128     try:
129         base = Path(sys._MEIPASS)
130     except (AttributeError, TypeError):
131         base = Path(__file__).resolve().parent
132     return base / filename
133
134 def set_windows_app_user_model_id():
135     """
136     Give Windows a stable application identity so the taskbar can associate
137     the Tk window with MBSniffer instead of a generic Python process.
138     """
139     if os.name != "nt":
140         return
141     try:
142         ctypes.windll.shell32.SetCurrentProcessExplicitAppUserModelID(
143             f"{APP_NAME}.{APP_VERSION}"
144         )
145     except Exception:
146         pass
147
148
149
150 def get_settings_file() -> Path:
151     """
152     Return the persistent per-user settings file.
153
154     Windows:
155         %APPDATA%\\MBSniffer\\settings.json
156
157     Fallback:
158         ~/.mbsniffer/settings.json
159
160     This location is independent from the EXE/BAT folder, so the selected
```

MBSniffer v3.2/mb_config.py — lines 161-200

```
161     Light/Dark mode survives application updates and also works when the
162     application folder is read-only.
163     """
164     try:
165         if os.name == "nt":
166             roaming_app_data = os.environ.get("APPDATA")
167             if roaming_app_data:
168                 folder = Path(roaming_app_data) / APP_NAME
169             else:
170                 folder = Path.home() / "AppData" / "Roaming" / APP_NAME
171         else:
172             folder = Path.home() / ".mbsniffer"
173 
174         folder.mkdir(parents=True, exist_ok=True)
175         return folder / "settings.json"
176     except Exception:
177         return Path.home() / ".mbsniffer_settings.json"
178 
179 
180 def _validated_choice(value, allowed, default):
181     return value if value in allowed else default
182 
183 
184 def _validated_bool(value, default):
185     return value if isinstance(value, bool) else default
186 
187 
188 def _validated_number_string(value, minimum, maximum, default):
189     try:
190         number = float(value)
191         if minimum <= number <= maximum:
192             return f"{number:g}"
193     except (TypeError, ValueError):
194         pass
195     return str(default)
196 
197 
198 def _validated_int_string(value, minimum, maximum, default):
199     try:
200         number = int(value)
```

MBSniffer v3.2/mb_config.py — lines 201-240

```
201         if minimum <= number <= maximum:
202             return str(number)
203     except (TypeError, ValueError):
204         pass
205     return str(default)
206 
207 
208 def load_ui_settings() -> dict:
209     """
210     Load validated user preferences from the single Roaming settings file.
211 
212     Machine-specific state such as COM-port names and window geometry is
213     deliberately excluded.
214     """
215     path = get_settings_file()
216     try:
217         data = json.loads(path.read_text(encoding="utf-8"))
218     except Exception:
219         return {}
220 
221     if not isinstance(data, dict):
222         return {}
223 
224     result = {}
225 
226     if isinstance(data.get("dark_mode"), bool):
227         result["dark_mode"] = data["dark_mode"]
228 
229     if data.get("language") in ("Português", "English"):
230         result["language"] = data["language"]
231 
232     if data.get("last_main_tab") in ("Sniffer", "Bus Slave Finder", "Master"):
233         result["last_main_tab"] = data["last_main_tab"]
234 
235     sniffer_in = data.get("sniffer")
236     if isinstance(sniffer_in, dict):
237         # v2.9 migration: the UI option was renamed from "Realçar anomalias"
238         # to "Realçar resultados". Accept the old key once so existing user
239         # preferences are not lost, but expose/save only the new name.
240         highlight_results_value = sniffer_in.get(
```

MBSniffer v3.2/mb_config.py — lines 241-280

```
            "highlight_results",
            sniffer_in.get("highlight_anomalies"),
        )

        result["sniffer"] = {
            "physical_mode": _validated_choice(
                sniffer_in.get("physical_mode"),
                ("RS485 2-wire", "RS232 single RX", "RS232 dual RX"),
                "RS485 2-wire",
            ),
            "baud": _validated_choice(
                str(sniffer_in.get("baud", "")),
                tuple(str(v) for v in DISCOVERY_BAUD_RATES),
                "9600",
            ),
            "data_bits": _validated_choice(
                str(sniffer_in.get("data_bits", "")),
                ("5", "6", "7", "8"),
                "8",
            ),
            "parity": _validated_choice(
                sniffer_in.get("parity"),
                ("None", "Even", "Odd", "Mark", "Space"),
                "None",
            ),
            "stop_bits": _validated_choice(
                str(sniffer_in.get("stop_bits", "")),
                ("1", "1.5", "2"),
                "1",
            ),
            "frame_gap_mode": _validated_choice(
                sniffer_in.get("frame_gap_mode"),
                ("Auto", "Manual"),
                "Auto",
            ),
            "frame_gap_ms": str(sniffer_in.get("frame_gap_ms", "")),
            "pending_timeout_s": _validated_number_string(
                sniffer_in.get("pending_timeout_s"),
                MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
                MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
```

MBSniffer v3.2/mb_config.py — lines 281-320

```
                DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS,
            ),
            "auto_scroll": _validated_bool(
                sniffer_in.get("auto_scroll"),
                True,
            ),
            "separate_transactions": _validated_bool(
                sniffer_in.get("separate_transactions"),
                True,
            ),
            "highlight_results": _validated_bool(
                highlight_results_value,
                False,
            ),
        }

    finder_in = data.get("bus_slave_finder")
    if isinstance(finder_in, dict):
        saved_bauds = finder_in.get("baud_rates")
        if isinstance(saved_bauds, list):
            baud_rates = [
                int(v) for v in saved_bauds
                if isinstance(v, (int, float, str))
                and str(v).isdigit()
                and int(v) in DISCOVERY_BAUD_RATES
            ]
        else:
            baud_rates = [9600, 19200]

        saved_parities = finder_in.get("parities")
        if isinstance(saved_parities, list):
            parities = [
                v for v in saved_parities
                if v in DISCOVERY_PARITIES
            ]
        else:
            parities = ["None", "Even"]

        saved_stops = finder_in.get("stop_bits")
        if isinstance(saved_stops, list):
```

MBSniffer v3.2/mb_config.py — lines 321-360

```
            stop_bits = [
                str(v) for v in saved_stops
                if str(v) in DISCOVERY_STOP_BITS
            ]
        else:
            stop_bits = ["1"]

        min_timeout_ms = _validated_int_string(
            finder_in.get("min_timeout_ms"),
            DISCOVERY_MIN_TIMEOUT_MS,
            DISCOVERY_MAX_TIMEOUT_MS,
            DISCOVERY_DEFAULT_MIN_TIMEOUT_MS,
        )
        # Settings written before v3.1 have no "attempts" key. Their 40 ms
        # value is the old automatic default, not a user choice.
        if (
            "attempts" not in finder_in
            and min_timeout_ms == str(DISCOVERY_LEGACY_DEFAULT_MIN_TIMEOUT_MS)
        ):
            min_timeout_ms = str(DISCOVERY_DEFAULT_MIN_TIMEOUT_MS)

        result["bus_slave_finder"] = {
            "baud_rates": baud_rates,
            "parities": parities,
            "stop_bits": stop_bits,
            "slave_start": _validated_int_string(
                finder_in.get("slave_start"),
                1,
                247,
                1,
            ),
            "slave_end": _validated_int_string(
                finder_in.get("slave_end"),
                1,
                247,
                247,
            ),
            "min_timeout_ms": min_timeout_ms,
            "attempts": _validated_int_string(
                finder_in.get("attempts"),
```

MBSniffer v3.2/mb_config.py — lines 361-400

```
361                 DISCOVERY_MIN_ATTEMPTS,
362                 DISCOVERY_MAX_ATTEMPTS,
363                 DISCOVERY_DEFAULT_ATTEMPTS,
364             ),
365             "fc04_fallback": _validated_bool(
366                 finder_in.get("fc04_fallback"),
367                 False,
368             ),
369             "device_identification": _validated_bool(
370                 finder_in.get("device_identification"),
371                 False,
372             ),
373         }
374
375     master_in = data.get("master")
376     if isinstance(master_in, dict):
377         result["master"] = {
378             "baud": _validated_choice(
379                 str(master_in.get("baud", "")),
380                 tuple(str(v) for v in DISCOVERY_BAUD_RATES),
381                 "9600",
382             ),
383             "parity": _validated_choice(
384                 master_in.get("parity"), ("None", "Even", "Odd"), "None"
385             ),
386             "stop_bits": _validated_choice(
387                 str(master_in.get("stop_bits", "")), ("1", "2"), "1"
388             ),
389             "response_timeout_ms": _validated_int_string(
390                 master_in.get("response_timeout_ms"),
391                 MASTER_MIN_RESPONSE_TIMEOUT_MS,
392                 MASTER_MAX_RESPONSE_TIMEOUT_MS,
393                 MASTER_DEFAULT_RESPONSE_TIMEOUT_MS,
394             ),
395             "delay_ms": _validated_int_string(
396                 master_in.get("delay_ms"),
397                 MASTER_MIN_DELAY_MS,
398                 MASTER_MAX_DELAY_MS,
399                 MASTER_DEFAULT_DELAY_MS,
400             ),
```

MBSniffer v3.2/mb_config.py — lines 401-440

```
401         }
402
403     return result
404
405
406 def save_ui_settings(settings=None, **updates) -> None:
407     """
408     Persist UI preferences atomically in the single Roaming settings file.
409
410     Partial updates merge with the existing file. This lets the Dark-mode
411     toggle update immediately without discarding the other remembered options.
412     """
413     path = get_settings_file()
414
415     current = load_ui_settings()
416     if isinstance(settings, dict):
417         current.update(settings)
418     current.update(updates)
419
420     try:
421         path.parent.mkdir(parents=True, exist_ok=True)
422         tmp = path.with_suffix(path.suffix + ".tmp")
423         tmp.write_text(
424             json.dumps(current, indent=2, sort_keys=True),
425             encoding="utf-8",
426         )
427         tmp.replace(path)
428     except Exception:
429         # Remembered UI state must never break application use.
430         pass
431
432
433 def get_application_directory() -> Path:
434     """
435     Return the user-facing application directory.
436
437     - PyInstaller EXE: directory containing MBSniffer.exe.
438     - Packaged Python/BAT mode: directory containing the root MBSniffer.bat.
439       The Python sources live inside the versioned child folder, e.g. ``MBSniffer v3.2``.
440     - Development Python mode: directory containing this module.
```

MBSniffer v3.2/mb_config.py — lines 441-464

```
441     """
442     try:
443         if getattr(sys, "frozen", False):
444             return Path(sys.executable).resolve().parent
445 
446         source_dir = Path(__file__).resolve().parent
447         launcher_dir = source_dir.parent
448 
449         if (launcher_dir / "MBSniffer.bat").is_file():
450             return launcher_dir
451 
452         return source_dir
453     except Exception:
454         return Path.cwd()
455 
456 def get_logs_folder() -> Path:
457     """
458     Return/create the log directory beside MBSniffer.bat or MBSniffer.exe:
459 
460         <launcher/exe directory>/MBSniffer Logs
461     """
462     folder = get_application_directory() / "MBSniffer Logs"
463     folder.mkdir(parents=True, exist_ok=True)
464     return folder
```

MBSniffer v3.2/mb_protocol.py

1007 source lines.

MBSniffer v3.2/mb_protocol.py — lines 1-40

```python
#!/usr/bin/env python3
"""Modbus RTU parsing, CRC, transaction metrics and simulation helpers."""

import math
import random
import time
from bisect import bisect_right
from collections import defaultdict, deque
from datetime import datetime

from mb_config import (
    MODBUS_EXCEPTION_CODES,
    PENDING_REQUEST_TIMEOUT_SECONDS,
    SIMULATION_FUNCTION_CODES,
    SIMULATION_EXCEPTION_CODES,
    SLOW_RESPONSE_THRESHOLD_MS,
)

def _build_crc16_table():
    table = []
    for value in range(256):
        crc = value
        for _ in range(8):
            if crc & 1:
                crc = (crc >> 1) ^ 0xA001
            else:
                crc >>= 1
        table.append(crc)
    return tuple(table)

# Table-driven CRC: one lookup per byte instead of eight shift/XOR steps. The
# resynchronisation scan computes CRCs at many offsets, so this matters on
# noisy buses.
_CRC16_TABLE = _build_crc16_table()

def crc16_modbus(data: bytes) -> int:
    crc = 0xFFFF
    table = _CRC16_TABLE
    for b in data:
        crc = (crc >> 8) ^ table[(crc ^ b) & 0xFF]
```

MBSniffer v3.2/mb_protocol.py — lines 41-80

```
41      return crc
42
43  def crc_ok(frame: bytes) -> bool:
44      if len(frame) < 4:
45          return False
46      expected = crc16_modbus(frame[:-2])
47      received = frame[-2] | (frame[-1] << 8)
48      return expected == received
49
50  def append_crc(payload: bytes) -> bytes:
51      crc = crc16_modbus(payload)
52      return payload + bytes((crc & 0xFF, (crc >> 8) & 0xFF))
53
54  def hex_bytes(data: bytes) -> str:
55      return " ".join(f"{b:02X}" for b in data)
56
57  def modbus_rtu_interframe_gap_seconds(baud: int, bits_per_char: float) -> float:
58      """Return the recommended Modbus RTU t3.5 inter-frame gap.
59
60      Modbus Serial Line V1.02 requires the character-time timers to be
61      respected at baud rates up to and including 19200 bit/s. Above 19200
62      bit/s it recommends a fixed t3.5 value of 1.750 ms.
63      """
64      baud = int(baud)
65      bits_per_char = float(bits_per_char)
66      if baud <= 0:
67          raise ValueError("Baud rate must be greater than zero")
68      if bits_per_char <= 0:
69          raise ValueError("Bits per character must be greater than zero")
70      if baud > 19200:
71          return 0.001750
72      return 3.5 * bits_per_char / float(baud)
73
74  def plausible_request_qty(fc: int, qty: int) -> bool:
75      if fc in (1, 2):
76          return 1 <= qty <= 2000
77      if fc in (3, 4):
78          return 1 <= qty <= 125
79      return True
80
```

MBSniffer v3.2/mb_protocol.py — lines 81-120

```python
# Function Codes whose RTU frame length the parser can derive.
KNOWN_FRAME_FUNCTION_CODES = (1, 2, 3, 4, 5, 6, 7, 8, 15, 16, 22, 23)

# Function Codes whose normal response is an exact copy of the request.
EXACT_ECHO_FUNCTION_CODES = (5, 6, 22)

BROADCAST_UNIT_ID = 0

def _plausible_unit_id(value: int) -> bool:
    """Return True for Modbus RTU unit addresses that can appear on the wire."""
    try:
        return 0 <= int(value) <= 247
    except (TypeError, ValueError):
        return False


def _candidate_prefixes_at(buf, start=0):
    """Return structurally plausible known-frame lengths at *start*."""
    remaining = len(buf) - int(start)
    if remaining < 2:
        return []

    unit = buf[start]
    if not _plausible_unit_id(unit):
        return []

    fc = buf[start + 1]
    out = []

    if fc & 0x80:
        # Exception responses use FC | 0x80 and have a fixed five-byte length.
        base_fc = fc & 0x7F
        if base_fc in KNOWN_FRAME_FUNCTION_CODES:
            out.append((5, "EXCEPTION"))
        return out

    if fc in (1, 2, 3, 4):
        if remaining >= 6:
            qty = (buf[start + 4] << 8) | buf[start + 5]
            if plausible_request_qty(fc, qty):
```

MBSniffer v3.2/mb_protocol.py — lines 121-160

```
121                 out.append((8, "REQUEST"))
122
123         if remaining >= 3:
124             byte_count = buf[start + 2]
125             resp_len = 5 + byte_count
126             if fc in (3, 4):
127                 if byte_count > 0 and byte_count % 2 == 0:
128                     out.append((resp_len, "RESPONSE"))
129             elif byte_count > 0:
130                 out.append((resp_len, "RESPONSE"))
131
132     elif fc in (5, 6, 8):
133         out.append((8, "REQ/RESP"))
134
135     elif fc == 7:
136         # FC07 request: Slave + FC + CRC = 4 bytes.
137         # FC07 response: Slave + FC + Status + CRC = 5 bytes.
138         if remaining >= 5:
139             out.append((5, "RESPONSE"))
140         out.append((4, "REQUEST"))
141
142     elif fc in (15, 16):
143         out.append((8, "RESPONSE"))
144         if remaining >= 7:
145             byte_count = buf[start + 6]
146             if byte_count > 0:
147                 out.append((9 + byte_count, "REQUEST"))
148
149     elif fc == 22:
150         out.append((10, "REQ/RESP"))
151
152     elif fc == 23:
153         if remaining >= 3:
154             bc = buf[start + 2]
155             if bc > 0 and bc % 2 == 0:
156                 out.append((5 + bc, "RESPONSE"))
157         if remaining >= 11:
158             bc = buf[start + 10]
159             if bc > 0:
160                 out.append((13 + bc, "REQUEST"))
```

MBSniffer v3.2/mb_protocol.py — lines 161-200

```
161
162     cleaned = []
163     seen = set()
164     for length, kind in out:
165         if 4 <= length <= 260 and (length, kind) not in seen:
166             seen.add((length, kind))
167             cleaned.append((length, kind))
168     return cleaned
169
170
171 def candidate_prefixes(buf: bytes):
172     """Compatibility wrapper for callers/tests that inspect offset zero."""
173     return _candidate_prefixes_at(buf, 0)
174
175
176 def _crc_ok_at(buf, start: int, length: int) -> bool:
177     """CRC check without copying the whole remaining capture buffer."""
178     if length < 4 or start < 0 or start + length > len(buf):
179         return False
180     end = start + length
181     expected = crc16_modbus(memoryview(buf)[start:end - 2])
182     received = buf[end - 2] | (buf[end - 1] << 8)
183     return expected == received
184
185
186 def find_valid_prefix(buf: bytes, start=0):
187     """Find a CRC-valid, structurally recognised Modbus frame at *start*."""
188     for length, kind in _candidate_prefixes_at(buf, start):
189         if start + length <= len(buf) and _crc_ok_at(buf, start, length):
190             return length, kind
191     return None
192
193
194 def guess_invalid_whole_frame(buf: bytes):
195     """Classify a complete known-shape frame whose CRC is invalid."""
196     if len(buf) < 4 or not _plausible_unit_id(buf[0]):
197         return None
198     for length, kind in _candidate_prefixes_at(buf, 0):
199         if len(buf) == length:
200             return kind
```

MBSniffer v3.2/mb_protocol.py — lines 201-240

```
201     return None
202 
203 
204 def _unknown_whole_frame_is_valid(buf: bytes) -> bool:
205     """Conservative fallback for one unsupported FC occupying the whole burst.
206 
207     Unknown frames cannot be length-validated from their FC, so CRC-only
208     recognition is deliberately restricted to the complete remaining burst.
209     It is never used while scanning arbitrary offsets for resynchronisation.
210     """
211     if not 4 <= len(buf) <= 260:
212         return False
213     if not _plausible_unit_id(buf[0]):
214         return False
215     if _candidate_prefixes_at(buf, 0):
216         return False
217     return crc_ok(buf)
218 
219 
220 def split_capture_buffer(data: bytes):
221     results = []
222     buf = bytearray(data)
223 
224     while buf:
225         match = find_valid_prefix(buf, 0)
226         if match:
227             length, kind = match
228             results.append((bytes(buf[:length]), kind, "OK"))
229             del buf[:length]
230             continue
231 
232         # Search the full remaining burst, but only for structurally recognised
233         # Modbus frames. The former generic CRC scan tried up to 256 candidate
234         # lengths at every offset and became very slow on long noise bursts.
235         # Restricting resynchronisation to known frame shapes makes the scan
236         # effectively linear in the burst size and strongly reduces accidental
237         # CRC-only "phantom" frames in random noise.
238         found_offset = None
239         search_limit = max(len(buf) - 4, 0)
240         for offset in range(1, search_limit + 1):
```

MBSniffer v3.2/mb_protocol.py — lines 241-280

```
241             if find_valid_prefix(buf, offset):
242                 found_offset = offset
243                 break
244 
245         if found_offset is not None:
246             prefix = bytes(buf[:found_offset])
247 
248             # If the bytes immediately before the recovered frame are exactly a
249             # recognised Modbus frame shape, preserve the useful CRC diagnosis
250             # instead of hiding it as RAW/UNSYNC. Otherwise they are genuinely
251             # unsynchronised bytes and remain RAW.
252             guessed = guess_invalid_whole_frame(prefix)
253             if guessed:
254                 results.append((prefix, guessed, "ERROR"))
255             else:
256                 results.append((prefix, "RAW/UNSYNC", "UNKNOWN"))
257             del buf[:found_offset]
258             continue
259 
260         guessed = guess_invalid_whole_frame(bytes(buf))
261         if guessed:
262             results.append((bytes(buf), guessed, "ERROR"))
263         elif _unknown_whole_frame_is_valid(bytes(buf)):
264             results.append((bytes(buf), "UNKNOWN", "OK"))
265         else:
266             results.append((bytes(buf), "RAW/UNPARSED", "UNKNOWN"))
267         break
268 
269     return results
270 
271 def burst_may_continue(data: bytes) -> bool:
272     """Return True when *data* ends with the start of an unfinished frame.
273 
274     USB-serial adapters deliver received bytes in chunks, so a silence longer
275     than t3.5 can occur in the middle of one RTU frame. The reader uses this
276     to wait a little longer before closing such a burst. Only the final
277     unvalidated segment is inspected, and only at its first byte.
278     """
279     segments = split_capture_buffer(data)
280     if not segments:
```

MBSniffer v3.2/mb_protocol.py — lines 281-320

```
281         return False
282
283     tail, _kind, state = segments[-1]
284     if state == "OK":
285         return False
286
287     if not _plausible_unit_id(tail[0]):
288         return False
289     if len(tail) == 1:
290         return True
291
292     fc = tail[1]
293     if (fc & 0x7F) not in KNOWN_FRAME_FUNCTION_CODES:
294         return False
295     if len(tail) < 8 and not fc & 0x80:
296         # Too short for the request/response length fields to be known.
297         return True
298
299     return any(
300         length > len(tail)
301         for length, _kind in _candidate_prefixes_at(tail, 0)
302     )
303
304
305 def read_bits_from_response(frame: bytes, qty=None):
306     """Return FC01/FC02 coil/input states, least-significant bit first.
307
308     Without the matching request quantity the padding bits of the last data
309     byte cannot be distinguished, so every transmitted bit is returned.
310     """
311     if len(frame) < 5:
312         return []
313     byte_count = frame[2]
314     data = frame[3:3 + byte_count]
315     bits = [(byte >> bit) & 1 for byte in data for bit in range(8)]
316     try:
317         qty = int(qty) if qty is not None else None
318     except (TypeError, ValueError):
319         qty = None
320     if qty is not None and 0 < qty <= len(bits):
```

MBSniffer v3.2/mb_protocol.py — lines 321-360

```
321         bits = bits[:qty]
322     return bits
323 
324 
325 # FC01/FC02 can return up to 2000 bits; the Traffic row shows only the first
326 # ones (the Frame Inspector lists them all with their addresses).
327 READ_BITS_DETAILS_LIMIT = 32
328 
329 
330 def read_bits_details(frame: bytes, qty=None) -> str:
331     """Compact Details text: bits in address order, grouped per data byte."""
332     byte_count = frame[2] if len(frame) >= 3 else 0
333     bits = read_bits_from_response(frame, qty)
334     shown = "".join(str(bit) for bit in bits[:READ_BITS_DETAILS_LIMIT])
335     grouped = " ".join(shown[i:i + 8] for i in range(0, len(shown), 8))
336     if len(bits) > READ_BITS_DETAILS_LIMIT:
337         grouped += f" …(+{len(bits) - READ_BITS_DETAILS_LIMIT})"
338     return f"ByteCount={byte_count}  Bits={grouped}"
339 
340 
341 def decode_frame(frame: bytes, kind_hint: str):
342     info = {
343         "kind": kind_hint,
344         "slave": "",
345         "fc": "",
346         "details": "",
347         "pdu_address": None,
348         "one_based": None,
349         "qty": None,
350     }
351 
352     if str(kind_hint).startswith("RAW"):
353         info["details"] = f"{len(frame)} raw bytes"
354         return info
355 
356     if len(frame) < 2:
357         info["details"] = "Raw bytes"
358         return info
359 
360     slave = frame[0]
```

MBSniffer v3.2/mb_protocol.py — lines 361-400

```
    fc = frame[1]
    info["slave"] = str(slave)
    info["fc"] = f"{fc:02X}"

    if fc & 0x80:
        info["kind"] = "EXCEPTION"
        code = frame[2] if len(frame) >= 3 else None

        if code is None:
            info["details"] = "Exception"
            return info

        exception_info = MODBUS_EXCEPTION_CODES.get(code)
        if exception_info is not None:
            name, _description = exception_info
            info["details"] = f"Exception 0x{code:02X} — {name}"
        else:
            info["details"] = f"Exception 0x{code:02X} — Unknown/Reserved Exception"
        return info

    if fc in (1, 2, 3, 4):
        if kind_hint == "REQUEST" and len(frame) >= 8:
            start_addr = (frame[2] << 8) | frame[3]
            qty = (frame[4] << 8) | frame[5]
            info["pdu_address"] = start_addr
            info["one_based"] = start_addr + 1
            info["qty"] = qty
            info["details"] = (
                f"PDU Address={start_addr}  1-based={start_addr + 1}  Qty={qty}"
            )
            return info

        if kind_hint == "RESPONSE" and len(frame) >= 5:
            bc = frame[2]
            details = f"ByteCount={bc}"
            if fc in (3, 4) and bc % 2 == 0 and len(frame) >= 5 + bc:
                data = frame[3:3+bc]
                regs = [(data[i] << 8) | data[i+1] for i in range(0, len(data), 2)]
                details += f"  Registos={regs}"
            elif fc in (1, 2) and len(frame) >= 5 + bc:
```

MBSniffer v3.2/mb_protocol.py — lines 401-440

```
401                 details = read_bits_details(frame)
402             info["details"] = details
403             return info
404
405     if fc in (5, 6, 8) and len(frame) >= 8:
406         addr = (frame[2] << 8) | frame[3]
407         value = (frame[4] << 8) | frame[5]
408         info["pdu_address"] = addr
409         info["one_based"] = addr + 1
410         info["details"] = (
411             f"PDU Address={addr}  1-based={addr + 1}  Value=0x{value:04X}"
412         )
413         return info
414
415     if fc == 7:
416         if kind_hint == "REQUEST":
417             info["details"] = "Read Exception Status"
418         elif kind_hint == "RESPONSE" and len(frame) >= 5:
419             info["details"] = f"Status=0x{frame[2]:02X}"
420         else:
421             info["details"] = "Read Exception Status"
422         return info
423
424     if fc in (15, 16):
425         if kind_hint == "REQUEST" and len(frame) >= 9:
426             start_addr = (frame[2] << 8) | frame[3]
427             qty = (frame[4] << 8) | frame[5]
428             bc = frame[6]
429             info["pdu_address"] = start_addr
430             info["one_based"] = start_addr + 1
431             info["qty"] = qty
432             info["details"] = (
433                 f"PDU Address={start_addr}  1-based={start_addr + 1}  "
434                 f"Qty={qty}  ByteCount={bc}"
435             )
436             return info
437         if len(frame) >= 8:
438             start_addr = (frame[2] << 8) | frame[3]
439             qty = (frame[4] << 8) | frame[5]
440             info["pdu_address"] = start_addr
```

MBSniffer v3.2/mb_protocol.py — lines 441-480

```
            info["one_based"] = start_addr + 1
            info["qty"] = qty
            info["details"] = (
                f"PDU Address={start_addr}  1-based={start_addr + 1}  Qty={qty}"
            )
            return info

    info["details"] = f"{len(frame)} bytes"
    return info

def format_clock_time(epoch: float) -> str:
    return datetime.fromtimestamp(float(epoch)).strftime("%H:%M:%S.%f")[:-3]


def parse_burst(data: bytes, timestamp: str, channel: str = "", event_time=None,
                chunk_times=None):
    """Split one received burst into decoded rows.

    *chunk_times* is an optional ascending list of ``(byte_offset, epoch)``
    pairs recording when each serial read arrived. When given, every frame
    gets the arrival time of the chunk holding its first byte instead of the
    burst start time, so frames merged into one burst keep distinct times.
    """
    rows = []
    if event_time is None:
        event_time = time.time()

    chunk_offsets = [int(o) for o, _t in chunk_times] if chunk_times else []
    offset = 0

    for frame, kind, state in split_capture_buffer(data):
        frame_time = float(event_time)
        frame_timestamp = timestamp
        if chunk_offsets:
            index = bisect_right(chunk_offsets, offset) - 1
            if index > 0:
                frame_time = float(chunk_times[index][1])
                frame_timestamp = format_clock_time(frame_time)
        offset += len(frame)

```

MBSniffer v3.2/mb_protocol.py — lines 481-520

```
481         info = decode_frame(frame, kind)
482         crc_status = "OK" if state == "OK" else ("ERROR" if state == "ERROR" else "?")
483         rows.append({
484             "time": frame_timestamp,
485             "channel": channel,
486             "type": info["kind"],
487             "slave": info["slave"],
488             "fc": info["fc"],
489             "details": info["details"],
490             "crc": crc_status,
491             "raw": hex_bytes(frame),
492             "pdu_address": info["pdu_address"],
493             "one_based": info["one_based"],
494             "qty": info["qty"],
495             "_frame_len": len(frame),
496             "_epoch": frame_time,
497         })
498     return rows
499 
500 class SessionMetrics:
501     """Tracks pairing, counters and bus-health diagnostics for one session."""
502 
503     def __init__(self, pending_timeout_seconds=PENDING_REQUEST_TIMEOUT_SECONDS):
504         self.pending_timeout_seconds = float(pending_timeout_seconds)
505         self.reset()
506 
507     def reset(self):
508         self.first_event_time = None
509         self.last_event_time = None
510         self.pending = {}
511         self.requests = 0
512         # Broadcast requests (Slave 0) never receive a response, so they are
513         # never pending and are excluded from the timeout-rate denominator.
514         self.broadcast_requests = 0
515         self.responses = 0
516         self.timeouts = 0
517         self.crc_errors = 0
518         self.exceptions = 0
519         self.raw = 0
520 
```

MBSniffer v3.2/mb_protocol.py — lines 521-560

```
521         # Bus Health.
522         self.total_bytes = 0
523         self.nonvalidated_bytes = 0
524         self.active_slaves = set()
525         self.exception_by_slave = defaultdict(int)
526         self.response_times_ms = deque(maxlen=10000)
527         self.response_times_by_slave = defaultdict(lambda: deque(maxlen=2000))
528 
529         # Timeout events are exposed to the view so the original request row can
530         # be marked/highlighted without inventing a fake serial frame.
531         self._expired_requests = []
532 
533     def _key(self, row):
534         try:
535             slave = int(row.get("slave", ""))
536             fc = int(row.get("fc", ""), 16)
537         except (ValueError, TypeError):
538             return None
539 
540         if row.get("type") == "EXCEPTION":
541             fc &= 0x7F
542 
543         return (slave, fc)
544 
545     def pending_count(self):
546         return sum(len(v) for v in self.pending.values())
547 
548     def pop_expired_requests(self):
549         items = self._expired_requests
550         self._expired_requests = []
551         return items
552 
553     def expire_pending(self, now=None):
554         """
555         Expire stale requests and retain their sequence IDs for the GUI.
556 
557         Returns the number of newly expired requests, preserving the original
558         public behavior used by CaptureMixin.
559         """
560         if now is None:
```

MBSniffer v3.2/mb_protocol.py — lines 561-600

```
561             now = time.time()
562         now = float(now)
563 
564         expired = 0
565         cutoff = now - self.pending_timeout_seconds
566 
567         for key in list(self.pending.keys()):
568             pending_list = self.pending[key]
569 
570             keep_from = 0
571             for item in pending_list:
572                 if item["time"] <= cutoff:
573                     expired += 1
574                     keep_from += 1
575                     event = dict(item)
576                     event["slave"] = key[0]
577                     event["fc"] = key[1]
578                     self._expired_requests.append(event)
579                 else:
580                     break
581 
582             if keep_from:
583                 del pending_list[:keep_from]
584 
585             if not pending_list:
586                 self.pending.pop(key, None)
587 
588         if expired:
589             self.timeouts += expired
590 
591         return expired
592 
593     def force_timeout(self, slave, fc):
594         """
595         Debug/simulation helper: expire the oldest pending request for one key.
596 
597         It uses the same pending bookkeeping and returns the expired request
598         descriptor so the GUI marks the real REQUEST row as timed out.
599         """
600         try:
```

MBSniffer v3.2/mb_protocol.py — lines 601-640

```
601             key = (int(slave), int(fc))
602         except (TypeError, ValueError):
603             return []
604 
605         pending_list = self.pending.get(key, [])
606         if not pending_list:
607             return []
608 
609         item = pending_list.pop(0)
610         if not pending_list:
611             self.pending.pop(key, None)
612 
613         event = dict(item)
614         event["slave"] = key[0]
615         event["fc"] = key[1]
616         self.timeouts += 1
617         return [event]
618 
619     def _resolve_echo_direction(self, row):
620         """Classify a REQ/RESP frame (FC05/06/08/22) as REQUEST or RESPONSE.
621 
622         These requests and their normal responses have the same shape, so the
623         parser cannot tell them apart. A frame is the RESPONSE when an older
624         pending request with the same (Slave, FC) exists; for exact-echo FCs
625         the bytes must also be identical. In RS232 dual RX a response must
626         arrive on the opposite channel to its request.
627         """
628         if row.get("crc") != "OK":
629             return row.get("type", "")
630 
631         key = self._key(row)
632         if key is None:
633             return row.get("type", "")
634         if key[0] == BROADCAST_UNIT_ID:
635             return "REQUEST"
636 
637         pending_list = self.pending.get(key)
638         if pending_list:
639             candidate = pending_list[0]
640             channel = row.get("channel", "")
```

MBSniffer v3.2/mb_protocol.py — lines 641-680

```
641             same_direction = (
642                 channel in ("A→B", "B→A")
643                 and candidate.get("channel") == channel
644             )
645             same_bytes = candidate.get("raw") == row.get("raw")
646             if not same_direction and (
647                 key[1] not in EXACT_ECHO_FUNCTION_CODES or same_bytes
648             ):
649                 return "RESPONSE"
650 
651         return "REQUEST"
652 
653     def _record_response_time(self, row, response_ms):
654         if response_ms is None or row.get("crc") != "OK":
655             return
656 
657         response_ms = float(response_ms)
658         self.response_times_ms.append(response_ms)
659 
660         try:
661             slave = int(row.get("slave", ""))
662         except (TypeError, ValueError):
663             return
664         self.response_times_by_slave[slave].append(response_ms)
665 
666     def process(self, row):
667         event_time = float(row.get("_epoch", time.time()))
668 
669         # Expire stale requests BEFORE pairing the current frame.
670         self.expire_pending(event_time)
671         expired_requests = self.pop_expired_requests()
672 
673         if self.first_event_time is None:
674             self.first_event_time = event_time
675 
676         delta_ms = None
677         if self.last_event_time is not None and event_time >= self.last_event_time:
678             delta_ms = (event_time - self.last_event_time) * 1000.0
679         if self.last_event_time is None or event_time >= self.last_event_time:
680             self.last_event_time = event_time
```

MBSniffer v3.2/mb_protocol.py — lines 681-720

```
681
682         if row.get("type") == "REQ/RESP":
683             row["type"] = self._resolve_echo_direction(row)
684
685         row_type = row.get("type", "")
686         crc = row.get("crc", "")
687
688         try:
689             frame_len = max(0, int(row.get("_frame_len", 0)))
690         except (TypeError, ValueError):
691             frame_len = 0
692         self.total_bytes += frame_len
693
694         try:
695             slave_num = int(row.get("slave", ""))
696             if 1 <= slave_num <= 247 and not str(row_type).startswith("RAW"):
697                 self.active_slaves.add(slave_num)
698         except (TypeError, ValueError):
699             slave_num = None
700
701         if crc == "ERROR":
702             self.crc_errors += 1
703
704         # RAW bytes cannot honestly be called CRC errors: there may be no
705         # complete frame on which to validate a CRC. For Bus Health we track
706         # them together with explicit CRC failures as traffic that could not be
707         # validated as a correct Modbus RTU frame.
708         if crc == "ERROR" or str(row_type).startswith("RAW"):
709             self.nonvalidated_bytes += frame_len
710
711         if row_type == "REQUEST":
712             self.requests += 1
713             key = self._key(row)
714             if key is not None and key[0] == BROADCAST_UNIT_ID:
715                 self.broadcast_requests += 1
716             elif key is not None and crc == "OK":
717                 self.pending.setdefault(key, []).append({
718                     "time": event_time,
719                     "seq": row.get("_view_seq"),
720                     "pdu_address": row.get("pdu_address"),
```

MBSniffer v3.2/mb_protocol.py — lines 721-760

```
                    "one_based": row.get("one_based"),
                    "qty": row.get("qty"),
                    "channel": row.get("channel", ""),
                    "raw": row.get("raw", ""),
                })

        elif row_type == "RESPONSE":
            self.responses += 1

        elif row_type == "EXCEPTION":
            self.responses += 1
            self.exceptions += 1
            if slave_num is not None:
                self.exception_by_slave[slave_num] += 1

        elif str(row_type).startswith("RAW"):
            self.raw += 1

        response_ms = None
        matched_request = None

        if row_type in ("RESPONSE", "EXCEPTION"):
            key = self._key(row)
            if key is not None:
                pending_list = self.pending.get(key, [])
                if pending_list:
                    matched_request = pending_list.pop(0)
                    if not pending_list:
                        self.pending.pop(key, None)

                    if event_time >= matched_request["time"]:
                        response_ms = (
                            event_time - matched_request["time"]
                        ) * 1000.0

        self._record_response_time(row, response_ms)

        return {
            "delta_ms": delta_ms,
            "response_ms": response_ms,
```

MBSniffer v3.2/mb_protocol.py — lines 761-800

```
            "matched_request": matched_request,
            "expired_requests": expired_requests,
            "pending": self.pending_count(),
            "timeouts": self.timeouts,
        }

    @staticmethod
    def _p95(values):
        if not values:
            return None
        ordered = sorted(float(v) for v in values)
        index = max(0, min(len(ordered) - 1, math.ceil(0.95 * len(ordered)) - 1))
        return ordered[index]

    def health_summary(
        self,
        *,
        baud=9600,
        data_bits=8,
        parity="None",
        stop_bits=1,
        now=None,
    ):
        """Return compact diagnostic metrics for the Bus Health view."""
        if now is None:
            now = time.time()

        if self.first_event_time is None:
            elapsed = 0.0
        else:
            end = max(
                float(now),
                float(self.last_event_time or self.first_event_time),
            )
            elapsed = max(0.0, end - float(self.first_event_time))

        responses = list(self.response_times_ms)
        response_avg = (
            sum(responses) / len(responses)
            if responses else None
```

MBSniffer v3.2/mb_protocol.py — lines 801-840

```
        )
        response_min = min(responses) if responses else None
        response_max = max(responses) if responses else None
        response_p95 = self._p95(responses)

        slowest_slave = None
        slowest_avg = None
        for slave, samples in self.response_times_by_slave.items():
            if not samples:
                continue
            avg = sum(samples) / len(samples)
            if slowest_avg is None or avg > slowest_avg:
                slowest_slave = slave
                slowest_avg = avg

        request_rate = (
            self.requests / elapsed
            if elapsed > 0.0 else 0.0
        )

        nonvalidated_rate = (
            100.0 * self.nonvalidated_bytes / self.total_bytes
            if self.total_bytes else 0.0
        )
        answerable_requests = self.requests - self.broadcast_requests
        timeout_rate = (
            100.0 * self.timeouts / answerable_requests
            if answerable_requests > 0 else 0.0
        )

        try:
            baud = max(1.0, float(baud))
        except (TypeError, ValueError):
            baud = 9600.0
        try:
            data_bits = float(data_bits)
        except (TypeError, ValueError):
            data_bits = 8.0
        try:
            stop_bits = float(stop_bits)
```

MBSniffer v3.2/mb_protocol.py — lines 841-880

```
841         except (TypeError, ValueError):
842             stop_bits = 1.0
843 
844         parity_bits = 0.0 if str(parity) == "None" else 1.0
845         bits_per_char = 1.0 + data_bits + parity_bits + stop_bits
846         bus_load = (
847             100.0 * (self.total_bytes * bits_per_char) / (baud * elapsed)
848             if elapsed > 0.0 else 0.0
849         )
850 
851         exception_summary = ", ".join(
852             f"S{slave}:{count}"
853             for slave, count in sorted(self.exception_by_slave.items())
854         ) or "-"
855 
856         return {
857             "elapsed_s": elapsed,
858             "request_rate": request_rate,
859             "response_avg_ms": response_avg,
860             "response_min_ms": response_min,
861             "response_max_ms": response_max,
862             "response_p95_ms": response_p95,
863             "nonvalidated_rate_pct": nonvalidated_rate,
864             "timeout_rate_pct": timeout_rate,
865             "active_slaves": len(self.active_slaves),
866             "bus_load_pct": bus_load,
867             "slowest_slave": slowest_slave,
868             "slowest_avg_ms": slowest_avg,
869             "exception_summary": exception_summary,
870             "total_bytes": self.total_bytes,
871             "nonvalidated_bytes": self.nonvalidated_bytes,
872         }
873 
874 def u16_bytes(value):
875     value = int(value) & 0xFFFF
876     return bytes(((value >> 8) & 0xFF, value & 0xFF))
877 
878 def corrupt_crc(frame):
879     damaged = bytearray(frame)
880     if len(damaged) >= 2:
```

MBSniffer v3.2/mb_protocol.py — lines 881-920

```
881         damaged[-1] ^= 0xFF
882     return bytes(damaged)
883 
884 def build_simulation_transaction(slave, fc, rng):
885     """
886     Build one syntactically valid Modbus request/response pair.
887 
888     Simulation rules:
889     - FC01..FC04 only.
890     - Address random 1..32.
891     - Qty random 1..8.
892     """
893     slave = int(slave)
894     fc = int(fc)
895     start_addr = rng.randint(1, 32)
896     qty = rng.randint(1, 8)
897 
898     if fc in (1, 2):
899         request = append_crc(
900             bytes((slave, fc)) + u16_bytes(start_addr) + u16_bytes(qty)
901         )
902 
903         byte_count = (qty + 7) // 8
904         data = bytes(rng.randrange(256) for _ in range(byte_count))
905         response = append_crc(bytes((slave, fc, byte_count)) + data)
906 
907     elif fc in (3, 4):
908         request = append_crc(
909             bytes((slave, fc)) + u16_bytes(start_addr) + u16_bytes(qty)
910         )
911 
912         values = [rng.randint(0, 65535) for _ in range(qty)]
913         payload = b"".join(u16_bytes(value) for value in values)
914         response = append_crc(
915             bytes((slave, fc, len(payload))) + payload
916         )
917 
918     else:
919         raise ValueError(f"Unsupported simulation FC: {fc}")
920 
```

MBSniffer v3.2/mb_protocol.py — lines 921-960

```
    return {
        "slave": slave,
        "fc": fc,
        "address": start_addr,
        "qty": qty,
        "request": request,
        "response": response,
    }

def build_random_simulation_plan(rng=None):
    """
    Build exactly four simulation transactions:
    FC01, FC02, FC03 and FC04, one each.

    Four unique random Slave IDs are used, one per transaction.
    """
    if rng is None:
        rng = random.Random()

    slaves = rng.sample(range(1, 248), 4)

    plan = []
    for slave, fc in zip(slaves, SIMULATION_FUNCTION_CODES):
        tx = build_simulation_transaction(slave, fc, rng)
        tx["outcome"] = "response"
        tx["reply"] = tx["response"]
        plan.append(tx)

    return {
        "slaves": slaves,
        "transactions": plan,
    }



def build_anomaly_simulation_plan(rng=None):
    """
    Build a deterministic set of anomaly categories for UI colour preview.

    Used whenever debug simulation is executed. "Realçar resultados" controls
```

MBSniffer v3.2/mb_protocol.py — lines 961-1000

```
    only whether these rows receive colours; it does not change the generated
    scenario.

    It demonstrates:
      - normal response
      - slow response
      - Modbus exception
      - CRC error
      - timeout
      - RAW/UNPARSED bytes
    """
    if rng is None:
        rng = random.Random()

    slaves = rng.sample(range(1, 248), 5)

    normal = build_simulation_transaction(slaves[0], 1, rng)
    normal["outcome"] = "response"
    normal["reply"] = normal["response"]

    slow = build_simulation_transaction(slaves[1], 2, rng)
    slow["outcome"] = "slow"
    slow["reply"] = slow["response"]

    exception = build_simulation_transaction(slaves[2], 3, rng)
    exception["outcome"] = "exception"
    exception["reply"] = append_crc(
        bytes((slaves[2], 0x83, 0x02))
    )

    crc_error = build_simulation_transaction(slaves[3], 4, rng)
    crc_error["outcome"] = "crc"
    crc_error["reply"] = corrupt_crc(crc_error["response"])

    timeout = build_simulation_transaction(slaves[4], 3, rng)
    timeout["outcome"] = "timeout"
    timeout["reply"] = None

    return {
        "slaves": slaves,
```

MBSniffer v3.2/mb_protocol.py — lines 1001-1007

```
1001        "transactions": [normal, slow, exception, crc_error, timeout],
1002        # Four bytes are deliberately too short to be accepted as the FC=0xFF
1003        # exception frame candidate, producing a visible RAW/UNPARSED row.
1004        "raw_noise": bytes((0x00, 0x00, 0x00, 0x00)),
1005        "slow_delay_seconds": (SLOW_RESPONSE_THRESHOLD_MS / 1000.0) + 0.15,
1006    }
1007
```

MBSniffer v3.2/mb_capture.py

883 source lines.

MBSniffer v3.2/mb_capture.py — lines 1-40

```python
#!/usr/bin/env python3
"""Serial-port management, passive capture, simulation and event processing."""

import queue
import random
import threading
import time
import traceback
from datetime import datetime

import tkinter as tk
from tkinter import messagebox

from mb_config import (
    APP_NAME,
    APP_VERSION,
    MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
    MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
    SNIFFER_SPLIT_FRAME_HOLD_MS,
    UI_QUEUE_BATCH_SIZE,
    UI_QUEUE_IDLE_INTERVAL_MS,
    UI_QUEUE_BUSY_INTERVAL_MS,
    debug_sim,
)
from mb_protocol import (
    parse_burst,
    burst_may_continue,
    build_anomaly_simulation_plan,
    modbus_rtu_interframe_gap_seconds,
)
from mb_slave_finder import cancel_serial_io

class CaptureMixin:
    """Methods for serial capture, COM management and simulation."""

    def set_sniffer_refresh_state(self, state):
        """Apply the same state to both Sniffer COM refresh buttons."""
        for name in ("refresh_a_btn", "refresh_b_btn"):
            button = getattr(self, name, None)
            if button is not None:
```

MBSniffer v3.2/mb_capture.py — lines 41-80

```
41                button.configure(state=state)
42
43    def update_mode_ui(self):
44        mode = self.mode_var.get()
45        dual = mode == "RS232 dual RX"
46
47        state = "readonly" if dual else "disabled"
48        self.port_b_combo.configure(state=state)
49
50        if dual:
51            self.mode_help_var.set(
52                self.tr("RS232 dual RX: COM A escuta uma direção; COM B escuta a direção oposta.")
53            )
54        elif mode == "RS232 single RX":
55            self.mode_help_var.set(
56                self.tr("RS232 single RX: uma COM escuta apenas uma direção da ligação RS232.")
57            )
58        else:
59            self.mode_help_var.set(
60                self.tr("RS485 2-wire: uma COM consegue observar pedidos e respostas no mesmo par diferencial.")
61            )
62
63        self.update_com_status()
64
65    def update_gap_state(self):
66        self.gap_entry.configure(
67            state="normal" if self.gap_mode_var.get() == "Manual" else "disabled"
68        )
69
70    def get_serial_modules(self):
71        try:
72            import serial
73            from serial.tools import list_ports
74            return serial, list_ports
75        except ImportError:
76            return None, None
77
78    def refresh_ports(self):
79        _, list_ports = self.get_serial_modules()
80        ports = [p.device for p in list_ports.comports()] if list_ports else []
```

MBSniffer v3.2/mb_capture.py — lines 81-120

```
81
82          self.port_a_combo["values"] = ports
83          self.port_b_combo["values"] = ports
84
85          if ports:
86              if self.port_a_var.get() not in ports:
87                  self.port_a_var.set(ports[0])
88              if len(ports) > 1:
89                  if self.port_b_var.get() not in ports or self.port_b_var.get() == self.port_a_var.get():
90                      self.port_b_var.set(ports[1])
91              elif not self.port_b_var.get():
92                  self.port_b_var.set("")
93          else:
94              self.port_a_var.set("")
95              self.port_b_var.set("")
96
97          if hasattr(self, "discovery_port_combo"):
98              self.discovery_port_combo["values"] = ports
99              current_discovery = self.discovery_port_var.get().strip()
100             if ports:
101                 if current_discovery not in ports:
102                     self.discovery_port_var.set(ports[0])
103             else:
104                 self.discovery_port_var.set("")
105             self.update_discovery_start_state()
106
107         if hasattr(self, "master_port_combo"):
108             self.master_port_combo["values"] = ports
109             if ports:
110                 if self.master_port_var.get().strip() not in ports:
111                     self.master_port_var.set(ports[0])
112             elif not getattr(self, "master_active", False):
113                 self.master_port_var.set("")
114
115         self.update_com_status()
116
117     def detected_ports(self):
118         """Return the set of COM device names currently reported by Windows/pyserial."""
119         _, list_ports = self.get_serial_modules()
120         if list_ports is None:
```

MBSniffer v3.2/mb_capture.py — lines 121-160

```
            return set()
        try:
            return {p.device for p in list_ports.comports()}
        except Exception:
            return set()

    def port_status_text(self, label, port, enabled=True):
        if not enabled:
            return f"{label} - {self.tr('Não Utilizada')}"
        if not port:
            return f"{label} - {self.tr('Não selecionada')}"

        ser = self.serial_by_port.get(port)
        if ser is not None:
            try:
                if ser.is_open:
                    return f"{label} - {self.tr('Aberta')}"
            except Exception:
                pass

        if port in self.detected_ports():
            return f"{label} - {self.tr('Detetada')}"
        return f"{label} - {self.tr('Não detetada')}"

    def update_com_status(self):
        """
        Update COM status without opening/probing a port.

        Status means:
          - Aberta: currently opened by MBSniffer
          - Detetada: reported by Windows/pyserial, but not opened by MBSniffer
          - Não detetada: selected name is no longer reported by the OS
        """
        if not hasattr(self, "port_a_status_var"):
            return

        mode = self.mode_var.get() if hasattr(self, "mode_var") else "RS485 2-wire"
        dual = mode == "RS232 dual RX"
        self.port_a_status_var.set(
            self.port_status_text("COM A", self.port_a_var.get().strip(), True)
```

MBSniffer v3.2/mb_capture.py — lines 161-200

```
161         )
162         self.port_b_status_var.set(
163             self.port_status_text("COM B", self.port_b_var.get().strip(), dual)
164         )
165 
166     def poll_com_status(self):
167         if self._closing:
168             return
169         try:
170             self.update_com_status()
171             if hasattr(self, "stat_requests_var"):
172                 self.update_stats_labels()
173         finally:
174             if not self._closing:
175                 try:
176                     self.after(1000, self.poll_com_status)
177                 except tk.TclError:
178                     pass
179 
180     def selected_port_plan(self):
181         """Return [(channel, port), ...] for the currently selected physical mode."""
182         mode = self.mode_var.get()
183         port_a = self.port_a_var.get().strip()
184         port_b = self.port_b_var.get().strip()
185 
186         if not port_a:
187             raise ValueError(self.tr("Seleciona a COM A."))
188 
189         plan = [("BUS" if mode == "RS485 2-wire" else "A→B", port_a)]
190 
191         if mode == "RS232 dual RX":
192             if not port_b:
193                 raise ValueError(self.tr("Seleciona também a COM B."))
194             if port_a == port_b:
195                 raise ValueError(self.tr("COM A e COM B têm de ser diferentes."))
196             plan.append(("B→A", port_b))
197 
198         return plan
199 
200     def open_selected_ports(self, serial, config):
```

MBSniffer v3.2/mb_capture.py — lines 201-240

```
201         """Open the selected capture ports and return [(channel, serial_obj), ...]."""
202         opened = []
203         try:
204             for channel, port in self.selected_port_plan():
205                 opened.append((channel, self.open_serial_port(serial, port, config)))
206             return opened
207         except Exception:
208             for _, ser_obj in opened:
209                 try:
210                     ser_obj.close()
211                 except Exception:
212                     pass
213             raise
214 
215     def install_opened_ports(self, opened):
216         self.serial_objects = [ser_obj for _, ser_obj in opened]
217         self.serial_by_port = {
218             ser_obj.port: ser_obj for _, ser_obj in opened if getattr(ser_obj, "port", None)
219         }
220         self.update_com_status()
221 
222     def start_reader_threads(self, opened, gap_ms):
223         self.reader_threads = []
224         generation = self.reader_generation
225         for channel, ser_obj in opened:
226             t = threading.Thread(
227                 target=self.reader_loop,
228                 args=(ser_obj, gap_ms, channel, generation),
229                 daemon=True
230             )
231             self.reader_threads.append(t)
232             t.start()
233 
234     def restart_com_port(self):
235         """
236         Close and reopen the selected COM port(s).
237 
238         During an active capture the capture continues in the same session/log.
239         When stopped, the selected COM port(s) are opened and closed once as a
240         restart/test of the application's serial handle.
```

MBSniffer v3.2/mb_capture.py — lines 241-280

```
241         """
242         if self.busy and not self.capture_active:
243             return
244         if self.restart_in_progress:
245             return
246 
247         serial, _ = self.get_serial_modules()
248         if serial is None:
249             messagebox.showerror(self.tr("pyserial em falta"), self.tr("pyserial não está instalado."))
250             return
251 
252         try:
253             config = self.serial_config(serial)
254             gap_ms = self.calculated_gap_ms()
255             self.selected_port_plan()
256         except Exception as exc:
257             messagebox.showerror(self.tr("Porta COM"), str(exc))
258             return
259 
260         self.restart_in_progress = True
261         self.restart_btn.configure(state="disabled")
262 
263         if not self.capture_active:
264             opened = []
265             try:
266                 opened = self.open_selected_ports(serial, config)
267                 self.install_opened_ports(opened)
268                 self.update_idletasks()
269             except Exception as exc:
270                 messagebox.showerror(self.tr("Reiniciar COM"), self.trf("Não foi possível reiniciar a porta COM:\n\n{error}", error=exc))
271             finally:
272                 for _, ser_obj in opened:
273                     try:
274                         ser_obj.close()
275                     except Exception:
276                         pass
277                 self.serial_objects = []
278                 self.serial_by_port = {}
279                 self.restart_in_progress = False
280                 self.restart_btn.configure(state="normal")
```

MBSniffer v3.2/mb_capture.py — lines 281-320

```
281                 self.refresh_ports()
282             return
283 
284         # Active capture: stop only the current serial-reader generation.
285         self.status_var.set(self.tr("● A reiniciar COM"))
286         self.reader_generation += 1
287         self.stop_event.set()
288 
289         for thread in list(self.reader_threads):
290             try:
291                 thread.join(timeout=0.5)
292             except Exception:
293                 pass
294 
295         for ser_obj in list(self.serial_objects):
296             try:
297                 ser_obj.close()
298             except Exception:
299                 pass
300 
301         self.reader_threads = []
302         self.serial_objects = []
303         self.serial_by_port = {}
304         self.stop_event.clear()
305 
306         try:
307             opened = self.open_selected_ports(serial, config)
308         except Exception as exc:
309             self.restart_in_progress = False
310             self.capture_active = False
311             self.busy = False
312             self.stop_event.set()
313             self.finish_capture_ui()
314             self.update_com_status()
315             messagebox.showerror(
316                 self.tr("Reiniciar COM"),
317                 self.trf("Não foi possível reabrir a porta COM. A captura foi parada.\n\n{error}", error=exc)
318             )
319             return
320 
```

MBSniffer v3.2/mb_capture.py — lines 321-360

```
321         self.install_opened_ports(opened)
322         self.stop_event.clear()
323         self.start_reader_threads(opened, gap_ms)
324         self.queue_log_note(
325             f"# COM restart: {datetime.now().isoformat(timespec='milliseconds')}\n"
326         )
327         self.status_var.set(self.tr("● A capturar"))
328         self.restart_in_progress = False
329         self.restart_btn.configure(state="normal")
330         self.update_com_status()
331 
332     def set_simulation_button_state(self, state):
333         """Change Simulation button state only when the optional button exists."""
334         if self.sim_btn is not None:
335             self.sim_btn.configure(state=state)
336 
337     def selected_pending_timeout_seconds(self):
338         try:
339             value = float(self.pending_timeout_var.get().strip().replace(",", "."))
340         except ValueError:
341             raise ValueError(self.tr("Pending timeout inválido."))
342 
343         if not (
344             MIN_PENDING_REQUEST_TIMEOUT_SECONDS
345             <= value
346             <= MAX_PENDING_REQUEST_TIMEOUT_SECONDS
347         ):
348             raise ValueError(
349                 self.trf(
350                     "Pending timeout deve estar entre {minimum:g} e {maximum:g} segundos.",
351                     minimum=MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
352                     maximum=MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
353                 )
354             )
355 
356         return value
357 
358     def calculated_gap_ms(self):
359         if self.gap_mode_var.get() == "Manual":
360             value = float(self.gap_ms_var.get().replace(",", "."))
```

MBSniffer v3.2/mb_capture.py — lines 361-400

```
361             if value <= 0:
362                 raise ValueError(self.tr("Frame gap manual inválido."))
363             return value
364 
365         baud = int(self.baud_var.get())
366         data_bits = int(self.data_var.get())
367         parity_bits = 0 if self.parity_var.get() == "None" else 1
368         stop_bits = float(self.stop_var.get())
369         bits_per_char = 1 + data_bits + parity_bits + stop_bits
370         return modbus_rtu_interframe_gap_seconds(baud, bits_per_char) * 1000.0
371 
372     def serial_config(self, serial):
373         byte_map = {
374             5: serial.FIVEBITS, 6: serial.SIXBITS,
375             7: serial.SEVENBITS, 8: serial.EIGHTBITS,
376         }
377         parity_map = {
378             "None": serial.PARITY_NONE,
379             "Even": serial.PARITY_EVEN,
380             "Odd": serial.PARITY_ODD,
381             "Mark": serial.PARITY_MARK,
382             "Space": serial.PARITY_SPACE,
383         }
384         stop_map = {
385             "1": serial.STOPBITS_ONE,
386             "1.5": serial.STOPBITS_ONE_POINT_FIVE,
387             "2": serial.STOPBITS_TWO,
388         }
389         return {
390             "baudrate": int(self.baud_var.get()),
391             "bytesize": byte_map[int(self.data_var.get())],
392             "parity": parity_map[self.parity_var.get()],
393             "stopbits": stop_map[self.stop_var.get()],
394         }
395 
396     def open_serial_port(self, serial, port, config):
397         ser = serial.Serial()
398         ser.port = port
399         ser.baudrate = config["baudrate"]
400         ser.bytesize = config["bytesize"]
```

MBSniffer v3.2/mb_capture.py — lines 401-440

```
401         ser.parity = config["parity"]
402         ser.stopbits = config["stopbits"]
403         ser.timeout = 0
404         ser.write_timeout = 0
405         ser.xonxoff = False
406         ser.rtscts = False
407         ser.dsrdtr = False
408 
409         try:
410             ser.rts = False
411             ser.dtr = False
412         except Exception:
413             pass
414 
415         ser.open()
416 
417         try:
418             ser.rts = False
419             ser.dtr = False
420         except Exception:
421             pass
422 
423         try:
424             ser.reset_input_buffer()
425         except Exception:
426             pass
427 
428         return ser
429 
430     def start_capture(self):
431         if self.busy:
432             return
433 
434         serial, _ = self.get_serial_modules()
435         if serial is None:
436             messagebox.showerror(
437                 self.tr("pyserial em falta"),
438                 self.tr("pyserial não está instalado.\n\nO build_exe.bat inclui-o automaticamente no EXE.")
439             )
440             return
```

MBSniffer v3.2/mb_capture.py — lines 441-480

```
441
442         mode = self.mode_var.get()
443         port_a = self.port_a_var.get().strip()
444         port_b = self.port_b_var.get().strip()
445
446         try:
447             gap_ms = self.calculated_gap_ms()
448             pending_timeout_s = self.selected_pending_timeout_seconds()
449             config = self.serial_config(serial)
450             opened = self.open_selected_ports(serial, config)
451         except Exception as exc:
452             messagebox.showerror(self.tr("Erro ao abrir porta"), str(exc))
453             self.update_com_status()
454             return
455
456         self.reader_generation += 1
457         self.install_opened_ports(opened)
458         self.reader_threads = []
459         self.stop_event.clear()
460         self.busy = True
461         self.capture_active = True
462         self.metrics.pending_timeout_seconds = pending_timeout_s
463         self.reset_session_metrics()
464
465         capture_started = datetime.now()
466         log_header = (
467             f"# {APP_NAME} v{APP_VERSION}\n"
468             f"# Start: {capture_started.isoformat(timespec='milliseconds')}\n"
469             f"# Mode: {mode}\n"
470             f"# COM A: {port_a}\n"
471             + (f"# COM B: {port_b}\n" if mode == "RS232 dual RX" else "")
472             + f"# Serial: baud={self.baud_var.get()} data={self.data_var.get()} "
473               f"parity={self.parity_var.get()} stop={self.stop_var.get()} "
474               f"frame_gap_ms={gap_ms:.3f}\n"
475             + f"# Pending timeout: {pending_timeout_s:g} s\n"
476             + "# Passive/read-only: no Serial.write() calls.\n#\n"
477         )
478         self.prepare_log_session("MBSniffer", capture_started, log_header)
479
480         self.status_var.set(self.tr("● A capturar"))
```

MBSniffer v3.2/mb_capture.py — lines 481-520

```
        self.start_btn.configure(state="disabled")
        self.stop_btn.configure(state="normal")
        self.restart_btn.configure(state="normal")
        self.set_simulation_button_state("disabled")
        self.set_sniffer_refresh_state("disabled")

        self.start_reader_threads(opened, gap_ms)
        self.update_com_status()

    def reader_loop(self, ser, gap_ms, channel, generation):
        gap_s = gap_ms / 1000.0
        hold_s = max(gap_s, SNIFFER_SPLIT_FRAME_HOLD_MS / 1000.0)
        poll_sleep = min(max(gap_s / 8.0, 0.0002), 0.001)

        buffer = bytearray()
        buffer_wall = None
        # (byte offset, epoch) of every read, so frames merged into one burst
        # by USB latency keep their own arrival times.
        chunk_times = []
        last_rx = None
        # Burst length already checked by burst_may_continue(); avoids
        # re-parsing an unchanged buffer on every poll of the hold window.
        hold_checked_len = None
        hold_active = False

        def flush_buffer():
            ts = (
                buffer_wall.strftime("%H:%M:%S.%f")[:-3]
                if buffer_wall else datetime.now().strftime("%H:%M:%S.%f")[:-3]
            )
            event_epoch = (
                buffer_wall.timestamp() if buffer_wall else time.time()
            )
            for row in parse_burst(
                bytes(buffer), ts, channel, event_epoch, chunk_times
            ):
                self.event_queue.put(("frame", row))

        try:
            while not self.stop_event.is_set():
```

MBSniffer v3.2/mb_capture.py — lines 521-560

```
                waiting = ser.in_waiting

                if waiting:
                    chunk = ser.read(waiting)
                    if chunk:
                        if not buffer:
                            buffer_wall = datetime.now()
                            chunk_times.append((0, buffer_wall.timestamp()))
                        else:
                            chunk_times.append((len(buffer), time.time()))
                        buffer.extend(chunk)
                        last_rx = time.perf_counter()

                elif buffer and last_rx is not None:
                    idle_s = time.perf_counter() - last_rx
                    if idle_s >= gap_s:
                        if idle_s < hold_s:
                            if hold_checked_len != len(buffer):
                                hold_checked_len = len(buffer)
                                hold_active = burst_may_continue(bytes(buffer))
                            if hold_active:
                                time.sleep(poll_sleep)
                                continue

                        flush_buffer()
                        buffer.clear()
                        chunk_times.clear()
                        buffer_wall = None
                        last_rx = None
                        hold_checked_len = None
                        hold_active = False

                time.sleep(poll_sleep)

        except Exception as exc:
            if not self.stop_event.is_set():
                self.event_queue.put(("error", (generation, channel, str(exc))))

        finally:
            if buffer:
```

MBSniffer v3.2/mb_capture.py — lines 561-600

```
561                 flush_buffer()
562 
563             try:
564                 ser.close()
565             except Exception:
566                 pass
567             self.event_queue.put(("reader_stopped", (generation, channel)))
568 
569     def stop_capture(self):
570         if self.capture_active:
571             self.status_var.set(self.tr("● A parar"))
572         self.stop_event.set()
573 
574     def run_simulation(self):
575         if debug_sim != 1:
576             return
577 
578         if self.busy:
579             return
580 
581         try:
582             pending_timeout_s = self.selected_pending_timeout_seconds()
583         except Exception as exc:
584             messagebox.showerror(self.tr("Configuração inválida"), str(exc))
585             return
586 
587         self.busy = True
588         self.metrics.pending_timeout_seconds = pending_timeout_s
589         self.reset_session_metrics()
590 
591         # Simulation is GUI-only. Never create or write a .txt log.
592         self.close_log()
593 
594         self.status_var.set(self.tr("● A simular"))
595         self.start_btn.configure(state="disabled")
596         self.restart_btn.configure(state="disabled")
597         self.set_simulation_button_state("disabled")
598         self.set_sniffer_refresh_state("disabled")
599 
600         def worker():
```

MBSniffer v3.2/mb_capture.py — lines 601-640

```
601             try:
602                 # Simulation always exercises the complete diagnostic scenario.
603                 # "Realçar resultados" affects only Treeview row colouring; it
604                 # never changes which simulated frames are generated.
605                 rng = random.Random()
606                 plan = build_anomaly_simulation_plan(rng=rng)
607 
608                 for tx in plan["transactions"]:
609                     time.sleep(rng.uniform(0.025, 0.085))
610 
611                     request_now = datetime.now()
612                     request_ts = request_now.strftime("%H:%M:%S.%f")[:-3]
613 
614                     request_channel = "SIM"
615 
616                     for row in parse_burst(
617                         tx["request"], request_ts, request_channel,
618                         request_now.timestamp()
619                     ):
620                         self.event_queue.put(("frame", row))
621 
622                     outcome = tx.get("outcome", "response")
623 
624                     if outcome == "timeout":
625                         # Do not sleep for the user-configured Pending timeout.
626                         # The queue event expires this exact pending request
627                         # through SessionMetrics and marks the REQUEST row.
628                         self.event_queue.put((
629                             "simulation_force_timeout",
630                             (tx["slave"], tx["fc"]),
631                         ))
632                         continue
633 
634                     if outcome == "slow":
635                         time.sleep(plan.get("slow_delay_seconds", 0.65))
636                     else:
637                         time.sleep(rng.uniform(0.018, 0.095))
638 
639                     reply_now = datetime.now()
640                     reply_ts = reply_now.strftime("%H:%M:%S.%f")[:-3]
```

MBSniffer v3.2/mb_capture.py — lines 641-680

```
                    reply_channel = "SIM"

                    reply = tx.get("reply")
                    if reply:
                        for row in parse_burst(
                            reply, reply_ts, reply_channel,
                            reply_now.timestamp()
                        ):
                            self.event_queue.put(("frame", row))

                # One RAW row completes the diagnostic scenario. With anomaly
                # highlighting OFF it remains a normal, uncoloured RAW row;
                # with highlighting ON the same row receives the RAW colour.
                raw_now = datetime.now()
                raw_ts = raw_now.strftime("%H:%M:%S.%f")[:-3]
                raw_channel = "SIM"
                for row in parse_burst(
                    plan["raw_noise"],
                    raw_ts,
                    raw_channel,
                    raw_now.timestamp(),
                ):
                    self.event_queue.put(("frame", row))

                self.event_queue.put(("simulation_done", None))

            except Exception as exc:
                # Never leave the GUI permanently busy if the simulation worker
                # fails unexpectedly.
                self.event_queue.put((
                    "simulation_error",
                    f"{type(exc).__name__}: {exc}"
                ))

        threading.Thread(target=worker, daemon=True).start()

    def process_queue(self):
        if self._closing:
            return

```

MBSniffer v3.2/mb_capture.py — lines 681-720

```
        delay = UI_QUEUE_IDLE_INTERVAL_MS
        try:
            events = []
            for _ in range(UI_QUEUE_BATCH_SIZE):
                try:
                    events.append(self.event_queue.get_nowait())
                except queue.Empty:
                    break

            frame_batch = []

            def flush_frames():
                nonlocal frame_batch
                if frame_batch:
                    self.add_frame_rows_batch(frame_batch)
                    frame_batch = []

            for event, payload in events:
                if event == "frame":
                    frame_batch.append(payload)
                    continue

                flush_frames()

                if event == "error":
                    generation, channel, error_text = payload
                    if generation != self.reader_generation:
                        continue
                    # Stop all readers in this capture generation after any
                    # reader failure; never leave a half-alive dual-RX capture.
                    self.stop_event.set()
                    self.update_com_status()
                    messagebox.showerror(
                        self.tr("Erro de captura"), f"{channel}: {error_text}"
                    )

                elif event == "reader_stopped":
                    generation, _channel = payload
                    if generation != self.reader_generation:
                        continue
```

MBSniffer v3.2/mb_capture.py — lines 721-760

```
                    if not any(t.is_alive() for t in self.reader_threads):
                        self.finish_capture_ui()

                elif event == "simulation_force_timeout":
                    slave, fc = payload
                    expired = self.metrics.force_timeout(slave, fc)
                    if expired:
                        self.mark_timed_out_records(expired, rerender=True)
                    self.update_stats_labels()

                elif event == "simulation_done":
                    self.busy = False
                    self.status_var.set(self.tr("● Simulação concluída"))
                    self.start_btn.configure(state="normal")
                    self.stop_btn.configure(state="disabled")
                    self.restart_btn.configure(state="normal")
                    self.set_simulation_button_state("normal")
                    self.set_sniffer_refresh_state("normal")
                    self.close_log()
                    self.update_com_status()

                elif event == "simulation_error":
                    self.busy = False
                    self.status_var.set(self.tr("● Erro na simulação"))
                    self.start_btn.configure(state="normal")
                    self.stop_btn.configure(state="disabled")
                    self.restart_btn.configure(state="normal")
                    self.set_simulation_button_state("normal")
                    self.set_sniffer_refresh_state("normal")
                    self.close_log()
                    self.update_com_status()
                    messagebox.showerror(self.tr("Erro na simulação"), str(payload))

                elif event == "discovery_status":
                    self.discovery_status_var.set(str(payload))

                elif event == "discovery_progress":
                    self.apply_discovery_progress(payload)

                elif event == "discovery_found":
```

MBSniffer v3.2/mb_capture.py — lines 761-800

```
                    self.add_discovery_result(payload)

                elif event == "discovery_done":
                    self.finish_discovery_ui(payload)

                elif event.startswith("master_"):
                    self.handle_master_event(event, payload)

            flush_frames()

            if self.metrics.expire_pending(time.time()):
                expired = self.metrics.pop_expired_requests()
                if expired:
                    self.mark_timed_out_records(expired, rerender=True)
                self.update_stats_labels()

            if (
                self.capture_active
                and not self.restart_in_progress
                and self.stop_event.is_set()
                and self.reader_threads
                and not any(t.is_alive() for t in self.reader_threads)
                and self.stop_btn["state"] != "disabled"
            ):
                self.finish_capture_ui()

            delay = (
                UI_QUEUE_BUSY_INTERVAL_MS
                if len(events) >= UI_QUEUE_BATCH_SIZE
                else UI_QUEUE_IDLE_INTERVAL_MS
            )

        except tk.TclError:
            if self._closing:
                return
            traceback.print_exc()
            delay = UI_QUEUE_IDLE_INTERVAL_MS

        except Exception as exc:
            # One unexpected event/UI error must not permanently kill the
```

MBSniffer v3.2/mb_capture.py — lines 801-840

```
            # periodic queue callback and make the application appear frozen.
            traceback.print_exc()
            if not self._process_queue_error_shown and not self._closing:
                self._process_queue_error_shown = True
                try:
                    messagebox.showerror(
                        self.tr("Erro interno"),
                        self.trf(
                            "Ocorreu um erro ao processar eventos da interface. O processamento continuará.\n\n{error}",
                            error=f"{type(exc).__name__}: {exc}",
                        )
                    )
                except Exception:
                    pass
            delay = UI_QUEUE_IDLE_INTERVAL_MS

        finally:
            if not self._closing:
                try:
                    self.after(delay, self.process_queue)
                except tk.TclError:
                    pass

    def on_close(self):
        if self._closing:
            return

        try:
            saver = getattr(self, "save_current_ui_settings", None)
            if callable(saver):
                saver()
        except Exception:
            pass

        self._closing = True
        self.busy = False
        self.capture_active = False
        self.stop_event.set()
        self.discovery_stop_event.set()

```

MBSniffer v3.2/mb_capture.py — lines 841-880

```
841         # Cancel driver I/O before joining workers so shutdown is not held up
842         # by a serial read/write timeout.
843         cancel_serial_io(self.discovery_serial)
844         stopper = getattr(self, "master_shutdown", None)
845         if callable(stopper):
846             stopper()
847 
848         if self.discovery_thread is not None:
849             try:
850                 if self.discovery_thread.is_alive():
851                     self.discovery_thread.join(timeout=0.8)
852             except Exception:
853                 pass
854 
855         if self.discovery_serial is not None:
856             try:
857                 self.discovery_serial.close()
858             except Exception:
859                 pass
860         self.discovery_serial = None
861 
862         for ser in list(self.serial_objects):
863             cancel_serial_io(ser)
864 
865         for thread in list(self.reader_threads):
866             try:
867                 if thread.is_alive():
868                     thread.join(timeout=0.5)
869             except Exception:
870                 pass
871 
872         for ser in list(self.serial_objects):
873             try:
874                 ser.close()
875             except Exception:
876                 pass
877 
878         self.close_log()
879         try:
880             self.destroy()
```

MBSniffer v3.2/mb_capture.py — lines 881-883

```
881         except tk.TclError:
882             pass
883 
```

MBSniffer v3.2/mb_diagnostics.py

286 source lines.

MBSniffer v3.2/mb_diagnostics.py — lines 1-40

```python
#!/usr/bin/env python3
"""Diagnostic helpers for MBSniffer traffic inspection and anomaly display."""

from mb_config import MODBUS_EXCEPTION_CODES, SLOW_RESPONSE_THRESHOLD_MS
from mb_protocol import crc16_modbus, read_bits_from_response
from mb_i18n import LANGUAGE_PORTUGUESE, translate_details, translate_text


FUNCTION_NAMES = {
    0x01: "Read Coils",
    0x02: "Read Discrete Inputs",
    0x03: "Read Holding Registers",
    0x04: "Read Input Registers",
    0x05: "Write Single Coil",
    0x06: "Write Single Register",
    0x07: "Read Exception Status",
    0x08: "Diagnostics",
    0x0F: "Write Multiple Coils",
    0x10: "Write Multiple Registers",
    0x16: "Mask Write Register",
    0x17: "Read/Write Multiple Registers",
    0x2B: "Encapsulated Interface Transport",
}


def raw_hex_to_bytes(raw):
    try:
        return bytes.fromhex(str(raw or ""))
    except (TypeError, ValueError):
        return b""


def fc_number(record):
    text = str(record.get("fc", "") or "").strip()
    try:
        value = int(text, 16)
    except (TypeError, ValueError):
        return None
    if record.get("type") == "EXCEPTION":
        value &= 0x7F
```

MBSniffer v3.2/mb_diagnostics.py — lines 41-80

```
41      return value
42
43
44  def function_label(record, language=LANGUAGE_PORTUGUESE):
45      fc = fc_number(record)
46      if fc is None:
47          return "—"
48      name = FUNCTION_NAMES.get(fc, translate_text("Desconhecida/outra", language))
49      return f"0x{fc:02X} — {name}"
50
51
52  def crc_fields(record):
53      if str(record.get("type", "")).startswith("RAW"):
54          return None, None, str(record.get("crc", "?"))
55
56      frame = raw_hex_to_bytes(record.get("raw"))
57      if len(frame) < 4:
58          return None, None, str(record.get("crc", "?"))
59
60      received = frame[-2] | (frame[-1] << 8)
61      calculated = crc16_modbus(frame[:-2])
62      return received, calculated, ("OK" if received == calculated else "ERROR")
63
64
65  def response_payload(record, language=LANGUAGE_PORTUGUESE):
66      """Return (byte_count, data_hex, decoded_data) where available."""
67      frame = raw_hex_to_bytes(record.get("raw"))
68      if len(frame) < 3:
69          return None, "", ""
70
71      kind = str(record.get("type", ""))
72      fc_raw = frame[1] if len(frame) >= 2 else 0
73      fc = fc_raw & 0x7F
74
75      if kind == "EXCEPTION":
76          code = frame[2] if len(frame) >= 3 else None
77          if code is None:
78              return None, "", "Exception"
79          info = MODBUS_EXCEPTION_CODES.get(code)
80          name = info[0] if info else "Unknown/Reserved Exception"
```

MBSniffer v3.2/mb_diagnostics.py — lines 81-120

```
        return None, "", f"0x{code:02X} — {name}"

    if kind == "RESPONSE" and fc in (1, 2, 3, 4) and len(frame) >= 5:
        byte_count = frame[2]
        data = frame[3:3 + byte_count]
        data_hex = " ".join(f"{b:02X}" for b in data)
        decoded = ""
        start = record.get("request_pdu_address")
        if fc in (3, 4) and byte_count % 2 == 0 and len(data) == byte_count:
            regs = [
                (data[i] << 8) | data[i + 1]
                for i in range(0, len(data), 2)
            ]
            if start is not None:
                # The matched request gives the address of every value.
                pairs = ", ".join(
                    f"{start + i}={value}" for i, value in enumerate(regs)
                )
                decoded = translate_details(
                    f"Registos (PDU Address=valor): {pairs}", language
                )
            else:
                decoded = translate_details(f"Registos={regs}", language)
        elif fc in (1, 2) and len(data) == byte_count:
            bits = read_bits_from_response(frame, record.get("request_qty"))
            if start is not None:
                pairs = ", ".join(
                    f"{start + i}={value}" for i, value in enumerate(bits)
                )
                decoded = translate_details(
                    f"Bits (PDU Address=valor): {pairs}", language
                )
            else:
                decoded = "Bits=" + "".join(str(bit) for bit in bits)
        return byte_count, data_hex, decoded

    if kind == "REQUEST" and fc in (15, 16) and len(frame) >= 9:
        byte_count = frame[6]
        data = frame[7:7 + byte_count]
        return byte_count, " ".join(f"{b:02X}" for b in data), ""
```

MBSniffer v3.2/mb_diagnostics.py — lines 121-160

```
121
122     if kind == "REQUEST" and fc == 23 and len(frame) >= 13:
123         byte_count = frame[10]
124         data = frame[11:11 + byte_count]
125         return byte_count, " ".join(f"{b:02X}" for b in data), ""
126
127     return record.get("byte_count"), str(record.get("data_hex", "") or ""), ""
128
129
130 def anomaly_kind(record, threshold_ms=SLOW_RESPONSE_THRESHOLD_MS):
131     """
132     Return one anomaly class or None.
133
134     Priority avoids ambiguous colouring when one frame satisfies several
135     conditions. CRC is strongest because the frame itself is invalid.
136     """
137     if str(record.get("crc", "")) == "ERROR":
138         return "crc"
139     if bool(record.get("timed_out")):
140         return "timeout"
141     if str(record.get("type", "")) == "EXCEPTION":
142         return "exception"
143     if str(record.get("type", "")).startswith("RAW"):
144         return "raw"
145
146     try:
147         response_ms = float(record.get("response_text", ""))
148     except (TypeError, ValueError):
149         response_ms = None
150
151     if response_ms is not None and response_ms >= float(threshold_ms):
152         return "slow"
153     return None
154
155
156 RESULT_LABELS = {
157     "success": ("✓ Sucesso", "Verde"),
158     "slow": ("⚠ Resposta lenta", "Amarelo"),
159     "exception": ("Exception", "Vermelho"),
160     "crc": ("× CRC Error", "Vermelho"),
```

MBSniffer v3.2/mb_diagnostics.py — lines 161-200

```
161     "timeout": ("⚠ Timeout", "Laranja"),
162     "raw": ("RAW", "Roxo"),
163     "pending": ("Pendente / sem resultado", "Neutro"),
164     "neutral": ("Sem classificação", "Neutro"),
165 }
166
167
168 def result_info(record, successful_request=False, language=LANGUAGE_PORTUGUESE):
169     """
170     Return (result_key, human_label, colour_name) for one traffic record.
171
172     The semantic result is always available to the Frame Inspector. Whether
173     the table/Inspector actually uses the category colour is controlled by
174     "Realçar resultados".
175     """
176     if not record:
177         return "neutral", "–", translate_text("Neutro", language)
178
179     kind = anomaly_kind(record, SLOW_RESPONSE_THRESHOLD_MS)
180     if kind:
181         label, colour = RESULT_LABELS[kind]
182         if kind == "raw":
183             # Preserve the exact parser state in the Inspector.
184             row_type = str(record.get("type", "RAW") or "RAW")
185             label = row_type
186         return kind, translate_text(label, language), translate_text(colour, language)
187
188     row_type = str(record.get("type", ""))
189
190     if (
191         row_type == "RESPONSE"
192         and record.get("crc") == "OK"
193         and record.get("matched")
194     ):
195         label, colour = RESULT_LABELS["success"]
196         return "success", translate_text(label, language), translate_text(colour, language)
197
198     if row_type == "REQUEST":
199         if successful_request and not record.get("timed_out"):
200             label, colour = RESULT_LABELS["success"]
```

MBSniffer v3.2/mb_diagnostics.py — lines 201-240

```
            return "success", translate_text(label, language), translate_text(colour, language)

        label, colour = RESULT_LABELS["pending"]
        return "pending", translate_text(label, language), translate_text(colour, language)

    label, colour = RESULT_LABELS["neutral"]
    return "neutral", translate_text(label, language), translate_text(colour, language)


def inspector_values(record, successful_request=False, language=LANGUAGE_PORTUGUESE):
    if not record:
        return {
            "slave": "Slave: —",
            "type": f"{translate_text('Tipo', language)}: —",
            "function": "Function: —",
            "response": "Resp.: —",
            "result": f"{translate_text('Resultado', language)}: —",
            "result_key": "neutral",
            "pdu": "PDU Address: —",
            "one_based": "1-based: —",
            "qty": "Qty: —",
            "byte_count": "ByteCount: —",
            "crc": "CRC: —",
            "data": f"{translate_text('Dados', language)}: —",
            "raw": "",
        }

    received, calculated, crc_state = crc_fields(record)
    if received is None:
        crc_text = f"CRC: {crc_state or '—'}"
    else:
        crc_text = (
            (f"CRC: received 0x{received:04X} / calculated 0x{calculated:04X} — {crc_state}" if language == "English" else f"CRC: recebido 0x{received:04X} / calculado 0x{calcul
ated:04X} — {crc_state}")
        )

    byte_count, data_hex, decoded = response_payload(record, language=language)
    payload = decoded or data_hex or str(record.get("details", "") or "—")

    response = str(record.get("response_text", "") or "")
    if response:
```

MBSniffer v3.2/mb_diagnostics.py — lines 241-280

```
        response += " ms"

    timeout_suffix = " — TIMEOUT" if record.get("timed_out") else ""
    result_key, result_label, _result_colour = result_info(
        record,
        successful_request=successful_request,
        language=language,
    )

    return {
        "slave": f"Slave: {record.get('slave') or '—'}",
        "type": f"{translate_text('Tipo', language)}: {record.get('type') or '—'}{timeout_suffix}",
        "function": f"Function: {function_label(record, language=language)}",
        "response": f"Resp.: {response or '—'}",
        "result": f"{translate_text('Resultado', language)}: {result_label}",
        "result_key": result_key,
        "pdu": f"PDU Address: {record.get('pdu_address') if record.get('pdu_address') is not None else '—'}",
        "one_based": f"1-based: {record.get('one_based') if record.get('one_based') is not None else '—'}",
        "qty": f"Qty: {record.get('qty') if record.get('qty') is not None else '—'}",
        "byte_count": f"ByteCount: {byte_count if byte_count is not None else '—'}",
        "crc": crc_text,
        "data": f"{translate_text('Dados', language)}: {translate_details(payload, language)}",
        "raw": str(record.get("raw", "") or ""),
    }


def decoded_record_text(record, successful_request=False, language=LANGUAGE_PORTUGUESE):
    values = inspector_values(
        record,
        successful_request=successful_request,
        language=language,
    )
    return "\n".join((
        values["slave"],
        values["type"],
        values["function"],
        values["response"],
        values["result"],
        values["pdu"],
        values["one_based"],
```

MBSniffer v3.2/mb_diagnostics.py — lines 281-286

```
        values["qty"],
        values["byte_count"],
        values["crc"],
        values["data"],
        f"Raw: {values['raw']}",
    ))
```

MBSniffer v3.2/mb_view.py

1853 source lines.

MBSniffer v3.2/mb_view.py — lines 1-40

```python
#!/usr/bin/env python3
"""Traffic rendering, filtering, Raw Hex view, statistics and log handling."""

import csv
import math
import os
import subprocess
import sys
import time
from datetime import datetime

import tkinter as tk
from tkinter import filedialog, messagebox

from mb_config import (
    MAX_UI_FRAMES,
    MAX_RAW_TEXT_LINES,
    UI_PRUNE_CHUNK_FRAMES,
    RAW_TEXT_PRUNE_CHUNK_LINES,
    SLOW_RESPONSE_THRESHOLD_MS,
    get_application_directory,
    get_logs_folder,
)
from mb_diagnostics import (
    anomaly_kind,
    decoded_record_text,
    inspector_values,
)
from mb_i18n import canonical_ui_text, translate_details
from mb_protocol import read_bits_details

class ViewMixin:
    """Methods for rendering, filtering, statistics and logging."""

    def reset_session_metrics(self):
        self.metrics.reset()
        self.update_stats_labels()

    def update_stats_labels(self):
        self.stat_requests_var.set(f"{self.tr('Pedidos')}: {self.metrics.requests}")
```

MBSniffer v3.2/mb_view.py — lines 41-80

```
        self.stat_responses_var.set(f"{self.tr('Respostas')}: {self.metrics.responses}")
        self.stat_pending_var.set(f"{self.tr('Pendentes')}: {self.metrics.pending_count()}")
        self.stat_timeouts_var.set(f"Timeouts: {self.metrics.timeouts}")
        self.stat_crc_var.set(f"{self.tr('Erros de CRC')}: {self.metrics.crc_errors}")
        self.stat_exceptions_var.set(f"{self.tr('Exceções')}: {self.metrics.exceptions}")
        self.stat_raw_var.set(f"RAW: {self.metrics.raw}")

        try:
            health_now = (
                time.time()
                if getattr(self, "capture_active", False) or getattr(self, "busy", False)
                else self.metrics.last_event_time
            )
            summary = self.metrics.health_summary(
                baud=self.baud_var.get(),
                data_bits=self.data_var.get(),
                parity=self.parity_var.get(),
                stop_bits=self.stop_var.get(),
                now=health_now,
            )
        except Exception:
            summary = self.metrics.health_summary()

        def ms(value):
            return "–" if value is None else f"{value:.1f} ms"

        if hasattr(self, "stat_slaves_var"):
            self.stat_slaves_var.set(f"Slaves: {summary['active_slaves']}")

        mappings = {
            "health_requests_var": str(self.metrics.requests),
            "health_responses_var": str(self.metrics.responses),
            "health_crc_count_var": str(self.metrics.crc_errors),
            "health_timeout_count_var": str(self.metrics.timeouts),
            "health_exception_count_var": str(self.metrics.exceptions),
            "health_req_rate_var": f"{summary['request_rate']:.2f} req/s",
            "health_resp_avg_var": ms(summary["response_avg_ms"]),
            "health_resp_min_var": ms(summary["response_min_ms"]),
            "health_resp_max_var": ms(summary["response_max_ms"]),
            "health_p95_var": ms(summary["response_p95_ms"]),
```

MBSniffer v3.2/mb_view.py — lines 81-120

```
            "health_nonvalidated_rate_var": f"{summary['nonvalidated_rate_pct']:.2f} %",
            "health_timeout_rate_var": f"{summary['timeout_rate_pct']:.2f} %",
            "health_bus_load_var": f"{summary['bus_load_pct']:.2f} %",
            "health_slaves_var": str(summary["active_slaves"]),
            "health_total_bytes_var": str(summary["total_bytes"]),
            "health_exception_slaves_var": summary["exception_summary"],
        }

        slowest_slave = summary["slowest_slave"]
        if slowest_slave is None:
            mappings["health_slowest_var"] = "—"
        else:
            mappings["health_slowest_var"] = (
                self.trf(
                    "Slave {slave} — média {average:.1f} ms",
                    slave=slowest_slave,
                    average=summary["slowest_avg_ms"],
                )
            )

        for attr, value in mappings.items():
            var = getattr(self, attr, None)
            if var is not None:
                var.set(value)

    # ------------------------------------------------------------------
    # Bus activity, filters, anomaly highlighting and Frame Inspector.
    # ------------------------------------------------------------------
    def pulse_bus_activity(self):
        var = getattr(self, "bus_activity_var", None)
        if var is None:
            return

        var.set("BUS ● RX")
        previous = getattr(self, "_bus_activity_after_id", None)
        if previous is not None:
            try:
                self.after_cancel(previous)
            except tk.TclError:
                pass
```

MBSniffer v3.2/mb_view.py — lines 121-160

```
121
122         try:
123             self._bus_activity_after_id = self.after(
124                 180,
125                 self._clear_bus_activity_pulse,
126             )
127         except tk.TclError:
128             self._bus_activity_after_id = None
129
130     def _clear_bus_activity_pulse(self):
131         self._bus_activity_after_id = None
132         var = getattr(self, "bus_activity_var", None)
133         if var is not None:
134             var.set("BUS ○")
135
136     def schedule_advanced_filter_refresh(self, *_args):
137         previous = getattr(self, "_advanced_filter_after_id", None)
138         if previous is not None:
139             try:
140                 self.after_cancel(previous)
141             except tk.TclError:
142                 pass
143         try:
144             self._advanced_filter_after_id = self.after(
145                 120,
146                 self.render_filtered_view,
147             )
148         except tk.TclError:
149             self._advanced_filter_after_id = None
150
151     def clear_advanced_filters(self):
152         pending = getattr(self, "_advanced_filter_after_id", None)
153         if pending is not None:
154             try:
155                 self.after_cancel(pending)
156             except tk.TclError:
157                 pass
158             self._advanced_filter_after_id = None
159
160         self.advanced_filter_type_var.set(self.tr("Todos"))
```

MBSniffer v3.2/mb_view.py — lines 161-200

```
161         self.advanced_filter_fc_var.set("")
162         self.advanced_filter_response_ms_var.set("")
163         self.advanced_filter_text_var.set("")
164         self._transaction_filter_seqs = None
165 
166         self.slave_filter_all_mode = True
167         for var in self.slave_filter_vars.values():
168             var.set(True)
169         self.update_slave_heading()
170 
171         self.render_filtered_view()
172 
173     @staticmethod
174     def normalized_fc_filter(value):
175         text = str(value or "").strip().upper()
176         if not text or text == "TODOS":
177             return ""
178         if text.startswith("0X"):
179             text = text[2:]
180         try:
181             number = int(text, 16)
182         except ValueError:
183             return "__INVALID__"
184         if not 0 <= number <= 0xFF:
185             return "__INVALID__"
186         return f"{number:02X}"
187 
188     def record_passes_advanced_filters(self, record):
189         transaction_filter = getattr(self, "_transaction_filter_seqs", None)
190         if transaction_filter is not None:
191             if record.get("seq") not in transaction_filter:
192                 return False
193 
194         kind = canonical_ui_text(self.advanced_filter_type_var.get())
195         row_type = str(record.get("type", ""))
196 
197         if kind == "Requests" and row_type != "REQUEST":
198             return False
199         if kind == "Responses" and row_type != "RESPONSE":
200             return False
```

MBSniffer v3.2/mb_view.py — lines 201-240

```
201         if kind == "Exceptions" and row_type != "EXCEPTION":
202             return False
203         if kind == "CRC errors" and record.get("crc") != "ERROR":
204             return False
205         if kind == "Timeouts" and not record.get("timed_out"):
206             return False
207         if kind == "RAW" and not row_type.startswith("RAW"):
208             return False
209 
210         fc_filter = self.normalized_fc_filter(
211             self.advanced_filter_fc_var.get()
212         )
213         if fc_filter == "__INVALID__":
214             return False
215         if fc_filter:
216             record_fc_text = str(record.get("fc", "")).upper()
217             candidates = {record_fc_text}
218             if str(record.get("type", "")) == "EXCEPTION":
219                 try:
220                     candidates.add(f"{int(record_fc_text, 16) & 0x7F:02X}")
221                 except ValueError:
222                     pass
223             if fc_filter not in candidates:
224                 return False
225 
226         response_limit = self.advanced_filter_response_ms_var.get().strip()
227         if response_limit:
228             try:
229                 minimum_ms = float(response_limit.replace(",", "."))
230                 response_ms = float(record.get("response_text", ""))
231             except (TypeError, ValueError):
232                 return False
233             if response_ms <= minimum_ms:
234                 return False
235 
236         query = self.advanced_filter_text_var.get().strip().casefold()
237         if query:
238             haystack = " ".join((
239                 str(record.get("details", "")),
240                 str(record.get("raw", "")),
```

MBSniffer v3.2/mb_view.py — lines 241-280

```
                str(record.get("slave", "")),
                str(record.get("fc", "")),
                str(record.get("type", "")),
                str(record.get("channel", "")),
            )).casefold()
            if query not in haystack:
                return False

        return True

    def configure_anomaly_tags(self, dark=None):
        if not hasattr(self, "tree"):
            return

        if dark is None:
            dark = bool(getattr(self, "dark_mode_var", tk.BooleanVar(value=False)).get())

        if dark:
            palette = {
                "result_success": ("#20372a", "#a7e4b8"),
                "anomaly_exception": ("#3b2728", "#ffc0c5"),
                "anomaly_crc": ("#482427", "#ffb3ba"),
                "anomaly_timeout": ("#42351f", "#ffd27a"),
                "anomaly_slow": ("#3b3422", "#ffe19a"),
                "anomaly_raw": ("#30283c", "#d8c5ff"),
            }
        else:
            palette = {
                "result_success": ("#e4f4e9", "#1f6b3a"),
                "anomaly_exception": ("#ffe8e9", "#8c1d24"),
                "anomaly_crc": ("#ffdfe1", "#8c1d24"),
                "anomaly_timeout": ("#fff0cf", "#774b00"),
                "anomaly_slow": ("#fff6dd", "#6b5200"),
                "anomaly_raw": ("#f0e9ff", "#5d3d8f"),
            }

        for tag, (background, foreground) in palette.items():
            self.tree.tag_configure(
                tag,
                background=background,
```

MBSniffer v3.2/mb_view.py — lines 281-320

```
281                 foreground=foreground,
282             )
283
284     def anomaly_tag_for_record(self, record):
285         """
286         Return the optional result-highlight tag for one logical traffic row.
287
288         "Realçar resultados" is a presentation option only:
289         - diagnostic outcomes retain their existing anomaly colors;
290         - a successfully completed normal REQUEST/RESPONSE transaction is green;
291         - unmatched/pending requests remain neutral.
292         """
293         if not getattr(self, "highlight_anomalies_var", None):
294             return ""
295         if not self.highlight_anomalies_var.get():
296             return ""
297
298         kind = anomaly_kind(record, SLOW_RESPONSE_THRESHOLD_MS)
299         if kind:
300             return f"anomaly_{kind}"
301
302         row_type = str(record.get("type", ""))
303
304         # A normal, CRC-valid matched RESPONSE is a successful result.
305         if (
306             row_type == "RESPONSE"
307             and record.get("crc") == "OK"
308             and record.get("matched")
309         ):
310             return "result_success"
311
312         # Once the normal response has arrived, also mark its REQUEST so the
313         # complete successful transaction pair reads as one result.
314         if (
315             row_type == "REQUEST"
316             and record.get("seq") in self._successful_request_seqs
317             and not record.get("timed_out")
318         ):
319             return "result_success"
320
```

MBSniffer v3.2/mb_view.py — lines 321-360

```
321         return ""
322
323     def on_highlight_anomalies_changed(self):
324         self.configure_anomaly_tags()
325         self.configure_inspector_result_styles()
326         self.render_filtered_view()
327         self.update_frame_inspector(self.selected_record())
328         saver = getattr(self, "_schedule_ui_settings_save", None)
329         if callable(saver):
330             saver()
331
332     def mark_timed_out_records(self, expired, rerender=True):
333         if not expired:
334             return 0
335
336         seqs = {
337             item.get("seq")
338             for item in expired
339             if item.get("seq") is not None
340         }
341         if not seqs:
342             return 0
343
344         changed_records = []
345         for record in self._frame_history:
346             if record.get("seq") in seqs and not record.get("timed_out"):
347                 record["timed_out"] = True
348                 changed_records.append(record)
349
350         if changed_records and rerender:
351             # The "Timeouts" filter changes which rows are visible, so only
352             # then is a full rebuild needed. Otherwise a timeout changes just
353             # the row colour: retag the already visible rows in place.
354             if canonical_ui_text(self.advanced_filter_type_var.get()) == "Timeouts":
355                 self.render_filtered_view()
356             else:
357                 self.refresh_record_tags(changed_records)
358         return len(changed_records)
359
360     def refresh_record_tags(self, records):
```

MBSniffer v3.2/mb_view.py — lines 361-400

```
361         """Re-apply the result tag of visible rows without rebuilding the table."""
362         for record in records:
363             row_tag = self.anomaly_tag_for_record(record)
364             for iid in self._tree_iids_by_seq.get(record.get("seq"), []):
365                 try:
366                     self.tree.item(iid, tags=((row_tag,) if row_tag else ()))
367                 except tk.TclError:
368                     pass
369 
370         selected = self.selected_record()
371         if selected is not None and any(selected is record for record in records):
372             self.update_frame_inspector(selected)
373 
374     def selected_record(self):
375         selection = self.tree.selection()
376         if not selection:
377             return None
378         return self._record_by_iid.get(selection[0])
379 
380     def update_frame_inspector(self, record=None):
381         if not hasattr(self, "inspector_vars"):
382             return
383 
384         successful_request = bool(
385             record
386             and record.get("type") == "REQUEST"
387             and record.get("seq") in self._successful_request_seqs
388         )
389         values = inspector_values(
390             record,
391             successful_request=successful_request,
392             language=self.current_language,
393         )
394         for key, var in self.inspector_vars.items():
395             if key in values:
396                 var.set(values[key])
397         self.selected_raw_var.set(values.get("raw", ""))
398 
399         result_label = getattr(self, "inspector_result_label", None)
400         if result_label is not None:
```

MBSniffer v3.2/mb_view.py — lines 401-440

```
            style_key = values.get("result_key", "neutral")
            if not self.highlight_anomalies_var.get():
                style_key = "neutral"
            try:
                result_label.configure(
                    style=f"MBS.Result.{style_key}.TLabel"
                )
            except tk.TclError:
                pass

    def toggle_frame_inspector(self):
        if getattr(self, "inspector_collapsed", False):
            self.inspector_body.grid()
            self.inspector_collapsed = False
            self.inspector_toggle_btn.configure(text=self.tr("Recolher"))
        else:
            self.inspector_body.grid_remove()
            self.inspector_collapsed = True
            self.inspector_toggle_btn.configure(text=self.tr("Expandir"))

        scheduler = getattr(self, "_schedule_bus_health_layout_sync", None)
        if callable(scheduler):
            scheduler()

    def copy_to_clipboard(self, text):
        try:
            self.clipboard_clear()
            self.clipboard_append(str(text))
            self.update_idletasks()
        except tk.TclError:
            pass

    def copy_selected_raw(self):
        record = self.selected_record()
        if record:
            self.copy_to_clipboard(record.get("raw", ""))

    def copy_selected_decoded(self):
        record = self.selected_record()
        if record:
```

MBSniffer v3.2/mb_view.py — lines 441-480

```
441             successful_request = bool(
442                 record.get("type") == "REQUEST"
443                 and record.get("seq") in self._successful_request_seqs
444             )
445             self.copy_to_clipboard(
446                 decoded_record_text(
447                     record,
448                     successful_request=successful_request,
449                     language=self.current_language,
450                 )
451             )
452 
453     def context_filter_selected_slave(self):
454         record = self.selected_record()
455         if not record:
456             return
457         slave = str(record.get("slave", "") or "")
458         if not slave or slave not in self.slave_filter_vars:
459             return
460 
461         self._transaction_filter_seqs = None
462         self.slave_filter_all_mode = False
463         for key, var in self.slave_filter_vars.items():
464             var.set(key == slave)
465         self.update_slave_heading()
466         self.render_filtered_view()
467 
468     def context_filter_selected_fc(self):
469         record = self.selected_record()
470         if not record:
471             return
472         fc = str(record.get("fc", "") or "")
473         if not fc:
474             return
475         if str(record.get("type", "")) == "EXCEPTION":
476             try:
477                 fc = f"{int(fc, 16) & 0x7F:02X}"
478             except ValueError:
479                 pass
480         self.advanced_filter_fc_var.set(fc)
```

MBSniffer v3.2/mb_view.py — lines 481-520

```
        self._transaction_filter_seqs = None
        self.render_filtered_view()

    def context_show_selected_transaction(self):
        record = self.selected_record()
        if not record:
            return

        seq = record.get("seq")
        seqs = {seq}

        if record.get("request_seq") is not None:
            seqs.add(record["request_seq"])
        else:
            for candidate in self._frame_history:
                if candidate.get("request_seq") == seq:
                    seqs.add(candidate.get("seq"))
                    break

        self.slave_filter_all_mode = True
        for var in self.slave_filter_vars.values():
            var.set(True)
        self.update_slave_heading()

        self.advanced_filter_type_var.set(self.tr("Todos"))
        self.advanced_filter_fc_var.set("")
        self.advanced_filter_response_ms_var.set("")
        self.advanced_filter_text_var.set("")

        self._transaction_filter_seqs = {
            value for value in seqs if value is not None
        }
        self.render_filtered_view()

    def clear_transaction_filter(self):
        self._transaction_filter_seqs = None
        self.render_filtered_view()

    def show_traffic_context_menu(self, event):
        row_iid = self.tree.identify_row(event.y)
```

MBSniffer v3.2/mb_view.py — lines 521-560

```
521         if not row_iid or row_iid not in self._record_by_iid:
522             return
523 
524         self.tree.selection_set(row_iid)
525         self.tree.focus(row_iid)
526         self.on_row_select()
527 
528         try:
529             self.traffic_context_menu.tk_popup(event.x_root, event.y_root)
530         finally:
531             try:
532                 self.traffic_context_menu.grab_release()
533             except tk.TclError:
534                 pass
535 
536     def export_session_csv(self, path=None):
537         interactive = path is None
538         records = list(self._frame_history)
539         if not records:
540             if path is None:
541                 messagebox.showinfo(
542                     self.tr("Exportar CSV"),
543                     self.tr("Não existem frames na sessão atual para exportar."),
544                 )
545             return None
546 
547         if path is None:
548             filename = datetime.now().strftime(
549                 "MBSniffer_session_%Y%m%d_%H%M%S.csv"
550             )
551             path = filedialog.asksaveasfilename(
552                 title=self.tr("Exportar sessão para CSV"),
553                 defaultextension=".csv",
554                 initialfile=filename,
555                 filetypes=[("CSV", "*.csv"), (self.tr("Todos os ficheiros"), "*.*")],
556             )
557             if not path:
558                 return None
559 
560         fieldnames = [
```

MBSniffer v3.2/mb_view.py — lines 561-600

```
            "Time",
            "Delta ms",
            "Response ms",
            "Channel",
            "Type",
            "Slave",
            "FC",
            "Details",
            "CRC",
            "Raw",
            "PDU Address",
            "1-based Address",
            "Qty",
            "Byte Count",
            "Timed out",
            "Anomaly",
        ]

        try:
            with open(path, "w", newline="", encoding="utf-8-sig") as handle:
                writer = csv.DictWriter(handle, fieldnames=fieldnames)
                writer.writeheader()

                for record in records:
                    inspector = inspector_values(record, language=self.current_language)
                    byte_count_text = inspector["byte_count"].split(
                        ":", 1
                    )[-1].strip()
                    writer.writerow({
                        "Time": record.get("time", ""),
                        "Delta ms": record.get("delta_text", ""),
                        "Response ms": record.get("response_text", ""),
                        "Channel": record.get("channel", ""),
                        "Type": record.get("type", ""),
                        "Slave": record.get("slave", ""),
                        "FC": record.get("fc", ""),
                        "Details": translate_details(record.get("details", ""), self.current_language),
                        "CRC": record.get("crc", ""),
                        "Raw": record.get("raw", ""),
                        "PDU Address": (
```

MBSniffer v3.2/mb_view.py — lines 601-640

```
                            "" if record.get("pdu_address") is None
                            else record.get("pdu_address")
                        ),
                        "1-based Address": (
                            "" if record.get("one_based") is None
                            else record.get("one_based")
                        ),
                        "Qty": (
                            "" if record.get("qty") is None
                            else record.get("qty")
                        ),
                        "Byte Count": (
                            "" if byte_count_text == "—"
                            else byte_count_text
                        ),
                        "Timed out": "Yes" if record.get("timed_out") else "No",
                        "Anomaly": anomaly_kind(
                            record,
                            SLOW_RESPONSE_THRESHOLD_MS,
                        ) or "",
                    })

            if interactive:
                try:
                    messagebox.showinfo(
                        self.tr("Exportar CSV"),
                        self.trf("Sessão exportada para:\n\n{path}", path=path),
                    )
                except tk.TclError:
                    pass
            return path

        except OSError as exc:
            messagebox.showerror(
                self.tr("Exportar CSV"),
                self.trf("Não foi possível guardar o CSV:\n\n{error}", error=exc),
            )
            return None

    def tree_record_column_values(self, record):
```

MBSniffer v3.2/mb_view.py — lines 641-680

```
641         return {
642             "time": record.get("time", ""),
643             "delta": record.get("delta_text", ""),
644             "response": record.get("response_text", ""),
645             "channel": record.get("channel", ""),
646             "type": record.get("type", ""),
647             "slave": record.get("slave", ""),
648             "fc": record.get("fc", ""),
649             "details": translate_details(record.get("details", ""), self.current_language),
650             "crc": record.get("crc", ""),
651         }
652 
653     def measure_tree_column_width(self, column, records, heading_text=None):
654         """
655         Measure the pixel width required by a non-wrapped Treeview column.
656 
657         Details is handled separately because it uses the remaining viewport
658         width and wraps instead of expanding horizontally without limit.
659         """
660         heading = (
661             heading_text
662             if heading_text is not None
663             else self.tree_headings.get(column, column)
664         )
665         max_px = self.tree_heading_font.measure(str(heading))
666 
667         for record in records:
668             value = self.tree_record_column_values(record).get(column, "")
669             max_px = max(max_px, self.tree_cell_font.measure(str(value)))
670 
671         return max_px + self.tree_column_padding_px
672 
673     def available_details_width(self):
674         """
675         Return the width that can be assigned to Details without intentionally
676         forcing horizontal scrolling in the current Treeview viewport.
677         """
678         tree_width = int(self.tree.winfo_width())
679         if tree_width <= 20:
680             tree_width = 1100  # sensible pre-layout fallback
```

MBSniffer v3.2/mb_view.py — lines 681-720

```
681
682         other_total = 0
683         for column in self.tree["columns"]:
684             if column == "details":
685                 continue
686             other_total += int(self.tree.column(column, "width"))
687
688         return max(
689             self.tree_detail_min_width_px,
690             tree_width - other_total - self.tree_detail_border_reserve_px
691         )
692
693     def wrap_text_to_pixels(self, value, max_px):
694         """
695         Word-wrap text using the actual Tk font width.
696
697         Long individual tokens are split character-by-character as a fallback,
698         so a cell can never force the Details column wider than its target.
699         """
700         text = str(value or "")
701         if not text:
702             return ""
703
704         max_px = max(40, int(max_px))
705         paragraphs = text.splitlines() or [text]
706         output_lines = []
707
708         for paragraph in paragraphs:
709             words = paragraph.split()
710             if not words:
711                 output_lines.append("")
712                 continue
713
714             line = ""
715             for word in words:
716                 candidate = word if not line else f"{line} {word}"
717                 if self.tree_cell_font.measure(candidate) <= max_px:
718                     line = candidate
719                     continue
720
```

MBSniffer v3.2/mb_view.py — lines 721-760

```
                if line:
                    output_lines.append(line)
                    line = ""

                # The token itself may be wider than the available space.
                if self.tree_cell_font.measure(word) <= max_px:
                    line = word
                    continue

                chunk = ""
                for char in word:
                    candidate_chunk = chunk + char
                    if chunk and self.tree_cell_font.measure(candidate_chunk) > max_px:
                        output_lines.append(chunk)
                        chunk = char
                    else:
                        chunk = candidate_chunk
                line = chunk

            if line:
                output_lines.append(line)

        return "\n".join(output_lines)

    def details_horizontal_rule(self):
        """
        Return a rule that fills the Details column as far as the Treeview cell
        permits. A slight overdraw is intentional; Treeview clips it at the
        column boundary.
        """
        details_width = max(
            40,
            int(self.tree.column("details", "width"))
        )
        char_px = max(1, self.tree_cell_font.measure("—"))
        count = max(8, int(math.ceil(details_width / char_px)) + 4)
        return "—" * count

    def completed_pairs_for_records(self, records):
        """
```

MBSniffer v3.2/mb_view.py — lines 761-800

```
        Return complete visible REQUEST/RESPONSE pairs in current display order.

        Each matched RESPONSE stores the exact request sequence id. This allows
        correct boundaries in both Hora ↑ and Hora ↓ without relying on row
        adjacency alone.
        """
        seq_to_index = {
            record.get("seq"): index
            for index, record in enumerate(records)
        }

        pairs = []
        for response in records:
            request_seq = response.get("request_seq")
            response_seq = response.get("seq")

            if request_seq is None or request_seq not in seq_to_index:
                continue

            request_index = seq_to_index[request_seq]
            response_index = seq_to_index.get(response_seq)
            if response_index is None:
                continue

            start_index = min(request_index, response_index)
            end_index = max(request_index, response_index)

            pairs.append({
                "request_seq": request_seq,
                "response_seq": response_seq,
                "start_index": start_index,
                "end_index": end_index,
                # Separator belongs after whichever logical frame is visually
                # the end of the pair.
                "end_seq": records[end_index].get("seq"),
            })

        pairs.sort(key=lambda item: item["start_index"])
        return pairs

```

MBSniffer v3.2/mb_view.py — lines 801-840

```
    def separator_after_sequences(self, records):
        """
        Return logical frame seq ids that need a separator AFTER them.

        There is intentionally no separator after the final complete visible
        pair because there is nothing below it to divide.
        """
        if not self.separate_transactions_var.get():
            return set()

        pairs = self.completed_pairs_for_records(records)
        if len(pairs) < 2:
            return set()

        return {
            pair["end_seq"]
            for pair in pairs[:-1]
        }

    def tree_separator_values(self):
        return (
            "", "", "", "", "", "", "",
            self.details_horizontal_rule(),
            ""
        )

    def insert_tree_separator_after_seq(self, seq):
        """Insert one Details-width separator after the visual rows of seq."""
        if seq in self._tree_separator_iid_by_seq:
            return

        iids = self._tree_iids_by_seq.get(seq, [])
        if not iids:
            return

        last_iid = iids[-1]
        insert_at = self.tree.index(last_iid) + 1

        iid = self.tree.insert(
            "",
```

MBSniffer v3.2/mb_view.py — lines 841-880

```
            insert_at,
            values=self.tree_separator_values()
        )
        self._raw_by_iid[iid] = ""
        self._tree_separator_iid_by_seq[seq] = iid

    def sync_tree_transaction_separators(self, records=None):
        """
        Synchronize standalone transaction separator rows without rebuilding
        normal traffic rows.
        """
        if records is None:
            records = self.records_for_display()

        desired = self.separator_after_sequences(records)
        existing = set(self._tree_separator_iid_by_seq)

        # Remove boundaries that are no longer valid (filter/order changes).
        for seq in existing - desired:
            iid = self._tree_separator_iid_by_seq.pop(seq, None)
            if iid and self.tree.exists(iid):
                self.tree.delete(iid)
            self._raw_by_iid.pop(iid, None)

        # Add newly required boundaries.
        for seq in desired - existing:
            self.insert_tree_separator_after_seq(seq)

    def detail_lines_for_record(self, record):
        """
        Return wrapped Details as separate visual Treeview rows.

        ttk.Treeview has one global row height for the entire widget. Putting
        newline-wrapped text in one item would therefore make every REQUEST and
        RESPONSE row as tall as the largest Details message. Continuation rows
        preserve wrapping while keeping normal rows compact.
        """
        content_width = max(
            40,
            int(self.tree.column("details", "width"))
```

MBSniffer v3.2/mb_view.py — lines 881-920

```
881             - self.tree_column_padding_px
882         )
883         wrapped = self.wrap_text_to_pixels(
884             translate_details(record.get("details", ""), self.current_language),
885             content_width
886         )
887 
888         if not wrapped:
889             return [""]
890 
891         return wrapped.splitlines()
892 
893     def set_tree_rowheight_for_records(self, records):
894         """Treeview row height stays fixed; wrapping uses continuation rows."""
895         self.tree_style.configure(
896             self.tree_style_name,
897             rowheight=self.tree_single_line_rowheight
898         )
899 
900     def update_tree_rowheight_for_record(self, record):
901         """No-op by design: one logical frame may use several compact rows."""
902         return
903 
904     def autofit_tree_columns(self, records=None):
905         """
906         Non-Details columns fit the longest visible value/title.
907 
908         Details deliberately behaves differently: it consumes the remaining
909         viewport width and wraps. This keeps the full row information visible
910         without letting long register lists create an extremely wide table.
911         """
912         if records is None:
913             records = self.records_for_display()
914 
915         # First compact every non-Details column.
916         for column in self.tree["columns"]:
917             if column == "details":
918                 continue
919 
920             width = self.measure_tree_column_width(column, records)
```

MBSniffer v3.2/mb_view.py — lines 921-960

```
            self.tree.column(
                column,
                width=width,
                minwidth=width,
                anchor="center",
                stretch=False
            )

        # Then allocate the remaining viewport to Details.
        heading_min = self.measure_tree_column_width(
            "details",
            [],
            heading_text=self.tree_headings["details"]
        )
        details_width = max(
            heading_min,
            self.available_details_width()
        )
        self.tree.column(
            "details",
            width=details_width,
            minwidth=heading_min,
            anchor="w",
            stretch=True
        )

        self.set_tree_rowheight_for_records(records)

    def expand_tree_columns_for_record(self, record):
        """
        Incrementally enlarge only the compact non-Details columns.

        Returns True when another column became wider. If that happens the
        remaining Details width changes, so the caller can rebuild/re-wrap the
        visible table once for the batch.
        """
        values = self.tree_record_column_values(record)
        layout_changed = False

        for column in self.tree["columns"]:
```

MBSniffer v3.2/mb_view.py — lines 961-1000

```
            if column == "details":
                continue

            heading_px = self.tree_heading_font.measure(
                str(self.tree_headings.get(column, column))
            )
            value_px = self.tree_cell_font.measure(str(values.get(column, "")))
            required = max(heading_px, value_px) + self.tree_column_padding_px
            current = int(self.tree.column(column, "width"))

            if required > current:
                self.tree.column(
                    column,
                    width=required,
                    minwidth=required,
                    anchor="center",
                    stretch=False
                )
                layout_changed = True

        if layout_changed:
            heading_min = self.measure_tree_column_width(
                "details",
                [],
                heading_text=self.tree_headings["details"]
            )
            details_width = max(
                heading_min,
                self.available_details_width()
            )
            self.tree.column(
                "details",
                width=details_width,
                minwidth=heading_min,
                anchor="w",
                stretch=True
            )

        return layout_changed

```

MBSniffer v3.2/mb_view.py — lines 1001-1040

```
1001    def refresh_tree_heading_width(self, column):
1002        """
1003        Dynamic Hora/Slave headings remain compact. Details is intentionally
1004        excluded because it is viewport-driven and wrapped.
1005        """
1006        if column == "details":
1007            return
1008
1009        records = self.records_for_display()
1010        width = self.measure_tree_column_width(column, records)
1011        self.tree.column(
1012            column,
1013            width=width,
1014            minwidth=width,
1015            anchor="center",
1016            stretch=False
1017        )
1018
1019    def schedule_tree_reflow(self, _event=None):
1020        """
1021        Debounce window/Treeview resizing and then re-wrap Details to the new
1022        available width.
1023        """
1024        if self._tree_reflow_after_id is not None:
1025            try:
1026                self.after_cancel(self._tree_reflow_after_id)
1027            except Exception:
1028                pass
1029
1030        self._tree_reflow_after_id = self.after(
1031            120,
1032            self.reflow_tree_after_resize
1033        )
1034
1035    def reflow_tree_after_resize(self):
1036        self._tree_reflow_after_id = None
1037        if not hasattr(self, "tree"):
1038            return
1039        self.render_filtered_view()
1040
```

MBSniffer v3.2/mb_view.py — lines 1041-1080

```
    def _slave_sort_key(self, slave):
        try:
            return (0, int(slave))
        except (TypeError, ValueError):
            return (1, str(slave))

    def rebuild_slave_filter_menu(self):
        self.slave_filter_menu.delete(0, "end")

        self.slave_filter_menu.add_command(
            label=self.tr("Todos"),
            command=self.select_all_slaves
        )
        self.slave_filter_menu.add_command(
            label=self.tr("Nenhum"),
            command=self.select_no_slaves
        )

        if self.slave_filter_vars:
            self.slave_filter_menu.add_separator()
            for slave in sorted(self.slave_filter_vars, key=self._slave_sort_key):
                self.slave_filter_menu.add_checkbutton(
                    label=f"Slave {slave}",
                    variable=self.slave_filter_vars[slave],
                    command=self.on_slave_filter_changed
                )
        else:
            self.slave_filter_menu.add_separator()
            self.slave_filter_menu.add_command(
                label=self.tr("Nenhum slave detetado"),
                state="disabled"
            )

        self.update_slave_heading()

    def update_slave_heading(self):
        if self.slave_filter_all_mode:
            label = "Slave ▾"
        else:
            selected = [
```

MBSniffer v3.2/mb_view.py — lines 1081-1120

```
                slave for slave, var in self.slave_filter_vars.items()
                if var.get()
            ]
            label = (self.tr("Slave (Filtro) ▾") if selected else self.tr("Slave (Nenhum) ▾"))

        self.tree_headings["slave"] = label
        self.tree.heading(
            "slave",
            text=label,
            anchor="center",
            command=self.show_slave_filter_menu
        )
        self.refresh_tree_heading_width("slave")

    def show_slave_filter_menu(self):
        """
        Open the slave filter menu at the position where the Slave heading
        was clicked.
        """
        try:
            x = self.winfo_pointerx()
            y = self.winfo_pointery() + 8
            self.slave_filter_menu.tk_popup(x, y)
        finally:
            try:
                self.slave_filter_menu.grab_release()
            except tk.TclError:
                pass

    def register_slave_for_filter(self, slave):
        slave = str(slave or "").strip()
        if not slave:
            return

        if slave in self.slave_filter_vars:
            return

        self.slave_filter_vars[slave] = tk.BooleanVar(
            value=self.slave_filter_all_mode
        )
```

MBSniffer v3.2/mb_view.py — lines 1121-1160

```
1121        self.rebuild_slave_filter_menu()
1122
1123    def select_all_slaves(self):
1124        self.slave_filter_all_mode = True
1125        for var in self.slave_filter_vars.values():
1126            var.set(True)
1127        self.update_slave_heading()
1128        self.render_filtered_view()
1129
1130    def select_no_slaves(self):
1131        self.slave_filter_all_mode = False
1132        for var in self.slave_filter_vars.values():
1133            var.set(False)
1134        self.update_slave_heading()
1135        self.render_filtered_view()
1136
1137    def on_slave_filter_changed(self):
1138        if not self.slave_filter_vars:
1139            self.slave_filter_all_mode = True
1140        else:
1141            values = [var.get() for var in self.slave_filter_vars.values()]
1142            # If every currently known slave is checked, remain in "All" mode
1143            # so future slaves are automatically displayed too.
1144            self.slave_filter_all_mode = bool(values) and all(values)
1145
1146        self.update_slave_heading()
1147        self.render_filtered_view()
1148
1149    def selected_slave_ids(self):
1150        return {
1151            slave for slave, var in self.slave_filter_vars.items()
1152            if var.get()
1153        }
1154
1155    def record_passes_slave_filter(self, record):
1156        # "Todos" includes RAW/unsynchronised data where a slave ID could not
1157        # be identified. "Nenhum" hides absolutely all traffic.
1158        if self.slave_filter_all_mode:
1159            return True
1160
```

MBSniffer v3.2/mb_view.py — lines 1161-1200

```
1161        selected = self.selected_slave_ids()
1162        if not selected:
1163            return False
1164
1165        return str(record.get("slave", "")) in selected
1166
1167    def time_order_descending(self):
1168        return canonical_ui_text(self.time_order_var.get()).startswith("Decrescente")
1169
1170    def on_time_order_changed(self):
1171        label = self.tr("Hora ↓") if self.time_order_descending() else self.tr("Hora ↑")
1172        self.tree_headings["time"] = label
1173        self.tree.heading(
1174            "time",
1175            text=label,
1176            anchor="center",
1177            command=self.toggle_time_order
1178        )
1179        self.render_filtered_view()
1180
1181    def toggle_time_order(self):
1182        if self.time_order_descending():
1183            self.time_order_var.set(self.tr("Crescente — antigo → recente"))
1184        else:
1185            self.time_order_var.set(self.tr("Decrescente — recente → antigo"))
1186        self.on_time_order_changed()
1187
1188    def records_for_display(self):
1189        records = [
1190            record for record in self._frame_history
1191            if (
1192                self.record_passes_slave_filter(record)
1193                and self.record_passes_advanced_filters(record)
1194            )
1195        ]
1196        if self.time_order_descending():
1197            records.reverse()
1198        return records
1199
1200    def raw_line_for_record(self, record):
```

MBSniffer v3.2/mb_view.py — lines 1201-1240

```
        # Keep fixed-width timing fields so the rest of every Raw Hex line
        # starts at exactly the same horizontal position. When a value does not
        # exist, show a completely blank field instead of "-", "dt=", "resp="
        # or "ms".
        delta_field = (
            f"dt={record['delta_text']:>7}ms "
            if record.get("delta_text")
            else " " * 13
        )
        response_field = (
            f"resp={record['response_text']:>7}ms "
            if record.get("response_text")
            else " " * 15
        )

        return (
            f"[{record['time']}] "
            f"{delta_field}"
            f"{response_field}"
            f"{record['channel']:<6} {record['type']:<10} "
            f"Slave={record['slave'] or '-':<3} FC={record['fc'] or '-':<2} "
            f"CRC={record['crc']:<5} | {record['raw']} | {translate_details(record['details'], self.current_language)}\n"
        )

    def raw_separator_width_chars(self, raw_lines=None):
        """
        Width of one Raw Hex separator in visible character cells.

        Raw Hex now wraps to the viewport, so the separator must span only the
        visible width. Using the longest unwrapped data line would make the
        separator wrap onto multiple lines.
        """
        try:
            # Leave a small pixel reserve for Text internal borders/padding so
            # the last separator character does not wrap onto a second line.
            widget_px = max(1, int(self.raw_text.winfo_width()) - 14)
            char_px = max(1, self.raw_text_font.measure("—"))
            viewport_chars = max(
                1,
                int(widget_px // char_px)
```

MBSniffer v3.2/mb_view.py — lines 1241-1280

```
            )
        except Exception:
            viewport_chars = 1

        return viewport_chars

    def raw_transaction_separator(self, width_chars=None):
        if width_chars is None:
            width_chars = self._raw_separator_width_chars
        return "—" * max(1, int(width_chars)) + "\n"

    def schedule_raw_separator_reflow(self, _event=None):
        if self._raw_reflow_after_id is not None:
            try:
                self.after_cancel(self._raw_reflow_after_id)
            except Exception:
                pass

        self._raw_reflow_after_id = self.after(
            120,
            self.reflow_raw_separator_after_resize
        )

    def reflow_raw_separator_after_resize(self):
        self._raw_reflow_after_id = None
        if self.separate_transactions_var.get():
            self.render_raw_text_only()

    def insert_tree_record(self, record, index="end", allow_separator=True):
        detail_lines = self.detail_lines_for_record(record)
        visual_rows = []

        # Main row: all columns + first Details line.
        visual_rows.append({
            "kind": "main",
            "values": (
                record["time"],
                record["delta_text"],
                record["response_text"],
                record["channel"],
```

MBSniffer v3.2/mb_view.py — lines 1281-1320

```
                record["type"],
                record["slave"],
                record["fc"],
                detail_lines[0],
                record["crc"],
            ),
            "raw": record["raw"],
        })

        # Wrapped continuation lines keep only the Details text.
        for continuation in detail_lines[1:]:
            visual_rows.append({
                "kind": "continuation",
                "values": (
                    "", "", "", "", "", "", "",
                    continuation,
                    ""
                ),
                "raw": record["raw"],
            })

        inserted_main_iid = None
        inserted_iids = []

        # index=0 requires reverse insertion to preserve visual block order.
        rows_to_insert = (
            list(reversed(visual_rows))
            if index == 0
            else visual_rows
        )

        row_tag = self.anomaly_tag_for_record(record)

        for visual in rows_to_insert:
            iid = self.tree.insert(
                "",
                index,
                values=visual["values"],
                tags=((row_tag,) if row_tag else ()),
            )
```

MBSniffer v3.2/mb_view.py — lines 1321-1360

```
1321
1322            self._raw_by_iid[iid] = visual["raw"]
1323            self._record_by_iid[iid] = record
1324
1325            if index == 0:
1326                inserted_iids.insert(0, iid)
1327                self._tree_history.appendleft((iid, True))
1328            else:
1329                inserted_iids.append(iid)
1330                self._tree_history.append((iid, True))
1331
1332            if visual["kind"] == "main":
1333                inserted_main_iid = iid
1334
1335        self._tree_iids_by_seq[record["seq"]] = inserted_iids
1336        self._ui_frame_count += 1
1337        return inserted_main_iid
1338
1339    def render_raw_text_only(self):
1340        records = self.records_for_display()
1341        separator_after = self.separator_after_sequences(records)
1342
1343        line_by_seq = {
1344            record["seq"]: self.raw_line_for_record(record)
1345            for record in records
1346        }
1347        all_raw_lines = list(line_by_seq.values())
1348        self._raw_separator_width_chars = self.raw_separator_width_chars(
1349            all_raw_lines
1350        )
1351        separator_line = self.raw_transaction_separator(
1352            self._raw_separator_width_chars
1353        )
1354
1355        # Respect the visual line cap, counting transaction separators as one
1356        # line each.
1357        if not self.time_order_descending():
1358            selected_reversed = []
1359            line_count = 0
1360
```

MBSniffer v3.2/mb_view.py — lines 1361-1400

```
            for record in reversed(records):
                cost = 1 + (1 if record["seq"] in separator_after else 0)
                if line_count + cost > MAX_RAW_TEXT_LINES:
                    break
                selected_reversed.append(record)
                line_count += cost

            raw_records = list(reversed(selected_reversed))
        else:
            selected = []
            line_count = 0

            for record in records:
                cost = 1 + (1 if record["seq"] in separator_after else 0)
                if line_count + cost > MAX_RAW_TEXT_LINES:
                    break
                selected.append(record)
                line_count += cost

            raw_records = selected

        parts = []
        for record in raw_records:
            parts.append(line_by_seq[record["seq"]])
            if record["seq"] in separator_after:
                parts.append(separator_line)

        block = "".join(parts)
        self.raw_text.delete("1.0", "end")
        if block:
            self.raw_text.insert("end", block)
        self._raw_text_line_count = block.count("\n")

    def render_filtered_view(self):
        # This is intentionally called on user-driven filter/sort changes and
        # infrequent history-prune events, not once per received frame.
        selected_record = self.selected_record()
        selected_seq = (
            selected_record.get("seq")
            if selected_record is not None
```

MBSniffer v3.2/mb_view.py — lines 1401-1440

```
            else None
        )

        for item in self.tree.get_children():
            self.tree.delete(item)

        self._raw_by_iid = {}
        self._record_by_iid = {}
        self._tree_history.clear()
        self._tree_iids_by_seq.clear()
        self._tree_separator_iid_by_seq.clear()
        self._ui_frame_count = 0
        self.selected_raw_var.set("")

        descending = self.time_order_descending()
        records = self.records_for_display()

        # Column sizing determines the wrap width, so it must happen before
        # inserting the visible rows.
        self.autofit_tree_columns(records)

        for record in records:
            self.insert_tree_record(
                record,
                index="end",
                allow_separator=False
            )

        self.sync_tree_transaction_separators(records)
        self.render_raw_text_only()

        if selected_seq is not None:
            iids = self._tree_iids_by_seq.get(selected_seq, [])
            if iids:
                try:
                    self.tree.selection_set(iids[0])
                    self.tree.focus(iids[0])
                    self.update_frame_inspector(
                        self._record_by_iid.get(iids[0])
                    )
```

MBSniffer v3.2/mb_view.py — lines 1441-1480

```
                except tk.TclError:
                    pass
            else:
                self.update_frame_inspector(None)

        if self.auto_scroll_var.get() and self._tree_history:
            if descending:
                iid = self._tree_history[0][0]
                self.tree.see(iid)
                self.raw_text.see("1.0")
            else:
                iid = self._tree_history[-1][0]
                self.tree.see(iid)
                self.raw_text.see("end")

    def add_frame_row(self, row):
        # Kept as a convenience wrapper for callers/tests.
        self.add_frame_rows_batch([row])

    def add_frame_rows_batch(self, rows):
        if not rows:
            return

        raw_lines_log = []
        any_visible = False
        tree_layout_changed = False
        pair_boundary_changed = False
        expired_requests = []

        self.pulse_bus_activity()

        for row in rows:
            seq = self._next_frame_seq
            self._next_frame_seq += 1
            row["_view_seq"] = seq

            timing = self.metrics.process(row)

            delta_text = (
                f"{timing['delta_ms']:.1f}" if timing["delta_ms"] is not None else ""
```

MBSniffer v3.2/mb_view.py — lines 1481-1520

```
            )
            response_text = (
                f"{timing['response_ms']:.1f}"
                if timing["response_ms"] is not None
                else ""
            )

            # Keep pairing for response-time/statistics, but do not copy the
            # request Address/1-based/Qty into RESPONSE/EXCEPTION details.
            # Those fields remain visible only on the REQUEST itself.
            matched = timing.get("matched_request")
            expired_requests.extend(
                timing.get("expired_requests", [])
            )

            self.register_slave_for_filter(row.get("slave", ""))

            # FC01/FC02: the request Qty tells which bits of the last data
            # byte are padding, so trim them from the response details.
            if (
                matched is not None
                and row.get("type") == "RESPONSE"
                and row.get("fc") in ("01", "02")
            ):
                try:
                    row["details"] = read_bits_details(
                        bytes.fromhex(row.get("raw", "")),
                        matched.get("qty"),
                    )
                except ValueError:
                    pass

            record = {
                "seq": seq,
                "epoch": float(row.get("_epoch", time.time())),
                "time": row["time"],
                "delta_text": delta_text,
                "response_text": response_text,
                "channel": row["channel"],
                "type": row["type"],
```

MBSniffer v3.2/mb_view.py — lines 1521-1560

```
                "slave": row["slave"],
                "fc": row["fc"],
                "details": row["details"],
                "crc": row["crc"],
                "raw": row["raw"],
                "pdu_address": row.get("pdu_address"),
                "one_based": row.get("one_based"),
                "qty": row.get("qty"),
                "frame_len": row.get("_frame_len", 0),
                "timed_out": False,
                "matched": matched is not None,
                "request_seq": (
                    matched.get("seq")
                    if matched is not None
                    else None
                ),
                # Used only by the Frame Inspector to show the address of
                # each returned value; RESPONSE Details stay response-native.
                "request_pdu_address": (
                    matched.get("pdu_address")
                    if matched is not None
                    else None
                ),
                "request_qty": (
                    matched.get("qty")
                    if matched is not None
                    else None
                ),
            }
            self._frame_history.append(record)

            # Track successful normal transaction pairs independently of whether
            # highlighting is currently enabled. This lets the user toggle
            # "Realçar resultados" on/off against the exact same session.
            if (
                record.get("type") == "RESPONSE"
                and record.get("crc") == "OK"
                and record.get("matched")
                and anomaly_kind(record, SLOW_RESPONSE_THRESHOLD_MS) is None
                and record.get("request_seq") is not None
```

MBSniffer v3.2/mb_view.py — lines 1561-1600

```
            ):
                self._successful_request_seqs.add(record["request_seq"])

                # If the REQUEST is already visible, update it immediately
                # without rebuilding the entire Treeview.
                if self.highlight_anomalies_var.get():
                    for request_iid in self._tree_iids_by_seq.get(
                        record["request_seq"], []
                    ):
                        try:
                            self.tree.item(
                                request_iid,
                                tags=("result_success",),
                            )
                        except tk.TclError:
                            pass

            # Disk log is always complete and ignores GUI filters/sorting.
            raw_lines_log.append(self.raw_line_for_record(record))

            if (
                self.record_passes_slave_filter(record)
                and self.record_passes_advanced_filters(record)
            ):
                any_visible = True
                if self.expand_tree_columns_for_record(record):
                    tree_layout_changed = True

                self.update_tree_rowheight_for_record(record)

                if self.time_order_descending():
                    self.insert_tree_record(
                        record,
                        index=0,
                        allow_separator=False
                    )

                    raw_block = self.raw_line_for_record(record)
                    self.raw_text.insert("1.0", raw_block)
                    self._raw_text_line_count += raw_block.count("\n")
```

MBSniffer v3.2/mb_view.py — lines 1601-1640

```
                else:
                    self.insert_tree_record(
                        record,
                        index="end",
                        allow_separator=False
                    )

                    raw_block = self.raw_line_for_record(record)
                    self.raw_text.insert("end", raw_block)
                    self._raw_text_line_count += raw_block.count("\n")

                if record.get("matched"):
                    pair_boundary_changed = True

        if expired_requests:
            # A new frame can itself cause older requests to cross the Pending
            # timeout. Mark those original REQUEST records in place.
            self.mark_timed_out_records(
                expired_requests,
                rerender=True,
            )

        log_block = "".join(raw_lines_log)

        # Only real traffic is allowed to create a .txt log. GUI filters do not
        # change the complete saved traffic.
        if self.capture_active and log_block:
            if self.ensure_log_started():
                self.log_line(log_block)

        self.update_stats_labels()
        rebuilt = self.prune_ui_history()

        if tree_layout_changed and not rebuilt:
            self.render_filtered_view()
            rebuilt = True

        if not rebuilt and pair_boundary_changed:
            visible_records = self.records_for_display()
            self.sync_tree_transaction_separators(visible_records)
```

MBSniffer v3.2/mb_view.py — lines 1641-1680

```
            self.render_raw_text_only()

        # Raw Hex has its own bounded line count. Rebuild it in the correct
        # chronological direction when its hysteresis threshold is reached.
        if (
            not rebuilt
            and self._raw_text_line_count
            > MAX_RAW_TEXT_LINES + RAW_TEXT_PRUNE_CHUNK_LINES
        ):
            self.render_raw_text_only()

        if self.auto_scroll_var.get() and any_visible and self._tree_history:
            if self.time_order_descending():
                self.tree.see(self._tree_history[0][0])
                self.raw_text.see("1.0")
            else:
                self.tree.see(self._tree_history[-1][0])
                self.raw_text.see("end")

    def prune_ui_history(self):
        """
        Keep the GUI source history bounded.

        The full disk log is independent and remains complete. Pruning happens
        by deque chunks and only triggers a full GUI rebuild occasionally.
        """
        trigger_frames = MAX_UI_FRAMES + UI_PRUNE_CHUNK_FRAMES

        if len(self._frame_history) <= trigger_frames:
            return False

        while len(self._frame_history) > MAX_UI_FRAMES:
            self._frame_history.popleft()

        retained_sequences = {
            record.get("seq")
            for record in self._frame_history
            if record.get("seq") is not None
        }
        self._successful_request_seqs.intersection_update(retained_sequences)
```

MBSniffer v3.2/mb_view.py — lines 1681-1720

```
1681
1682        self.render_filtered_view()
1683        return True
1684
1685    def on_row_select(self, _event=None):
1686        record = self.selected_record()
1687        self.update_frame_inspector(record)
1688
1689    def finish_capture_ui(self):
1690        self.busy = False
1691        self.capture_active = False
1692        self.status_var.set(self.tr("● Parado"))
1693        self.start_btn.configure(state="normal")
1694        self.stop_btn.configure(state="disabled")
1695        self.restart_btn.configure(state="normal")
1696        self.set_simulation_button_state("normal")
1697        self.set_sniffer_refresh_state("normal")
1698        self.serial_objects = []
1699        self.serial_by_port = {}
1700        self.reader_threads = []
1701        self.close_log()
1702        self.update_com_status()
1703
1704    def app_directory(self):
1705        return get_application_directory()
1706
1707    def prepare_log_session(self, prefix, started_at, header):
1708        """
1709        Prepare a real-capture log session without creating a file.
1710
1711        The .txt file is created only after the first received frame/RAW block.
1712        """
1713        self.close_log()
1714        self.pending_log_filename = started_at.strftime(
1715            prefix + "_%Y%m%d_%H%M%S.txt"
1716        )
1717        self.pending_log_header = header
1718        self.pending_log_notes = []
1719        self.log_start_failed = False
1720        self.log_path = None
```

MBSniffer v3.2/mb_view.py — lines 1721-1760

```
1721
1722    def ensure_log_started(self):
1723        """
1724        Create the pending real-capture log on first actual traffic.
1725
1726        Returns True when the log is open. A creation failure is reported only
1727        once for the current capture session.
1728        """
1729        if self.log_file:
1730            return True
1731
1732        if not self.pending_log_filename or self.log_start_failed:
1733            return False
1734
1735        try:
1736            target = get_logs_folder() / self.pending_log_filename
1737            self.log_file = target.open("w", encoding="utf-8", buffering=1)
1738            self.log_path = target
1739
1740            if self.pending_log_header:
1741                self.log_file.write(self.pending_log_header)
1742
1743            if self.pending_log_notes:
1744                self.log_file.write("".join(self.pending_log_notes))
1745
1746            self.pending_log_notes = []
1747            return True
1748
1749        except OSError as exc:
1750            self.log_start_failed = True
1751            self.log_file = None
1752            self.log_path = None
1753            messagebox.showerror(
1754                self.tr("Erro ao criar log"),
1755                self.trf(
1756                    "Foi detetado tráfego, mas não foi possível criar o ficheiro de log em:\n\n{folder}\n\n{error}",
1757                    folder=get_logs_folder(),
1758                    error=exc,
1759                )
1760            )
```

MBSniffer v3.2/mb_view.py — lines 1761-1800

```
1761            return False
1762
1763    def queue_log_note(self, text):
1764        """
1765        Write a capture note to an already-open log, or keep it in memory until
1766        the first traffic frame creates the log.
1767        """
1768        if self.log_file:
1769            self.log_line(text)
1770        elif self.capture_active and self.pending_log_filename:
1771            self.pending_log_notes.append(text)
1772
1773    def log_line(self, text):
1774        if self.log_file:
1775            try:
1776                self.log_file.write(text)
1777            except Exception:
1778                pass
1779
1780    def close_log(self):
1781        if self.log_file:
1782            try:
1783                self.log_file.write(
1784                    f"# Stop: {datetime.now().isoformat(timespec='milliseconds')}\n"
1785                )
1786                self.log_file.close()
1787            except Exception:
1788                pass
1789
1790        self.log_file = None
1791        self.log_path = None
1792        self.pending_log_filename = None
1793        self.pending_log_header = ""
1794        self.pending_log_notes = []
1795        self.log_start_failed = False
1796
1797    def open_log_folder(self):
1798        try:
1799            folder = get_logs_folder()
1800            if os.name == "nt":
```

MBSniffer v3.2/mb_view.py — lines 1801-1840

```
1801                os.startfile(str(folder))
1802            elif sys.platform == "darwin":
1803                subprocess.Popen(["open", str(folder)])
1804            else:
1805                subprocess.Popen(["xdg-open", str(folder)])
1806        except Exception as exc:
1807            messagebox.showerror(self.tr("Abrir pasta"), str(exc))
1808
1809    def clear_view(self):
1810        for item in self.tree.get_children():
1811            self.tree.delete(item)
1812        self.raw_text.delete("1.0", "end")
1813        self.selected_raw_var.set("")
1814        self._raw_by_iid = {}
1815        self._record_by_iid = {}
1816        self._tree_history.clear()
1817        self._tree_iids_by_seq.clear()
1818        self._tree_separator_iid_by_seq.clear()
1819        self._frame_history.clear()
1820        self._next_frame_seq = 1
1821        self._ui_frame_count = 0
1822        self._raw_text_line_count = 0
1823        self._raw_separator_width_chars = 1
1824
1825        self.slave_filter_vars.clear()
1826        self.slave_filter_all_mode = True
1827        self.rebuild_slave_filter_menu()
1828
1829        self._transaction_filter_seqs = None
1830        self._successful_request_seqs.clear()
1831        if hasattr(self, "advanced_filter_type_var"):
1832            self.advanced_filter_type_var.set(self.tr("Todos"))
1833            self.advanced_filter_fc_var.set("")
1834            self.advanced_filter_response_ms_var.set("")
1835            self.advanced_filter_text_var.set("")
1836        self.update_frame_inspector(None)
1837        if hasattr(self, "bus_activity_var"):
1838            self.bus_activity_var.set("BUS ○")
1839
1840        self.time_order_var.set(self.tr("Crescente — antigo → recente"))
```

MBSniffer v3.2/mb_view.py — lines 1841-1853

```
1841        self.tree_headings["time"] = self.tr("Hora ↑")
1842        self.tree.heading(
1843            "time",
1844            text=self.tree_headings["time"],
1845            anchor="center",
1846            command=self.toggle_time_order
1847        )
1848
1849        # With no traffic, widths/row height return to their default layout.
1850        self.autofit_tree_columns([])
1851
1852        self.reset_session_metrics()
1853
```

MBSniffer v3.2/mb_slave_finder.py

1587 source lines.

MBSniffer v3.2/mb_slave_finder.py — lines 1-40

```python
#!/usr/bin/env python3
"""Active Modbus RTU Bus Slave Finder logic and GUI mixin."""

import threading
import time
import tkinter as tk
from tkinter import ttk, messagebox

from mb_config import (
    DISCOVERY_BAUD_RATES,
    DISCOVERY_PARITIES,
    DISCOVERY_STOP_BITS,
    DISCOVERY_DEFAULT_MIN_TIMEOUT_MS,
    DISCOVERY_MIN_TIMEOUT_MS,
    DISCOVERY_MAX_TIMEOUT_MS,
    DISCOVERY_PROBE_ADDRESS,
    DISCOVERY_PROBE_QTY,
    DISCOVERY_WRITE_TIMEOUT_SECONDS,
    DISCOVERY_WRITE_RETRIES,
    DISCOVERY_RX_BUFFER_LIMIT,
    DISCOVERY_MIN_ATTEMPTS,
    DISCOVERY_MAX_ATTEMPTS,
    DISCOVERY_DEFAULT_ATTEMPTS,
    DISCOVERY_IDLE_CHECK_SECONDS,
    MODBUS_EXCEPTION_CODES,
)
from mb_protocol import (
    append_crc,
    crc_ok,
    hex_bytes,
    modbus_rtu_interframe_gap_seconds,
)
from mb_widgets import RefreshButton

def build_discovery_probe(slave: int, fc: int) -> bytes:
    """Build a read-only discovery probe: FC03/FC04, Address 0, Qty 1."""
    if not (1 <= int(slave) <= 247):
        raise ValueError("Slave ID fora do intervalo 1..247")
    if int(fc) not in (3, 4):
        raise ValueError("A descoberta ativa suporta apenas FC03/FC04")
```

MBSniffer v3.2/mb_slave_finder.py — lines 41-80

```
41
42      payload = bytes((
43          int(slave),
44          int(fc),
45          0x00, DISCOVERY_PROBE_ADDRESS & 0xFF,
46          0x00, DISCOVERY_PROBE_QTY & 0xFF,
47      ))
48      return append_crc(payload)
49
50  DEVICE_ID_FC = 0x2B
51  DEVICE_ID_MEI_TYPE = 0x0E
52  DEVICE_ID_READ_BASIC = 0x01
53  DEVICE_ID_MORE_FOLLOWS = 0xFF
54  DEVICE_ID_BASIC_OBJECTS = (0x00, 0x01, 0x02)
55  DEVICE_ID_MAX_PAGES = 8
56
57  def build_device_identification_probe(slave: int, object_id: int = 0x00) -> bytes:
58      """Build FC43/MEI 14 Basic Device Identification request."""
59      slave = int(slave)
60      object_id = int(object_id)
61      if not (1 <= slave <= 247):
62          raise ValueError("Slave ID fora do intervalo 1..247")
63      if not (0 <= object_id <= 0xFF):
64          raise ValueError("Object ID fora do intervalo 0..255")
65      return append_crc(bytes((
66          slave,
67          DEVICE_ID_FC,
68          DEVICE_ID_MEI_TYPE,
69          DEVICE_ID_READ_BASIC,
70          object_id,
71      )))
72
73  def _decode_device_id_value(value: bytes) -> str:
74      """Decode a Modbus Device Identification ASCII object for display."""
75      return bytes(value).decode("ascii", errors="replace").strip("\x00 ")
76
77  def find_device_identification_response(buffer: bytes, slave: int):
78      """Find one CRC-valid FC43/14 response or Modbus exception in *buffer*."""
79      data = bytes(buffer)
80      slave = int(slave)
```

MBSniffer v3.2/mb_slave_finder.py — lines 81-120

```
81
82      for offset in range(max(0, len(data) - 3)):
83          if data[offset] != slave or offset + 2 > len(data):
84              continue
85
86          response_fc = data[offset + 1]
87
88          # FC43 exception response: Slave + 0xAB + Exception + CRC.
89          if response_fc == (DEVICE_ID_FC | 0x80):
90              end = offset + 5
91              if end <= len(data):
92                  frame = data[offset:end]
93                  if crc_ok(frame):
94                      return {
95                          "kind": "EXCEPTION",
96                          "frame": frame,
97                          "exception_code": frame[2],
98                          "objects": {},
99                          "more_follows": False,
100                         "next_object_id": 0,
101                     }
102             continue
103
104         if response_fc != DEVICE_ID_FC:
105             continue
106
107         # Normal response fixed header before the object list:
108         # Slave, FC, MEI, ReadCode, Conformity, MoreFollows, NextObject, Count.
109         if offset + 8 > len(data):
110             continue
111         if data[offset + 2] != DEVICE_ID_MEI_TYPE:
112             continue
113         if data[offset + 3] != DEVICE_ID_READ_BASIC:
114             continue
115
116         more_follows = data[offset + 5] == DEVICE_ID_MORE_FOLLOWS
117         next_object_id = data[offset + 6]
118         object_count = data[offset + 7]
119         pos = offset + 8
120         objects = {}
```

MBSniffer v3.2/mb_slave_finder.py — lines 121-160

```
121         complete = True
122
123         for _ in range(object_count):
124             if pos + 2 > len(data):
125                 complete = False
126                 break
127             object_id = data[pos]
128             object_len = data[pos + 1]
129             pos += 2
130             if pos + object_len > len(data):
131                 complete = False
132                 break
133             objects[object_id] = _decode_device_id_value(
134                 data[pos:pos + object_len]
135             )
136             pos += object_len
137
138         if not complete:
139             continue
140
141         end = pos + 2
142         if end > len(data):
143             continue
144         frame = data[offset:end]
145         if not crc_ok(frame):
146             continue
147
148         return {
149             "kind": "DEVICE_ID",
150             "frame": frame,
151             "exception_code": None,
152             "objects": objects,
153             "more_follows": more_follows,
154             "next_object_id": next_object_id,
155             "conformity_level": data[offset + 4],
156         }
157
158     return None
159
160 def read_device_identification_response(ser, slave, timeout_s, stop_event):
```

MBSniffer v3.2/mb_slave_finder.py — lines 161-200

```
161     """Read until a valid FC43/14 response/exception is received or timed out."""
162     deadline = time.perf_counter() + max(0.001, float(timeout_s))
163     buffer = bytearray()
164 
165     while time.perf_counter() < deadline:
166         if stop_event.is_set():
167             return None
168 
169         waiting = int(getattr(ser, "in_waiting", 0) or 0)
170         if waiting:
171             chunk = ser.read(waiting)
172             if chunk:
173                 buffer.extend(chunk)
174                 if len(buffer) > DISCOVERY_RX_BUFFER_LIMIT:
175                     del buffer[:-DISCOVERY_RX_BUFFER_LIMIT]
176                 found = find_device_identification_response(buffer, slave)
177                 if found is not None:
178                     return found
179         elif stop_event.wait(0.001):
180             return None
181 
182     return find_device_identification_response(buffer, slave)
183 
184 def device_identification_timeout_seconds(
185     baud: int,
186     parity: str,
187     stopbits: str,
188     minimum_timeout_ms: float = DISCOVERY_DEFAULT_MIN_TIMEOUT_MS,
189 ) -> float:
190     """Conservative timeout for one FC43/14 response, up to one RTU frame."""
191     baud = max(1, int(baud))
192     bits_per_char = discovery_bits_per_char(parity, stopbits)
193     char_s = bits_per_char / float(baud)
194     # A Modbus RTU ADU is at most 256 bytes. Allow one full response frame,
195     # the recommended t3.5 gap and a small device-processing margin.
196     calculated = (
197         256.0 * char_s
198         + modbus_rtu_interframe_gap_seconds(baud, bits_per_char)
199         + 0.020
200     )
```

MBSniffer v3.2/mb_slave_finder.py — lines 201-240

```
    return max(float(minimum_timeout_ms) / 1000.0, calculated)

def device_identification_request_transmit_seconds(
    baud: int, parity: str, stopbits: str
) -> float:
    """Estimated on-wire duration of the 7-byte FC43/14 request."""
    return 7.0 * discovery_bits_per_char(parity, stopbits) / float(max(1, int(baud)))

def format_device_identification(result) -> str:
    """Return compact Device Identification text for the Finder result table."""
    if result is None:
        return "No response"
    if result.get("kind") == "EXCEPTION":
        code = int(result.get("exception_code", 0))
        name = MODBUS_EXCEPTION_CODES.get(
            code, ("Unknown/Reserved Exception", "")
        )[0]
        return f"Exception 0x{code:02X} – {name}"

    objects = result.get("objects", {}) or {}
    values = [objects.get(obj_id, "") for obj_id in DEVICE_ID_BASIC_OBJECTS]
    if any(values):
        return " | ".join(value or "–" for value in values)
    return "Response – no Basic Device Identification objects"

def read_basic_device_identification(
    ser, slave, baud, parity, stopbits, minimum_timeout_ms, stop_event, serial_module
):
    """Read Basic Device Identification, following More Follows when required."""
    object_id = 0x00
    merged_objects = {}
    frames = []
    conformity_level = None

    for _page in range(DEVICE_ID_MAX_PAGES):
        if stop_event.is_set():
            return None

        try:
            ser.reset_input_buffer()
```

MBSniffer v3.2/mb_slave_finder.py — lines 241-280

```
        except Exception:
            raise

        request = build_device_identification_probe(slave, object_id)
        sent = write_discovery_request(
            ser, request, baud, parity, stopbits, stop_event, serial_module
        )
        if not sent or stop_event.is_set():
            return None

        result = read_device_identification_response(
            ser,
            slave,
            device_identification_timeout_seconds(
                baud, parity, stopbits, minimum_timeout_ms
            ),
            stop_event,
        )
        if result is None or result.get("kind") == "EXCEPTION":
            return result

        frames.append(result["frame"])
        merged_objects.update(result.get("objects", {}))
        conformity_level = result.get("conformity_level")

        # Basic identification is complete as soon as VendorName, ProductCode
        # and MajorMinorRevision have all been received.
        if all(obj_id in merged_objects for obj_id in DEVICE_ID_BASIC_OBJECTS):
            return {
                "kind": "DEVICE_ID",
                "frame": b"".join(frames),
                "exception_code": None,
                "objects": merged_objects,
                "more_follows": bool(result.get("more_follows")),
                "next_object_id": int(result.get("next_object_id", 0)),
                "conformity_level": conformity_level,
            }

        if not result.get("more_follows"):
            break
```

MBSniffer v3.2/mb_slave_finder.py — lines 281-320

```
281
282         next_object_id = int(result.get("next_object_id", 0))
283         if next_object_id == object_id:
284             break
285         object_id = next_object_id
286
287         gap_s = modbus_rtu_interframe_gap_seconds(
288             baud, discovery_bits_per_char(parity, stopbits)
289         )
290         if stop_event.wait(gap_s):
291             return None
292
293     return {
294         "kind": "DEVICE_ID",
295         "frame": b"".join(frames),
296         "exception_code": None,
297         "objects": merged_objects,
298         "more_follows": False,
299         "next_object_id": object_id,
300         "conformity_level": conformity_level,
301     } if frames else None
302
303 def discovery_bits_per_char(parity: str, stopbits: str) -> float:
304     """Return serial bits/character: start + 8 data + parity + stop."""
305     parity_bits = 0 if parity == "None" else 1
306     stop_value = float(stopbits)
307     return 1.0 + 8.0 + parity_bits + stop_value
308
309 def discovery_response_timeout_seconds(
310     baud: int,
311     parity: str,
312     stopbits: str,
313     minimum_timeout_ms: float = DISCOVERY_DEFAULT_MIN_TIMEOUT_MS,
314 ) -> float:
315     """
316     Adaptive response deadline after the request has left the serial driver.
317
318     The deadline includes the on-wire duration of a normal 7-byte FC03/FC04
319     response plus processing/inter-frame margin, with a user-configurable
320     minimum.
```

MBSniffer v3.2/mb_slave_finder.py — lines 321-360

```
321     """
322     baud = max(1, int(baud))
323     bits_per_char = discovery_bits_per_char(parity, stopbits)
324     char_s = bits_per_char / float(baud)
325     normal_response_s = 7.0 * char_s
326     serial_margin_s = max(
327         0.012, modbus_rtu_interframe_gap_seconds(baud, bits_per_char)
328     )
329     calculated = normal_response_s + serial_margin_s
330     return max(float(minimum_timeout_ms) / 1000.0, calculated)
331
332 def discovery_request_transmit_seconds(baud: int, parity: str, stopbits: str) -> float:
333     """Estimated on-wire duration of the fixed 8-byte discovery request."""
334     return 8.0 * discovery_bits_per_char(parity, stopbits) / float(max(1, int(baud)))
335
336 def find_discovery_response(buffer: bytes, slave: int, fc: int):
337     """
338     Find a valid normal or exception response for one discovery probe.
339
340     Any CRC-valid Modbus exception counts as proof that the slave exists.
341     The scanner intentionally ignores other traffic and request echoes.
342     """
343     data = bytes(buffer)
344     slave = int(slave)
345     fc = int(fc)
346
347     for offset in range(max(0, len(data) - 3)):
348         if data[offset] != slave or offset + 2 > len(data):
349             continue
350
351         response_fc = data[offset + 1]
352
353         if response_fc == (fc | 0x80):
354             end = offset + 5
355             if end <= len(data):
356                 frame = data[offset:end]
357                 if crc_ok(frame):
358                     code = frame[2]
359                     return {
360                         "kind": "EXCEPTION",
```

MBSniffer v3.2/mb_slave_finder.py — lines 361-400

```
                        "frame": frame,
                        "exception_code": code,
                    }

        if response_fc == fc and offset + 3 <= len(data):
            byte_count = data[offset + 2]
            # FC03/04 register responses must contain an even positive byte count.
            if byte_count > 0 and byte_count % 2 == 0 and byte_count <= 250:
                end = offset + 5 + byte_count
                if end <= len(data):
                    frame = data[offset:end]
                    if crc_ok(frame):
                        return {
                            "kind": "RESPONSE",
                            "frame": frame,
                            "exception_code": None,
                        }

    return None

def read_discovery_response(ser, slave, fc, timeout_s, stop_event):
    """Read non-blocking serial bytes until a valid probe response or timeout."""
    deadline = time.perf_counter() + max(0.001, float(timeout_s))
    buffer = bytearray()

    while time.perf_counter() < deadline:
        if stop_event.is_set():
            return None

        # Do not hide serial-driver failures as ordinary slave timeouts.
        waiting = int(getattr(ser, "in_waiting", 0) or 0)

        if waiting:
            chunk = ser.read(waiting)
            if chunk:
                buffer.extend(chunk)
                if len(buffer) > DISCOVERY_RX_BUFFER_LIMIT:
                    del buffer[:-DISCOVERY_RX_BUFFER_LIMIT]
                found = find_discovery_response(buffer, slave, fc)
                if found is not None:
```

MBSniffer v3.2/mb_slave_finder.py — lines 401-440

```
401                     return found
402         else:
403             # Event.wait() keeps the Stop button responsive.
404             if stop_event.wait(0.001):
405                 return None
406 
407     return find_discovery_response(buffer, slave, fc)
408 
409 def bus_activity_detected(ser, listen_s, stop_event):
410     """Listen passively for *listen_s* seconds; return the bytes received.
411 
412     Any byte, even one framed at another baud rate, means something else is
413     transmitting, so the active scan must not start.
414     """
415     deadline = time.perf_counter() + max(0.0, float(listen_s))
416     received = bytearray()
417     while time.perf_counter() < deadline:
418         if stop_event.is_set():
419             break
420         waiting = int(getattr(ser, "in_waiting", 0) or 0)
421         if waiting:
422             received.extend(ser.read(waiting) or b"")
423             if received:
424                 break
425         elif stop_event.wait(0.005):
426             break
427     return bytes(received)
428 
429 
430 def cancel_serial_io(ser):
431     """Best-effort cancellation of pending serial I/O."""
432     if ser is None:
433         return
434     for method_name in ("cancel_read", "cancel_write"):
435         method = getattr(ser, method_name, None)
436         if callable(method):
437             try:
438                 method()
439             except Exception:
440                 pass
```

MBSniffer v3.2/mb_slave_finder.py — lines 441-480

```
441
442 def recover_discovery_write(ser):
443     """Best-effort cleanup after a discovery write timeout."""
444     cancel = getattr(ser, "cancel_write", None)
445     if callable(cancel):
446         try:
447             cancel()
448         except Exception:
449             pass
450     try:
451         ser.reset_output_buffer()
452     except Exception:
453         pass
454     try:
455         ser.reset_input_buffer()
456     except Exception:
457         pass
458
459 def write_discovery_request(
460     ser, request, baud, parity, stopbits, stop_event, serial_module
461 ):
462     """
463     Transmit one discovery request robustly.
464
465     A write timeout is retried once after cancelling/clearing TX. ser.flush()
466     is deliberately avoided because some USB-serial drivers can block there.
467     The helper also waits for the request's estimated physical on-wire time so
468     the slave response timeout starts at a sensible point on slow baud rates.
469     """
470     request = bytes(request)
471     timeout_exc = getattr(serial_module, "SerialTimeoutException", TimeoutError)
472     serial_exc = getattr(serial_module, "SerialException", OSError)
473     attempts = DISCOVERY_WRITE_RETRIES + 1
474     last_error = None
475     bits_per_char = discovery_bits_per_char(parity, stopbits)
476     char_time_s = bits_per_char / float(max(1, int(baud)))
477     request_wire_s = len(request) * char_time_s
478     retry_silence_s = max(
479         0.050,
480         request_wire_s + modbus_rtu_interframe_gap_seconds(baud, bits_per_char),
```

MBSniffer v3.2/mb_slave_finder.py — lines 481-520

```
481     )
482
483     for attempt in range(attempts):
484         if stop_event.is_set():
485             return False
486
487         started_write = time.perf_counter()
488         try:
489             written_total = 0
490             while written_total < len(request):
491                 if stop_event.is_set():
492                     return False
493                 written = ser.write(request[written_total:])
494                 written = 0 if written is None else int(written)
495                 if written <= 0:
496                     raise timeout_exc(
497                         "A porta série não aceitou bytes para transmissão."
498                     )
499                 written_total += written
500
501             elapsed_write_s = time.perf_counter() - started_write
502             remaining_wire_s = max(0.0, request_wire_s - elapsed_write_s)
503             if remaining_wire_s > 0 and stop_event.wait(remaining_wire_s):
504                 return False
505             return True
506
507         except timeout_exc as exc:
508             last_error = exc
509             recover_discovery_write(ser)
510             if attempt + 1 >= attempts or stop_event.is_set():
511                 raise
512             # If any fragment reached the wire before the driver timed out,
513             # leave enough silence for that fragment to finish and for a full
514             # Modbus RTU frame gap before retrying.
515             if stop_event.wait(retry_silence_s):
516                 return False
517
518         except serial_exc:
519             raise
520
```

MBSniffer v3.2/mb_slave_finder.py — lines 521-560

```
521     if last_error is not None:
522         raise last_error
523     return False
524 
525 def format_duration(seconds: float) -> str:
526     total = max(0, int(round(float(seconds))))
527     hours, rem = divmod(total, 3600)
528     minutes, secs = divmod(rem, 60)
529     if hours:
530         return f"{hours:d}:{minutes:02d}:{secs:02d}"
531     return f"{minutes:02d}:{secs:02d}"
532 
533 
534 class SlaveFinderMixin:
535     """Methods that implement the active Bus Slave Finder tab."""
536 
537     def build_discovery_tab(self, page):
538         """Build the full-window active Modbus RTU slave-discovery workspace."""
539         finder_saved = getattr(self, "ui_settings", {}).get(
540             "bus_slave_finder", {}
541         )
542         if not isinstance(finder_saved, dict):
543             finder_saved = {}
544 
545         saved_bauds = set(
546             finder_saved.get("baud_rates", [9600, 19200])
547         )
548         saved_parities = set(
549             finder_saved.get("parities", ["None", "Even"])
550         )
551         saved_stops = set(
552             finder_saved.get("stop_bits", ["1"])
553         )
554 
555         page.columnconfigure(0, weight=1)
556         page.rowconfigure(3, weight=1)
557 
558         warning = ttk.LabelFrame(page, text="Pesquisa ativa — transmite no barramento")
559         warning.grid(row=0, column=0, sticky="ew", padx=12, pady=(12, 6))
560         warning.columnconfigure(0, weight=1)
```

MBSniffer v3.2/mb_slave_finder.py — lines 561-600

```
561         ttk.Label(
562             warning,
563             text=(
564                 "Esta ferramenta atua temporariamente como master Modbus RTU e envia apenas "
565                 "pedidos de leitura. Não a utilizes com outro master ativo no mesmo barramento."
566             ),
567             wraplength=1100,
568         ).grid(row=0, column=0, sticky="w", padx=10, pady=(8, 4))
569 
570         self.discovery_safety_var = tk.BooleanVar(value=False)
571         safety_cb = ttk.Checkbutton(
572             warning,
573             text="Confirmo que não existe outro master ativo no barramento.",
574             variable=self.discovery_safety_var,
575             command=self.update_discovery_start_state,
576         )
577         safety_cb.grid(row=1, column=0, sticky="w", padx=10, pady=(0, 8))
578 
579         options = ttk.LabelFrame(page, text="Configuração da pesquisa")
580         self._discovery_options_frame = options
581         options.grid(row=1, column=0, sticky="ew", padx=12, pady=6)
582         options.columnconfigure(0, weight=1)
583 
584         top_row = ttk.Frame(options)
585         top_row.grid(row=0, column=0, sticky="ew", padx=8, pady=(8, 4))
586         top_row.columnconfigure(3, weight=1)
587 
588         ttk.Label(top_row, text="Porta COM").grid(
589             row=0, column=0, sticky="w", padx=(0, 4)
590         )
591         self.discovery_port_var = tk.StringVar()
592         self.discovery_port_combo = ttk.Combobox(
593             top_row,
594             textvariable=self.discovery_port_var,
595             width=13,
596             state="readonly",
597         )
598         self.discovery_port_combo.grid(
599             row=0, column=1, sticky="w", padx=(0, 0)
600         )
```

MBSniffer v3.2/mb_slave_finder.py — lines 601-640

```
601         self.discovery_port_combo.bind(
602             "<<ComboboxSelected>>",
603             lambda _e: self.update_discovery_start_state(),
604         )
605 
606         self.discovery_refresh_btn = RefreshButton(
607             top_row,
608             command=self.refresh_ports,
609             size=24,
610         )
611         self.discovery_refresh_btn.grid(
612             row=0, column=2, sticky="w", padx=(4, 0)
613         )
614 
615         def _sync_discovery_refresh_square(_event=None):
616             height = self.discovery_port_combo.winfo_height()
617             if height > 1:
618                 self.discovery_refresh_btn.configure(
619                     width=height, height=height
620                 )
621 
622         self.discovery_port_combo.bind(
623             "<Configure>", _sync_discovery_refresh_square, add="+"
624         )
625 
626         ttk.Label(
627             top_row,
628             text="Data bits: 8 (fixo)",
629         ).grid(row=0, column=4, sticky="w", padx=(28, 20))
630         ttk.Label(
631             top_row,
632             text="Probe: FC03 / Address 0 / Qty 1",
633         ).grid(row=0, column=5, sticky="w")
634 
635         ttk.Label(top_row, text="Tentativas").grid(
636             row=0, column=6, sticky="w", padx=(28, 4)
637         )
638         self.discovery_attempts_var = tk.StringVar(
639             value=finder_saved.get("attempts", str(DISCOVERY_DEFAULT_ATTEMPTS))
640         )
```

MBSniffer v3.2/mb_slave_finder.py — lines 641-680

```
        self.discovery_attempts_spin = ttk.Spinbox(
            top_row,
            from_=DISCOVERY_MIN_ATTEMPTS,
            to=DISCOVERY_MAX_ATTEMPTS,
            increment=1,
            width=4,
            textvariable=self.discovery_attempts_var,
        )
        self.discovery_attempts_spin.grid(row=0, column=7, sticky="w")

        # Baud-rate checkboxes. Defaults keep a normal industrial scan quick.
        baud_frame = ttk.LabelFrame(options, text="Baud rate")
        baud_frame.grid(row=1, column=0, sticky="ew", padx=8, pady=4)
        baud_frame.rowconfigure(0, weight=1)
        self.discovery_baud_vars = {}
        for idx in range(len(DISCOVERY_BAUD_RATES)):
            baud_frame.columnconfigure(idx, weight=1, uniform="baud")
        for idx, baud in enumerate(DISCOVERY_BAUD_RATES):
            var = tk.BooleanVar(value=baud in saved_bauds)
            self.discovery_baud_vars[baud] = var
            cb = ttk.Checkbutton(baud_frame, text=str(baud), variable=var)
            cb.grid(row=0, column=idx, sticky="", padx=4, pady=(8, 7))
            var.trace_add("write", lambda *_a: self.on_discovery_options_changed())

        serial_frame = ttk.Frame(options)
        serial_frame.grid(row=2, column=0, sticky="ew", padx=8, pady=4)
        # Equal thirds independent of translated child captions.
        for column in range(3):
            serial_frame.columnconfigure(
                column, weight=1, uniform="finder_serial_columns"
            )

        parity_box = ttk.LabelFrame(serial_frame, text="Parity")
        parity_box.grid(row=0, column=0, sticky="nsew", padx=(0, 5))
        parity_box.rowconfigure(0, weight=1)
        for idx in range(len(DISCOVERY_PARITIES)):
            parity_box.columnconfigure(idx, weight=1, uniform="parity")
        self.discovery_parity_vars = {}
        for idx, parity in enumerate(DISCOVERY_PARITIES):
            var = tk.BooleanVar(value=parity in saved_parities)
```

MBSniffer v3.2/mb_slave_finder.py — lines 681-720

```
            self.discovery_parity_vars[parity] = var
            ttk.Checkbutton(parity_box, text=parity, variable=var).grid(
                row=0, column=idx, sticky="", padx=4, pady=(8, 7)
            )
            var.trace_add("write", lambda *_a: self.on_discovery_options_changed())
        stop_box = ttk.LabelFrame(serial_frame, text="Stop bits")
        stop_box.grid(row=0, column=1, sticky="nsew", padx=5)
        stop_box.rowconfigure(0, weight=1)
        for idx in range(len(DISCOVERY_STOP_BITS)):
            stop_box.columnconfigure(idx, weight=1, uniform="stop")
        self.discovery_stop_vars = {}
        for idx, stopbits in enumerate(DISCOVERY_STOP_BITS):
            var = tk.BooleanVar(value=stopbits in saved_stops)
            self.discovery_stop_vars[stopbits] = var
            ttk.Checkbutton(stop_box, text=stopbits, variable=var).grid(
                row=0, column=idx, sticky="", padx=4, pady=(8, 7)
            )
            var.trace_add("write", lambda *_a: self.on_discovery_options_changed())
        range_box = ttk.LabelFrame(serial_frame, text="Slave IDs / velocidade")
        range_box.grid(row=0, column=2, sticky="nsew", padx=(5, 0))
        range_box.columnconfigure(0, weight=1)
        range_box.rowconfigure(0, weight=1)

        # Keep the range and timeout controls on one centred line so this
        # LabelFrame stays as compact as the neighbouring Parity/Stop bits
        # boxes.  The option checkboxes keep their own second line so the long
        # Device Identification caption remains fully inside the rectangle.
        range_content = ttk.Frame(range_box)
        range_content.grid(
            row=0, column=0, sticky="nsew", padx=8, pady=(8, 7)
        )
        range_content.columnconfigure(0, weight=1)
        for row in range(2):
            range_content.rowconfigure(row, weight=1, uniform="range_rows")

        id_row = ttk.Frame(range_content)
        id_row.grid(row=0, column=0, sticky="")
        ttk.Label(id_row, text="De").grid(row=0, column=0, padx=(0, 3))
        self.discovery_slave_start_var = tk.StringVar(value=finder_saved.get("slave_start", "1"))
        self.discovery_slave_start_spin = ttk.Spinbox(
```

MBSniffer v3.2/mb_slave_finder.py — lines 721-760

```
            id_row, from_=1, to=247, width=5, textvariable=self.discovery_slave_start_var
        )
        self.discovery_slave_start_spin.grid(row=0, column=1, padx=3)
        ttk.Label(id_row, text="até").grid(row=0, column=2, padx=3)
        self.discovery_slave_end_var = tk.StringVar(value=finder_saved.get("slave_end", "247"))
        self.discovery_slave_end_spin = ttk.Spinbox(
            id_row, from_=1, to=247, width=5, textvariable=self.discovery_slave_end_var
        )
        self.discovery_slave_end_spin.grid(row=0, column=3, padx=3)

        ttk.Label(id_row, text="Timeout mín.").grid(
            row=0, column=4, padx=(14, 3)
        )
        self.discovery_min_timeout_var = tk.StringVar(
            value=finder_saved.get(
                "min_timeout_ms",
                str(DISCOVERY_DEFAULT_MIN_TIMEOUT_MS),
            )
        )
        self.discovery_min_timeout_spin = ttk.Spinbox(
            id_row,
            from_=DISCOVERY_MIN_TIMEOUT_MS,
            to=DISCOVERY_MAX_TIMEOUT_MS,
            increment=5,
            width=6,
            textvariable=self.discovery_min_timeout_var,
        )
        self.discovery_min_timeout_spin.grid(row=0, column=5, padx=3)
        ttk.Label(id_row, text="ms").grid(row=0, column=6, padx=3)

        toggles = ttk.Frame(range_content)
        # Small vertical separation from the Slave range/timeout row.
        # Keep this deliberately modest so the panel remains compact.
        toggles.grid(row=1, column=0, sticky="", pady=(3, 0))

        self.discovery_fc04_fallback_var = tk.BooleanVar(value=finder_saved.get("fc04_fallback", False))
        fallback_cb = ttk.Checkbutton(
            toggles,
            text="FC04 fallback",
            variable=self.discovery_fc04_fallback_var,
```

MBSniffer v3.2/mb_slave_finder.py — lines 761-800

```
        )
        fallback_cb.grid(row=0, column=0, sticky="w", padx=(0, 10))

        self.discovery_device_identification_var = tk.BooleanVar(
            value=finder_saved.get("device_identification", False)
        )
        device_id_cb = ttk.Checkbutton(
            toggles,
            text="Device Identification (FC43/14)",
            variable=self.discovery_device_identification_var,
        )
        device_id_cb.grid(row=0, column=1, sticky="w")

        for var in (
            self.discovery_slave_start_var,
            self.discovery_slave_end_var,
            self.discovery_min_timeout_var,
            self.discovery_attempts_var,
            self.discovery_fc04_fallback_var,
            self.discovery_device_identification_var,
        ):
            var.trace_add("write", lambda *_a: self.on_discovery_options_changed())

        ttk.Label(
            options,
            text=(
                "FC03 é o modo rápido. Uma resposta normal OU uma Modbus Exception com CRC válido "
                "conta como slave encontrado. Ativa FC04 fallback apenas para dispositivos que "
                "possam ignorar FC03 sem devolver Exception. Device Identification envia FC43/14 "
                "apenas aos slaves encontrados e tenta ler VendorName, ProductCode e Revision. "
                "Tentativas repete cada pedido sem resposta. Antes de transmitir, a pesquisa escuta "
                "o barramento durante 1 s e não arranca se detetar tráfego."
            ),
            wraplength=1120,
        ).grid(row=3, column=0, sticky="w", padx=10, pady=(4, 8))

        controls = ttk.LabelFrame(page, text="Pesquisa")
        controls.grid(row=2, column=0, sticky="ew", padx=12, pady=6)
        controls.columnconfigure(4, weight=1)

```

MBSniffer v3.2/mb_slave_finder.py — lines 801-840

```
        self.discovery_start_btn = ttk.Button(
            controls, text="Iniciar pesquisa", command=self.start_discovery_scan
        )
        self.discovery_start_btn.grid(row=0, column=0, padx=(10, 5), pady=8)
        self.discovery_stop_btn = ttk.Button(
            controls, text="Parar", command=self.stop_discovery_scan, state="disabled"
        )
        self.discovery_stop_btn.grid(row=0, column=1, padx=5, pady=8)
        self.discovery_clear_btn = ttk.Button(
            controls, text="Limpar resultados", command=self.clear_discovery_results
        )
        self.discovery_clear_btn.grid(row=0, column=2, padx=5, pady=8)

        self.discovery_found_var = tk.StringVar(value="Encontrados: 0")
        ttk.Label(controls, textvariable=self.discovery_found_var).grid(
            row=0, column=3, padx=(18, 8), pady=8, sticky="w"
        )

        self.discovery_status_var = tk.StringVar(value="Parado")
        ttk.Label(controls, textvariable=self.discovery_status_var).grid(
            row=0, column=4, padx=8, pady=8, sticky="w"
        )

        self.discovery_progress_var = tk.DoubleVar(value=0.0)
        self.discovery_progress = ttk.Progressbar(
            controls, variable=self.discovery_progress_var, maximum=100.0
        )
        self.discovery_progress.grid(
            row=1, column=0, columnspan=5, sticky="ew", padx=10, pady=(0, 5)
        )

        self.discovery_progress_text_var = tk.StringVar(value="0 / 0")
        self.discovery_estimate_var = tk.StringVar(value="")
        ttk.Label(controls, textvariable=self.discovery_progress_text_var).grid(
            row=2, column=0, columnspan=2, sticky="w", padx=10, pady=(0, 8)
        )
        ttk.Label(controls, textvariable=self.discovery_estimate_var).grid(
            row=2, column=2, columnspan=3, sticky="e", padx=10, pady=(0, 8)
        )

```

MBSniffer v3.2/mb_slave_finder.py — lines 841-880

```
841         results = ttk.LabelFrame(page, text="Slaves encontrados")
842         results.grid(row=3, column=0, sticky="nsew", padx=12, pady=(6, 12))
843         results.columnconfigure(0, weight=1)
844         results.rowconfigure(0, weight=1)
845 
846         columns = (
847             "slave", "baud", "config", "fc", "result", "resp", "device_id", "raw"
848         )
849         self.discovery_tree = ttk.Treeview(
850             results, columns=columns, show="headings", selectmode="browse"
851         )
852         headings = {
853             "slave": "Slave",
854             "baud": "Baud",
855             "config": "Config.",
856             "fc": "FC",
857             "result": "Resultado",
858             "resp": "Resp. (ms)",
859             "device_id": "Device Identification",
860             "raw": "Raw Hex",
861         }
862         # Match the Traffic table behaviour: no horizontal scrollbar. Keep
863         # short fields compact and let the three text-heavy columns share the
864         # remaining viewport width. Column widths are recalculated whenever
865         # the Treeview is resized so the complete table remains visible.
866         self.discovery_fixed_widths = {
867             "slave": 70,
868             "baud": 90,
869             "config": 85,
870             "fc": 55,
871             "resp": 95,
872         }
873         self.discovery_flex_columns = ("result", "device_id", "raw")
874         self.discovery_flex_weights = {
875             "result": 0.27,
876             "device_id": 0.38,
877             "raw": 0.35,
878         }
879         self.discovery_flex_min_width = 90
880 
```

MBSniffer v3.2/mb_slave_finder.py — lines 881-920

```
        initial_widths = {
            **self.discovery_fixed_widths,
            "result": 220,
            "device_id": 330,
            "raw": 300,
        }
        for col in columns:
            self.discovery_tree.heading(col, text=headings[col], anchor="center")
            self.discovery_tree.column(
                col,
                width=initial_widths[col],
                minwidth=(
                    self.discovery_flex_min_width
                    if col in self.discovery_flex_columns
                    else self.discovery_fixed_widths[col]
                ),
                anchor="w" if col in self.discovery_flex_columns else "center",
                stretch=False,
            )

        discovery_y = ttk.Scrollbar(
            results, orient="vertical", command=self.discovery_tree.yview
        )
        self.discovery_tree.configure(yscrollcommand=discovery_y.set)
        self.discovery_tree.grid(row=0, column=0, sticky="nsew")
        discovery_y.grid(row=0, column=1, sticky="ns")
        self.discovery_tree.bind(
            "<Configure>", self.fit_discovery_columns_to_viewport, add="+"
        )

        self.update_discovery_estimate()
        self.update_discovery_start_state()

    def fit_discovery_columns_to_viewport(self, _event=None):
        """Fit Slave Finder columns to the visible Treeview width."""
        if not hasattr(self, "discovery_tree"):
            return

        viewport_width = int(self.discovery_tree.winfo_width())
        if viewport_width <= 1:
```

MBSniffer v3.2/mb_slave_finder.py — lines 921-960

```
            return

        # Reserve a few pixels for Treeview borders so Tk never needs a hidden
        # horizontal overflow area. The vertical scrollbar is a sibling and is
        # therefore already excluded from winfo_width().
        usable_width = max(1, viewport_width - 4)
        fixed_total = sum(self.discovery_fixed_widths.values())
        flex_count = len(self.discovery_flex_columns)
        min_flex_total = self.discovery_flex_min_width * flex_count

        flex_total = max(min_flex_total, usable_width - fixed_total)
        allocated = 0
        for index, col in enumerate(self.discovery_flex_columns):
            if index == flex_count - 1:
                width = flex_total - allocated
            else:
                width = int(round(flex_total * self.discovery_flex_weights[col]))
                allocated += width
            width = max(self.discovery_flex_min_width, width)
            self.discovery_tree.column(col, width=width)

        for col, width in self.discovery_fixed_widths.items():
            self.discovery_tree.column(col, width=width)

    def on_discovery_options_changed(self):
        if hasattr(self, "discovery_estimate_var"):
            self.update_discovery_estimate()
            self.update_discovery_start_state()

    def selected_discovery_configs(self):
        bauds = [baud for baud in DISCOVERY_BAUD_RATES if self.discovery_baud_vars[baud].get()]
        parities = [p for p in DISCOVERY_PARITIES if self.discovery_parity_vars[p].get()]
        stops = [s for s in DISCOVERY_STOP_BITS if self.discovery_stop_vars[s].get()]
        return [
            (baud, parity, stopbits)
            for baud in bauds
            for parity in parities
            for stopbits in stops
        ]

```

MBSniffer v3.2/mb_slave_finder.py — lines 961-1000

```
    def selected_discovery_slaves(self):
        start = int(self.discovery_slave_start_var.get())
        end = int(self.discovery_slave_end_var.get())
        if not (1 <= start <= 247 and 1 <= end <= 247):
            raise ValueError(self.tr("Os Slave IDs têm de estar entre 1 e 247."))
        if start > end:
            raise ValueError(self.tr("O Slave ID inicial não pode ser superior ao final."))
        return list(range(start, end + 1))

    def selected_discovery_min_timeout_ms(self):
        value = float(self.discovery_min_timeout_var.get())
        if not (DISCOVERY_MIN_TIMEOUT_MS <= value <= DISCOVERY_MAX_TIMEOUT_MS):
            raise ValueError(
                self.trf(
                    "O timeout mínimo tem de estar entre {minimum} e {maximum} ms.",
                    minimum=DISCOVERY_MIN_TIMEOUT_MS,
                    maximum=DISCOVERY_MAX_TIMEOUT_MS,
                )
            )
        return value

    def selected_discovery_attempts(self):
        value = int(self.discovery_attempts_var.get())
        if not (DISCOVERY_MIN_ATTEMPTS <= value <= DISCOVERY_MAX_ATTEMPTS):
            raise ValueError(
                self.trf(
                    "Tentativas tem de estar entre {minimum} e {maximum}.",
                    minimum=DISCOVERY_MIN_ATTEMPTS,
                    maximum=DISCOVERY_MAX_ATTEMPTS,
                )
            )
        return value

    def discovery_config_label(self, parity, stopbits):
        parity_letter = {"None": "N", "Even": "E", "Odd": "O"}[parity]
        return f"8{parity_letter}{stopbits}"

    def update_discovery_estimate(self):
        try:
            configs = self.selected_discovery_configs()
```

MBSniffer v3.2/mb_slave_finder.py — lines 1001-1040

```
            slaves = self.selected_discovery_slaves()
            minimum_timeout_ms = self.selected_discovery_min_timeout_ms()
            attempts = self.selected_discovery_attempts()
        except Exception:
            self.discovery_estimate_var.set(self.tr("Máx. estimado: —"))
            self.discovery_progress_text_var.set("0 / 0")
            return

        total_units = len(configs) * len(slaves)
        probes_per_unit = 2 if self.discovery_fc04_fallback_var.get() else 1
        identify_devices = bool(self.discovery_device_identification_var.get())

        seconds = 0.0
        for baud, parity, stopbits in configs:
            per_probe = (
                discovery_request_transmit_seconds(baud, parity, stopbits)
                + discovery_response_timeout_seconds(
                    baud, parity, stopbits, minimum_timeout_ms
                )
            )
            # Worst case: no Slave answers, so every probe uses all attempts.
            seconds += len(slaves) * probes_per_unit * attempts * per_probe
            if identify_devices:
                # Worst case: every probed Slave exists and therefore receives
                # one optional Basic Device Identification query.
                seconds += len(slaves) * (
                    device_identification_request_transmit_seconds(
                        baud, parity, stopbits
                    )
                    + device_identification_timeout_seconds(
                        baud, parity, stopbits, minimum_timeout_ms
                    )
                )
            seconds += 0.025  # small reopen/reconfigure allowance
        seconds += DISCOVERY_IDLE_CHECK_SECONDS if configs else 0.0

        self.discovery_progress_text_var.set(f"0 / {total_units}")
        suffix_parts = []
        if probes_per_unit == 2:
            suffix_parts.append("FC04 fallback")
```

MBSniffer v3.2/mb_slave_finder.py — lines 1041-1080

```
        if identify_devices:
            suffix_parts.append("Device Identification")
        suffix = (
            self.trf(" (com {options})", options=" + ".join(suffix_parts))
            if suffix_parts else ""
        )
        self.discovery_estimate_var.set(
            self.trf(
                "{configs} config. × {slaves} slaves — máx. estimado: {duration}",
                configs=len(configs),
                slaves=len(slaves),
                duration=format_duration(seconds),
            )
            + suffix
        )

    def update_discovery_start_state(self):
        if not hasattr(self, "discovery_start_btn"):
            return
        enabled = (
            not self.discovery_active
            and not self.busy
            and bool(self.discovery_port_var.get().strip())
            and bool(self.discovery_safety_var.get())
        )
        try:
            enabled = enabled and bool(self.selected_discovery_configs())
            self.selected_discovery_slaves()
            self.selected_discovery_min_timeout_ms()
            self.selected_discovery_attempts()
        except Exception:
            enabled = False
        self.discovery_start_btn.configure(state="normal" if enabled else "disabled")

    def clear_discovery_results(self):
        if self.discovery_active:
            return
        for iid in self.discovery_tree.get_children():
            self.discovery_tree.delete(iid)
        self.discovery_found_count = 0
```

MBSniffer v3.2/mb_slave_finder.py — lines 1081-1120

```
1081        self.discovery_found_var.set(self.trf("Encontrados: {count}", count=0))
1082        self.discovery_progress_var.set(0.0)
1083        self.discovery_status_var.set(self.tr("Parado"))
1084        self.update_discovery_estimate()
1085
1086    def open_discovery_serial_port(self, serial, port, baud, parity, stopbits):
1087        parity_map = {
1088            "None": serial.PARITY_NONE,
1089            "Even": serial.PARITY_EVEN,
1090            "Odd": serial.PARITY_ODD,
1091        }
1092        stop_map = {
1093            "1": serial.STOPBITS_ONE,
1094            "2": serial.STOPBITS_TWO,
1095        }
1096
1097        ser = serial.Serial()
1098        ser.port = port
1099        ser.baudrate = int(baud)
1100        ser.bytesize = serial.EIGHTBITS
1101        ser.parity = parity_map[parity]
1102        ser.stopbits = stop_map[stopbits]
1103        ser.timeout = 0
1104
1105        ser.write_timeout = DISCOVERY_WRITE_TIMEOUT_SECONDS
1106        ser.xonxoff = False
1107        ser.rtscts = False
1108        ser.dsrdtr = False
1109
1110        # Do not force RTS/DTR levels. Many USB-RS485 adapters manage direction
1111        # internally and forcing modem-control lines can interfere with some
1112        # adapters/drivers. Hardware flow control is disabled above.
1113        ser.open()
1114
1115        try:
1116            ser.reset_input_buffer()
1117        except Exception:
1118            pass
1119        try:
1120            ser.reset_output_buffer()
```

MBSniffer v3.2/mb_slave_finder.py — lines 1121-1160

```
1121        except Exception:
1122            pass
1123
1124        return ser
1125
1126    def start_discovery_scan(self):
1127        if self.busy or self.discovery_active:
1128            return
1129
1130        serial, _ = self.get_serial_modules()
1131        if serial is None:
1132            messagebox.showerror(
1133                self.tr("pyserial em falta"),
1134                self.tr("pyserial não está instalado. O build_exe.bat inclui-o automaticamente no EXE.")
1135            )
1136            return
1137
1138        if not self.discovery_safety_var.get():
1139            messagebox.showwarning(
1140                self.tr("Pesquisa ativa"),
1141                self.tr("Confirma primeiro que não existe outro master ativo no barramento.")
1142            )
1143            return
1144
1145        try:
1146            port = self.discovery_port_var.get().strip()
1147            if not port:
1148                raise ValueError(self.tr("Seleciona uma porta COM."))
1149            configs = self.selected_discovery_configs()
1150            if not configs:
1151                raise ValueError(self.tr("Seleciona pelo menos uma configuração série."))
1152            slaves = self.selected_discovery_slaves()
1153            minimum_timeout_ms = self.selected_discovery_min_timeout_ms()
1154            attempts = self.selected_discovery_attempts()
1155        except Exception as exc:
1156            messagebox.showerror(self.tr("Configuração inválida"), str(exc))
1157            return
1158
1159        fallback_fc04 = bool(self.discovery_fc04_fallback_var.get())
1160        identify_devices = bool(self.discovery_device_identification_var.get())
```

MBSniffer v3.2/mb_slave_finder.py — lines 1161-1200

```
1161
1162        self.clear_discovery_results()
1163        self.discovery_active = True
1164        self.busy = True
1165        self.discovery_stop_event.clear()
1166        self.discovery_found_count = 0
1167        self.discovery_progress_var.set(0.0)
1168        self.discovery_status_var.set(self.tr("A iniciar pesquisa…"))
1169
1170        self.discovery_start_btn.configure(state="disabled")
1171        self.discovery_stop_btn.configure(state="normal")
1172        self.discovery_clear_btn.configure(state="disabled")
1173        self.discovery_refresh_btn.configure(state="disabled")
1174        self.discovery_port_combo.configure(state="disabled")
1175
1176        # Prevent passive capture/simulation from being started concurrently.
1177        self.start_btn.configure(state="disabled")
1178        self.stop_btn.configure(state="disabled")
1179        self.restart_btn.configure(state="disabled")
1180        self.set_sniffer_refresh_state("disabled")
1181        self.set_simulation_button_state("disabled")
1182
1183        immutable_configs = list(configs)
1184        immutable_slaves = list(slaves)
1185
1186        self.discovery_thread = threading.Thread(
1187            target=self.discovery_worker,
1188            args=(
1189                serial, port, immutable_configs, immutable_slaves,
1190                minimum_timeout_ms, fallback_fc04, identify_devices, attempts
1191            ),
1192            daemon=True,
1193        )
1194        try:
1195            self.discovery_thread.start()
1196        except Exception as exc:
1197            # Restore the interface if the worker cannot even be started.
1198            self.discovery_active = False
1199            self.busy = False
1200            self.discovery_thread = None
```

MBSniffer v3.2/mb_slave_finder.py — lines 1201-1240

```
            self.discovery_stop_btn.configure(state="disabled")
            self.discovery_clear_btn.configure(state="normal")
            self.discovery_refresh_btn.configure(state="normal")
            self.discovery_port_combo.configure(state="readonly")
            self.start_btn.configure(state="normal")
            self.restart_btn.configure(state="normal")
            self.set_sniffer_refresh_state("normal")
            self.set_simulation_button_state("normal")
            self.discovery_status_var.set(self.tr("Erro ao iniciar pesquisa"))
            self.update_discovery_start_state()
            messagebox.showerror(
                self.tr("Erro na Bus Slave Finder"),
                self.trf(
                    "Não foi possível iniciar a thread de pesquisa:\n\n{error}",
                    error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                )
            )

    def stop_discovery_scan(self):
        if self.discovery_active:
            self.discovery_status_var.set(self.tr("A parar…"))
            self.discovery_stop_event.set()
            self.discovery_stop_btn.configure(state="disabled")
            # Wake/cancel pending driver I/O without closing the handle from
            # the Tk thread while the discovery worker may still be using it.
            cancel_serial_io(self.discovery_serial)

    def discovery_worker(
        self, serial, port, configs, slaves, minimum_timeout_ms, fallback_fc04,
        identify_devices, attempts=DISCOVERY_DEFAULT_ATTEMPTS
    ):
        started = time.perf_counter()
        total_units = len(configs) * len(slaves)
        completed = 0
        found_count = 0
        fatal_error = None
        context = str(port)
        idle_checked = False

        try:
```

MBSniffer v3.2/mb_slave_finder.py — lines 1241-1280

```
            for baud, parity, stopbits in configs:
                if self.discovery_stop_event.is_set():
                    break

                config_label = self.discovery_config_label(parity, stopbits)
                context = f"{port} — {baud} / {config_label}"
                self.event_queue.put((
                    "discovery_status",
                    self.trf(
                        "A configurar {baud} / {config}…",
                        baud=baud,
                        config=config_label,
                    )
                ))

                ser = None
                try:
                    try:
                        ser = self.open_discovery_serial_port(
                            serial, port, baud, parity, stopbits
                        )
                    except Exception as exc:
                        raise RuntimeError(
                            self.trf(
                                "Não foi possível abrir {port} em {baud} / {config}: {error}",
                                port=port,
                                baud=baud,
                                config=config_label,
                                error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                            )
                        ) from exc

                    self.discovery_serial = ser
                    if self.discovery_stop_event.wait(0.025):
                        break

                    if not idle_checked:
                        # Last line of defence behind the safety checkbox:
                        # never transmit onto a bus that is already busy.
                        idle_checked = True
```

MBSniffer v3.2/mb_slave_finder.py — lines 1281-1320

```
                        self.event_queue.put((
                            "discovery_status",
                            self.tr("A verificar se o barramento está livre…"),
                        ))
                        traffic = bus_activity_detected(
                            ser,
                            DISCOVERY_IDLE_CHECK_SECONDS,
                            self.discovery_stop_event,
                        )
                        if traffic:
                            raise RuntimeError(
                                self.trf(
                                    "Foi detetado tráfego em {port} antes de transmitir "
                                    "({count} bytes: {raw}). Pode existir outro master "
                                    "ativo no barramento; a pesquisa não foi iniciada.",
                                    port=port,
                                    count=len(traffic),
                                    raw=hex_bytes(traffic[:16]),
                                )
                            )
                        if self.discovery_stop_event.is_set():
                            break
                        self.event_queue.put((
                            "discovery_status",
                            self.trf(
                                "A configurar {baud} / {config}…",
                                baud=baud,
                                config=config_label,
                            ),
                        ))

                    response_timeout_s = discovery_response_timeout_seconds(
                        baud, parity, stopbits, minimum_timeout_ms
                    )
                    char_gap_s = modbus_rtu_interframe_gap_seconds(
                        baud, discovery_bits_per_char(parity, stopbits)
                    )

                    for slave in slaves:
                        if self.discovery_stop_event.is_set():
```

MBSniffer v3.2/mb_slave_finder.py — lines 1321-1360

```
                            break

                    result = None
                    used_fc = None
                    elapsed_ms = None
                    probe_fcs = [3, 4] if fallback_fc04 else [3]

                    probe_plan = [
                        (fc, attempt)
                        for fc in probe_fcs
                        for attempt in range(1, attempts + 1)
                    ]
                    for fc, attempt in probe_plan:
                        if self.discovery_stop_event.is_set():
                            break

                        context = (
                            f"{port} — {baud} / {config_label} / "
                            f"Slave {slave} / FC{fc:02X}"
                        )
                        if attempts > 1:
                            context += f" / {attempt}/{attempts}"

                        # If this fails, the COM/driver is unhealthy. Do not
                        # disguise it as a normal no-response slave timeout.
                        try:
                            ser.reset_input_buffer()
                        except Exception as exc:
                            raise RuntimeError(
                                self.trf(
                                    "Falha ao preparar a receção em {context}: {error}",
                                    context=context,
                                    error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                                )
                            ) from exc

                        request = build_discovery_probe(slave, fc)
                        started_probe = time.perf_counter()
                        try:
                            sent = write_discovery_request(
```

MBSniffer v3.2/mb_slave_finder.py — lines 1361-1400

```
                                    ser, request, baud, parity, stopbits,
                                    self.discovery_stop_event, serial
                                )
                            except Exception as exc:
                                raise RuntimeError(
                                    self.trf(
                                        "Falha ao transmitir em {context}: {error}",
                                        context=context,
                                        error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                                    )
                                ) from exc

                            if not sent or self.discovery_stop_event.is_set():
                                break

                            try:
                                result = read_discovery_response(
                                    ser, slave, fc, response_timeout_s,
                                    self.discovery_stop_event
                                )
                            except Exception as exc:
                                raise RuntimeError(
                                    self.trf(
                                        "Falha ao receber em {context}: {error}",
                                        context=context,
                                        error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                                    )
                                ) from exc

                            if result is not None:
                                elapsed_ms = (
                                    time.perf_counter() - started_probe
                                ) * 1000.0
                                used_fc = fc
                                if char_gap_s > 0:
                                    self.discovery_stop_event.wait(
                                        min(char_gap_s, 0.05)
                                    )
                                break

```

MBSniffer v3.2/mb_slave_finder.py — lines 1401-1440

```
                    if result is not None and used_fc is not None:
                        found_count += 1
                        if result["kind"] == "EXCEPTION":
                            code = result["exception_code"]
                            name = MODBUS_EXCEPTION_CODES.get(
                                code, ("Unknown/Reserved Exception", "")
                            )[0]
                            result_text = f"Exception 0x{code:02X} – {name}"
                        else:
                            result_text = "Response"

                        device_id_text = ""
                        if identify_devices and not self.discovery_stop_event.is_set():
                            context = (
                                f"{port} – {baud} / {config_label} / "
                                f"Slave {slave} / FC43/14"
                            )
                            try:
                                device_result = read_basic_device_identification(
                                    ser,
                                    slave,
                                    baud,
                                    parity,
                                    stopbits,
                                    minimum_timeout_ms,
                                    self.discovery_stop_event,
                                    serial,
                                )
                            except Exception as exc:
                                raise RuntimeError(
                                    self.trf(
                                        "Falha em Device Identification em {context}: {error}",
                                        context=context,
                                        error=f"{type(exc).__name__}: {self.tr(str(exc))}",
                                    )
                                ) from exc
                            device_id_text = format_device_identification(
                                device_result
                            )
                            if char_gap_s > 0:
```

MBSniffer v3.2/mb_slave_finder.py — lines 1441-1480

```
                                    self.discovery_stop_event.wait(
                                        min(char_gap_s, 0.05)
                                    )

                                self.event_queue.put((
                                    "discovery_found",
                                    {
                                        "slave": slave,
                                        "baud": baud,
                                        "config": config_label,
                                        "fc": f"{used_fc:02X}",
                                        "result": result_text,
                                        "response_ms": elapsed_ms,
                                        "device_id": device_id_text,
                                        "raw": hex_bytes(result["frame"]),
                                    }
                                ))

                            completed += 1
                            elapsed = time.perf_counter() - started
                            eta = (
                                max(
                                    0.0,
                                    (elapsed / completed) * (total_units - completed)
                                )
                                if completed > 0 else 0.0
                            )

                            self.event_queue.put((
                                "discovery_progress",
                                {
                                    "completed": completed,
                                    "total": total_units,
                                    "baud": baud,
                                    "config": config_label,
                                    "slave": slave,
                                    "elapsed": elapsed,
                                    "eta": eta,
                                    "found": found_count,
                                }
```

MBSniffer v3.2/mb_slave_finder.py — lines 1481-1520

```
                        ))

                finally:
                    if ser is not None:
                        cancel_serial_io(ser)
                        try:
                            ser.close()
                        except Exception:
                            pass
                    self.discovery_serial = None

        except Exception as exc:
            if not self.discovery_stop_event.is_set():
                fatal_error = str(exc) or f"{context}: {type(exc).__name__}"

        elapsed = time.perf_counter() - started
        self.event_queue.put((
            "discovery_done",
            {
                "stopped": self.discovery_stop_event.is_set(),
                "error": fatal_error,
                "elapsed": elapsed,
                "completed": completed,
                "total": total_units,
                "found": found_count,
            }
        ))

    def apply_discovery_progress(self, payload):
        completed = int(payload["completed"])
        total = max(1, int(payload["total"]))
        self.discovery_progress_var.set(100.0 * completed / total)
        self.discovery_progress_text_var.set(
            f"{completed} / {payload['total']} — "
            f"{payload['baud']} / {payload['config']} / Slave {payload['slave']}"
        )
        self.discovery_found_var.set(self.trf("Encontrados: {count}", count=payload["found"]))
        self.discovery_status_var.set(
            self.trf(
                "Decorrido {elapsed} — restante ~{remaining}",
```

MBSniffer v3.2/mb_slave_finder.py — lines 1521-1560

```
                elapsed=format_duration(payload["elapsed"]),
                remaining=format_duration(payload["eta"]),
            )
        )

    def add_discovery_result(self, payload):
        self.discovery_found_count += 1
        self.discovery_found_var.set(
            self.trf("Encontrados: {count}", count=self.discovery_found_count)
        )
        self.discovery_tree.insert(
            "", "end",
            values=(
                payload["slave"],
                payload["baud"],
                payload["config"],
                payload["fc"],
                payload["result"],
                f"{payload['response_ms']:.1f}" if payload.get("response_ms") is not None else "",
                payload.get("device_id", ""),
                payload["raw"],
            )
        )

    def finish_discovery_ui(self, payload):
        self.discovery_active = False
        self.busy = False
        self.discovery_stop_event.clear()
        self.discovery_serial = None
        self.discovery_thread = None

        self.discovery_stop_btn.configure(state="disabled")
        self.discovery_clear_btn.configure(state="normal")
        self.discovery_refresh_btn.configure(state="normal")
        self.discovery_port_combo.configure(state="readonly")

        self.start_btn.configure(state="normal")
        self.stop_btn.configure(state="disabled")
        self.restart_btn.configure(state="normal")
        self.set_sniffer_refresh_state("normal")
```

MBSniffer v3.2/mb_slave_finder.py — lines 1561-1587

```
1561        self.set_simulation_button_state("normal")
1562
1563        if payload.get("error"):
1564            self.discovery_status_var.set(self.tr("Erro na pesquisa"))
1565            messagebox.showerror(self.tr("Erro na Bus Slave Finder"), payload["error"])
1566        elif payload.get("stopped"):
1567            self.discovery_status_var.set(
1568                self.trf(
1569                    "Pesquisa parada — {found} encontrados — {elapsed}",
1570                    found=payload["found"],
1571                    elapsed=format_duration(payload["elapsed"]),
1572                )
1573            )
1574        else:
1575            self.discovery_progress_var.set(100.0)
1576            self.discovery_status_var.set(
1577                self.trf(
1578                    "Pesquisa concluída — {found} encontrados — {elapsed}",
1579                    found=payload["found"],
1580                    elapsed=format_duration(payload["elapsed"]),
1581                )
1582            )
1583
1584        self.update_discovery_estimate()
1585        self.update_discovery_start_state()
1586        self.update_com_status()
1587
```

MBSniffer v3.2/mb_master.py

516 source lines.

MBSniffer v3.2/mb_master.py — lines 1-40

```python
#!/usr/bin/env python3
"""Modbus RTU master (poll) engine: frames, data formats and scheduling.

This module has no widgets. ``MasterEngine`` owns one open serial port and
serves every poll window over it: Modbus RTU allows a single request on the
bus at a time, so windows are polled in turn according to their scan rate,
with queued writes and "read once" requests served first. Results are
posted to the application's ``event_queue`` like the other workers.
"""

import math
import struct
import threading
import time
from collections import deque
from dataclasses import dataclass

from mb_config import (
    MASTER_MAX_READ_BITS,
    MASTER_MAX_READ_REGISTERS,
    MASTER_MAX_WRITE_BITS,
    MASTER_MAX_WRITE_REGISTERS,
)
from mb_protocol import append_crc, crc_ok, hex_bytes, modbus_rtu_interframe_gap_seconds
from mb_slave_finder import (
    bus_activity_detected,
    cancel_serial_io,
    discovery_bits_per_char,
    write_discovery_request,
)

READ_FUNCTION_CODES = (1, 2, 3, 4)
BIT_FUNCTION_CODES = (1, 2)
WRITABLE_READ_FCS = {1: (5, 15), 3: (6, 16)}  # read window FC -> write FCs

# ---------------------------------------------------------------------------
# Data formats
# ---------------------------------------------------------------------------

BYTE_ORDERS = ("ABCD", "CDAB", "BADC", "DCBA")
```

MBSniffer v3.2/mb_master.py — lines 41-80

```
41
42 # (key, label, registers per value).  Labels are technical terms and are not
43 # translated.  Byte order letters follow the wire order of two registers:
44 # register n = A B, register n+1 = C D; "ABCD" is big-endian.
45 REGISTER_FORMATS = (
46     ("int16", "Signed 16-bit", 1),
47     ("uint16", "Unsigned 16-bit", 1),
48     ("hex16", "Hex 16-bit", 1),
49     ("bin16", "Binary 16-bit", 1),
50 ) + tuple(
51     (f"{kind}_{order}", f"{label} {order}", 2)
52     for kind, label in (
53         ("int32", "Signed 32-bit"),
54         ("uint32", "Unsigned 32-bit"),
55         ("float32", "Float 32-bit"),
56     )
57     for order in BYTE_ORDERS
58 )
59 FORMAT_LABELS = {key: label for key, label, _size in REGISTER_FORMATS}
60 FORMAT_KEYS_BY_LABEL = {label: key for key, label, _size in REGISTER_FORMATS}
61 FORMAT_SIZES = {key: size for key, _label, size in REGISTER_FORMATS}
62 DEFAULT_FORMAT = "int16"
63
64
65 def _value_bytes_from_registers(high, low, order):
66     wire = (high >> 8 & 0xFF, high & 0xFF, low >> 8 & 0xFF, low & 0xFF)
67     return bytes(wire["ABCD".index(letter)] for letter in order)
68
69
70 def _registers_from_value_bytes(value, order):
71     wire = [0, 0, 0, 0]
72     for index, letter in enumerate(order):
73         wire["ABCD".index(letter)] = value[index]
74     return [(wire[0] << 8) | wire[1], (wire[2] << 8) | wire[3]]
75
76
77 def format_float(value):
78     if math.isnan(value):
79         return "NaN"
80     if math.isinf(value):
```

MBSniffer v3.2/mb_master.py — lines 81-120

```
81          return "-Inf" if value < 0 else "Inf"
82      return f"{value:.7g}"
83
84
85  def format_register_values(registers, fmt):
86      """Return one display string per register.
87
88      32-bit formats show the value on the first register of each pair and an
89      empty cell on the second; an unpaired final register shows "—".
90      """
91      fmt = fmt if fmt in FORMAT_SIZES else DEFAULT_FORMAT
92      out = []
93      if FORMAT_SIZES[fmt] == 1:
94          for value in registers:
95              value &= 0xFFFF
96              if fmt == "int16":
97                  out.append(str(value - 0x10000 if value & 0x8000 else value))
98              elif fmt == "uint16":
99                  out.append(str(value))
100             elif fmt == "hex16":
101                 out.append(f"0x{value:04X}")
102             else:
103                 out.append(f"{value >> 8:08b} {value & 0xFF:08b}")
104         return out
105
106     kind, order = fmt.split("_")
107     for index in range(0, len(registers), 2):
108         if index + 1 >= len(registers):
109             out.append("—")
110             break
111         raw = _value_bytes_from_registers(registers[index], registers[index + 1], order)
112         if kind == "int32":
113             text = str(struct.unpack(">i", raw)[0])
114         elif kind == "uint32":
115             text = str(struct.unpack(">I", raw)[0])
116         else:
117             text = format_float(struct.unpack(">f", raw)[0])
118         out.extend((text, ""))
119     return out
120
```

MBSniffer v3.2/mb_master.py — lines 121-160

```
121
122 def _parse_int(text):
123     text = str(text).strip().replace(" ", "").replace("_", "")
124     lowered = text.lower()
125     if lowered.startswith(("0x", "-0x")):
126         return int(text, 16)
127     if lowered.startswith(("0b", "-0b")):
128         return int(text, 2)
129     return int(text, 10)
130
131
132 def encode_register_value(text, fmt):
133     """Convert one user-entered value to the registers it occupies.
134
135     Raises ValueError with a message key (PT canonical text) when the value
136     does not fit the format.
137     """
138     fmt = fmt if fmt in FORMAT_SIZES else DEFAULT_FORMAT
139     text = str(text).strip()
140     if not text:
141         raise ValueError("Valor vazio.")
142     try:
143         if fmt == "hex16":
144             value = int(text[2:] if text.lower().startswith("0x") else text, 16)
145             if not 0 <= value <= 0xFFFF:
146                 raise ValueError
147             return [value]
148         if fmt == "bin16":
149             value = int(text.replace(" ", "")[2:] if text.lower().startswith("0b")
150                         else text.replace(" ", ""), 2)
151             if not 0 <= value <= 0xFFFF:
152                 raise ValueError
153             return [value]
154         if fmt == "int16":
155             value = _parse_int(text)
156             if not -0x8000 <= value <= 0x7FFF:
157                 raise ValueError
158             return [value & 0xFFFF]
159         if fmt == "uint16":
160             value = _parse_int(text)
```

MBSniffer v3.2/mb_master.py — lines 161-200

```
            if not 0 <= value <= 0xFFFF:
                raise ValueError
            return [value]

        kind, order = fmt.split("_")
        if kind == "float32":
            number = float(text.replace(",", "."))
            raw = struct.pack(">f", number)
            if math.isinf(struct.unpack(">f", raw)[0]) and not math.isinf(number):
                raise ValueError
        elif kind == "int32":
            raw = struct.pack(">i", _parse_int(text))
        else:
            raw = struct.pack(">I", _parse_int(text))
        return _registers_from_value_bytes(raw, order)
    except (ValueError, OverflowError, struct.error):
        raise ValueError("Valor fora do formato selecionado.") from None


def encode_coil_value(text):
    value = str(text).strip().lower()
    if value in ("1", "on", "true"):
        return 1
    if value in ("0", "off", "false"):
        return 0
    raise ValueError("Uma coil só aceita 0 ou 1.")


# ---------------------------------------------------------------------------
# Request frames
# ---------------------------------------------------------------------------
def _u16(value):
    return bytes(((value >> 8) & 0xFF, value & 0xFF))


def read_limit(fc):
    return MASTER_MAX_READ_BITS if fc in BIT_FUNCTION_CODES else MASTER_MAX_READ_REGISTERS

```

MBSniffer v3.2/mb_master.py — lines 201-240

```
201 def validate_read_definition(slave, fc, address, qty):
202     """Raise ValueError (PT canonical message) for an invalid poll definition."""
203     if not 1 <= int(slave) <= 247:
204         raise ValueError("O Slave ID de leitura tem de estar entre 1 e 247.")
205     if int(fc) not in READ_FUNCTION_CODES:
206         raise ValueError("Função de leitura não suportada.")
207     if not 0 <= int(address) <= 0xFFFF:
208         raise ValueError("O endereço tem de estar entre 0 e 65535.")
209     if not 1 <= int(qty) <= read_limit(int(fc)):
210         raise ValueError("Quantidade fora do limite desta função.")
211     if int(address) + int(qty) > 0x10000:
212         raise ValueError("O intervalo de endereços ultrapassa 65535.")
213 
214 
215 def build_read_request(slave, fc, address, qty):
216     validate_read_definition(slave, fc, address, qty)
217     return append_crc(bytes((int(slave), int(fc))) + _u16(int(address)) + _u16(int(qty)))
218 
219 
220 def build_write_request(slave, fc, address, values):
221     """Build an FC05/06/15/16 request. *values* are 0/1 bits or 16-bit registers."""
222     slave, fc, address = int(slave), int(fc), int(address)
223     values = [int(v) for v in values]
224     if not 0 <= slave <= 247:
225         raise ValueError("O Slave ID de escrita tem de estar entre 0 (broadcast) e 247.")
226     if not 0 <= address <= 0xFFFF or address + max(1, len(values)) > 0x10000:
227         raise ValueError("O endereço tem de estar entre 0 e 65535.")
228     if fc == 5:
229         if len(values) != 1:
230             raise ValueError("FC05 escreve exatamente uma coil.")
231         payload = _u16(address) + (b"\xFF\x00" if values[0] else b"\x00\x00")
232     elif fc == 6:
233         if len(values) != 1:
234             raise ValueError("FC06 escreve exatamente um registo.")
235         payload = _u16(address) + _u16(values[0] & 0xFFFF)
236     elif fc == 15:
237         if not 1 <= len(values) <= MASTER_MAX_WRITE_BITS:
238             raise ValueError("Quantidade fora do limite desta função.")
239         data = bytearray((len(values) + 7) // 8)
240         for index, bit in enumerate(values):
```

MBSniffer v3.2/mb_master.py — lines 241-280

```
            if bit:
                data[index // 8] |= 1 << (index % 8)
        payload = _u16(address) + _u16(len(values)) + bytes((len(data),)) + bytes(data)
    elif fc == 16:
        if not 1 <= len(values) <= MASTER_MAX_WRITE_REGISTERS:
            raise ValueError("Quantidade fora do limite desta função.")
        data = b"".join(_u16(v & 0xFFFF) for v in values)
        payload = _u16(address) + _u16(len(values)) + bytes((len(data),)) + data
    else:
        raise ValueError("Função de escrita não suportada.")
    return append_crc(bytes((slave, fc)) + payload)


def expected_response_length(request):
    fc = request[1]
    if fc in BIT_FUNCTION_CODES:
        qty = (request[4] << 8) | request[5]
        return 5 + (qty + 7) // 8
    if fc in (3, 4):
        qty = (request[4] << 8) | request[5]
        return 5 + 2 * qty
    return 8  # FC05/06/15/16


# ---------------------------------------------------------------------------
# Response matching
# ---------------------------------------------------------------------------

def find_master_response(buffer, request):
    """Find the response to *request* in the received bytes.

    Returns a dict with ``kind`` RESPONSE or EXCEPTION, the ``frame`` and,
    for reads, the decoded ``values`` (bits or registers); None when no
    complete, CRC-valid, matching frame is present yet. A request echoed by
    an RS485 adapter is skipped for reads because its length and CRC do not
    fit the expected response. For FC05/06 a slave reply is itself an exact
    echo, so an adapter echo cannot be distinguished from it.
    """
    data = bytes(buffer)
    slave, fc = request[0], request[1]
```

MBSniffer v3.2/mb_master.py — lines 281-320

```
281     length = expected_response_length(request)
282 
283     for offset in range(len(data)):
284         if data[offset] != slave or offset + 2 > len(data):
285             continue
286         response_fc = data[offset + 1]
287 
288         if response_fc == (fc | 0x80) and offset + 5 <= len(data):
289             frame = data[offset:offset + 5]
290             if crc_ok(frame):
291                 return {"kind": "EXCEPTION", "frame": frame, "exception_code": frame[2],
292                         "values": None}
293 
294         if response_fc != fc or offset + length > len(data):
295             continue
296         frame = data[offset:offset + length]
297         if not crc_ok(frame):
298             continue
299 
300         if fc in READ_FUNCTION_CODES:
301             if frame[2] != length - 5:
302                 continue
303             payload = frame[3:-2]
304             qty = (request[4] << 8) | request[5]
305             if fc in BIT_FUNCTION_CODES:
306                 values = [(payload[i // 8] >> (i % 8)) & 1 for i in range(qty)]
307             else:
308                 values = [(payload[i] << 8) | payload[i + 1] for i in range(0, len(payload), 2)]
309             return {"kind": "RESPONSE", "frame": frame, "exception_code": None, "values": values}
310 
311         if fc in (5, 6) and frame != request:
312             continue
313         if fc in (15, 16) and frame[2:6] != request[2:6]:
314             continue
315         return {"kind": "RESPONSE", "frame": frame, "exception_code": None, "values": None}
316 
317     return None
318 
319 
320 # ---------------------------------------------------------------------------
```

MBSniffer v3.2/mb_master.py — lines 321-360

```
321 # Poll definitions and engine
322 # ---------------------------------------------------------------------------
323
324 @dataclass
325 class PollDefinition:
326     window_id: int
327     slave: int
328     fc: int
329     address: int
330     qty: int
331     scan_ms: int
332     enabled: bool = True
333
334     def request(self):
335         return build_read_request(self.slave, self.fc, self.address, self.qty)
336
337
338 class MasterEngine:
339     """Serve all poll windows over one serial port, one transaction at a time.
340
341     Order of service: queued writes, then "read once" requests, then the
342     enabled poll whose next due time is earliest. A poll becomes due again
343     ``scan_ms`` after its previous transaction started (0 = continuously).
344     After every transaction the bus is left silent for at least the
345     configured delay between polls, and never less than Modbus t3.5.
346     """
347
348     def __init__(self, ser, event_queue, *, baud, parity, stopbits,
349                  response_timeout_ms, delay_ms, serial_module=None,
350                  idle_check_seconds=0.0):
351         self.ser = ser
352         self.event_queue = event_queue
353         self.baud = int(baud)
354         self.parity = parity
355         self.stopbits = stopbits
356         self.response_timeout_s = max(0.001, float(response_timeout_ms) / 1000.0)
357         gap_s = modbus_rtu_interframe_gap_seconds(
358             self.baud, discovery_bits_per_char(parity, stopbits)
359         )
360         self.delay_s = max(gap_s, float(delay_ms) / 1000.0)
```

MBSniffer v3.2/mb_master.py — lines 361-400

```
361         self.serial_module = serial_module
362         self.idle_check_seconds = float(idle_check_seconds)
363 
364         self._lock = threading.Lock()
365         self._polls = {}            # window_id -> [PollDefinition, next_due]
366         self._writes = deque()      # (window_id, request, description)
367         self._once = deque()        # window_id
368         self._wake = threading.Event()
369         self.stop_event = threading.Event()
370         self.thread = None
371 
372     # -- control (any thread) ------------------------------------------------
373     def set_poll(self, definition):
374         with self._lock:
375             self._polls[definition.window_id] = [definition, 0.0]
376         self._wake.set()
377 
378     def remove_poll(self, window_id):
379         with self._lock:
380             self._polls.pop(window_id, None)
381             self._once = deque(w for w in self._once if w != window_id)
382 
383     def read_once(self, window_id):
384         with self._lock:
385             if window_id in self._polls and window_id not in self._once:
386                 self._once.append(window_id)
387         self._wake.set()
388 
389     def queue_write(self, window_id, request, description=""):
390         with self._lock:
391             self._writes.append((window_id, bytes(request), description))
392         self._wake.set()
393 
394     def start(self):
395         self.thread = threading.Thread(target=self.run, daemon=True)
396         self.thread.start()
397 
398     def stop(self, join_timeout=1.0):
399         self.stop_event.set()
400         self._wake.set()
```

MBSniffer v3.2/mb_master.py — lines 401-440

```
401         cancel_serial_io(self.ser)
402         thread = self.thread
403         if thread is not None and thread.is_alive() and thread is not threading.current_thread():
404             thread.join(timeout=join_timeout)
405 
406     # -- scheduling ------------------------------------------------------------
407     def next_job(self, now=None):
408         """Return ('write', item), ('read', definition, once) or ('wait', seconds)."""
409         now = time.perf_counter() if now is None else now
410         with self._lock:
411             if self._writes:
412                 return ("write", self._writes.popleft())
413             while self._once:
414                 window_id = self._once.popleft()
415                 entry = self._polls.get(window_id)
416                 if entry is not None:
417                     return ("read", entry[0], True)
418             due = [
419                 (entry[1], window_id)
420                 for window_id, entry in self._polls.items()
421                 if entry[0].enabled
422             ]
423             if not due:
424                 return ("wait", 0.5)
425             next_due, window_id = min(due)
426             if next_due > now:
427                 return ("wait", min(0.5, next_due - now))
428             entry = self._polls[window_id]
429             entry[1] = now + max(0, entry[0].scan_ms) / 1000.0
430             return ("read", entry[0], False)
431 
432     # -- worker ----------------------------------------------------------------
433     def run(self):
434         error = None
435         try:
436             if self.idle_check_seconds > 0:
437                 traffic = bus_activity_detected(
438                     self.ser, self.idle_check_seconds, self.stop_event
439                 )
440                 if traffic:
```

MBSniffer v3.2/mb_master.py — lines 441-480

```
                    error = ("traffic", hex_bytes(traffic[:16]), len(traffic))
                    return
            self.event_queue.put(("master_ready", None))

            while not self.stop_event.is_set():
                job = self.next_job()
                if job[0] == "wait":
                    self._wake.wait(job[1])
                    self._wake.clear()
                    continue

                if job[0] == "write":
                    window_id, request, description = job[1]
                    result = self.transact(request)
                    result.update(kind="write", window_id=window_id, description=description)
                else:
                    definition, once = job[1], job[2]
                    result = self.transact(definition.request())
                    result.update(kind="read", window_id=definition.window_id, once=once,
                                  fc=definition.fc, address=definition.address)
                if self.stop_event.is_set():
                    break
                self.event_queue.put(("master_result", result))
                if self.stop_event.wait(self.delay_s):
                    break
        except Exception as exc:  # serial/driver failure ends the session
            if not self.stop_event.is_set():
                error = ("serial", f"{type(exc).__name__}: {exc}", 0)
        finally:
            self.event_queue.put(("master_stopped", {"error": error}))

    def transact(self, request):
        """Send one request and wait for its response (no wait for broadcast)."""
        try:
            self.ser.reset_input_buffer()
        except Exception:
            pass
        sent = write_discovery_request(
            self.ser, request, self.baud, self.parity, self.stopbits,
            self.stop_event, self.serial_module,
```

MBSniffer v3.2/mb_master.py — lines 481-516

```
        )
        # write_discovery_request() returns once the request has left the
        # wire, so response time and timeout count from the end of the request.
        started = time.perf_counter()
        result = {"tx": hex_bytes(request), "rx": "", "status": "stopped",
                  "values": None, "exception_code": None, "response_ms": None}
        if not sent:
            return result
        if request[0] == 0:
            result["status"] = "broadcast"
            return result

        deadline = time.perf_counter() + self.response_timeout_s
        buffer = bytearray()
        while time.perf_counter() < deadline and not self.stop_event.is_set():
            waiting = int(getattr(self.ser, "in_waiting", 0) or 0)
            if waiting:
                buffer.extend(self.ser.read(waiting) or b"")
                found = find_master_response(buffer, request)
                if found is not None:
                    result.update(
                        status="exception" if found["kind"] == "EXCEPTION" else "ok",
                        values=found["values"],
                        exception_code=found["exception_code"],
                        rx=hex_bytes(found["frame"]),
                        response_ms=(time.perf_counter() - started) * 1000.0,
                    )
                    return result
            elif self.stop_event.wait(0.001):
                break

        if self.stop_event.is_set():
            return result
        result["rx"] = hex_bytes(buffer)
        result["status"] = "invalid" if buffer else "timeout"
        return result
```

MBSniffer v3.2/mb_master_ui.py

1150 source lines.

MBSniffer v3.2/mb_master_ui.py — lines 1-40

```python
#!/usr/bin/env python3
"""Master (poll) tab: one window per Slave / data area, polled over one COM.

The protocol and scheduling live in ``mb_master.py``; this mixin owns the
widgets. Each poll window has a model (``MasterWindow``) and a pane that can
sit in the tab's grid or be popped out into its own top-level window.
"""

import json
import time
import tkinter as tk
from tkinter import filedialog, messagebox, ttk

from mb_config import (
    DISCOVERY_BAUD_RATES,
    DISCOVERY_IDLE_CHECK_SECONDS,
    MASTER_DEFAULT_DELAY_MS,
    MASTER_DEFAULT_RESPONSE_TIMEOUT_MS,
    MASTER_DEFAULT_SCAN_MS,
    MASTER_MAX_DELAY_MS,
    MASTER_MAX_RESPONSE_TIMEOUT_MS,
    MASTER_MAX_SCAN_MS,
    MASTER_MAX_WINDOWS,
    MASTER_MIN_DELAY_MS,
    MASTER_MIN_RESPONSE_TIMEOUT_MS,
    MASTER_WORKSPACE_FORMAT,
    MASTER_WORKSPACE_VERSION,
    MODBUS_EXCEPTION_CODES,
)
from mb_diagnostics import FUNCTION_NAMES
from mb_master import (
    BIT_FUNCTION_CODES,
    DEFAULT_FORMAT,
    FORMAT_KEYS_BY_LABEL,
    FORMAT_LABELS,
    FORMAT_SIZES,
    READ_FUNCTION_CODES,
    WRITABLE_READ_FCS,
    MasterEngine,
    PollDefinition,
```

MBSniffer v3.2/mb_master_ui.py — lines 41-80

```
    build_write_request,
    encode_coil_value,
    encode_register_value,
    format_register_values,
    validate_read_definition,
)
from mb_theme import theme_colors
from mb_widgets import RefreshButton

MASTER_PARITIES = ("None", "Even", "Odd")
MASTER_STOP_BITS = ("1", "2")
MASTER_PANE_MIN_WIDTH = 600
MASTER_PANE_ROWS = 10
MASTER_OK_MESSAGE_SECONDS = 3.0
WHEEL_GUARD_TAG = "MBSniffer.MouseWheelGuard"
MASTER_WHEEL_TAG = "MBSniffer.MasterWheel"


def fc_choice_label(fc):
    return f"{fc:02d} {FUNCTION_NAMES[fc]}"


FC_CHOICES = tuple(fc_choice_label(fc) for fc in READ_FUNCTION_CODES)


def parse_int_field(text, minimum, maximum, message):
    try:
        value = int(str(text).strip())
    except ValueError:
        raise ValueError(message) from None
    if not minimum <= value <= maximum:
        raise ValueError(message)
    return value


class MasterWindow:
    """State of one poll window (independent of where its pane is shown)."""

    def __init__(self, window_id, number, slave=1, fc=3, address=0, qty=10,
                 scan_ms=MASTER_DEFAULT_SCAN_MS, base=0, fmt=DEFAULT_FORMAT,
```

MBSniffer v3.2/mb_master_ui.py — lines 81-120

```
                names=None, enabled=True):
        self.window_id = window_id
        self.number = number
        self.slave = slave
        self.fc = fc
        self.address = address
        self.qty = qty
        self.scan_ms = scan_ms
        self.base = base
        self.fmt = fmt
        self.names = dict(names or {})  # PDU address -> label
        self.enabled = enabled
        self.popped = False
        self.reset_counters()

    def reset_counters(self):
        self.tx = 0
        self.errors = 0
        self.timeouts = 0
        self.response_ms = None
        self.values = None
        # (canonical PT template, values, style) shown on the status line.
        self.message = ("", {}, "normal")
        self.message_at = 0.0

    def definition(self):
        return PollDefinition(self.window_id, self.slave, self.fc, self.address,
                              self.qty, self.scan_ms, self.enabled)

    def to_json(self):
        return {
            "slave": self.slave, "fc": self.fc, "address": self.address,
            "qty": self.qty, "scan_ms": self.scan_ms, "base": self.base,
            "format": self.fmt, "enabled": self.enabled,
            "names": {str(address): name for address, name in sorted(self.names.items())},
        }

    @staticmethod
    def validated_kwargs(data):
        """Validate one workspace window entry; raise ValueError if invalid."""
```

MBSniffer v3.2/mb_master_ui.py — lines 121-160

```
        if not isinstance(data, dict):
            raise ValueError("Janela inválida.")
        slave, fc = int(data.get("slave", 1)), int(data.get("fc", 3))
        address, qty = int(data.get("address", 0)), int(data.get("qty", 10))
        validate_read_definition(slave, fc, address, qty)
        scan_ms = int(data.get("scan_ms", MASTER_DEFAULT_SCAN_MS))
        if not 0 <= scan_ms <= MASTER_MAX_SCAN_MS:
            raise ValueError("Scan rate fora do intervalo.")
        fmt = data.get("format", DEFAULT_FORMAT)
        names = data.get("names", {})
        return {
            "slave": slave, "fc": fc, "address": address, "qty": qty,
            "scan_ms": scan_ms, "base": 1 if data.get("base") == 1 else 0,
            "fmt": fmt if fmt in FORMAT_SIZES else DEFAULT_FORMAT,
            "enabled": bool(data.get("enabled", True)),
            "names": {
                int(k): str(v) for k, v in (names.items() if isinstance(names, dict) else ())
                if str(k).isdigit() and str(v).strip()
            },
        }


class MasterMixin:
    """Master (poll) tab GUI."""

    # -- construction -------------------------------------------------------
    def build_master_tab(self, page):
        saved = getattr(self, "ui_settings", {}).get("master", {})
        if not isinstance(saved, dict):
            saved = {}

        self.master_windows = {}
        self._master_panes = {}
        self._master_next_id = 1
        self.master_engine = None
        self.master_serial = None
        self.master_active = False
        self._master_reflow_after_id = None

        page.columnconfigure(0, weight=1)
```

MBSniffer v3.2/mb_master_ui.py — lines 161-200

```
        page.rowconfigure(2, weight=1)

        connection = ttk.LabelFrame(page, text="Ligação — o Master transmite no barramento")
        connection.grid(row=0, column=0, sticky="ew", padx=12, pady=(12, 6))
        self._master_connection_frame = connection
        row = ttk.Frame(connection)
        row.grid(row=0, column=0, sticky="w", padx=8, pady=(8, 4))

        ttk.Label(row, text="Porta COM").grid(row=0, column=0, padx=(0, 4))
        self.master_port_var = tk.StringVar()
        self.master_port_combo = ttk.Combobox(
            row, textvariable=self.master_port_var, width=13, state="readonly"
        )
        self.master_port_combo.grid(row=0, column=1)
        self.master_refresh_btn = RefreshButton(row, command=self.refresh_ports, size=24)
        self.master_refresh_btn.grid(row=0, column=2, padx=(4, 18))

        def _sync_refresh_square(_event=None):
            height = self.master_port_combo.winfo_height()
            if height > 1:
                self.master_refresh_btn.configure(width=height, height=height)

        self.master_port_combo.bind("<Configure>", _sync_refresh_square, add="+")

        self.master_baud_var = tk.StringVar(value=str(saved.get("baud", "9600")))
        self.master_parity_var = tk.StringVar(value=saved.get("parity", "None"))
        self.master_stop_var = tk.StringVar(value=str(saved.get("stop_bits", "1")))
        self.master_timeout_var = tk.StringVar(
            value=str(saved.get("response_timeout_ms", MASTER_DEFAULT_RESPONSE_TIMEOUT_MS))
        )
        self.master_delay_var = tk.StringVar(
            value=str(saved.get("delay_ms", MASTER_DEFAULT_DELAY_MS))
        )
        fields = (
            ("Baud rate", ttk.Combobox(row, textvariable=self.master_baud_var, width=8,
                                       state="readonly",
                                       values=[str(b) for b in DISCOVERY_BAUD_RATES])),
            ("Parity", ttk.Combobox(row, textvariable=self.master_parity_var, width=6,
                                    state="readonly", values=MASTER_PARITIES)),
            ("Stop bits", ttk.Combobox(row, textvariable=self.master_stop_var, width=3,
```

MBSniffer v3.2/mb_master_ui.py — lines 201-240

```
                                    state="readonly", values=MASTER_STOP_BITS)),
            ("Timeout resposta (ms)", ttk.Spinbox(
                row, textvariable=self.master_timeout_var, width=6, increment=50,
                from_=MASTER_MIN_RESPONSE_TIMEOUT_MS, to=MASTER_MAX_RESPONSE_TIMEOUT_MS)),
            ("Atraso entre pedidos (ms)", ttk.Spinbox(
                row, textvariable=self.master_delay_var, width=6, increment=5,
                from_=MASTER_MIN_DELAY_MS, to=MASTER_MAX_DELAY_MS)),
        )
        column = 3
        self._master_setting_widgets = [self.master_port_combo]
        for caption, widget in fields:
            ttk.Label(row, text=caption).grid(row=0, column=column, padx=(0, 4))
            widget.grid(row=0, column=column + 1, padx=(0, 16))
            self._master_setting_widgets.append(widget)
            column += 2
        ttk.Label(row, text="Data bits: 8").grid(row=0, column=column)

        controls = ttk.Frame(connection)
        controls.grid(row=1, column=0, sticky="w", padx=8, pady=(0, 8))
        self.master_safety_var = tk.BooleanVar(value=False)
        ttk.Checkbutton(
            controls,
            text="Confirmo que não existe outro master ativo no barramento.",
            variable=self.master_safety_var,
        ).grid(row=0, column=0, padx=(0, 18))
        self.master_connect_btn = ttk.Button(
            controls, text="Ligar", command=self.master_connect, style="MBS.Primary.TButton"
        )
        self.master_connect_btn.grid(row=0, column=1, padx=(0, 6))
        self.master_disconnect_btn = ttk.Button(
            controls, text="Desligar", command=self.master_disconnect, state="disabled"
        )
        self.master_disconnect_btn.grid(row=0, column=2, padx=(0, 18))
        self.master_status_var = tk.StringVar(value=self.tr("Desligado"))
        ttk.Label(controls, textvariable=self.master_status_var).grid(row=0, column=3)

        toolbar = ttk.Frame(page)
        toolbar.grid(row=1, column=0, sticky="ew", padx=12, pady=(0, 6))
        toolbar.columnconfigure(3, weight=1)
        ttk.Button(toolbar, text="Nova janela", command=self.master_add_window).grid(
```

MBSniffer v3.2/mb_master_ui.py — lines 241-280

```
            row=0, column=0, padx=(0, 6))
        ttk.Button(toolbar, text="Abrir workspace", command=self.master_open_workspace).grid(
            row=0, column=1, padx=(0, 6))
        ttk.Button(toolbar, text="Guardar workspace", command=self.master_save_workspace).grid(
            row=0, column=2, padx=(0, 6))
        ttk.Label(
            toolbar,
            text=("Uma janela por Slave / zona de memória. Duplo clique num valor para "
                  "escrever; duplo clique num nome para o editar."),
        ).grid(row=0, column=3, sticky="w", padx=(12, 0))

        holder = ttk.Frame(page)
        holder.grid(row=2, column=0, sticky="nsew", padx=12, pady=(0, 12))
        holder.columnconfigure(0, weight=1)
        holder.rowconfigure(0, weight=1)
        self.master_canvas = tk.Canvas(holder, highlightthickness=0, borderwidth=0)
        self.master_canvas.grid(row=0, column=0, sticky="nsew")
        scrollbar = ttk.Scrollbar(holder, orient="vertical", command=self.master_canvas.yview)
        scrollbar.grid(row=0, column=1, sticky="ns")
        self.master_canvas.configure(yscrollcommand=scrollbar.set)
        self.master_grid = ttk.Frame(self.master_canvas)
        self._master_grid_item = self.master_canvas.create_window(
            (0, 0), window=self.master_grid, anchor="nw"
        )
        self.master_grid.bind(
            "<Configure>",
            lambda _e: self.master_canvas.configure(scrollregion=self.master_canvas.bbox("all")),
        )
        self.master_canvas.bind("<Configure>", self._master_schedule_reflow, add="+")
        self.master_empty_label = ttk.Label(
            self.master_grid,
            text="Sem janelas. Usa \"Nova janela\" para começar a ler um Slave.",
        )

        for sequence in ("<MouseWheel>", "<Button-4>", "<Button-5>"):
            self.bind_class(MASTER_WHEEL_TAG, sequence, self._master_wheel)

        for variable in (self.master_baud_var, self.master_parity_var, self.master_stop_var,
                         self.master_timeout_var, self.master_delay_var):
            variable.trace_add("write", lambda *_a: self._schedule_ui_settings_save())
```

MBSniffer v3.2/mb_master_ui.py — lines 281-320

```
281
282         self.master_add_window()
283
284     # -- helpers -----------------------------------------------------------
285     def _master_install_wheel_tags(self, root_widget):
286         """Give new widgets the global wheel guard; plain panels scroll the grid."""
287         stack = [root_widget]
288         while stack:
289             widget = stack.pop()
290             try:
291                 stack.extend(widget.winfo_children())
292                 tags = tuple(t for t in widget.bindtags()
293                              if t not in (WHEEL_GUARD_TAG, MASTER_WHEEL_TAG))
294                 if isinstance(widget, (ttk.Treeview, ttk.Scrollbar, tk.Toplevel)):
295                     widget.bindtags((WHEEL_GUARD_TAG,) + tags)
296                 else:
297                     widget.bindtags((MASTER_WHEEL_TAG, WHEEL_GUARD_TAG) + tags)
298             except tk.TclError:
299                 continue
300
301     def _master_wheel(self, event):
302         canvas = getattr(self, "master_canvas", None)
303         if canvas is None:
304             return "break"
305         try:
306             top, bottom = canvas.yview()
307             if top <= 0.0 and bottom >= 1.0:
308                 return "break"
309             if getattr(event, "num", None) == 4:
310                 step = -1
311             elif getattr(event, "num", None) == 5:
312                 step = 1
313             else:
314                 step = -1 if event.delta > 0 else 1
315             canvas.yview_scroll(step * 3, "units")
316         except tk.TclError:
317             pass
318         return "break"
319
320     def _master_window_title(self, window):
```

MBSniffer v3.2/mb_master_ui.py — lines 321-360

```
321         shown = window.address + window.base
322         return self.trf(
323             "Janela {number} — Slave {slave} · FC{fc:02d} · {address} · {qty}",
324             number=window.number, slave=window.slave, fc=window.fc,
325             address=shown, qty=window.qty,
326         )
327 
328     def _master_counters_text(self, window):
329         response = f"{window.response_ms:.1f} ms" if window.response_ms is not None else "—"
330         return (f"Tx: {window.tx}   Err: {window.errors}   "
331                 f"Timeouts: {window.timeouts}   Resp.: {response}")
332 
333     def _master_set_message(self, window, template, style="normal", **values):
334         window.message = (template, values, style)
335         window.message_at = time.monotonic()
336         self._master_render_status(window)
337 
338     def _master_render_status(self, window):
339         pane = self._master_panes.get(window.window_id)
340         if pane is None:
341             return
342         template, values, style = window.message
343         if not window.enabled and style != "error":
344             template, values, style = "Pausada", {}, "normal"
345         text = self.trf(template, **values) if template else ""
346         try:
347             pane["counters_var"].set(self._master_counters_text(window))
348             pane["message_var"].set(text)
349             pane["message_label"].configure(style={
350                 "error": "MBS.Master.Error.TLabel",
351                 "ok": "MBS.Master.Ok.TLabel",
352             }.get(style, "TLabel"))
353             pane["frame"].configure(text=self._master_window_title(window))
354             pane["pause_btn"].configure(text=self.tr("Retomar" if not window.enabled else "Pausar"))
355             pane["popout_btn"].configure(text=self.tr("Ancorar" if window.popped else "Destacar"))
356             if window.popped and pane.get("toplevel") is not None:
357                 pane["toplevel"].title(f"MBSniffer — {self._master_window_title(window)}")
358         except tk.TclError:
359             pass
360 
```

MBSniffer v3.2/mb_master_ui.py — lines 361-400

```
    # -- panes ---------------------------------------------------------------
    def _master_build_pane(self, window, parent):
        frame = ttk.LabelFrame(parent, text=self._master_window_title(window))
        frame.columnconfigure(0, weight=1)
        frame.rowconfigure(4, weight=1)
        pane = {"frame": frame}

        definition = ttk.Frame(frame)
        definition.grid(row=0, column=0, sticky="w", padx=6, pady=(6, 2))
        pane["slave_var"] = tk.StringVar(value=str(window.slave))
        pane["fc_var"] = tk.StringVar(value=fc_choice_label(window.fc))
        pane["address_var"] = tk.StringVar(value=str(window.address + window.base))
        pane["qty_var"] = tk.StringVar(value=str(window.qty))
        pane["scan_var"] = tk.StringVar(value=str(window.scan_ms))
        pane["base_var"] = tk.StringVar(value=str(window.base))
        pane["format_var"] = tk.StringVar(value=FORMAT_LABELS[window.fmt])

        layout = (
            ("Slave", ttk.Spinbox(definition, from_=1, to=247, width=4,
                                  textvariable=pane["slave_var"])),
            ("FC", ttk.Combobox(definition, values=FC_CHOICES, width=24, state="readonly",
                                textvariable=pane["fc_var"])),
            ("Endereço", ttk.Entry(definition, width=6, textvariable=pane["address_var"])),
            ("Qtd.", ttk.Entry(definition, width=5, textvariable=pane["qty_var"])),
        )
        for index, (caption, widget) in enumerate(layout):
            ttk.Label(definition, text=caption).grid(row=0, column=index * 2, padx=(0, 3))
            widget.grid(row=0, column=index * 2 + 1, padx=(0, 10))

        definition2 = ttk.Frame(frame)
        definition2.grid(row=1, column=0, sticky="w", padx=6, pady=2)
        format_combo = ttk.Combobox(
            definition2, values=[FORMAT_LABELS[k] for k in FORMAT_LABELS], width=20,
            state="readonly", textvariable=pane["format_var"],
        )
        layout2 = (
            ("Scan (ms)", ttk.Entry(definition2, width=7, textvariable=pane["scan_var"])),
            ("Base", ttk.Combobox(definition2, values=("0", "1"), width=2, state="readonly",
                                  textvariable=pane["base_var"])),
            ("Formato", format_combo),
```

MBSniffer v3.2/mb_master_ui.py — lines 401-440

```
        )
        for index, (caption, widget) in enumerate(layout2):
            ttk.Label(definition2, text=caption).grid(row=0, column=index * 2, padx=(0, 3))
            widget.grid(row=0, column=index * 2 + 1, padx=(0, 10))
        pane["format_combo"] = format_combo

        buttons = ttk.Frame(frame)
        buttons.grid(row=2, column=0, sticky="w", padx=6, pady=(2, 2))
        window_id = window.window_id
        actions = (
            ("Aplicar", lambda: self.master_apply_window(window_id)),
            ("Pausar", lambda: self.master_toggle_pause(window_id)),
            ("Ler 1x", lambda: self.master_read_once(window_id)),
            ("Duplicar", lambda: self.master_duplicate_window(window_id)),
            ("Destacar", lambda: self.master_toggle_popout(window_id)),
            ("Fechar", lambda: self.master_close_window(window_id)),
        )
        for index, (caption, command) in enumerate(actions):
            button = ttk.Button(buttons, text=caption, command=command, width=10)
            button.grid(row=0, column=index, padx=(0, 3))
            if caption == "Pausar":
                pane["pause_btn"] = button
            elif caption == "Destacar":
                pane["popout_btn"] = button

        status = ttk.Frame(frame)
        status.grid(row=3, column=0, sticky="ew", padx=6, pady=(2, 2))
        status.columnconfigure(1, weight=1)
        pane["counters_var"] = tk.StringVar()
        pane["message_var"] = tk.StringVar()
        ttk.Label(status, textvariable=pane["counters_var"]).grid(row=0, column=0, sticky="w")
        pane["message_label"] = ttk.Label(status, textvariable=pane["message_var"])
        pane["message_label"].grid(row=0, column=1, sticky="w", padx=(14, 0))

        table = ttk.Frame(frame)
        table.grid(row=4, column=0, sticky="nsew", padx=6, pady=(2, 6))
        table.columnconfigure(0, weight=1)
        table.rowconfigure(0, weight=1)
        tree = ttk.Treeview(table, columns=("address", "name", "value"), show="headings",
                            height=MASTER_PANE_ROWS, selectmode="browse")
```

MBSniffer v3.2/mb_master_ui.py — lines 441-480

```
        for column, caption, width, anchor in (
            ("address", "Endereço", 90, "center"),
            ("name", "Nome", 200, "w"),
            ("value", "Valor", 160, "e"),
        ):
            tree.heading(column, text=self.tr(caption), anchor="center")
            tree.column(column, width=width, minwidth=60, anchor=anchor, stretch=column != "address")
        tree.grid(row=0, column=0, sticky="nsew")
        scrollbar = ttk.Scrollbar(table, orient="vertical", command=tree.yview)
        scrollbar.grid(row=0, column=1, sticky="ns")
        tree.configure(yscrollcommand=scrollbar.set)
        tree.bind("<Double-1>", lambda e: self._master_on_double_click(window_id, e))
        tree.bind("<Return>", lambda e: self._master_edit_value(window_id))
        pane["tree"] = tree

        for widget in (definition, definition2):
            for child in widget.winfo_children():
                if isinstance(child, (ttk.Entry, ttk.Spinbox)):
                    child.bind("<Return>", lambda _e: self.master_apply_window(window_id))

        # Captions were written in canonical PT; show them in the current language.
        for widget in (definition, definition2, buttons):
            for child in widget.winfo_children():
                try:
                    if "text" in child.keys() and child.cget("text"):
                        child.configure(text=self.tr(child.cget("text")))
                except tk.TclError:
                    pass

        self._master_panes[window_id] = pane
        self._master_fill_rows(window)
        self._master_update_format_state(window)
        self._master_render_status(window)
        self._master_install_wheel_tags(frame)
        return frame

    def _master_update_format_state(self, window):
        pane = self._master_panes.get(window.window_id)
        if pane is not None:
            state = "disabled" if window.fc in BIT_FUNCTION_CODES else "readonly"
```

MBSniffer v3.2/mb_master_ui.py — lines 481-520

```
481             pane["format_combo"].configure(state=state)
482
483     def _master_fill_rows(self, window):
484         pane = self._master_panes.get(window.window_id)
485         if pane is None:
486             return
487         tree = pane["tree"]
488         tree.delete(*tree.get_children())
489         for index in range(window.qty):
490             pdu = window.address + index
491             tree.insert("", "end", iid=f"r{index}",
492                         values=(pdu + window.base, window.names.get(pdu, ""), ""))
493         self._master_render_values(window)
494
495     def _master_render_values(self, window):
496         pane = self._master_panes.get(window.window_id)
497         if pane is None or window.values is None:
498             return
499         if window.fc in BIT_FUNCTION_CODES:
500             texts = [str(v) for v in window.values]
501         else:
502             texts = format_register_values(window.values, window.fmt)
503         tree = pane["tree"]
504         for index, text in enumerate(texts[:window.qty]):
505             try:
506                 tree.set(f"r{index}", "value", text)
507             except tk.TclError:
508                 pass
509
510     def _master_schedule_reflow(self, _event=None):
511         if self._master_reflow_after_id is not None:
512             try:
513                 self.after_cancel(self._master_reflow_after_id)
514             except tk.TclError:
515                 pass
516         self._master_reflow_after_id = self.after(80, self._master_reflow)
517
518     def _master_reflow(self):
519         """Arrange docked panes in as many equal columns as the width allows."""
520         self._master_reflow_after_id = None
```

MBSniffer v3.2/mb_master_ui.py — lines 521-560

```
        canvas = self.master_canvas
        width = max(1, canvas.winfo_width())
        canvas.itemconfigure(self._master_grid_item, width=width)
        docked = [w for w in sorted(self.master_windows.values(), key=lambda w: w.number)
                  if not w.popped]
        columns = max(1, width // MASTER_PANE_MIN_WIDTH)
        for column in range(8):
            self.master_grid.columnconfigure(column, weight=1 if column < columns else 0,
                                             uniform="master_panes" if column < columns else "")
        if not docked:
            self.master_empty_label.grid(row=0, column=0, sticky="w", padx=4, pady=8)
        else:
            self.master_empty_label.grid_remove()
        for index, window in enumerate(docked):
            pane = self._master_panes.get(window.window_id)
            if pane is not None:
                pane["frame"].grid(row=index // columns, column=index % columns,
                                   sticky="nsew", padx=(0, 8), pady=(0, 8))

    # -- window actions ------------------------------------------------------
    def master_add_window(self, kwargs=None):
        if len(self.master_windows) >= MASTER_MAX_WINDOWS:
            messagebox.showwarning(
                self.tr("Master"),
                self.trf("Máximo de {count} janelas.", count=MASTER_MAX_WINDOWS),
            )
            return None
        window_id = self._master_next_id
        self._master_next_id += 1
        numbers = {w.number for w in self.master_windows.values()}
        number = next(n for n in range(1, MASTER_MAX_WINDOWS + 2) if n not in numbers)
        window = MasterWindow(window_id, number, **(kwargs or {}))
        self.master_windows[window_id] = window
        self._master_build_pane(window, self.master_grid)
        self._master_reflow()
        if self.master_engine is not None:
            self.master_engine.set_poll(window.definition())
        return window

    def master_duplicate_window(self, window_id):
```

MBSniffer v3.2/mb_master_ui.py — lines 561-600

```
561         window = self.master_windows.get(window_id)
562         if window is not None:
563             data = MasterWindow.validated_kwargs(window.to_json())
564             self.master_add_window(data)
565 
566     def master_close_window(self, window_id):
567         window = self.master_windows.pop(window_id, None)
568         if window is None:
569             return
570         if self.master_engine is not None:
571             self.master_engine.remove_poll(window_id)
572         pane = self._master_panes.pop(window_id, None)
573         if pane is not None:
574             top = pane.get("toplevel")
575             try:
576                 (top if top is not None else pane["frame"]).destroy()
577             except tk.TclError:
578                 pass
579         self._master_reflow()
580 
581     def master_apply_window(self, window_id):
582         window = self.master_windows.get(window_id)
583         pane = self._master_panes.get(window_id)
584         if window is None or pane is None:
585             return False
586         try:
587             base = 1 if pane["base_var"].get() == "1" else 0
588             slave = parse_int_field(pane["slave_var"].get(), 1, 247,
589                                     "O Slave ID de leitura tem de estar entre 1 e 247.")
590             fc = int(pane["fc_var"].get()[:2])
591             address = parse_int_field(pane["address_var"].get(), base, 0xFFFF + base,
592                                       "O endereço tem de estar entre 0 e 65535.") - base
593             qty = parse_int_field(pane["qty_var"].get(), 1, 2000,
594                                   "Quantidade fora do limite desta função.")
595             scan_ms = parse_int_field(pane["scan_var"].get(), 0, MASTER_MAX_SCAN_MS,
596                                       "Scan rate fora do intervalo.")
597             validate_read_definition(slave, fc, address, qty)
598         except ValueError as exc:
599             messagebox.showerror(self.tr("Janela inválida"), self.tr(str(exc)))
600             return False
```

MBSniffer v3.2/mb_master_ui.py — lines 601-640

```
601
602         changed_area = (slave, fc, address, qty) != (window.slave, window.fc,
603                                                      window.address, window.qty)
604         window.slave, window.fc, window.address, window.qty = slave, fc, address, qty
605         window.scan_ms, window.base = scan_ms, base
606         window.fmt = FORMAT_KEYS_BY_LABEL.get(pane["format_var"].get(), DEFAULT_FORMAT)
607         if changed_area:
608             window.reset_counters()
609         self._master_fill_rows(window)
610         self._master_update_format_state(window)
611         self._master_render_status(window)
612         if self.master_engine is not None:
613             self.master_engine.set_poll(window.definition())
614         return True
615
616     def master_toggle_pause(self, window_id):
617         window = self.master_windows.get(window_id)
618         if window is None:
619             return
620         window.enabled = not window.enabled
621         if window.enabled and window.message[0] == "Pausada":
622             window.message = ("", {}, "normal")
623         self._master_render_status(window)
624         if self.master_engine is not None:
625             self.master_engine.set_poll(window.definition())
626
627     def master_read_once(self, window_id):
628         if self.master_engine is None:
629             messagebox.showinfo(self.tr("Master"), self.tr("Liga primeiro o Master."))
630             return
631         self.master_engine.read_once(window_id)
632
633     def master_toggle_popout(self, window_id):
634         window = self.master_windows.get(window_id)
635         pane = self._master_panes.pop(window_id, None)
636         if window is None or pane is None:
637             return
638         old_top = pane.get("toplevel")
639         try:
640             pane["frame"].destroy()
```

MBSniffer v3.2/mb_master_ui.py — lines 641-680

```
641             if old_top is not None:
642                 old_top.destroy()
643         except tk.TclError:
644             pass
645 
646         window.popped = not window.popped
647         if window.popped:
648             top = tk.Toplevel(self)
649             top.geometry("660x420")
650             top.minsize(MASTER_PANE_MIN_WIDTH, 300)
651             top.columnconfigure(0, weight=1)
652             top.rowconfigure(0, weight=1)
653             top.protocol("WM_DELETE_WINDOW", lambda: self.master_toggle_popout(window_id))
654             frame = self._master_build_pane(window, top)
655             frame.grid(row=0, column=0, sticky="nsew", padx=8, pady=8)
656             self._master_panes[window_id]["toplevel"] = top
657             self._master_apply_toplevel_palette(top)
658         else:
659             self._master_build_pane(window, self.master_grid)
660         self._master_render_status(window)
661         self._master_reflow()
662 
663     # -- value / name editing ------------------------------------------------
664     def _master_on_double_click(self, window_id, event):
665         pane = self._master_panes.get(window_id)
666         if pane is None:
667             return "break"
668         tree = pane["tree"]
669         row = tree.identify_row(event.y)
670         if not row:
671             return "break"
672         tree.selection_set(row)
673         if tree.identify_column(event.x) == "#2":
674             self._master_edit_name(window_id, row)
675         else:
676             self._master_edit_value(window_id)
677         return "break"
678 
679     def _master_dialog(self, title):
680         dialog = tk.Toplevel(self)
```

MBSniffer v3.2/mb_master_ui.py — lines 681-720

```
        dialog.title(title)
        dialog.transient(self)
        dialog.resizable(False, False)
        self._master_apply_toplevel_palette(dialog)
        body = ttk.Frame(dialog, padding=12)
        body.grid(row=0, column=0, sticky="nsew")
        return dialog, body

    def _master_edit_name(self, window_id, row):
        window = self.master_windows.get(window_id)
        if window is None:
            return
        pdu = window.address + int(row[1:])
        dialog, body = self._master_dialog(self.tr("Nome"))
        ttk.Label(body, text=self.trf("Nome do endereço {address}",
                                      address=pdu + window.base)).grid(row=0, column=0, sticky="w")
        name_var = tk.StringVar(value=window.names.get(pdu, ""))
        entry = ttk.Entry(body, textvariable=name_var, width=36)
        entry.grid(row=1, column=0, pady=(4, 8))

        def save(_event=None):
            name = name_var.get().strip()
            if name:
                window.names[pdu] = name
            else:
                window.names.pop(pdu, None)
            pane = self._master_panes.get(window_id)
            if pane is not None:
                pane["tree"].set(row, "name", name)
            dialog.destroy()

        ttk.Button(body, text=self.tr("Guardar"), command=save).grid(row=2, column=0, sticky="e")
        entry.bind("<Return>", save)
        entry.focus_set()
        dialog.grab_set()

    def _master_edit_value(self, window_id):
        window = self.master_windows.get(window_id)
        pane = self._master_panes.get(window_id)
        if window is None or pane is None:
```

MBSniffer v3.2/mb_master_ui.py — lines 721-760

```
            return
        selection = pane["tree"].selection()
        if not selection:
            return
        index = int(selection[0][1:])
        write_fcs = WRITABLE_READ_FCS.get(window.fc)
        if write_fcs is None:
            messagebox.showinfo(
                self.tr("Escrita"),
                self.tr("FC02 e FC04 são só de leitura: abre uma janela FC01 ou FC03 para escrever."),
            )
            return
        bits = window.fc in BIT_FUNCTION_CODES
        size = 1 if bits else FORMAT_SIZES[window.fmt]
        index -= index % size  # 32-bit values start on the first register of the pair
        current = ""
        try:
            current = pane["tree"].set(f"r{index}", "value")
        except tk.TclError:
            pass

        dialog, body = self._master_dialog(self.tr("Escrever"))
        slave_var = tk.StringVar(value=str(window.slave))
        address_var = tk.StringVar(value=str(window.address + index + window.base))
        value_var = tk.StringVar(value=current)
        single_fc, multiple_fc = write_fcs
        fc_var = tk.IntVar(value=single_fc if size == 1 else multiple_fc)

        rows = (
            ("Slave (0 = broadcast)", ttk.Spinbox(body, from_=0, to=247, width=6,
                                                  textvariable=slave_var)),
            ("Endereço", ttk.Entry(body, width=8, textvariable=address_var)),
            ("Valor(es)", ttk.Entry(body, width=30, textvariable=value_var)),
        )
        for row_index, (caption, widget) in enumerate(rows):
            ttk.Label(body, text=self.tr(caption)).grid(row=row_index, column=0, sticky="w",
                                                        pady=2, padx=(0, 8))
            widget.grid(row=row_index, column=1, sticky="w", pady=2)
        functions = ttk.Frame(body)
        functions.grid(row=3, column=0, columnspan=2, sticky="w", pady=(6, 2))
```

MBSniffer v3.2/mb_master_ui.py — lines 761-800

```
        single = ttk.Radiobutton(functions, text=fc_choice_label_write(single_fc),
                                 variable=fc_var, value=single_fc)
        single.grid(row=0, column=0, sticky="w")
        if size != 1:
            single.configure(state="disabled")
        ttk.Radiobutton(functions, text=fc_choice_label_write(multiple_fc),
                        variable=fc_var, value=multiple_fc).grid(row=1, column=0, sticky="w")
        format_text = "0 / 1" if bits else FORMAT_LABELS[window.fmt]
        ttk.Label(
            body,
            text=self.trf("Formato: {fmt}. Vários valores (FC15/FC16): separar com ;",
                          fmt=format_text),
        ).grid(row=4, column=0, columnspan=2, sticky="w", pady=(6, 8))
        message_var = tk.StringVar()
        ttk.Label(body, textvariable=message_var, style="MBS.Master.Error.TLabel").grid(
            row=5, column=0, columnspan=2, sticky="w")

        def send(_event=None):
            try:
                request, description = self.master_build_write(
                    window, slave_var.get(), address_var.get(), value_var.get(), fc_var.get()
                )
            except ValueError as exc:
                message_var.set(self.tr(str(exc)))
                return
            if self.master_engine is None:
                message_var.set(self.tr("Liga primeiro o Master."))
                return
            self.master_engine.queue_write(window.window_id, request, description)
            dialog.destroy()

        buttons = ttk.Frame(body)
        buttons.grid(row=6, column=0, columnspan=2, sticky="e", pady=(8, 0))
        ttk.Button(buttons, text=self.tr("Enviar"), command=send,
                   style="MBS.Primary.TButton").grid(row=0, column=0, padx=(0, 6))
        ttk.Button(buttons, text=self.tr("Cancelar"), command=dialog.destroy).grid(row=0, column=1)
        rows[2][1].bind("<Return>", send)
        rows[2][1].focus_set()
        rows[2][1].select_range(0, "end")
        dialog.grab_set()
```

MBSniffer v3.2/mb_master_ui.py — lines 801-840

```
801
802     def master_build_write(self, window, slave_text, address_text, values_text, fc):
803         """Validate the write dialog fields and return (request, description)."""
804         slave = parse_int_field(slave_text, 0, 247,
805                                 "O Slave ID de escrita tem de estar entre 0 (broadcast) e 247.")
806         address = parse_int_field(address_text, window.base, 0xFFFF + window.base,
807                                   "O endereço tem de estar entre 0 e 65535.") - window.base
808         parts = [p for p in str(values_text).split(";") if p.strip()]
809         if not parts:
810             raise ValueError("Valor vazio.")
811         if window.fc in BIT_FUNCTION_CODES:
812             values = [encode_coil_value(p) for p in parts]
813         else:
814             values = []
815             for part in parts:
816                 values.extend(encode_register_value(part, window.fmt))
817         if fc in (5, 6) and len(values) != 1:
818             raise ValueError("FC05/FC06 escrevem um único valor; usa FC15/FC16 para vários.")
819         request = build_write_request(slave, fc, address, values)
820         description = f"FC{fc:02d} @ {address + window.base} × {len(values)}"
821         return request, description
822
823     # -- connection ---------------------------------------------------------
824     def _master_connection_settings(self):
825         timeout = parse_int_field(
826             self.master_timeout_var.get(), MASTER_MIN_RESPONSE_TIMEOUT_MS,
827             MASTER_MAX_RESPONSE_TIMEOUT_MS,
828             "O timeout de resposta tem de estar entre 10 e 10000 ms.")
829         delay = parse_int_field(
830             self.master_delay_var.get(), MASTER_MIN_DELAY_MS, MASTER_MAX_DELAY_MS,
831             "O atraso entre pedidos tem de estar entre 0 e 10000 ms.")
832         return {
833             "port": self.master_port_var.get().strip(),
834             "baud": int(self.master_baud_var.get()),
835             "parity": self.master_parity_var.get(),
836             "stopbits": self.master_stop_var.get(),
837             "response_timeout_ms": timeout,
838             "delay_ms": delay,
839         }
840
```

MBSniffer v3.2/mb_master_ui.py — lines 841-880

```
841     def master_connect(self):
842         if self.master_active:
843             return
844         if self.busy:
845             messagebox.showwarning(
846                 self.tr("Master"),
847                 self.tr("Para a captura ou a pesquisa em curso antes de ligar o Master."),
848             )
849             return
850         if not self.master_safety_var.get():
851             messagebox.showwarning(
852                 self.tr("Master"),
853                 self.tr("Confirma primeiro que não existe outro master ativo no barramento."),
854             )
855             return
856         serial, _ = self.get_serial_modules()
857         if serial is None:
858             messagebox.showerror(self.tr("pyserial em falta"), self.tr("pyserial não está instalado."))
859             return
860         try:
861             settings = self._master_connection_settings()
862             if not settings["port"]:
863                 raise ValueError("Seleciona uma porta COM.")
864         except ValueError as exc:
865             messagebox.showerror(self.tr("Configuração inválida"), self.tr(str(exc)))
866             return
867         for window_id in list(self.master_windows):
868             if not self.master_apply_window(window_id):
869                 return
870         try:
871             ser = self.open_discovery_serial_port(
872                 serial, settings["port"], settings["baud"], settings["parity"],
873                 settings["stopbits"],
874             )
875         except Exception as exc:
876             messagebox.showerror(
877                 self.tr("Master"),
878                 self.trf("Não foi possível abrir {port}: {error}",
879                          port=settings["port"], error=f"{type(exc).__name__}: {exc}"),
880             )
```

MBSniffer v3.2/mb_master_ui.py — lines 881-920

```
881             return
882 
883         self.master_serial = ser
884         self.master_engine = MasterEngine(
885             ser, self.event_queue, baud=settings["baud"], parity=settings["parity"],
886             stopbits=settings["stopbits"],
887             response_timeout_ms=settings["response_timeout_ms"],
888             delay_ms=settings["delay_ms"], serial_module=serial,
889             idle_check_seconds=DISCOVERY_IDLE_CHECK_SECONDS,
890         )
891         for window in self.master_windows.values():
892             window.reset_counters()
893             self._master_render_status(window)
894             self.master_engine.set_poll(window.definition())
895         self.master_active = True
896         self.busy = True
897         self._master_lock_other_features(True)
898         self.master_status_var.set(self.tr("A verificar se o barramento está livre…"))
899         self.master_engine.start()
900 
901     def master_disconnect(self):
902         if self.master_engine is None:
903             return
904         self.master_status_var.set(self.tr("A desligar…"))
905         self.master_disconnect_btn.configure(state="disabled")
906         self.master_engine.stop()
907 
908     def _master_lock_other_features(self, locked):
909         state = "disabled" if locked else "normal"
910         self.master_connect_btn.configure(state=state)
911         self.master_disconnect_btn.configure(state="normal" if locked else "disabled")
912         for widget in self._master_setting_widgets:
913             try:
914                 widget.configure(state="disabled" if locked else "readonly"
915                                  if isinstance(widget, ttk.Combobox) else "normal")
916             except tk.TclError:
917                 pass
918         self.master_refresh_btn.configure(state=state)
919         self.start_btn.configure(state=state)
920         self.restart_btn.configure(state=state)
```

MBSniffer v3.2/mb_master_ui.py — lines 921-960

```
        self.set_sniffer_refresh_state(state)
        self.set_simulation_button_state(state)
        self.update_discovery_start_state()

    def _master_finish_session(self, payload):
        engine = self.master_engine
        self.master_engine = None
        if engine is not None:
            engine.stop(join_timeout=0.5)
        if self.master_serial is not None:
            try:
                self.master_serial.close()
            except Exception:
                pass
        self.master_serial = None
        self.master_active = False
        self.busy = False
        self._master_lock_other_features(False)
        self.master_status_var.set(self.tr("Desligado"))

        error = (payload or {}).get("error")
        if error:
            kind, detail, count = error
            if kind == "traffic":
                text = self.trf(
                    "Foi detetado tráfego em {port} antes de transmitir "
                    "({count} bytes: {raw}). Pode existir outro master "
                    "ativo no barramento; a pesquisa não foi iniciada.",
                    port=self.master_port_var.get(), count=count, raw=detail,
                )
            else:
                text = self.trf("Erro na porta série: {error}", error=detail)
            messagebox.showerror(self.tr("Master"), text)

    # -- worker events ------------------------------------------------------
    def handle_master_event(self, event, payload):
        if event == "master_ready":
            self.master_status_var.set(self.tr("● Ligado"))
        elif event == "master_stopped":
            self._master_finish_session(payload)
```

MBSniffer v3.2/mb_master_ui.py — lines 961-1000

```
961         elif event == "master_result":
962             self._master_apply_result(payload)
963 
964     def _master_apply_result(self, result):
965         window = self.master_windows.get(result.get("window_id"))
966         if window is None:
967             return
968         status = result["status"]
969         if status == "stopped":
970             return
971         window.tx += 1
972         if result.get("response_ms") is not None:
973             window.response_ms = result["response_ms"]
974 
975         if result["kind"] == "read":
976             if (result.get("fc"), result.get("address")) != (window.fc, window.address):
977                 return  # definition changed while the request was on the bus
978             if status == "ok":
979                 window.values = result["values"]
980                 self._master_render_values(window)
981                 # A successful read clears errors; a write confirmation stays
982                 # visible for a few seconds so the user can see it.
983                 if (window.message[2] != "ok"
984                         or time.monotonic() - window.message_at > MASTER_OK_MESSAGE_SECONDS):
985                     self._master_set_message(window, "")
986                 return
987         elif status == "ok":
988             self._master_set_message(window, "Escrita OK: {what}", "ok",
989                                      what=result.get("description", ""))
990             if self.master_engine is not None:
991                 self.master_engine.read_once(window.window_id)
992             return
993         elif status == "broadcast":
994             self._master_set_message(window, "Escrita broadcast enviada: {what}", "ok",
995                                      what=result.get("description", ""))
996             return
997 
998         window.errors += 1
999         if status == "timeout":
1000             window.timeouts += 1
```

MBSniffer v3.2/mb_master_ui.py — lines 1001-1040

```
            self._master_set_message(window, "Timeout", "error")
        elif status == "exception":
            code = result.get("exception_code") or 0
            name = MODBUS_EXCEPTION_CODES.get(code, ("Unknown/Reserved Exception", ""))[0]
            self._master_set_message(window, "Exception 0x{code:02X} — {name}", "error",
                                     code=code, name=name)
        else:
            self._master_set_message(window, "Resposta inválida (CRC ou formato): {rx}",
                                     "error", rx=result.get("rx", "")[:47])

    # -- workspace ------------------------------------------------------------
    def master_workspace_data(self):
        return {
            "format": MASTER_WORKSPACE_FORMAT,
            "version": MASTER_WORKSPACE_VERSION,
            "connection": {
                "port": self.master_port_var.get(),
                "baud": self.master_baud_var.get(),
                "parity": self.master_parity_var.get(),
                "stop_bits": self.master_stop_var.get(),
                "response_timeout_ms": self.master_timeout_var.get(),
                "delay_ms": self.master_delay_var.get(),
            },
            "windows": [w.to_json() for w in sorted(self.master_windows.values(),
                                                    key=lambda w: w.number)],
        }

    def master_save_workspace(self, path=None):
        if path is None:
            path = filedialog.asksaveasfilename(
                parent=self, title=self.tr("Guardar workspace"), defaultextension=".mbsw",
                filetypes=[("MBSniffer workspace", "*.mbsw"), ("JSON", "*.json")],
            )
        if not path:
            return False
        try:
            with open(path, "w", encoding="utf-8") as handle:
                json.dump(self.master_workspace_data(), handle, indent=2, ensure_ascii=False)
            return True
        except OSError as exc:
```

MBSniffer v3.2/mb_master_ui.py — lines 1041-1080

```
            messagebox.showerror(self.tr("Guardar workspace"), str(exc))
            return False

    def master_open_workspace(self, path=None):
        if path is None:
            path = filedialog.askopenfilename(
                parent=self, title=self.tr("Abrir workspace"),
                filetypes=[("MBSniffer workspace", "*.mbsw"), ("JSON", "*.json")],
            )
        if not path:
            return False
        try:
            with open(path, encoding="utf-8") as handle:
                data = json.load(handle)
            if not isinstance(data, dict) or data.get("format") != MASTER_WORKSPACE_FORMAT:
                raise ValueError(self.tr("O ficheiro não é um workspace do MBSniffer."))
        except (OSError, ValueError) as exc:
            messagebox.showerror(self.tr("Abrir workspace"), str(exc))
            return False

        entries, skipped = [], 0
        for item in data.get("windows", [])[:MASTER_MAX_WINDOWS]:
            try:
                entries.append(MasterWindow.validated_kwargs(item))
            except (ValueError, TypeError):
                skipped += 1

        for window_id in list(self.master_windows):
            self.master_close_window(window_id)
        connection = data.get("connection", {})
        if isinstance(connection, dict) and not self.master_active:
            if str(connection.get("baud")) in [str(b) for b in DISCOVERY_BAUD_RATES]:
                self.master_baud_var.set(str(connection["baud"]))
            if connection.get("parity") in MASTER_PARITIES:
                self.master_parity_var.set(connection["parity"])
            if str(connection.get("stop_bits")) in MASTER_STOP_BITS:
                self.master_stop_var.set(str(connection["stop_bits"]))
            for key, var in (("response_timeout_ms", self.master_timeout_var),
                             ("delay_ms", self.master_delay_var)):
                if str(connection.get(key, "")).isdigit():
```

MBSniffer v3.2/mb_master_ui.py — lines 1081-1120

```
                    var.set(str(connection[key]))
            if connection.get("port") in tuple(self.master_port_combo["values"]):
                self.master_port_var.set(connection["port"])
        for kwargs in entries:
            self.master_add_window(kwargs)
        if skipped:
            messagebox.showwarning(
                self.tr("Abrir workspace"),
                self.trf("{count} janela(s) inválida(s) foram ignoradas.", count=skipped),
            )
        return True

    # -- language / palette / shutdown ------------------------------------------------
    def master_apply_language(self):
        if not hasattr(self, "master_windows"):
            return
        for window in self.master_windows.values():
            pane = self._master_panes.get(window.window_id)
            if pane is None:
                continue
            for column, caption in (("address", "Endereço"), ("name", "Nome"), ("value", "Valor")):
                try:
                    pane["tree"].heading(column, text=self.tr(caption))
                except tk.TclError:
                    pass
            self._master_render_status(window)
        if not self.master_active:
            self.master_status_var.set(self.tr("Desligado"))
        elif self.master_engine is not None:
            self.master_status_var.set(self.tr("● Ligado"))

    def _master_apply_toplevel_palette(self, top):
        dark = bool(getattr(self, "dark_mode_var", tk.BooleanVar(value=False)).get())
        try:
            top.configure(background=theme_colors(self, dark)["root"])
        except tk.TclError:
            pass

    def master_apply_palette(self, dark):
        colors = theme_colors(self, dark)
```

MBSniffer v3.2/mb_master_ui.py — lines 1121-1150

```
1121        if hasattr(self, "master_canvas"):
1122            self.master_canvas.configure(background=colors["root"])
1123        if hasattr(self, "ui_style"):
1124            self.ui_style.configure("MBS.Master.Error.TLabel",
1125                                    foreground="#ff8a80" if dark else "#b42318")
1126            self.ui_style.configure("MBS.Master.Ok.TLabel",
1127                                    foreground="#a7e4b8" if dark else "#1f6b3a")
1128        for pane in getattr(self, "_master_panes", {}).values():
1129            if pane.get("toplevel") is not None:
1130                self._master_apply_toplevel_palette(pane["toplevel"])
1131        button = getattr(self, "master_refresh_btn", None)
1132        if button is not None:
1133            button.set_dark_mode(dark)
1134
1135    def master_shutdown(self):
1136        engine = getattr(self, "master_engine", None)
1137        if engine is not None:
1138            engine.stop(join_timeout=0.8)
1139        ser = getattr(self, "master_serial", None)
1140        if ser is not None:
1141            try:
1142                ser.close()
1143            except Exception:
1144                pass
1145        self.master_engine = None
1146        self.master_serial = None
1147
1148
1149 def fc_choice_label_write(fc):
1150     return f"FC{fc:02d} {FUNCTION_NAMES[fc]}"
```

MBSniffer v3.2/mb_i18n.py

555 source lines.

MBSniffer v3.2/mb_i18n.py — lines 1-40

```
#!/usr/bin/env python3
"""Runtime language support for MBSniffer.

Portuguese (Portugal) is the canonical UI language.  English translations are
kept here so switching language never mutates protocol data or application
logic.
"""

from mb_config import DISCOVERY_IDLE_CHECK_SECONDS, SNIFFER_SPLIT_FRAME_HOLD_MS

LANGUAGE_PORTUGUESE = "Português"
LANGUAGE_ENGLISH = "English"
LANGUAGE_CHOICES = (LANGUAGE_PORTUGUESE, LANGUAGE_ENGLISH)

# Static UI text. Technical terms that are already English are intentionally
# left out: translate_text() returns unknown strings unchanged.
PT_TO_EN = {
    "Linguagem": "Language",
    "Modo escuro": "Dark mode",
    "Configuração": "Configuration",
    "Modo físico": "Physical mode",
    "RS232 dual RX: COM A escuta uma direção; COM B escuta a direção oposta.": "RS232 dual RX: COM A monitors one direction; COM B monitors the opposite direction.",
    "RS232 single RX: uma COM escuta apenas uma direção da ligação RS232.": "RS232 single RX: one COM port monitors only one direction of the RS232 link.",
    "RS485 2-wire: uma COM consegue observar pedidos e respostas no mesmo par diferencial.": "RS485 2-wire: one COM port can monitor requests and responses on the same differential pair.",
    "Estado": "Status",
    "Visualização": "Display",
    "Gap manual (ms)": "Manual gap (ms)",
    "Estado COM": "COM status",
    "Controlos": "Controls",
    "Iniciar Captura": "Start Capture",
    "Parar": "Stop",
    "Reiniciar COM": "Restart COM",
    "Limpar": "Clear",
    "Abrir pasta de logs": "Open logs folder",
    "Simulação": "Simulation",
    "Separar transações": "Separate transactions",
    "Realçar resultados": "Highlight results",
    "Estatísticas da sessão": "Session statistics",
    "Tráfego": "Traffic",
    "Ajuda / Ligações": "Help / Connections",
```

MBSniffer v3.2/mb_i18n.py — lines 41-80

```
    "Filtros": "Filters",
    "Todos": "All",
    "Pesquisa": "Search",
    "Limpar filtros": "Clear filters",
    "Exportar CSV": "Export CSV",
    "Hora ↑": "Time ↑",
    "Hora ↓": "Time ↓",
    "Tipo": "Type",
    "Detalhes": "Details",
    "Copiar Raw Hex": "Copy Raw Hex",
    "Copiar frame descodificado": "Copy decoded frame",
    "Filtrar por este Slave": "Filter by this Slave",
    "Filtrar por este FC": "Filter by this FC",
    "Mostrar apenas esta transação": "Show only this transaction",
    "Limpar filtro de transação": "Clear transaction filter",
    "Tempos de resposta": "Response times",
    "Qualidade": "Quality",
    "Pedidos": "Requests",
    "Pendentes": "Pending",
    "Slave (Filtro) ▾": "Slave (Filter) ▾",
    "Slave (Nenhum) ▾": "Slave (None) ▾",
    "✓ Sucesso": "✓ Success",
    "⚠ Resposta lenta": "⚠ Slow response",
    "● Parado": "● Stopped",
    "● A reiniciar COM": "● Restarting COM",
    "● A capturar": "● Capturing",
    "● A parar": "● Stopping",
    "● A simular": "● Simulating",
    "● Simulação concluída": "● Simulation complete",
    "● Erro na simulação": "● Simulation error",
    "Parado": "Stopped",
    "A iniciar pesquisa…": "Starting scan…",
    "A parar…": "Stopping…",
    "Erro ao iniciar pesquisa": "Error starting scan",
    "Respostas": "Responses",
    "Taxa de pedidos": "Request rate",
    "Slaves ativos": "Active Slaves",
    "Bytes observados": "Observed bytes",
    "Utilização do bus ~": "Bus utilization ~",
    "Média": "Average",
```

MBSniffer v3.2/mb_i18n.py — lines 81-120

```
    "Mínimo": "Minimum",
    "Máximo": "Maximum",
    "Slave mais lento": "Slowest Slave",
    "Erros de CRC": "CRC errors",
    "Bytes não validados": "Unvalidated bytes",
    "Taxa de timeout": "Timeout rate",
    "Exceções": "Exceptions",
    "Exceções / Slave": "Exceptions / Slave",
    "Seleciona uma linha no Tráfego para ver a descodificação estruturada do frame.":
        "Select a row in Traffic to view the structured frame decoding.",
    "Recolher": "Collapse",
    "Expandir": "Expand",
    "Resultado": "Result",
    "Dados": "Data",
    "Sucesso": "Success",
    "Resposta lenta": "Slow response",
    "Pendente / sem resultado": "Pending / no result",
    "Sem classificação": "Unclassified",
    "Verde": "Green",
    "Amarelo": "Yellow",
    "Vermelho": "Red",
    "Laranja": "Orange",
    "Roxo": "Purple",
    "Neutro": "Neutral",
    "Desconhecida/outra": "Unknown/other",
    "Não Utilizada": "Not Used",
    "Não selecionada": "Not selected",
    "Aberta": "Open",
    "Detetada": "Detected",
    "Não detetada": "Not detected",
    "Configuração inválida": "Invalid configuration",
    "pyserial em falta": "pyserial missing",
    "Porta COM": "COM port",
    "A porta série não aceitou bytes para transmissão.": "The serial port did not accept bytes for transmission.",
    "Erro ao abrir porta": "Error opening port",
    "Erro de captura": "Capture error",
    "Erro na simulação": "Simulation error",
    "Erro interno": "Internal error",
    "Abrir pasta": "Open folder",
    "Pesquisa ativa — transmite no barramento": "Active scan — transmits on the bus",
```

MBSniffer v3.2/mb_i18n.py — lines 121-160

```
    "Esta ferramenta atua temporariamente como master Modbus RTU e envia apenas pedidos de leitura. Não a utilizes com outro master ativo no mesmo barramento.": "This tool temporarily acts as a Modbus RTU master and sends read requests only. Do not use it while another master is active on the same bus.",
    "Confirmo que não existe outro master ativo no barramento.": "I confirm that no other master is active on the bus.",
    "FC03 é o modo rápido. Uma resposta normal OU uma Modbus Exception com CRC válido conta como slave encontrado. Ativa FC04 fallback apenas para dispositivos que possam ignorar FC03 sem devolver Exception. Device Identification envia FC43/14 apenas aos slaves encontrados e tenta ler VendorName, ProductCode e Revision. Tentativas repete cada pedido sem resposta. Antes de transmitir, a pesquisa escuta o barramento durante 1 s e não arranca se detetar tráfego.": "FC03 is the fast mode. A normal response OR a CRC-valid Modbus Exception counts as a found Slave. Enable FC04 fallback only for devices that may ignore FC03 without returning an Exception. Device Identification sends FC43/14 only to found Slaves and attempts to read VendorName, ProductCode and Revision. Attempts repeats every unanswered request. Before transmitting, the scan listens to the bus for 1 s and does not start if traffic is detected.",
    "Configuração da pesquisa": "Scan configuration",
    "Porta COM": "COM port",
    "A porta série não aceitou bytes para transmissão.": "The serial port did not accept bytes for transmission.",
    "Data bits: 8 (fixo)": "Data bits: 8 (fixed)",
    "Slave IDs / velocidade": "Slave IDs / speed",
    "De": "From",
    "até": "to",
    "Timeout mín.": "Min. timeout",
    "Iniciar pesquisa": "Start scan",
    "Limpar resultados": "Clear results",
    "Slaves encontrados": "Slaves found",
    "Encontrados: 0": "Found: 0",
    "Resultado": "Result",
    "Pesquisa ativa": "Active scan",
    "Erro ao iniciar pesquisa": "Error starting scan",
    "Erro na Bus Slave Finder": "Bus Slave Finder error",
    "Erro na pesquisa": "Scan error",
    "Nenhum": "None",
    "Nenhum slave detetado": "No Slave detected",
    "Crescente — antigo → recente": "Ascending — oldest → newest",
    "Decrescente — recente → antigo": "Descending — newest → oldest",
    "Todos os ficheiros": "All files",
    "Exportar sessão para CSV": "Export session to CSV",
    # Messages with live values (templates used with SnifferApp.trf()).
    'Seleciona a COM A.': 'Select COM A.',
    'Seleciona também a COM B.': 'Select COM B as well.',
    'COM A e COM B têm de ser diferentes.': 'COM A and COM B must be different.',
    'pyserial não está instalado.': 'pyserial is not installed.',
    'Não foi possível reiniciar a porta COM:\n\n{error}': 'Could not restart the COM port:\n\n{error}',
    'Não foi possível reabrir a porta COM. A captura foi parada.\n\n{error}': 'Could not reopen the COM port. Capture was stopped.\n\n{error}',
    'Pending timeout inválido.': 'Invalid Pending timeout.',
    'Pending timeout deve estar entre {minimum:g} e {maximum:g} segundos.': 'Pending timeout must be between {minimum:g} and {maximum:g} seconds.',
    'Frame gap manual inválido.': 'Invalid manual Frame gap.',
    'pyserial não está instalado.\n\nO build_exe.bat inclui-o automaticamente no EXE.': 'pyserial is not installed.\n\nbuild_exe.bat includes it automatically in the EXE.',
    'Ocorreu um erro ao processar eventos da interface. O processamento continuará.\n\n{error}': 'An error occurred while processing interface events. Processing will continue.\n\n{error}',
    'Slave {slave} — média {average:.1f} ms': 'Slave {slave} — average {average:.1f} ms',
    'Não existem frames na sessão atual para exportar.': 'There are no frames in the current session to export.',
```

MBSniffer v3.2/mb_i18n.py — lines 161-200

```
    'Sessão exportada para:\n\n{path}': 'Session exported to:\n\n{path}',
    'Não foi possível guardar o CSV:\n\n{error}': 'Could not save the CSV:\n\n{error}',
    'Erro ao criar log': 'Error creating log',
    'Foi detetado tráfego, mas não foi possível criar o ficheiro de log em:\n\n{folder}\n\n{error}': 'Traffic was detected, but the log file could not be created at:\n\n{folder}\n\n{error}',
    'Os Slave IDs têm de estar entre 1 e 247.': 'Slave IDs must be between 1 and 247.',
    'O Slave ID inicial não pode ser superior ao final.': 'The starting Slave ID cannot be greater than the ending Slave ID.',
    'O timeout mínimo tem de estar entre {minimum} e {maximum} ms.': 'The minimum timeout must be between {minimum} and {maximum} ms.',
    'Tentativas tem de estar entre {minimum} e {maximum}.': 'Attempts must be between {minimum} and {maximum}.',
    'Tentativas': 'Attempts',
    'Máx. estimado: —': 'Estimated max.: —',
    ' (com {options})': ' (with {options})',
    '{configs} config. × {slaves} slaves — máx. estimado: {duration}': '{configs} config. × {slaves} slaves — estimated max.: {duration}',
    'Encontrados: {count}': 'Found: {count}',
    'pyserial não está instalado. O build_exe.bat inclui-o automaticamente no EXE.': 'pyserial is not installed. build_exe.bat includes it automatically in the EXE.',
    'Confirma primeiro que não existe outro master ativo no barramento.': 'First confirm that no other master is active on the bus.',
    'Seleciona uma porta COM.': 'Select a COM port.',
    'Seleciona pelo menos uma configuração série.': 'Select at least one serial configuration.',
    'Não foi possível iniciar a thread de pesquisa:\n\n{error}': 'Could not start the scan thread:\n\n{error}',
    'A configurar {baud} / {config}…': 'Configuring {baud} / {config}…',
    'Foi detetado tráfego em {port} antes de transmitir ({count} bytes: {raw}). Pode existir outro master ativo no barramento; a pesquisa não foi iniciada.': 'Traffic was detected on {port} before transmitting ({count} bytes: {raw}). Another master may be active on the bus; the scan was not started.',
    'A verificar se o barramento está livre…': 'Checking that the bus is idle…',
    'Não foi possível abrir {port} em {baud} / {config}: {error}': 'Could not open {port} at {baud} / {config}: {error}',
    'Falha ao preparar a receção em {context}: {error}': 'Failed to prepare reception at {context}: {error}',
    'Falha ao transmitir em {context}: {error}': 'Failed to transmit at {context}: {error}',
    'Falha ao receber em {context}: {error}': 'Failed to receive at {context}: {error}',
    'Falha em Device Identification em {context}: {error}': 'Device Identification failed at {context}: {error}',
    'Decorrido {elapsed} — restante ~{remaining}': 'Elapsed {elapsed} — remaining ~{remaining}',
    'Pesquisa parada — {found} encontrados — {elapsed}': 'Scan stopped — {found} found — {elapsed}',
    'Pesquisa concluída — {found} encontrados — {elapsed}': 'Scan complete — {found} found — {elapsed}',
    # Master (poll) tab — v3.2.
    'Ligação — o Master transmite no barramento': 'Connection — the Master transmits on the bus',
    'Timeout resposta (ms)': 'Response timeout (ms)',
    'Atraso entre pedidos (ms)': 'Delay between polls (ms)',
    'Ligar': 'Connect',
    'Desligar': 'Disconnect',
    'Desligado': 'Disconnected',
    '● Ligado': '● Connected',
    'A desligar…': 'Disconnecting…',
    'Nova janela': 'New window',
    'Abrir workspace': 'Open workspace',
```

MBSniffer v3.2/mb_i18n.py — lines 201-240

```
    'Guardar workspace': 'Save workspace',
    'Uma janela por Slave / zona de memória. Duplo clique num valor para escrever; duplo clique num nome para o editar.': 'One window per Slave / data area. Double-click a value to write it; double-click a name to edit it.',
    'Sem janelas. Usa "Nova janela" para começar a ler um Slave.': 'No windows. Use "New window" to start reading a Slave.',
    'Endereço': 'Address',
    'Qtd.': 'Qty',
    'Formato': 'Format',
    'Aplicar': 'Apply',
    'Pausar': 'Pause',
    'Retomar': 'Resume',
    'Pausada': 'Paused',
    'Ler 1x': 'Read once',
    'Duplicar': 'Duplicate',
    'Destacar': 'Pop out',
    'Ancorar': 'Dock',
    'Fechar': 'Close',
    'Nome': 'Name',
    'Valor': 'Value',
    'Janela {number} — Slave {slave} · FC{fc:02d} · {address} · {qty}': 'Window {number} — Slave {slave} · FC{fc:02d} · {address} · {qty}',
    'Máximo de {count} janelas.': 'Maximum of {count} windows.',
    'Janela inválida': 'Invalid window',
    'Janela inválida.': 'Invalid window.',
    'Liga primeiro o Master.': 'Connect the Master first.',
    'Escrita': 'Write',
    'FC02 e FC04 são só de leitura: abre uma janela FC01 ou FC03 para escrever.': 'FC02 and FC04 are read-only: open an FC01 or FC03 window to write.',
    'Escrever': 'Write value',
    'Slave (0 = broadcast)': 'Slave (0 = broadcast)',
    'Valor(es)': 'Value(s)',
    'Formato: {fmt}. Vários valores (FC15/FC16): separar com ;': 'Format: {fmt}. Several values (FC15/FC16): separate with ;',
    'Enviar': 'Send',
    'Cancelar': 'Cancel',
    'Nome do endereço {address}': 'Name of address {address}',
    'Guardar': 'Save',
    'Para a captura ou a pesquisa em curso antes de ligar o Master.': 'Stop the running capture or scan before connecting the Master.',
    'Não foi possível abrir {port}: {error}': 'Could not open {port}: {error}',
    'Erro na porta série: {error}': 'Serial port error: {error}',
    'Escrita OK: {what}': 'Write OK: {what}',
    'Escrita broadcast enviada: {what}': 'Broadcast write sent: {what}',
    'Resposta inválida (CRC ou formato): {rx}': 'Invalid response (CRC or format): {rx}',
    'O ficheiro não é um workspace do MBSniffer.': 'The file is not an MBSniffer workspace.',
    '{count} janela(s) inválida(s) foram ignoradas.': '{count} invalid window(s) were skipped.',
```

MBSniffer v3.2/mb_i18n.py — lines 241-280

```
    'O timeout de resposta tem de estar entre 10 e 10000 ms.': 'The response timeout must be between 10 and 10000 ms.',
    'O atraso entre pedidos tem de estar entre 0 e 10000 ms.': 'The delay between polls must be between 0 and 10000 ms.',
    'Scan rate fora do intervalo.': 'Scan rate out of range.',
    'Valor vazio.': 'Empty value.',
    'Valor fora do formato selecionado.': 'Value does not fit the selected format.',
    'Uma coil só aceita 0 ou 1.': 'A coil only accepts 0 or 1.',
    'O Slave ID de leitura tem de estar entre 1 e 247.': 'The read Slave ID must be between 1 and 247.',
    'Função de leitura não suportada.': 'Unsupported read function.',
    'O endereço tem de estar entre 0 e 65535.': 'The address must be between 0 and 65535.',
    'Quantidade fora do limite desta função.': 'Quantity outside the limit of this function.',
    'O intervalo de endereços ultrapassa 65535.': 'The address range goes beyond 65535.',
    'O Slave ID de escrita tem de estar entre 0 (broadcast) e 247.': 'The write Slave ID must be between 0 (broadcast) and 247.',
    'FC05 escreve exatamente uma coil.': 'FC05 writes exactly one coil.',
    'FC06 escreve exatamente um registo.': 'FC06 writes exactly one register.',
    'Função de escrita não suportada.': 'Unsupported write function.',
    'FC05/FC06 escrevem um único valor; usa FC15/FC16 para vários.': 'FC05/FC06 write a single value; use FC15/FC16 for several.',
}

EN_TO_PT = {english: portuguese for portuguese, english in PT_TO_EN.items()}
# "None" is also a canonical technical serial-parity value in the Portuguese UI.
# Do not reverse-translate it to "Nenhum" when restoring Portuguese; menu entries
# that need "Nenhum" are rendered from their Portuguese canonical source text.
EN_TO_PT.pop("None", None)

EXCEPTION_DESCRIPTION_EN = {
    0x01: "The requested Modbus function is not supported or permitted by the device.",
    0x02: "The requested address or address range is not valid for this device.",
    0x03: "A value, quantity or request field is invalid for this function.",
    0x04: "The device encountered an internal error while attempting to execute the request.",
    0x05: "The request was accepted but requires more time to be processed.",
    0x06: "The device is busy and cannot process the request at this time.",
    0x08: "A parity error was detected while accessing the device memory.",
    0x0A: "The gateway could not establish a path to the target device.",
    0x0B: "The gateway did not receive a response from the target device.",
}


def normalize_language(value):
    return value if value in LANGUAGE_CHOICES else LANGUAGE_PORTUGUESE

```

MBSniffer v3.2/mb_i18n.py — lines 281-320

```
281
282 def canonical_ui_text(text):
283     """Return the canonical PT-PT form of a known static UI string."""
284     value = str(text)
285     return EN_TO_PT.get(value, value)
286
287
288 def translate_text(text, language):
289     """Translate a known static string; leave technical/unknown text intact."""
290     canonical = canonical_ui_text(text)
291     if normalize_language(language) == LANGUAGE_ENGLISH:
292         return PT_TO_EN.get(canonical, canonical)
293     return canonical
294
295
296 # Human-readable fragments embedded in protocol Details / Inspector data.
297 DETAIL_FRAGMENTS_PT_TO_EN = (
298     ("Registos (PDU Address=valor):", "Registers (PDU Address=value):"),
299     ("Bits (PDU Address=valor):", "Bits (PDU Address=value):"),
300     ("Registos=", "Registers="),
301 )
302
303
304 def translate_details(text, language):
305     """Translate only human-readable fragments embedded in protocol details."""
306     value = str(text or "")
307     if normalize_language(language) == LANGUAGE_ENGLISH:
308         for portuguese, english in DETAIL_FRAGMENTS_PT_TO_EN:
309             value = value.replace(portuguese, english)
310     else:
311         for portuguese, english in DETAIL_FRAGMENTS_PT_TO_EN:
312             value = value.replace(english, portuguese)
313     return value
314
315
316 def exception_description(code, portuguese_text, language):
317     if normalize_language(language) == LANGUAGE_ENGLISH:
318         return EXCEPTION_DESCRIPTION_EN.get(code, portuguese_text)
319     return portuguese_text
320
```

MBSniffer v3.2/mb_i18n.py — lines 321-360

```
321
322 # English Help content. Tags match the tags configured by mb_gui.py.
323 def english_help_blocks(max_ui_frames_text, max_raw_lines_text,
324                         min_pending, max_pending, debug_sim=False):
325     blocks = [
326         ("title", "MBSniffer — Help and connections\n"),
327         (None, "MBSniffer is designed for Modbus RTU communications diagnostics. The Sniffer tab is passive: it opens the COM port(s) for reception and does not transmit request
    s. Bus Slave Finder is an active scan tool and must only be used when no other master is active on the same bus.\n"),
328         ("heading", "Quick start — Sniffer\n"),
329         ("bullet", "1. Select the correct physical mode: RS485 2-wire, RS232 single RX or RS232 dual RX.\n"),
330         ("bullet", "2. Select the COM port(s) and serial parameters used by the communication being observed.\n"),
331         ("bullet", "3. Keep Frame gap set to Auto unless you have a specific reason to change it.\n"),
332         ("bullet", "4. Click “Start Capture” and confirm activity through BUS ● RX.\n"),
333         ("bullet", "5. Use Traffic for quick diagnostics and Raw Hex / Log to confirm the actual bytes.\n"),
334         ("bullet", "6. Select a frame to inspect its detailed analysis in Frame Inspector.\n"),
335
336         ("heading", "1. RS485 2-wire\n"),
337         (None, "In this mode, one COM port observes requests and responses on the same differential pair. The sniffer adapter must be connected in parallel and must not act as a
     master.\n"),
338         ("subheading", "Typical connection:\n"),
339         ("mono", "        Device A              Device B\n            D+ ---------------- D+\n            D- ---------------- D-\n              \\              /\n
       \\             /\n                 D+          D-\n                USB-RS485 sniffer\n\n"),
340         ("bullet", "• Connect sniffer D+ in parallel with bus D+.\n"),
341         ("bullet", "• Connect sniffer D− in parallel with bus D−.\n"),
342         ("bullet", "• If a C/COM/0 V reference is available, connect it only when you know it is the correct RS485 interface reference.\n"),
343         ("warning", "• Do not add a third 120 Ω termination resistor if the bus is already terminated at both ends.\n"),
344         ("bullet", "• If the USB-RS485 adapter has selectable termination or bias, leave them disabled during passive capture unless they are intentionally required by the insta
    llation.\n"),
345         ("bullet", "• A/B naming is not universal: different manufacturers may use opposite polarities. Prefer D+/D− and the equipment documentation.\n"),
346
347         ("heading", "2. RS232 single RX\n"),
348         (None, "Observes only one RS232 direction. RS232 is full-duplex: each direction uses an independent TX line, so one RX input cannot observe both directions simultaneousl
    y.\n"),
349         ("mono", "        Device A                      Device B\n            TX ----------------------------> RX\n             \\\n              \\----> USB-RS232 sniffer R
    X (COM A)\n\n             GND --------------------------- GND\n              \\--------------------------> sniffer GND\n\n        Sniffer TX: DO NOT CONNECT\n\n"),
350         ("bullet", "• Connect sniffer RX to the TX line you want to observe.\n"),
351         ("bullet", "• Connect the common GND as well.\n"),
352         ("warning", "• Leave the TX pin of the adapter used as a sniffer physically disconnected.\n"),
353
354         ("heading", "3. RS232 dual RX\n"),
355         (None, "Allows both RS232 directions to be observed at the same time. Two independent RX channels are required, normally two USB-RS232 adapters.\n"),
356         ("mono", "        Device A                      Device B\n            TX ----------------------------> RX\n             \\----> sniffer RX COM A   (A → B)\n\n
        RX <---------------------------- TX\n                                   \\----> sniffer RX COM B   (B → A)\n\n             GND ---------------------------- GND\n
            \\---------------------------- COM A GND\n                \\---------------------------- COM B GND\n\n        TX on both sniffers: DO NOT CONNECT\n\n"),
357         ("bullet", "• COM A is shown as A→B and COM B as B→A.\n"),
358         ("bullet", "• Both COM ports must use the same serial parameters as the observed link.\n"),
359
360         ("heading", "RS232 is not TTL\n"),
```

MBSniffer v3.2/mb_i18n.py — lines 361-400

```
361         ("warning", "Do not connect RS232 signals directly to 3.3 V or 5 V USB-TTL/UART interfaces. Use a USB-RS232 adapter with a transceiver suitable for RS232 voltage levels.\n"),
362 
363         ("heading", "Serial communication parameters\n"),
364         (None, "Baud rate, Data bits, Parity and Stop bits must match the actual communication. Incorrect parameters may produce apparently random bytes, frames with invalid CRC, or no recognizable frames at all.\n"),
365         ("bullet", "• Baud rate — speed in bit/s, for example 9600 or 19200.\n"),
366         ("bullet", "• Data bits — normally 8 for Modbus RTU.\n"),
367         ("bullet", "• Parity — None, Even, Odd, Mark or Space, according to the installation.\n"),
368         ("bullet", "• Stop bits — 1, 1.5 or 2, according to the observed configuration.\n"),
369 
370         ("heading", "Frame gap\n"),
371         (None, "In Auto mode, MBSniffer uses 3.5 character times up to and including 19200 bit/s. Above 19200 bit/s it uses the fixed 1.750 ms t3.5 recommended for Modbus RTU. Manual mode allows another millisecond value to be used when diagnosing equipment or drivers with particular behavior.\n"),
372         (None, "USB adapters and Windows may group bytes and introduce latency. MBSniffer timestamps are therefore useful for sequence diagnostics and relative timing, but they do not replace a logic analyzer when high-precision timing measurements are required.\n"),
373         (None, f"When a burst ends with the start of a known frame that is still incomplete, MBSniffer waits up to {SNIFFER_SPLIT_FRAME_HOLD_MS:g} ms more before closing it. A frame split in two by the USB adapter therefore does not appear as RAW. Frames merged into one burst keep the arrival time of their own block of bytes.\n"),
374 
375         ("heading", "Pending timeout\n"),
376         (None, "Defines how long a CRC-valid REQUEST may remain waiting for a matchable RESPONSE/EXCEPTION. When it expires, the original REQUEST is marked as timeout and the Timeout/Timeout rate counters increase.\n"),
377         (None, f"Allowed range: {min_pending:g} to {max_pending:g} s. The value should be greater than the normal maximum response time of the installation to avoid false timeouts.\n"),
378         (None, "Broadcast requests (Slave 0) never have a response by definition: they are not kept pending, never count as a timeout and are excluded from the Timeout rate.\n"),
379 
380         ("heading", "How to read the Traffic table\n"),
381         ("bullet", "• REQUEST — Modbus RTU request recognized by the parser.\n"),
382         ("bullet", "• RESPONSE — recognized Modbus RTU response that is not an EXCEPTION.\n"),
383         ("bullet", "• FC05/FC06/FC08/FC22 — the request and the response have the same format. The frame is a RESPONSE when a pending request from the same Slave/FC exists (for FC05/06/22 the bytes must be identical; in RS232 dual RX it must come from the opposite channel). Otherwise it is a REQUEST.\n"),
384         ("bullet", "• FC01/FC02 responses show the bits in address order, grouped per byte. In Frame Inspector, matched read responses (FC01–FC04) show the PDU Address of every value.\n"),
385         ("bullet", "• EXCEPTION — Modbus response with an exception Function Code (bit 7 set). Details shows the code and normalized description.\n"),
386         ("bullet", "• CRC OK — received CRC matches the calculated CRC.\n"),
387         ("bullet", "• CRC ERROR — the structure resembles a supported frame, but the received CRC does not match the calculated CRC.\n"),
388         ("bullet", "• RAW/UNSYNC — bytes discarded during resynchronization before a structurally recognized, CRC-valid frame is found later in the same burst.\n"),
389         ("bullet", "• RAW/UNPARSED — remaining bytes that could not be interpreted or resynchronized as a valid Modbus frame. This is the state normally shown for a noise block with no later valid frame in the same burst.\n"),
390         (None, "RAW/UNSYNC and RAW/UNPARSED are different conditions and both may appear in the Type column. Neither represents a valid Modbus response.\n"),
391 
392         ("heading", "Traffic table columns\n"),
393         ("bullet", "• Time — timestamp for the start of the processed burst/frame.\n"),
394         ("bullet", "• Δt — interval since the previous observed frame.\n"),
395         ("bullet", "• Resp. — time between a REQUEST and its matched RESPONSE/EXCEPTION. Blank when no valid pair exists.\n"),
396         ("bullet", "• COM — capture channel: BUS in RS485, A→B/B→A in RS232 dual RX and SIM in development mode. It does not necessarily represent the Windows COM number.\n"),
397         ("bullet", "• Type — REQUEST, RESPONSE, EXCEPTION or a RAW state.\n"),
398         ("bullet", "• Slave — Modbus Unit/Slave ID when identifiable.\n"),
399         ("bullet", "• FC — hexadecimal Function Code observed in the frame.\n"),
400         ("bullet", "• Details — address, quantity, data or operation description.\n"),
```

MBSniffer v3.2/mb_i18n.py — lines 401-440

```
401         ("bullet", "• CRC — OK, ERROR or ? when a complete Modbus frame is not available.\n"),
402 
403         ("heading", "REQUEST/RESPONSE pairing\n"),
404         (None, "Modbus RTU has no Transaction ID. MBSniffer pairs frames by (Slave ID, Function Code) in FIFO order. This is appropriate for normal RTU traffic, but can become ambiguous when requests using the same Slave and FC overlap before their corresponding responses arrive.\n"),
405         (None, "An EXCEPTION uses its base Function Code for pairing. For example, response 0x83 is associated with a pending FC03 REQUEST when a compatible request exists.\n"),
406 
407         ("heading", "PDU Address, 1-based and Qty\n"),
408         (None, "PDU Address is the address encoded in the Modbus request. 1-based is simply PDU Address + 1 and helps compare with manuals that number the first register/coil as 1. Qty is the requested quantity.\n"),
409         (None, "These fields belong to the REQUEST. A RESPONSE does not automatically repeat Address or Qty in the protocol, so MBSniffer does not invent those fields on the response row.\n"),
410 
411         ("heading", "Modbus exception codes\n"),
412         ("bullet", "• 0x01 — Illegal Function: function not supported or not permitted.\n"),
413         ("bullet", "• 0x02 — Illegal Data Address: invalid address/address range.\n"),
414         ("bullet", "• 0x03 — Illegal Data Value: invalid value, quantity or field.\n"),
415         ("bullet", "• 0x04 — Server Device Failure: device-side execution failure.\n"),
416         ("bullet", "• 0x05 — Acknowledge.\n"),
417         ("bullet", "• 0x06 — Server Device Busy.\n"),
418         ("bullet", "• 0x08 — Memory Parity Error.\n"),
419         ("bullet", "• 0x0A — Gateway Path Unavailable.\n"),
420         ("bullet", "• 0x0B — Gateway Target Device Failed to Respond.\n"),
421 
422         ("heading", "Advanced filters\n"),
423         (None, "The bar at the top of Traffic can combine several filters without changing the original session data:\n"),
424         ("bullet", "• Type — All, Requests, Responses, Exceptions, CRC errors, Timeouts or RAW.\n"),
425         ("bullet", "• FC (hex) — Function Code, with or without the 0x prefix.\n"),
426         ("bullet", "• Resp. > — only completed responses above the specified time in ms.\n"),
427         ("bullet", "• Search — searches Details, Raw Hex, Slave, FC, Type and COM/channel.\n"),
428         (None, "The Slave filter remains available in the column header and combines with the advanced filters. “Clear filters” resets all display filters. Filters do not change statistics or the capture log.\n"),
429 
430         ("heading", "Highlight results\n"),
431         (None, "This option is disabled by default to keep the table neutral. When enabled, results are differentiated by color: a completed normal transaction with a CRC-valid RESPONSE is green; CRC error, timeout, Exception, RAW and slow response retain their respective diagnostic colors.\n"),
432         (None, "In a successfully completed normal transaction, both the REQUEST and its RESPONSE are green. A still-pending REQUEST remains neutral. A slow response keeps the slow-response color even if valid so that the timing result is not hidden.\n"),
433 
434         ("heading", "Frame Inspector\n"),
435         (None, "Select a Traffic row to analyze the frame without manually interpreting every byte. Frame Inspector shows Slave, Type, Function, response time, Result, PDU Address, 1-based address, Qty, ByteCount, CRC, decoded Data and Raw Hex.\n"),
436         (None, "The CRC field shows both received and calculated values for a complete frame. Result uses the same diagnostic classification as Traffic, independently of whether Highlight results is enabled.\n"),
437 
438         ("heading", "Traffic context menu\n"),
439         (None, "Right-click a Traffic row to:\n"),
440         ("bullet", "• Copy Raw Hex.\n"),
```

MBSniffer v3.2/mb_i18n.py — lines 441-480

```
441         ("bullet", "• Copy the decoded frame.\n"),
442         ("bullet", "• Filter by that frame's Slave.\n"),
443         ("bullet", "• Filter by that frame's Function Code.\n"),
444         ("bullet", "• Show only the matching REQUEST/RESPONSE transaction.\n"),
445         ("bullet", "• Clear the temporary transaction filter.\n"),
446 
447         ("heading", "Bus Health\n"),
448         (None, "Bus Health summarizes session quality and performance without replacing the frame table.\n"),
449         ("bullet", "• Requests / Responses — session counters.\n"),
450         ("bullet", "• Request rate — average number of observed requests per second.\n"),
451         ("bullet", "• Active Slaves — valid Modbus IDs observed in the session.\n"),
452         ("bullet", "• Observed bytes — sum of bytes delivered to the parser.\n"),
453         ("bullet", "• Bus utilization ~ — estimate based on observed bytes, baud rate, data bits, parity and stop bits. It is approximate and depends on the capture point.\n"),
454         ("bullet", "• Average / Minimum / Maximum — paired response times.\n"),
455         ("bullet", "• P95 — 95% of measured responses were at or below this value.\n"),
456         ("bullet", "• Slowest Slave — Slave with the highest average response time among observed samples.\n"),
457         ("bullet", "• Unvalidated bytes — percentage of observed bytes belonging to invalid-CRC frames or RAW blocks. By itself this does not prove electrical noise; incorrect s
    erial parameters, incomplete capture or parser-unsupported functions can also produce RAW.\n"),
458         ("bullet", "• Timeout rate — percentage of REQUESTs that expired without a matched response.\n"),
459         ("bullet", "• Exceptions / Slave — distribution of Exception responses by Slave ID.\n"),
460 
461         ("heading", "BUS RX indicator\n"),
462         (None, "BUS ● RX pulses briefly when a batch of frames reaches the interface. At rest, BUS ○ is displayed. It is only a visual confirmation of received activity; it does
     not indicate electrical quality or guarantee that frames are valid.\n"),
463 
464         ("heading", "Raw Hex / Log\n"),
465         (None, "Shows captured bytes in hexadecimal with context information. Long lines wrap to the available width. The view follows the same display filters used in Traffic;
    the on-disk TXT file, when present, keeps the complete capture and is not truncated by filters.\n"),
466 
467         ("heading", "Export CSV\n"),
468         (None, "Export CSV saves the frames currently retained in the UI history using Excel-compatible UTF-8. It includes Time, Δt, Response time, Channel, Type, Slave, FC, Det
    ails, CRC, Raw, PDU Address, 1-based, Qty, ByteCount, Timeout and anomaly class.\n"),
469         (None, f"The interface retains approximately the latest {max_ui_frames_text} frames. In longer sessions, the CSV therefore represents the history still retained by the G
    UI, while the TXT file from a real capture remains the complete on-disk record.\n"),
470 
471         ("heading", "Capture logs\n"),
472         (None, "The TXT log is created only after real traffic arrives. Starting and stopping without receiving frames does not create an empty file.\n"),
473         (None, "Logs folder:\n"),
474         ("mono", "        <application folder>\\MBSniffer Logs\n"),
475         (None, "Filters, sorting, Highlight results and Frame Inspector are display-only features and do not change frames written to the log.\n"),
476 
477         ("heading", "Visible history limits\n"),
478         (None, f"To keep the application responsive, the interface retains approximately {max_ui_frames_text} frames and {max_raw_lines_text} Raw Hex lines. Pruning is performed
     in blocks. The capture TXT log, when active, is independent of this limit.\n"),
479 
480         ("heading", "COM status and ports\n"),
```

MBSniffer v3.2/mb_i18n.py — lines 481-520

```
        (None, "COM status distinguishes a port opened by MBSniffer, a port merely detected by Windows and a port that is no longer detected. Status is refreshed periodically.\n"),
        (None, "Restart COM closes and reopens the port(s) used by Sniffer. This restarts the serial-port handle inside MBSniffer; it does not restart the Windows USB/COM driver.\n"),
        (None, "The ↻ buttons beside the COM selectors refresh the available port enumeration in Sniffer and Bus Slave Finder.\n"),

        ("heading", "Bus Slave Finder — active scan\n"),
        (None, "Bus Slave Finder transmits Modbus RTU requests, so it is not passive. Before starting, confirm that no other master is active on the same bus.\n"),
        (None, "Select COM, baud rates, parities, stop bits and the Slave ID range. Data bits is fixed at 8. The scan tries FC03 Address 0 Qty 1; either a normal response or a CRC-valid Modbus Exception confirms that the Slave exists.\n"),
        (None, "FC04 fallback is optional and increases the maximum number of attempts. The minimum timeout is adapted to baud rate. A write timeout means transmission through the port/driver failed and must not be interpreted simply as “slave did not respond”.\n"),
        (None, f"Attempts (1 to 3) repeats every unanswered request before moving on to the next Slave. Before the first transmission, the scan listens to the bus for {DISCOVERY_IDLE_CHECK_SECONDS:g} s; if any byte is received, it does not transmit and reports that another master may be active.\n"),
        (None, "Device Identification (FC43/14) is optional. When enabled, it is sent only after a Slave has been found and attempts to read Basic Device Identification: VendorName, ProductCode and MajorMinorRevision. An Exception or no response to FC43/14 does not invalidate Slave discovery.\n"),
        (None, "Finder and Sniffer share the global activity lock so the same application cannot actively scan and capture at the same time.\n"),

        ("heading", "Master — read and write as a Modbus RTU master\n"),
        (None, f"The Master tab transmits requests: use it only when no other master is active on the bus. When connecting, the application listens to the bus for {DISCOVERY_IDLE_CHECK_SECONDS:g} s and does not transmit if traffic is detected. While the Master is connected, Sniffer and Bus Slave Finder are locked.\n"),
        (None, "Each window reads one data area of a Slave: Slave ID, function (FC01 Coils, FC02 Discrete Inputs, FC03 Holding Registers or FC04 Input Registers), address, quantity and scan rate. Base 0 shows the PDU Address; Base 1 numbers from 1, as in many manuals. Changes take effect only after “Apply”.\n"),
        (None, "All windows share the same COM port: Modbus RTU allows only one request at a time, so windows are served in turn according to their scan rate. Writes and “Read once” go ahead of cyclic reads. The “Delay between polls” (at least t3.5) is always left between two requests.\n"),
        (None, "Formats: signed and unsigned 16-bit, hex and binary; signed and unsigned 32-bit and float, in byte orders ABCD (big-endian), CDAB (word swap), BADC (byte swap) and DCBA (little-endian). 32-bit values use two registers and appear on the row of the first one.\n"),
        (None, "Writing: double-click a value in an FC01 or FC03 window. Coils with FC05 or FC15, registers with FC06 or FC16; several values separated by “;” are written to consecutive addresses with FC15/FC16. Slave 0 sends a broadcast, with no response. After an accepted write, the window is read again.\n"),
        (None, "Each window shows Tx, errors, timeouts, the last response time and the last error (Timeout, Exception or invalid response). “Pop out” opens the window outside the tab; “Dock” or closing that window returns it to the tab. “Save workspace” stores the connection, every window, formats and names in an .mbsw file.\n"),

        ("heading", "Saved preferences\n"),
        (None, "User preferences are stored in %APPDATA%\\MBSniffer\\settings.json. They include Light/Dark, Language, last tab, serial parameters, Frame gap, Pending timeout, Auto-scroll, Separate transactions, Highlight results, Bus Slave Finder options and Master connection parameters. Master windows are saved in a workspace (.mbsw).\n"),
        (None, "COM ports are not saved because they are specific to each computer. The Bus Slave Finder safety confirmation is not saved either.\n"),

        ("heading", "Diagnostic best practices\n"),
        ("bullet", "• Confirm wiring, polarity and serial parameters first.\n"),
        ("bullet", "• On RS485, keep the sniffer stub short and avoid changing the existing termination.\n"),
        ("bullet", "• Make sure no other program has the same COM port open.\n"),
        ("bullet", "• Treat CRC ERROR and RAW as indicators of framing, noise, incorrect parameters or incomplete capture.\n"),
        ("bullet", "• Treat Exceptions as valid protocol responses: they are not equivalent to CRC errors.\n"),
        ("bullet", "• Compare response times with normal equipment behavior, not only with a generic threshold.\n"),
        ("bullet", "• Keep the TXT log when you need later analysis of the complete capture.\n"),
    ]
    if debug_sim:
        blocks.extend([
            ("heading", "Simulation mode\n"),
            (None, "This development mode appears only when debug_sim = 1 in MBSniffer.py. It does not open serial ports and never creates TXT logs.\n"),
            (None, "Each run uses the same diagnostic scenario types: one normal response, one slow response, one Modbus Exception, one CRC error, one timeout and one RAW/UNPARSED block. This makes it possible to compare the same information directly with and without visual highlighting.\n"),
            (None, "Highlight results does not change Simulation data. When disabled, all rows use normal table colors; when enabled, the completed normal transaction is green and Exception, CRC error, timeout, slow response and RAW rows use their respective diagnostic colors.\n"),
            (None, "Frames in this scenario use the SIM channel so they are not confused with real physical capture.\n"),
```

MBSniffer v3.2/mb_i18n.py — lines 521-555

```
        ])
    blocks.extend([
        ("heading", "Important limitation\n"),
        (None, "MBSniffer interprets bytes, CRC, Function Code, addresses, quantities, pairing and timing observed by the operating system. It does not directly measure electric
    al levels, ringing, reflections, bias, common-mode, rise/fall times or analog signal integrity. Those problems still require an oscilloscope or an analyzer suitable for the phys
    ical interface.\n"),
    ])
    return blocks


def translate_runtime_text(text, language):
    """Translate status strings that contain live values."""
    import re

    value = str(text or "")
    target_en = normalize_language(language) == LANGUAGE_ENGLISH

    # Normalize either language to canonical PT where practical, then render.
    patterns = [
        (r"^Encontrados: (\d+)$", r"Found: \1", target_en),
        (r"^Found: (\d+)$", r"Encontrados: \1", not target_en),
        (r"^Decorrido (.+) — restante ~(.+)$", r"Elapsed \1 — remaining ~\2", target_en),
        (r"^Elapsed (.+) — remaining ~(.+)$", r"Decorrido \1 — restante ~\2", not target_en),
        (r"^A configurar (.+)…$", r"Configuring \1…", target_en),
        (r"^Configuring (.+)…$", r"A configurar \1…", not target_en),
        (r"^Pesquisa parada — (\d+) encontrados — (.+)$", r"Scan stopped — \1 found — \2", target_en),
        (r"^Scan stopped — (\d+) found — (.+)$", r"Pesquisa parada — \1 encontrados — \2", not target_en),
        (r"^Pesquisa concluída — (\d+) encontrados — (.+)$", r"Scan complete — \1 found — \2", target_en),
        (r"^Scan complete — (\d+) found — (.+)$", r"Pesquisa concluída — \1 encontrados — \2", not target_en),
        (r"^(\d+) config\. × (\d+) slaves — máx\. estimado: (.+)$", r"\1 config. × \2 slaves — estimated max.: \3", target_en),
        (r"^(\d+) config\. × (\d+) slaves — estimated max\.: (.+)$", r"\1 config. × \2 slaves — máx. estimado: \3", not target_en),
    ]
    for pattern, replacement, enabled in patterns:
        if enabled and re.match(pattern, value):
            return re.sub(pattern, replacement, value)

    return translate_text(value, language)
```

MBSniffer v3.2/mb_theme.py

692 source lines.

MBSniffer v3.2/mb_theme.py — lines 1-40

```
#!/usr/bin/env python3
"""MBSniffer fixed-geometry Light/Dark theme.

The application uses one ttk construction for its entire lifetime. Light/Dark
switching changes colours only; it never changes ttk theme engine, layouts,
padding, borders, margins, widget metrics or row heights.
"""

import tkinter as tk
from tkinter import ttk


BOOTSTRAP_ACTIVE = False
THEME_ENGINE = "clam"

LIGHT = {
    "root": "#f3f4f6",
    "panel": "#f3f4f6",
    "panel_alt": "#ffffff",
    "input": "#ffffff",
    "input_readonly": "#f5f6f7",
    "text": "#1f2328",
    "muted": "#59636e",
    "disabled": "#8b949e",
    "border": "#c3c9d0",
    "divider": "#d6dbe0",
    "button": "#ffffff",
    "button_hover": "#f0f5fa",
    "button_pressed": "#e3edf6",
    "accent": "#2f80d1",
    "accent_hover": "#3b8edb",
    "accent_pressed": "#246eB8",
    "select": "#2f80d1",
    "select_text": "#ffffff",
    "scroll_trough": "#e8ebee",
    "scroll_thumb": "#b9c2ca",
    "scroll_thumb_hover": "#929faa",
}

DARK = {
```

MBSniffer v3.2/mb_theme.py — lines 41-80

```
    "root": "#171a1f",
    "panel": "#171a1f",
    "panel_alt": "#1f242a",
    "input": "#20262c",
    "input_readonly": "#252c33",
    "text": "#e8eaed",
    "muted": "#aab2ba",
    "disabled": "#707982",
    "border": "#46525d",
    "divider": "#46525d",
    "button": "#252c33",
    "button_hover": "#303942",
    "button_pressed": "#1f252b",
    "accent": "#2f8fd8",
    "accent_hover": "#43a2e8",
    "accent_pressed": "#287bb8",
    "select": "#285f80",
    "select_text": "#ffffff",
    "scroll_trough": "#1d2329",
    "scroll_thumb": "#65717c",
    "scroll_thumb_hover": "#7d8994",
}

_CHECK_IMAGES = None
_CHECK_ELEMENT = "MBS.Check.indicator"
_FIXED_GEOMETRY_READY = False
_INTERP_ID = None


def theme_colors(root=None, dark=False):
    return dict(DARK if dark else LIGHT)


def _ensure_engine(style):
    """Select/configure the fixed engine once per Tk interpreter."""
    global _FIXED_GEOMETRY_READY, _CHECK_IMAGES, _INTERP_ID

    # A normal MBSniffer process owns one Tk root. Handling a replaced Tk
    # interpreter as well makes tests/relaunches in the same Python process
    # safe and prevents stale PhotoImage command references.
```

MBSniffer v3.2/mb_theme.py — lines 81-120

```
    current_interp = id(style.master.tk)
    if _INTERP_ID != current_interp:
        _INTERP_ID = current_interp
        _FIXED_GEOMETRY_READY = False
        _CHECK_IMAGES = None

    try:
        if THEME_ENGINE in style.theme_names() and style.theme_use() != THEME_ENGINE:
            style.theme_use(THEME_ENGINE)
    except Exception:
        pass

    if not _FIXED_GEOMETRY_READY:
        _configure_fixed_geometry(style)
        _FIXED_GEOMETRY_READY = True


def _configure_fixed_geometry(style):
    """
    Define every geometry/state rule once.

    Nothing in this function is called as part of a Light/Dark toggle after
    startup. This is what guarantees that the page construction cannot move.
    """

    # ------------------------------------------------------------------
    # LabelFrames: titles above the rectangular border in both modes.
    # ------------------------------------------------------------------
    style.configure(
        "TLabelframe",
        borderwidth=1,
        relief="solid",
        padding=0,
        labeloutside=True,
        labelmargins=(0, 0, 0, 4),
    )
    style.configure(
        "TLabelframe.Label",
        borderwidth=0,
        padding=0,
```

MBSniffer v3.2/mb_theme.py — lines 121-160

```
        font=("Segoe UI", 9, "bold"),
    )

    # ------------------------------------------------------------------
    # Notebook/tabs: fixed Vista-like top geometry.
    # ------------------------------------------------------------------
    style.configure(
        "TNotebook",
        borderwidth=1,
        padding=0,
        tabmargins=(2, 2, 2, 0),
    )
    style.configure(
        "TNotebook.Tab",
        borderwidth=1,
        padding=(12, 6),
        font=("Segoe UI", 9),
    )
    style.map(
        "TNotebook.Tab",
        expand=[("selected", (2, 2, 2, 2))],
        padding=[("selected", (12, 6))],
        relief=[
            ("selected", "flat"),
            ("active", "flat"),
            ("pressed", "flat"),
        ],
    )

    # ------------------------------------------------------------------
    # Buttons.
    # ------------------------------------------------------------------
    style.configure(
        "TButton",
        borderwidth=1,
        relief="solid",
        padding=(7, 4),
        font=("Segoe UI", 9),
    )
    style.configure(
```

MBSniffer v3.2/mb_theme.py — lines 161-200

```
        "MBS.Toolbar.TButton",
        borderwidth=1,
        relief="solid",
        padding=(9, 5),
        font=("Segoe UI", 9),
    )
    style.configure(
        "MBS.Primary.TButton",
        borderwidth=1,
        relief="solid",
        padding=(9, 5),
        font=("Segoe UI", 9, "bold"),
    )
    style.configure(
        "MBS.Compact.TButton",
        borderwidth=1,
        relief="solid",
        padding=(7, 4),
        font=("Segoe UI", 9),
    )

    # ------------------------------------------------------------------
    # Input controls.
    # ------------------------------------------------------------------
    style.configure("TEntry", borderwidth=1, relief="solid", padding=1)
    style.configure("TCombobox", borderwidth=1, relief="solid", padding=1, arrowsize=10)
    style.configure("TSpinbox", borderwidth=1, relief="solid", padding=1, arrowsize=7)

    # ------------------------------------------------------------------
    # Checkboxes: fixed 15x15 image indicator. The image pixels are recoloured
    # on toggle; the element/layout never changes.
    # ------------------------------------------------------------------
    style.configure(
        "TCheckbutton",
        borderwidth=0,
        relief="flat",
        padding=(1, 1),
        font=("Segoe UI", 9),
    )

```

MBSniffer v3.2/mb_theme.py — lines 201-240

```
    # Create the indicator element from the active clam layout once.
    _install_fixed_checkbutton_element(style)

    # ------------------------------------------------------------------
    # Treeviews.
    # ------------------------------------------------------------------
    style.configure(
        "Treeview",
        borderwidth=1,
        relief="solid",
        font=("Segoe UI", 9),
    )
    style.configure(
        "MBSniffer.Treeview",
        borderwidth=1,
        relief="solid",
        font=("Segoe UI", 9),
    )

    # Important: right + bottom border only. This reproduces Light mode's
    # vertical column divider lines without making every heading look like a
    # separate raised rectangular button.
    for heading in ("Treeview.Heading", "MBSniffer.Treeview.Heading"):
        style.configure(
            heading,
            borderwidth=(0, 0, 1, 1),
            relief="solid",
            padding=(5, 4),
            font=("Segoe UI", 9),
        )
        style.map(
            heading,
            relief=[
                ("active", "solid"),
                ("pressed", "solid"),
            ],
        )

    # Keep MBSniffer.Treeview on the same heading layout as Treeview.
    try:
```

MBSniffer v3.2/mb_theme.py — lines 241-280

```
        heading_layout = style.layout("Treeview.Heading")
        if heading_layout:
            style.layout("MBSniffer.Treeview.Heading", heading_layout)
    except Exception:
        pass

    # ------------------------------------------------------------------
    # Scrollbars and named labels.
    # ------------------------------------------------------------------
    # Minimal scrollbars: slim track + thumb only, without arrow buttons or extra controls.
    style.layout(
        "Vertical.TScrollbar",
        [(
            "Vertical.Scrollbar.trough",
            {
                "sticky": "ns",
                "children": [
                    ("Vertical.Scrollbar.thumb", {"sticky": "nswe"})
                ],
            },
        )],
    )
    style.layout(
        "Horizontal.TScrollbar",
        [(
            "Horizontal.Scrollbar.trough",
            {
                "sticky": "we",
                "children": [
                    ("Horizontal.Scrollbar.thumb", {"sticky": "nswe"})
                ],
            },
        )],
    )
    style.configure(
        "Vertical.TScrollbar",
        width=6,
        borderwidth=0,
        relief="flat",
    )
```

MBSniffer v3.2/mb_theme.py — lines 281-320

```
    style.configure(
        "Horizontal.TScrollbar",
        width=6,
        borderwidth=0,
        relief="flat",
    )

    style.configure(
        "Horizontal.TProgressbar",
        borderwidth=0,
        thickness=8,
        relief="flat",
    )

    style.configure("MBS.Stat.TLabel", font=("Segoe UI", 9))
    style.configure("MBS.Status.TLabel", font=("Segoe UI", 9, "bold"))


def _paint_checkbox_image(image, *, fill, border, check=None):
    """Repaint one fixed-size checkbox image without changing its dimensions."""
    size = 15
    image.put(border, to=(0, 0, size, size))
    image.put(fill, to=(1, 1, size - 1, size - 1))

    if check:
        # White/black check mark. Coordinates remain identical in both modes.
        points = (
            (3, 7), (4, 8), (5, 9), (6, 10),
            (7, 9), (8, 8), (9, 7), (10, 6), (11, 5), (12, 4),
        )
        for x, y in points:
            image.put(check, to=(x, y, x + 1, y + 2))
            if x + 1 < size - 1:
                image.put(check, to=(x + 1, y, x + 2, y + 1))


def _install_fixed_checkbutton_element(style):
    global _CHECK_IMAGES

    root = style.master
```

MBSniffer v3.2/mb_theme.py — lines 321-360

```
    if _CHECK_IMAGES is None:
        _CHECK_IMAGES = {
            "off": tk.PhotoImage(master=root, width=15, height=15),
            "on": tk.PhotoImage(master=root, width=15, height=15),
            "disabled_off": tk.PhotoImage(master=root, width=15, height=15),
            "disabled_on": tk.PhotoImage(master=root, width=15, height=15),
        }

    try:
        if _CHECK_ELEMENT not in style.element_names():
            style.element_create(
                _CHECK_ELEMENT,
                "image",
                _CHECK_IMAGES["off"],
                ("disabled selected", _CHECK_IMAGES["disabled_on"]),
                ("disabled", _CHECK_IMAGES["disabled_off"]),
                ("selected", _CHECK_IMAGES["on"]),
                sticky="",
            )
    except tk.TclError:
        pass

    try:
        layout = style.layout("TCheckbutton")
    except Exception:
        layout = None

    if not layout:
        return

    def replace_indicator(nodes):
        result = []
        for name, options in nodes:
            options = dict(options)
            if "children" in options:
                options["children"] = replace_indicator(options["children"])
            if name == "Checkbutton.indicator":
                name = _CHECK_ELEMENT
            result.append((name, options))
        return result
```

MBSniffer v3.2/mb_theme.py — lines 361-400

```
361
362     # Avoid repeatedly replacing an already-custom element.
363     if _CHECK_ELEMENT not in repr(layout):
364         try:
365             style.layout("TCheckbutton", replace_indicator(layout))
366         except tk.TclError:
367             pass
368
369
370 def _repaint_checkboxes(colors, dark):
371     if not _CHECK_IMAGES:
372         return
373
374     if dark:
375         face = colors["input_readonly"]
376         border = colors["disabled"]
377         check = colors["select_text"]
378         disabled_face = colors["panel_alt"]
379         disabled_check = colors["disabled"]
380     else:
381         face = colors["input"]
382         border = "#7a7a7a"
383         check = "#111111"
384         disabled_face = "#e4e4e4"
385         disabled_check = "#999999"
386
387     _paint_checkbox_image(
388         _CHECK_IMAGES["off"],
389         fill=face,
390         border=border,
391     )
392     _paint_checkbox_image(
393         _CHECK_IMAGES["on"],
394         fill=face,
395         border=border,
396         check=check,
397     )
398     _paint_checkbox_image(
399         _CHECK_IMAGES["disabled_off"],
400         fill=disabled_face,
```

MBSniffer v3.2/mb_theme.py — lines 401-440

```
401         border=colors["disabled"],
402     )
403     _paint_checkbox_image(
404         _CHECK_IMAGES["disabled_on"],
405         fill=disabled_face,
406         border=colors["disabled"],
407         check=disabled_check,
408     )
409
410
411 def _apply_palette(root, style, colors, dark):
412     """
413     Apply colours only.
414
415     No call in this function changes:
416     - theme engine
417     - layout
418     - padding
419     - border width
420     - relief
421     - margins
422     - font
423     - widget dimensions
424     """
425     c = colors
426
427     style.configure(".", background=c["root"], foreground=c["text"])
428     style.configure("TFrame", background=c["root"])
429     style.configure("TLabel", background=c["root"], foreground=c["text"])
430     style.configure(
431         "TLabelframe",
432         background=c["root"],
433         foreground=c["text"],
434         bordercolor=c["border"],
435         lightcolor=c["border"],
436         darkcolor=c["border"],
437     )
438     style.configure(
439         "TLabelframe.Label",
440         background=c["root"],
```

MBSniffer v3.2/mb_theme.py — lines 441-480

```
        foreground=c["text"],
    )

    # Notebook.
    style.configure(
        "TNotebook",
        background=c["root"],
        bordercolor=c["border"],
        lightcolor=c["border"],
        darkcolor=c["border"],
    )
    style.configure(
        "TNotebook.Tab",
        background=c["panel"],
        foreground=c["text"],
        bordercolor=c["border"],
        lightcolor=c["border"],
        darkcolor=c["border"],
    )
    style.map(
        "TNotebook.Tab",
        background=[
            ("selected", c["panel_alt"]),
            ("active", c["panel_alt"]),
        ],
        foreground=[
            ("selected", c["text"]),
            ("active", c["text"]),
        ],
    )

    # Buttons: neutral controls have clear contrast; the primary action uses
    # the same restrained accent in Light and Dark.
    for name in (
        "TButton",
        "MBS.Toolbar.TButton",
        "MBS.Compact.TButton",
    ):
        style.configure(
            name,
```

MBSniffer v3.2/mb_theme.py — lines 481-520

```
            background=c["button"],
            foreground=c["text"],
            bordercolor=c["border"],
            lightcolor=c["border"],
            darkcolor=c["border"],
            focuscolor=c["accent"],
        )
        style.map(
            name,
            background=[
                ("pressed", c["button_pressed"]),
                ("active", c["button_hover"]),
                ("disabled", c["panel"]),
            ],
            foreground=[("disabled", c["disabled"])],
        )

    style.configure(
        "MBS.Primary.TButton",
        background=c["accent"],
        foreground=c["select_text"],
        bordercolor=c["accent"],
        lightcolor=c["accent"],
        darkcolor=c["accent"],
        focuscolor=c["accent"],
    )
    style.map(
        "MBS.Primary.TButton",
        background=[
            ("pressed", c["accent_pressed"]),
            ("active", c["accent_hover"]),
            ("disabled", c["panel"]),
        ],
        foreground=[
            ("disabled", c["disabled"]),
        ],
        bordercolor=[
            ("pressed", c["accent_pressed"]),
            ("active", c["accent_hover"]),
            ("disabled", c["border"]),
```

MBSniffer v3.2/mb_theme.py — lines 521-560

```
        ],
    )

    # Entry / Spinbox.
    for name in ("TEntry", "TSpinbox"):
        style.configure(
            name,
            fieldbackground=c["input"],
            background=c["input"],
            foreground=c["text"],
            bordercolor=c["border"],
            lightcolor=c["border"],
            darkcolor=c["border"],
            insertcolor=c["text"],
        )
        style.map(
            name,
            fieldbackground=[
                ("readonly", c["input_readonly"]),
                ("disabled", c["panel"]),
            ],
            foreground=[("disabled", c["disabled"])],
        )

    # Combobox.
    style.configure(
        "TCombobox",
        fieldbackground=c["input"],
        background=c["button"],
        foreground=c["text"],
        arrowcolor=c["text"],
        bordercolor=c["border"],
        lightcolor=c["border"],
        darkcolor=c["border"],
    )
    style.map(
        "TCombobox",
        fieldbackground=[
            ("readonly", c["input_readonly"]),
            ("disabled", c["panel"]),
```

MBSniffer v3.2/mb_theme.py — lines 561-600

```
        ],
        foreground=[
            ("readonly", c["text"]),
            ("disabled", c["disabled"]),
        ],
        background=[
            ("active", c["button_hover"]),
            ("disabled", c["panel"]),
        ],
        arrowcolor=[("disabled", c["disabled"])],
    )

    # Checkbox label/background only; indicator pixels are repainted separately.
    style.configure(
        "TCheckbutton",
        background=c["root"],
        foreground=c["text"],
    )
    style.map(
        "TCheckbutton",
        background=[("active", c["root"])],
        foreground=[("disabled", c["disabled"])],
    )
    _repaint_checkboxes(c, dark)

    # Treeview body and flat heading.
    for name in ("Treeview", "MBSniffer.Treeview"):
        style.configure(
            name,
            background=c["input"],
            fieldbackground=c["input"],
            foreground=c["text"],
            bordercolor=c["border"],
            lightcolor=c["border"],
            darkcolor=c["border"],
        )
        style.map(
            name,
            background=[("selected", c["select"])],
            foreground=[("selected", c["select_text"])],
```

MBSniffer v3.2/mb_theme.py — lines 601-640

```
        )

    for name in ("Treeview.Heading", "MBSniffer.Treeview.Heading"):
        style.configure(
            name,
            background=c["input"],
            foreground=c["text"],
            bordercolor=c["divider"],
            lightcolor=c["divider"],
            darkcolor=c["divider"],
        )
        style.map(
            name,
            background=[
                ("active", c["input_readonly"]),
                ("pressed", c["input_readonly"]),
            ],
            foreground=[("active", c["text"])],
        )

    # Scrollbars: only the slim thumb should be visually apparent.
    # The required trough element remains for ttk mechanics, but is painted
    # exactly like the content background so it disappears visually.
    for name in ("Vertical.TScrollbar", "Horizontal.TScrollbar", "TScrollbar"):
        style.configure(
            name,
            background=c["scroll_thumb"],
            troughcolor=c["input"],
            bordercolor=c["input"],
            darkcolor=c["scroll_thumb"],
            lightcolor=c["scroll_thumb"],
        )
        style.map(
            name,
            background=[
                ("active", c["scroll_thumb_hover"]),
                ("pressed", c["scroll_thumb_hover"]),
            ],
            darkcolor=[
                ("active", c["scroll_thumb_hover"]),
```

MBSniffer v3.2/mb_theme.py — lines 641-680

```
641                 ("pressed", c["scroll_thumb_hover"]),
642             ],
643             lightcolor=[
644                 ("active", c["scroll_thumb_hover"]),
645                 ("pressed", c["scroll_thumb_hover"]),
646             ],
647         )
648
649     style.configure(
650         "Horizontal.TProgressbar",
651         background=c["accent"],
652         troughcolor=c["scroll_trough"],
653         bordercolor=c["scroll_trough"],
654         lightcolor=c["accent"],
655         darkcolor=c["accent"],
656     )
657
658     style.configure("MBS.Stat.TLabel", background=c["root"], foreground=c["text"])
659     style.configure("MBS.Status.TLabel", background=c["root"], foreground=c["text"])
660
661     try:
662         root.configure(background=c["root"])
663     except Exception:
664         pass
665
666     return style
667
668
669 def apply_theme(root, dark=False, style=None):
670     """
671     Change palette only.
672
673     The ttk engine and all geometry were fixed before the UI was built.
674     """
675     style = style or ttk.Style(root)
676     _ensure_engine(style)
677     return _apply_palette(root, style, DARK if dark else LIGHT, bool(dark))
678
679
680 def apply_light_theme(root, style=None):
```

MBSniffer v3.2/mb_theme.py — lines 681-692

```
681     return apply_theme(root, dark=False, style=style)
682 
683 
684 def apply_dark_theme(root, style=None):
685     return apply_theme(root, dark=True, style=style)
686 
687 
688 def apply_modern_theme(root, dark=False):
689     """Initial application setup with the same engine used for both modes."""
690     style = ttk.Style(root)
691     _ensure_engine(style)
692     return _apply_palette(root, style, DARK if dark else LIGHT, bool(dark))
```

MBSniffer v3.2/mb_widgets.py

418 source lines.

MBSniffer v3.2/mb_widgets.py — lines 1-40

```
#!/usr/bin/env python3
"""Small reusable Tk widgets for MBSniffer."""

import tkinter as tk


class RefreshButton(tk.Canvas):
    # The fixed clam UI gives the refresh control stable dimensions, so the ↻
    # can now use its measured rendered-bbox centre without manual offsets.
    OPTICAL_OFFSET_X = 0
    OPTICAL_OFFSET_Y = 0

    """
    Square refresh button using the Unicode ↻ glyph.

    The glyph is centered from its actual rendered Tk bounding box. No manual
    X/Y optical offset is applied.
    """

    def __init__(
        self,
        master,
        command,
        size=26,
        cursor="hand2",
        **kwargs,
    ):
        self._command = command
        self._hover = False
        self._pressed = False
        self._state = "normal"
        self._dark_mode = False
        self._final_redraw_after_id = None

        super().__init__(
            master,
            width=size,
            height=size,
            bd=0,
            highlightthickness=0,
```

MBSniffer v3.2/mb_widgets.py — lines 41-80

```
            relief="flat",
            cursor=cursor,
            takefocus=0,
            **kwargs,
        )

        self.bind("<Configure>", self._redraw, add="+")
        self.bind("<Enter>", self._on_enter, add="+")
        self.bind("<Leave>", self._on_leave, add="+")
        self.bind("<ButtonPress-1>", self._on_press, add="+")
        self.bind("<ButtonRelease-1>", self._on_release, add="+")
        self._redraw()
        self._schedule_final_redraw()

    @staticmethod
    def _system_color(widget, name, fallback):
        try:
            widget.winfo_rgb(name)
            return name
        except Exception:
            return fallback

    def _schedule_final_redraw(self):
        """
        Redraw after Tk has settled requested widget dimensions.

        Some themed comboboxes report their final height only after the first
        layout pass. A short deferred redraw prevents the Unicode glyph from
        staying centered for an earlier 24 px square after the control becomes
        26 px high.
        """
        try:
            if self._final_redraw_after_id is not None:
                self.after_cancel(self._final_redraw_after_id)
            self._final_redraw_after_id = self.after(20, self._final_redraw)
        except tk.TclError:
            self._final_redraw_after_id = None

    def _final_redraw(self):
        self._final_redraw_after_id = None
```

MBSniffer v3.2/mb_widgets.py — lines 81-120

```
        try:
            self._redraw()
        except tk.TclError:
            pass

    def configure(self, cnf=None, **kwargs):
        state = kwargs.pop("state", None)
        if state is not None:
            self._state = str(state)

        result = super().configure(cnf, **kwargs)
        try:
            self._redraw()
            self._schedule_final_redraw()
        except tk.TclError:
            pass
        return result

    config = configure

    def cget(self, key):
        if key == "state":
            return self._state
        return super().cget(key)

    def invoke(self):
        if self._state != "disabled" and callable(self._command):
            return self._command()
        return None

    def set_dark_mode(self, enabled):
        self._dark_mode = bool(enabled)
        self._redraw()
        self._schedule_final_redraw()

    def _palette(self):
        if self._dark_mode:
            face = "#2a3138"
            hover = "#36404a"
            pressed = "#20262c"
```

MBSniffer v3.2/mb_widgets.py — lines 121-160

```
            text = "#e8eaed"
            gray = "#707982"
            border = "#4a5560"
            highlight = "#2f8fd8"

            if self._state == "disabled":
                return face, gray, border
            if self._pressed:
                return pressed, text, highlight
            if self._hover:
                return hover, text, highlight
            return face, text, border

        face = self._system_color(self, "SystemButtonFace", "#f0f0f0")
        text = self._system_color(self, "SystemButtonText", "#202020")
        gray = self._system_color(self, "SystemGrayText", "#8a8a8a")
        shadow = self._system_color(self, "System3DShadow", "#adadad")
        highlight = self._system_color(self, "SystemHighlight", "#7a9fc2")

        if self._state == "disabled":
            return face, gray, shadow
        if self._pressed:
            return face, text, highlight
        if self._hover:
            return face, text, highlight
        return face, text, shadow

    def _center_item_bbox(self, item, width, height):
        """Center the visible Tk bbox of one Canvas item on the square."""
        target_x = width / 2.0
        target_y = height / 2.0

        # Tk text bboxes land on integer pixels. Repeat because a fractional
        # first move may be rounded by the Canvas implementation.
        for _ in range(4):
            bbox = self.bbox(item)
            if not bbox:
                return
            current_x = (bbox[0] + bbox[2]) / 2.0
            current_y = (bbox[1] + bbox[3]) / 2.0
```

MBSniffer v3.2/mb_widgets.py — lines 161-200

```
            dx = target_x - current_x
            dy = target_y - current_y

            if abs(dx) <= 0.5 and abs(dy) <= 0.5:
                break

            # Move by integral pixels when a full-pixel correction is needed.
            move_x = round(dx) if abs(dx) > 0.5 else 0
            move_y = round(dy) if abs(dy) > 0.5 else 0
            if move_x == 0 and move_y == 0:
                break
            self.move(item, move_x, move_y)

        # Keep the explicit offset hook at zero so all three refresh buttons
        # use the same measured centre.
        if self.OPTICAL_OFFSET_X or self.OPTICAL_OFFSET_Y:
            self.move(
                item,
                self.OPTICAL_OFFSET_X,
                self.OPTICAL_OFFSET_Y,
            )

    def _redraw(self, _event=None):
        if not self.winfo_exists():
            return

        # During a <Configure> callback winfo_width()/height() may still
        # report the previous geometry on some Tk builds. Prefer the event's
        # new dimensions so centering is calculated for the final square.
        if _event is not None and hasattr(_event, "width") and hasattr(_event, "height"):
            width = max(2, int(_event.width))
            height = max(2, int(_event.height))
        else:
            width = max(2, int(self.winfo_width()))
            height = max(2, int(self.winfo_height()))

        background, foreground, border = self._palette()

        self.delete("all")
        self.create_rectangle(
```

MBSniffer v3.2/mb_widgets.py — lines 201-240

```
            0,
            0,
            width - 1,
            height - 1,
            fill=background,
            outline=border,
            width=1,
            tags=("background",),
        )

        glyph = self.create_text(
            width / 2.0,
            height / 2.0,
            text="↻",
            fill=foreground,
            font=("Segoe UI Symbol", 13),
            anchor="center",
            tags=("glyph",),
        )
        self._center_item_bbox(glyph, width, height)

    def _on_enter(self, _event):
        if self._state != "disabled":
            self._hover = True
            self._redraw()

    def _on_leave(self, _event):
        self._hover = False
        self._pressed = False
        self._redraw()

    def _on_press(self, _event):
        if self._state != "disabled":
            self._pressed = True
            self._redraw()

    def _on_release(self, event):
        if self._state == "disabled":
            return

```

MBSniffer v3.2/mb_widgets.py — lines 241-280

```
241         was_pressed = self._pressed
242         self._pressed = False
243         self._redraw()
244 
245         inside = (
246             0 <= event.x < self.winfo_width()
247             and 0 <= event.y < self.winfo_height()
248         )
249         if was_pressed and inside and callable(self._command):
250             self._command()
251 
252 
253 
254 class ToggleSwitch(tk.Canvas):
255     """Compact on/off slider used for the Light/Dark mode control."""
256 
257     def __init__(
258         self,
259         master,
260         variable,
261         command=None,
262         width=42,
263         height=22,
264         cursor="hand2",
265         **kwargs,
266     ):
267         self.variable = variable
268         self.command = command
269         self._dark_mode = False
270         self._hover = False
271         self._final_redraw_after_id = None
272 
273         super().__init__(
274             master,
275             width=width,
276             height=height,
277             bd=0,
278             highlightthickness=0,
279             relief="flat",
280             cursor=cursor,
```

MBSniffer v3.2/mb_widgets.py — lines 281-320

```
            takefocus=0,
            **kwargs,
        )

        self.bind("<Configure>", self._redraw, add="+")
        self.bind("<ButtonPress-1>", self._press, add="+")
        self.bind("<ButtonRelease-1>", self._toggle, add="+")
        self.bind("<Enter>", self._enter, add="+")
        self.bind("<Leave>", self._leave, add="+")
        self.variable.trace_add("write", lambda *_a: self._redraw())
        self._redraw()
        self._schedule_final_redraw()

    def _schedule_final_redraw(self):
        try:
            if self._final_redraw_after_id is not None:
                self.after_cancel(self._final_redraw_after_id)
            self._final_redraw_after_id = self.after(20, self._final_redraw)
        except tk.TclError:
            self._final_redraw_after_id = None

    def _final_redraw(self):
        self._final_redraw_after_id = None
        try:
            self._redraw()
        except tk.TclError:
            pass

    def set_dark_mode(self, enabled):
        self._dark_mode = bool(enabled)
        self._redraw()
        self._schedule_final_redraw()

    def _parent_bg(self):
        if self._dark_mode:
            return "#171a1f"
        try:
            self.winfo_rgb("SystemButtonFace")
            return "SystemButtonFace"
        except Exception:
```

MBSniffer v3.2/mb_widgets.py — lines 321-360

```
321             return "#f0f0f0"
322
323     def _press(self, _event=None):
324         # Deliberately do not focus the Canvas. This removes the white focus
325         # rectangle that appeared around the switch after mouse clicks.
326         return "break"
327
328     def _toggle(self, _event=None):
329         self.variable.set(not bool(self.variable.get()))
330         if callable(self.command):
331             self.command()
332         return "break"
333
334     def _enter(self, _event=None):
335         self._hover = True
336         self._redraw()
337
338     def _leave(self, _event=None):
339         self._hover = False
340         self._redraw()
341
342     def _redraw(self, _event=None):
343         if not self.winfo_exists():
344             return
345
346         # Use the final dimensions delivered by <Configure>. Without this, the
347         # first draw can happen while Tk still reports the initial 1x1 Canvas,
348         # which made the switch look small until the first mouse hover.
349         if _event is not None and hasattr(_event, "width") and hasattr(_event, "height"):
350             width = max(22, int(_event.width))
351             height = max(14, int(_event.height))
352         else:
353             width = max(22, int(self.winfo_width()))
354             height = max(14, int(self.winfo_height()))
355
356         on = bool(self.variable.get())
357
358         self.configure(background=self._parent_bg())
359         self.delete("all")
360
```

MBSniffer v3.2/mb_widgets.py — lines 361-400

```
361         pad = 2
362         radius = (height - 2 * pad) / 2.0
363         x0, y0 = pad, pad
364         x1, y1 = width - pad, height - pad
365 
366         if self._dark_mode:
367             off_track = "#5a646e"
368             on_track = "#2f8fd8"
369             hover_track = "#43a2e8"
370             knob = "#f4f6f8"
371             outline = "#68737d"
372         else:
373             off_track = "#a8a8a8"
374             on_track = "#2689d9"
375             hover_track = "#3b9be5"
376             knob = "#ffffff"
377             outline = "#8f8f8f"
378 
379         track = hover_track if self._hover and on else (on_track if on else off_track)
380 
381         # Rounded track: rectangle + two circles.
382         self.create_rectangle(
383             x0 + radius,
384             y0,
385             x1 - radius,
386             y1,
387             fill=track,
388             outline=track,
389         )
390         self.create_oval(
391             x0,
392             y0,
393             x0 + 2 * radius,
394             y1,
395             fill=track,
396             outline=track,
397         )
398         self.create_oval(
399             x1 - 2 * radius,
400             y0,
```

MBSniffer v3.2/mb_widgets.py — lines 401-418

```
401             x1,
402             y1,
403             fill=track,
404             outline=track,
405         )
406
407         knob_radius = max(4.0, radius - 2)
408         knob_cx = (x1 - radius) if on else (x0 + radius)
409         knob_cy = height / 2.0
410         self.create_oval(
411             knob_cx - knob_radius,
412             knob_cy - knob_radius,
413             knob_cx + knob_radius,
414             knob_cy + knob_radius,
415             fill=knob,
416             outline=outline,
417             width=1,
418         )
```

MBSniffer v3.2/mb_gui.py

2529 source lines.

MBSniffer v3.2/mb_gui.py — lines 1-40

```python
#!/usr/bin/env python3
"""Main Tkinter window composition for MBSniffer."""

import queue
import threading
from collections import deque

import tkinter as tk
from tkinter import ttk
import tkinter.font as tkfont

from mb_config import (
    APP_NAME,
    APP_VERSION,
    DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS,
    DISCOVERY_IDLE_CHECK_SECONDS,
    MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
    MAX_RAW_TEXT_LINES,
    MAX_UI_FRAMES,
    MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
    SLOW_RESPONSE_THRESHOLD_MS,
    SNIFFER_SPLIT_FRAME_HOLD_MS,
    debug_sim,
    load_ui_settings,
    resource_path,
    save_ui_settings,
)
from mb_protocol import SessionMetrics
from mb_slave_finder import SlaveFinderMixin
from mb_master_ui import MasterMixin
from mb_capture import CaptureMixin
from mb_view import ViewMixin
from mb_widgets import RefreshButton, ToggleSwitch
from mb_theme import apply_modern_theme, apply_theme, theme_colors
from mb_i18n import (
    LANGUAGE_CHOICES,
    LANGUAGE_ENGLISH,
    LANGUAGE_PORTUGUESE,
    canonical_ui_text,
    english_help_blocks,
```

MBSniffer v3.2/mb_gui.py — lines 41-80

```
    normalize_language,
    translate_text,
    translate_runtime_text,
)


class SnifferApp(MasterMixin, SlaveFinderMixin, CaptureMixin, ViewMixin, tk.Tk):
    def __init__(self):
        super().__init__()
        self.title(f"{APP_NAME} v{APP_VERSION}")

        # Load the remembered mode before any ttk widgets are constructed.
        # Light and Dark use one fixed ttk construction; only colours differ.
        self.ui_settings = load_ui_settings()
        self.current_language = normalize_language(
            self.ui_settings.get("language", LANGUAGE_PORTUGUESE)
        )
        self.initial_dark_mode = bool(self.ui_settings.get("dark_mode", False))
        self.ui_style = apply_modern_theme(
            self,
            dark=self.initial_dark_mode,
        )

        # Explicit application/window icon. This works when launched through
        # MBSniffer.bat (Python) and when bundled as a PyInstaller EXE.
        try:
            icon_file = resource_path("MBSniffer.ico")
            if icon_file.exists():
                self.iconbitmap(default=str(icon_file))
        except Exception:
            pass

        self.geometry("1280x780")
        self.minsize(1050, 650)

        self.serial_objects = []
        self.serial_by_port = {}
        self.reader_threads = []
        self.reader_generation = 0
        self.stop_event = threading.Event()
```

MBSniffer v3.2/mb_gui.py — lines 81-120

```
        self.event_queue = queue.Queue()
        self.busy = False
        self.capture_active = False
        self.restart_in_progress = False

        # Active slave-discovery state. This is deliberately independent from
        # passive capture/simulation state, but shares the global busy lock so
        # the application can never act as scanner and sniffer at the same time.
        self.discovery_active = False
        self.discovery_stop_event = threading.Event()
        self.discovery_thread = None
        self.discovery_serial = None
        self.discovery_found_count = 0
        self._closing = False
        self._process_queue_error_shown = False

        self.log_file = None
        self.log_path = None

        # Real-capture logs are created lazily: starting/stopping a capture with
        # zero received traffic must not create an empty .txt file.
        self.pending_log_filename = None
        self.pending_log_header = ""
        self.pending_log_notes = []
        self.log_start_failed = False
        self._raw_by_iid = {}
        self._record_by_iid = {}
        self._transaction_filter_seqs = None
        self._successful_request_seqs = set()
        self._advanced_filter_after_id = None
        self._bus_activity_after_id = None

        # Bounded GUI-history bookkeeping.
        self._tree_history = deque()   # current visible Treeview rows
        self._ui_frame_count = 0
        self._raw_text_line_count = 0

        # Source-of-truth for the bounded GUI history. Filtering/sorting only
        # changes the view; it never changes statistics or the disk log.
        self._frame_history = deque()
```

MBSniffer v3.2/mb_gui.py — lines 121-160

```
121         self._next_frame_seq = 1
122
123         # Visible Treeview bookkeeping by logical frame sequence.
124         self._tree_iids_by_seq = {}
125         self._tree_separator_iid_by_seq = {}
126
127         # Raw Hex separator width/reflow bookkeeping.
128         self._raw_separator_width_chars = 1
129         self._raw_reflow_after_id = None
130
131         # Slave filter state. "All mode" also controls whether newly discovered
132         # slaves are automatically selected.
133         self.slave_filter_vars = {}
134         self.slave_filter_all_mode = True
135
136         self.metrics = SessionMetrics()
137
138         self._build_ui()
139         self._freeze_language_layout_geometry()
140         self._capture_portuguese_help()
141         self.apply_language(save=False)
142         self._install_mouse_wheel_guard()
143         self.rebuild_slave_filter_menu()
144         self.refresh_ports()
145         self.update_mode_ui()
146         self.update_gap_state()
147         self.update_com_status()
148         self._restore_saved_main_tab()
149         self._install_ui_settings_persistence()
150
151         self.protocol("WM_DELETE_WINDOW", self.on_close)
152         self.after(50, self.process_queue)
153         self.after(1000, self.poll_com_status)
154
155
156     def _freeze_language_layout_geometry(self):
157         """Freeze language-independent container geometry using the PT layout.
158
159         The UI is constructed from the canonical Portuguese captions. Capture
160         the natural geometry before any runtime translation and use those
```

MBSniffer v3.2/mb_gui.py — lines 161-200

```
        dimensions as layout constraints. Language changes may replace text,
        but must never move or resize the surrounding controls/panels.
        """
        try:
            self.update_idletasks()
        except tk.TclError:
            return

        # The global language/theme overlay is anchored to the top-right.
        # Freeze the canonical PT grid-cell widths. English captions are shorter
        # here, so this preserves the exact existing PT positions and prevents
        # the Dark Mode switch / language combobox from sliding left or right.
        overlay = getattr(self, "theme_overlay", None)
        if overlay is not None:
            try:
                self.update_idletasks()
                canonical_column_widths = [
                    max(1, int(overlay.grid_bbox(column, 0)[2]))
                    for column in range(4)
                ]
                for column, width in enumerate(canonical_column_widths):
                    overlay.columnconfigure(column, minsize=width)
                self.update_idletasks()
                self._language_overlay_width = max(1, overlay.winfo_reqwidth())
                self._language_overlay_height = max(1, overlay.winfo_reqheight())
                overlay.place_configure(
                    width=self._language_overlay_width,
                    height=self._language_overlay_height,
                )
            except (tk.TclError, IndexError, TypeError):
                pass

        # The Finder configuration includes wrapped explanatory text. It must
        # be measured while the Finder page is actually laid out; measuring a
        # hidden Notebook page can under-estimate the PT height by one line.
        page = getattr(self, "discovery_page", None)
        options = getattr(self, "_discovery_options_frame", None)
        if page is not None and options is not None:
            try:
                selected_tab = self.main_notebook.select()
```

MBSniffer v3.2/mb_gui.py — lines 201-240

```
                self.main_notebook.select(page)
                self.update_idletasks()
                height = max(options.winfo_height(), options.winfo_reqheight(), 1)
                # grid row height also includes the widget's external pady.
                # Preserve the complete PT cell, not only the LabelFrame body.
                try:
                    cell_height = max(height, int(page.grid_bbox(0, 1)[3]))
                except (tk.TclError, TypeError, IndexError):
                    cell_height = height + 12
                self._language_discovery_options_height = height
                page.rowconfigure(1, minsize=cell_height)
                options.grid_configure(sticky="nsew")
                self.main_notebook.select(selected_tab)
                self.update_idletasks()
            except tk.TclError:
                pass

    def _guard_mouse_wheel(self, event):
        """
        Keep wheel input local to genuinely scrollable content widgets.

        Treeviews and Text widgets retain their normal vertical wheel scrolling,
        as do the scrollbar widgets themselves.  Everywhere else the event is
        consumed before ttk class bindings see it; in particular, this prevents
        Windows ttk Notebook tabs from changing when the user turns the wheel
        over a tab or over non-scrollable page content.
        """
        widget = getattr(event, "widget", None)
        if isinstance(
            widget,
            (ttk.Treeview, tk.Text, tk.Listbox, ttk.Scrollbar, tk.Scrollbar),
        ):
            return None
        return "break"

    def _install_mouse_wheel_guard(self):
        """
        Prepend one application bindtag to every existing widget.

        The UI is statically constructed before this method is called, so this
```

MBSniffer v3.2/mb_gui.py — lines 241-280

```
        covers the Sniffer, Bus Health, Help and Bus Slave Finder pages without
        changing individual widget behavior or notebook keyboard navigation.
        """
        tag = "MBSniffer.MouseWheelGuard"

        for sequence in (
            "<MouseWheel>",
            "<Shift-MouseWheel>",
            "<Button-4>",
            "<Button-5>",
        ):
            try:
                self.bind_class(
                    tag,
                    sequence,
                    self._guard_mouse_wheel,
                )
            except tk.TclError:
                pass

        stack = [self]
        while stack:
            widget = stack.pop()
            try:
                tags = tuple(widget.bindtags())
                if tag not in tags:
                    widget.bindtags((tag,) + tags)
                stack.extend(widget.winfo_children())
            except tk.TclError:
                continue

    def _apply_classic_widget_palette(self, dark):
        """Apply palette to the small set of classic Tk widgets."""
        colors = theme_colors(self, dark)

        if hasattr(self, "raw_text"):
            self.raw_text.configure(
                background=colors["input"],
                foreground=colors["text"],
                insertbackground=colors["text"],
```

MBSniffer v3.2/mb_gui.py — lines 281-320

```
                selectbackground=colors["select"],
                selectforeground=colors["select_text"],
            )

        if hasattr(self, "help_text"):
            self.help_text.configure(
                background=colors["input"],
                foreground=colors["text"],
                insertbackground=colors["text"],
                selectbackground=colors["select"],
                selectforeground=colors["select_text"],
            )

        for menu_name in ("slave_filter_menu", "traffic_context_menu"):
            menu = getattr(self, menu_name, None)
            if menu is None:
                continue
            try:
                menu.configure(
                    background=colors["panel"],
                    foreground=colors["text"],
                    activebackground=colors["select"],
                    activeforeground=colors["select_text"],
                    selectcolor=colors["text"],
                )
            except tk.TclError:
                pass

        if hasattr(self, "tree"):
            self.configure_anomaly_tags(dark)
            self.configure_inspector_result_styles()

        for name in (
            "refresh_a_btn",
            "refresh_b_btn",
            "discovery_refresh_btn",
            "dark_mode_switch",
        ):
            widget = getattr(self, name, None)
            setter = getattr(widget, "set_dark_mode", None)
```

MBSniffer v3.2/mb_gui.py — lines 321-360

```
            if callable(setter):
                setter(dark)

        if hasattr(self, "master_canvas"):
            self.master_apply_palette(dark)

    def configure_inspector_result_styles(self):
        """
        Match the Inspector Result foreground to the Traffic result palette.

        Only colour values are configured here; no padding/font/layout metrics
        are changed, so Light/Dark construction remains identical.
        """
        if not hasattr(self, "ui_style"):
            return

        colors = theme_colors(
            self,
            bool(getattr(self, "dark_mode_var", tk.BooleanVar(value=False)).get()),
        )

        tag_map = {
            "success": "result_success",
            "slow": "anomaly_slow",
            "exception": "anomaly_exception",
            "crc": "anomaly_crc",
            "timeout": "anomaly_timeout",
            "raw": "anomaly_raw",
        }

        self.ui_style.configure(
            "MBS.Result.neutral.TLabel",
            background=colors["root"],
            foreground=colors["text"],
        )
        self.ui_style.configure(
            "MBS.Result.pending.TLabel",
            background=colors["root"],
            foreground=colors["text"],
        )
```

MBSniffer v3.2/mb_gui.py — lines 361-400

```
361
362         if hasattr(self, "tree"):
363             for key, tag in tag_map.items():
364                 try:
365                     foreground = self.tree.tag_configure(tag)["foreground"]
366                 except Exception:
367                     foreground = colors["text"]
368
369                 self.ui_style.configure(
370                     f"MBS.Result.{key}.TLabel",
371                     background=colors["root"],
372                     foreground=foreground,
373                 )
374
375     def tr(self, text):
376         """Translate one canonical UI string to the currently selected language."""
377         return translate_text(text, self.current_language)
378
379     def trf(self, text, **values):
380         """Translate a format template and substitute values afterwards."""
381         return self.tr(text).format(**values)
382
383     def _capture_portuguese_help(self):
384         """Store the original PT-PT Help text and tag ranges exactly as built."""
385         if not hasattr(self, "help_text"):
386             return
387         try:
388             self._help_pt_text = self.help_text.get("1.0", "end-1c")
389             ranges = {}
390             for tag in self.help_text.tag_names():
391                 if tag == "sel":
392                     continue
393                 tag_ranges = list(self.help_text.tag_ranges(tag))
394                 ranges[tag] = [
395                     (str(tag_ranges[i]), str(tag_ranges[i + 1]))
396                     for i in range(0, len(tag_ranges), 2)
397                 ]
398             self._help_pt_tag_ranges = ranges
399         except tk.TclError:
400             self._help_pt_text = ""
```

MBSniffer v3.2/mb_gui.py — lines 401-440

```
            self._help_pt_tag_ranges = {}

    def _render_help_language(self):
        """Render Help in English or restore the original PT-PT text byte-for-byte."""
        if not hasattr(self, "help_text"):
            return
        try:
            self.help_text.configure(state="normal")
            self.help_text.delete("1.0", "end")

            if self.current_language == LANGUAGE_ENGLISH:
                max_ui_frames_text = f"{MAX_UI_FRAMES:_}".replace("_", " ")
                max_raw_lines_text = f"{MAX_RAW_TEXT_LINES:_}".replace("_", " ")
                for tag, text in english_help_blocks(
                    max_ui_frames_text,
                    max_raw_lines_text,
                    MIN_PENDING_REQUEST_TIMEOUT_SECONDS,
                    MAX_PENDING_REQUEST_TIMEOUT_SECONDS,
                    debug_sim=(debug_sim == 1),
                ):
                    self.help_text.insert("end", text, tag)
            else:
                self.help_text.insert("1.0", getattr(self, "_help_pt_text", ""))
                for tag, ranges in getattr(self, "_help_pt_tag_ranges", {}).items():
                    for start, end in ranges:
                        self.help_text.tag_add(tag, start, end)

            self.help_text.configure(state="disabled")
            self.help_text.see("1.0")
        except tk.TclError:
            pass

    def _translate_widget_tree(self):
        """Translate static widget captions without changing widget geometry."""
        stack = [self]
        while stack:
            widget = stack.pop()
            try:
                stack.extend(widget.winfo_children())
            except tk.TclError:
```

MBSniffer v3.2/mb_gui.py — lines 441-480

```
441                 continue
442
443             # Text widgets contain documents, not captions.
444             if isinstance(widget, tk.Text):
445                 continue
446
447             try:
448                 keys = widget.keys()
449             except Exception:
450                 keys = ()
451             if "text" in keys:
452                 try:
453                     current = widget.cget("text")
454                     if current:
455                         widget.configure(text=self.tr(current))
456                 except (tk.TclError, TypeError):
457                     pass
458
459             if isinstance(widget, ttk.Notebook):
460                 try:
461                     for tab_id in widget.tabs():
462                         current = widget.tab(tab_id, "text")
463                         widget.tab(tab_id, text=self.tr(current))
464                 except tk.TclError:
465                     pass
466
467     def _translate_traffic_context_menu(self):
468         menu = getattr(self, "traffic_context_menu", None)
469         if menu is None:
470             return
471         labels = (
472             (0, "Copiar Raw Hex"),
473             (1, "Copiar frame descodificado"),
474             (3, "Filtrar por este Slave"),
475             (4, "Filtrar por este FC"),
476             (5, "Mostrar apenas esta transação"),
477             (6, "Limpar filtro de transação"),
478         )
479         for index, canonical in labels:
480             try:
```

MBSniffer v3.2/mb_gui.py — lines 481-520

```
                menu.entryconfigure(index, label=self.tr(canonical))
            except tk.TclError:
                pass

    def _apply_table_language(self):
        """Refresh headings and filter values that are not normal widget captions."""
        if hasattr(self, "advanced_filter_type_combo"):
            current = canonical_ui_text(self.advanced_filter_type_var.get())
            canonical_values = (
                "Todos", "Requests", "Responses", "Exceptions",
                "CRC errors", "Timeouts", "RAW",
            )
            self.advanced_filter_type_combo.configure(
                values=tuple(self.tr(v) for v in canonical_values)
            )
            self.advanced_filter_type_var.set(self.tr(current))

        if hasattr(self, "time_order_var"):
            canonical_order = canonical_ui_text(self.time_order_var.get())
            self.time_order_var.set(self.tr(canonical_order))

        if hasattr(self, "tree") and hasattr(self, "tree_headings"):
            canonical_headings = {
                "time": "Hora ↓" if self.time_order_descending() else "Hora ↑",
                "delta": "Δt (ms)",
                "response": "Resp. (ms)",
                "channel": "COM",
                "type": "Tipo",
                "slave": canonical_ui_text(self.tree_headings.get("slave", "Slave ▾")),
                "fc": "FC",
                "details": "Detalhes",
                "crc": "CRC",
            }
            for col, canonical in canonical_headings.items():
                translated = self.tr(canonical)
                self.tree_headings[col] = translated
                try:
                    self.tree.heading(col, text=translated)
                except tk.TclError:
                    pass
```

MBSniffer v3.2/mb_gui.py — lines 521-560

```
521
522         if hasattr(self, "discovery_tree"):
523             canonical = {
524                 "slave": "Slave",
525                 "baud": "Baud",
526                 "config": "Config.",
527                 "fc": "FC",
528                 "result": "Resultado",
529                 "resp": "Resp. (ms)",
530                 "device_id": "Device Identification",
531                 "raw": "Raw Hex",
532             }
533             for col, value in canonical.items():
534                 try:
535                     self.discovery_tree.heading(col, text=self.tr(value))
536                 except tk.TclError:
537                     pass
538
539     def apply_language(self, save=True):
540         """Apply the selected language immediately to the complete UI."""
541         selected = getattr(self, "language_var", None)
542         if selected is not None:
543             self.current_language = normalize_language(selected.get())
544             if selected.get() != self.current_language:
545                 selected.set(self.current_language)
546
547         self._translate_widget_tree()
548         self._apply_table_language()
549         self._translate_traffic_context_menu()
550         self._render_help_language()
551
552         # Rebuild dynamic captions/values in the selected language.
553         try:
554             self.update_mode_ui()
555             self.update_com_status()
556             self.update_stats_labels()
557             self.rebuild_slave_filter_menu()
558             self.update_slave_heading()
559             self.update_frame_inspector(self.selected_record())
560             self.update_discovery_estimate()
```

MBSniffer v3.2/mb_gui.py — lines 561-600

```
            self.master_apply_language()
        except (AttributeError, tk.TclError):
            pass

        # Translate idle/current status variables without altering state.
        for name in ("status_var", "discovery_status_var", "discovery_found_var"):
            var = getattr(self, name, None)
            if var is not None:
                try:
                    var.set(translate_runtime_text(var.get(), self.current_language))
                except tk.TclError:
                    pass

        # Refresh translated row contents without allowing the language change
        # itself to alter table/card geometry.  Column widths are restored after
        # rendering; normal viewport resizing and new traffic may still use the
        # existing responsive/autofit behaviour.
        traffic_widths = {}
        discovery_widths = {}
        try:
            if hasattr(self, "tree"):
                traffic_widths = {
                    col: int(self.tree.column(col, "width"))
                    for col in self.tree["columns"]
                }
            if hasattr(self, "discovery_tree"):
                discovery_widths = {
                    col: int(self.discovery_tree.column(col, "width"))
                    for col in self.discovery_tree["columns"]
                }
            self.render_filtered_view()
            for col, width in traffic_widths.items():
                self.tree.column(col, width=width)
            for col, width in discovery_widths.items():
                self.discovery_tree.column(col, width=width)
        except (AttributeError, tk.TclError):
            pass

        if save:
            self.save_current_ui_settings()
```

MBSniffer v3.2/mb_gui.py — lines 601-640

```
601
602     def on_language_changed(self, _event=None):
603         self.apply_language(save=True)
604
605     def toggle_dark_mode(self):
606         """
607         Switch Light/Dark colours only.
608
609         No ttk theme replacement, layout change, padding change or row-height
610         change occurs here. The page construction is immutable after startup.
611         """
612         dark = bool(self.dark_mode_var.get())
613
614         self.ui_style = apply_theme(
615             self,
616             dark=dark,
617             style=self.ui_style,
618         )
619         self._apply_classic_widget_palette(dark)
620
621         # Remember Dark mode together with the rest of the user's preferences.
622         self.save_current_ui_settings()
623
624     def _restore_saved_main_tab(self):
625         saved = self.ui_settings.get("last_main_tab", "Sniffer")
626         pages = {"Bus Slave Finder": self.discovery_page, "Master": self.master_page}
627         try:
628             self.main_notebook.select(pages.get(saved, self.sniffer_page))
629         except tk.TclError:
630             pass
631
632     def _current_main_tab_name(self):
633         try:
634             selected = self.main_notebook.select()
635             if selected == str(self.discovery_page):
636                 return "Bus Slave Finder"
637             if selected == str(self.master_page):
638                 return "Master"
639             return "Sniffer"
640         except Exception:
```

MBSniffer v3.2/mb_gui.py — lines 641-680

```
641             return "Sniffer"
642
643     def _schedule_ui_settings_save(self, *_args):
644         if getattr(self, "_closing", False):
645             return
646
647         previous = getattr(self, "_settings_save_after_id", None)
648         if previous is not None:
649             try:
650                 self.after_cancel(previous)
651             except tk.TclError:
652                 pass
653
654         try:
655             self._settings_save_after_id = self.after(
656                 300,
657                 self.save_current_ui_settings,
658             )
659         except tk.TclError:
660             self._settings_save_after_id = None
661
662     def save_current_ui_settings(self):
663         """Persist all roaming-safe user preferences in one settings.json."""
664         pending = getattr(self, "_settings_save_after_id", None)
665         if pending is not None:
666             try:
667                 self.after_cancel(pending)
668             except tk.TclError:
669                 pass
670         self._settings_save_after_id = None
671
672         try:
673             finder_bauds = [
674                 int(baud)
675                 for baud, var in self.discovery_baud_vars.items()
676                 if bool(var.get())
677             ]
678             finder_parities = [
679                 parity
680                 for parity, var in self.discovery_parity_vars.items()
```

MBSniffer v3.2/mb_gui.py — lines 681-720

```
                if bool(var.get())
            ]
            finder_stops = [
                stop
                for stop, var in self.discovery_stop_vars.items()
                if bool(var.get())
            ]

            payload = {
                "dark_mode": bool(self.dark_mode_var.get()),
                "language": normalize_language(self.current_language),
                "last_main_tab": self._current_main_tab_name(),
                "sniffer": {
                    "physical_mode": self.mode_var.get(),
                    "baud": self.baud_var.get(),
                    "data_bits": self.data_var.get(),
                    "parity": self.parity_var.get(),
                    "stop_bits": self.stop_var.get(),
                    "frame_gap_mode": self.gap_mode_var.get(),
                    "frame_gap_ms": self.gap_ms_var.get(),
                    "pending_timeout_s": self.pending_timeout_var.get(),
                    "auto_scroll": bool(self.auto_scroll_var.get()),
                    "separate_transactions": bool(
                        self.separate_transactions_var.get()
                    ),
                    "highlight_results": bool(
                        self.highlight_anomalies_var.get()
                    ),
                },
                "bus_slave_finder": {
                    "baud_rates": finder_bauds,
                    "parities": finder_parities,
                    "stop_bits": finder_stops,
                    "slave_start": self.discovery_slave_start_var.get(),
                    "slave_end": self.discovery_slave_end_var.get(),
                    "min_timeout_ms": self.discovery_min_timeout_var.get(),
                    "attempts": self.discovery_attempts_var.get(),
                    "fc04_fallback": bool(
                        self.discovery_fc04_fallback_var.get()
                    ),
```

MBSniffer v3.2/mb_gui.py — lines 721-760

```
                    "device_identification": bool(
                        self.discovery_device_identification_var.get()
                    ),
                },
                "master": {
                    "baud": self.master_baud_var.get(),
                    "parity": self.master_parity_var.get(),
                    "stop_bits": self.master_stop_var.get(),
                    "response_timeout_ms": self.master_timeout_var.get(),
                    "delay_ms": self.master_delay_var.get(),
                },
            }
            save_ui_settings(settings=payload)
        except Exception:
            # Persistence is convenience state only.
            pass

    def _install_ui_settings_persistence(self):
        """Debounced persistence for all roaming-safe user preferences."""
        self._settings_save_after_id = None

        variables = (
            self.mode_var,
            self.baud_var,
            self.data_var,
            self.parity_var,
            self.stop_var,
            self.gap_mode_var,
            self.gap_ms_var,
            self.pending_timeout_var,
            self.auto_scroll_var,
            self.separate_transactions_var,
            self.highlight_anomalies_var,
            self.discovery_slave_start_var,
            self.discovery_slave_end_var,
            self.discovery_min_timeout_var,
            self.discovery_fc04_fallback_var,
            self.discovery_device_identification_var,
        )

```

MBSniffer v3.2/mb_gui.py — lines 761-800

```
        for variable in variables:
            variable.trace_add(
                "write",
                lambda *_a: self._schedule_ui_settings_save(),
            )

        for mapping in (
            self.discovery_baud_vars,
            self.discovery_parity_vars,
            self.discovery_stop_vars,
        ):
            for variable in mapping.values():
                variable.trace_add(
                    "write",
                    lambda *_a: self._schedule_ui_settings_save(),
                )

        self.main_notebook.bind(
            "<<NotebookTabChanged>>",
            lambda _e: self._schedule_ui_settings_save(),
            add="+",
        )

    def _schedule_bus_health_layout_sync(self, _event=None):
        """Debounce Bus Health reflow after notebook/window/Inspector resizing."""
        previous = getattr(self, "_bus_health_layout_after_id", None)
        if previous is not None:
            try:
                self.after_cancel(previous)
            except tk.TclError:
                pass

        try:
            self._bus_health_layout_after_id = self.after_idle(
                self._sync_bus_health_layout
            )
        except tk.TclError:
            self._bus_health_layout_after_id = None

    def _initialize_bus_health_layout(self):
```

MBSniffer v3.2/mb_gui.py — lines 801-840

```
        """Measure the natural full Bus Health height before adaptive reflow."""
        self._bus_health_layout_after_id = None

        cards = getattr(self, "_health_cards", ())
        if not cards:
            return

        try:
            self.update_idletasks()
        except tk.TclError:
            return

        if getattr(self, "_health_full_required_card_height", None) is None:
            requested = [
                card.winfo_reqheight()
                for card in cards
                if card.winfo_exists()
            ]
            if requested:
                self._health_full_required_card_height = max(requested)

        self._sync_bus_health_layout()

    def _apply_bus_health_layout(self, compact):
        """
        Re-grid Bus Health metrics inside equal vertical bands.

        Full mode uses one metric/value pair per row.  Compact mode uses two
        pairs per row.  In both modes every used row receives the same weight,
        so the metrics are distributed evenly over the card height instead of
        accumulating at the top.  Compact columns use proportional uniform
        widths (label/value + label/value), which keeps both halves aligned and
        prevents one metric group from consuming the neighbouring group.
        """
        groups = getattr(self, "_health_metric_groups", ())
        if not groups:
            return

        for card, pairs in groups:
            for label_widget, value_widget in pairs:
```

MBSniffer v3.2/mb_gui.py — lines 841-880

```
                label_widget.grid_forget()
                value_widget.grid_forget()

            # Clear weights left by the other layout before applying the new
            # one.  Uniform groups are local to each LabelFrame.
            for column in range(4):
                card.columnconfigure(column, weight=0, minsize=0, uniform="")
            for row in range(max(6, len(pairs))):
                card.rowconfigure(row, weight=0, minsize=0, uniform="")

            if compact:
                # Size the four compact columns from their real content.
                # The two label columns absorb any spare width equally; value
                # columns keep their natural size.  This avoids the clipping
                # caused by forcing every card into one fixed percentage split.
                for column in (0, 2):
                    card.columnconfigure(column, weight=1, minsize=0, uniform="")
                for column in (1, 3):
                    card.columnconfigure(column, weight=0, minsize=0, uniform="")

                used_rows = max(1, (len(pairs) + 1) // 2)
                for row in range(used_rows):
                    card.rowconfigure(
                        row,
                        weight=1,
                        minsize=0,
                        uniform="health_metric_rows",
                    )

                pair_count = len(pairs)
                card_width = max(320, int(card.winfo_width()))
                # At normal widths the technical labels/values remain on one
                # line.  Near the application's minimum width labels may wrap
                # so the pair stays inside the LabelFrame instead of colliding
                # with the neighbouring pair.
                compact_label_wrap = (
                    0 if card_width >= 390 else max(80, int(card_width * 0.29))
                )

                for index, (label_widget, value_widget) in enumerate(pairs):
```

MBSniffer v3.2/mb_gui.py — lines 881-920

```
                    row = index // 2
                    group = index % 2

                    try:
                        label_widget.configure(wraplength=compact_label_wrap)
                        value_widget.configure(wraplength=0)
                    except tk.TclError:
                        pass

                    # Odd final metrics use the complete last row, leaving the
                    # value the whole right-hand side of the card.
                    if index == pair_count - 1 and pair_count % 2:
                        label_widget.grid(
                            row=row,
                            column=0,
                            sticky="w",
                            padx=(6, 3),
                            pady=1,
                        )
                        value_widget.grid(
                            row=row,
                            column=1,
                            columnspan=3,
                            sticky="e",
                            padx=(3, 6),
                            pady=1,
                        )
                        continue

                    base_column = group * 2
                    label_widget.grid(
                        row=row,
                        column=base_column,
                        sticky="w",
                        padx=((6 if base_column == 0 else 5), 3),
                        pady=1,
                    )
                    value_widget.grid(
                        row=row,
                        column=base_column + 1,
```

MBSniffer v3.2/mb_gui.py — lines 921-960

```
                        sticky="e",
                        padx=(3, (5 if base_column == 0 else 6)),
                        pady=1,
                    )
            else:
                # One pair per row, with every row sharing the card height
                # equally.  Proportional columns keep labels and values aligned
                # while still allowing long values to use the right-hand side.
                card.columnconfigure(0, weight=1, minsize=0, uniform="")
                card.columnconfigure(1, weight=0, minsize=0, uniform="")

                used_rows = max(1, len(pairs))
                for row in range(used_rows):
                    card.rowconfigure(
                        row,
                        weight=1,
                        minsize=0,
                        uniform="health_metric_rows",
                    )

                card_width = max(320, int(card.winfo_width()))
                full_label_wrap = (
                    0 if card_width >= 330 else max(100, int(card_width * 0.55))
                )

                for row, (label_widget, value_widget) in enumerate(pairs):
                    try:
                        label_widget.configure(wraplength=full_label_wrap)
                        value_widget.configure(wraplength=0)
                    except tk.TclError:
                        pass
                    label_widget.grid(
                        row=row,
                        column=0,
                        sticky="w",
                        padx=(10, 6),
                        pady=1,
                    )
                    value_widget.grid(
                        row=row,
```

MBSniffer v3.2/mb_gui.py — lines 961-1000

```
                        column=1,
                        sticky="e",
                        padx=(6, 10),
                        pady=1,
                    )

        self._health_layout_compact = bool(compact)

    def _sync_bus_health_layout(self):
        """
        Keep Bus Health metrics inside their rectangles at every available height.

        The natural/full card height is measured from the real widgets. If the
        Frame Inspector expands and the notebook becomes too short, the metrics
        reflow automatically. When enough vertical room returns, the original
        layout is restored. Hysteresis prevents resize oscillation.
        """
        self._bus_health_layout_after_id = None

        health_frame = getattr(self, "health_frame", None)
        cards = getattr(self, "_health_cards", ())
        full_required = getattr(
            self,
            "_health_full_required_card_height",
            None,
        )
        if health_frame is None or not cards:
            return

        try:
            if not health_frame.winfo_exists():
                return

            available_frame_height = health_frame.winfo_height()
            if available_frame_height <= 1:
                return

            # Each card has 6 px external top and bottom grid padding.
            available_card_height = max(0, available_frame_height - 12)

```

MBSniffer v3.2/mb_gui.py — lines 1001-1040

```
            if full_required is None:
                self.update_idletasks()
                full_required = max(card.winfo_reqheight() for card in cards)
                self._health_full_required_card_height = full_required

            compact = bool(getattr(self, "_health_layout_compact", False))
            if compact:
                should_compact = available_card_height < (full_required + 8)
            else:
                should_compact = available_card_height < (full_required + 2)

            if should_compact != compact:
                self._apply_bus_health_layout(should_compact)
                self.update_idletasks()
            else:
                # Width can change without crossing the vertical compact/full
                # threshold. Reapply the current layout so wrap limits and the
                # equal-width grid continue to match the real card size.
                self._apply_bus_health_layout(compact)

        except tk.TclError:
            return

    def _build_ui(self):
        sniffer_saved = self.ui_settings.get("sniffer", {})
        if not isinstance(sniffer_saved, dict):
            sniffer_saved = {}

        self.columnconfigure(0, weight=1)
        self.rowconfigure(0, weight=1)

        # Refresh controls use a custom Canvas widget whose arrow geometry is
        # centered explicitly; this avoids ttk theme/font/image alignment drift.

        self.main_notebook = ttk.Notebook(self)
        self.main_notebook.grid(row=0, column=0, sticky="nsew")

        self.sniffer_page = ttk.Frame(self.main_notebook)
        self.sniffer_page.columnconfigure(0, weight=1)
        self.sniffer_page.rowconfigure(1, weight=1)
```

MBSniffer v3.2/mb_gui.py — lines 1041-1080

```
        self.main_notebook.add(self.sniffer_page, text="Sniffer")

        self.discovery_page = ttk.Frame(self.main_notebook)
        self.discovery_page.columnconfigure(0, weight=1)
        self.discovery_page.rowconfigure(3, weight=1)
        self.main_notebook.add(self.discovery_page, text="Bus Slave Finder")

        self.master_page = ttk.Frame(self.main_notebook)
        self.main_notebook.add(self.master_page, text="Master")

        # Global Light/Dark control: root-level overlay in the top-right corner.
        # It is outside both tabs and therefore applies equally to Sniffer and
        # Bus Slave Finder. `place()` means it consumes no layout height.
        self.theme_overlay = ttk.Frame(self)
        self.theme_overlay.place(
            relx=1.0,
            x=-12,
            y=4,
            anchor="ne",
        )

        ttk.Label(
            self.theme_overlay,
            text="Modo escuro",
        ).grid(row=0, column=0, sticky="e", padx=(0, 6))

        self.dark_mode_var = tk.BooleanVar(value=self.initial_dark_mode)
        self.dark_mode_switch = ToggleSwitch(
            self.theme_overlay,
            variable=self.dark_mode_var,
            command=self.toggle_dark_mode,
            width=42,
            height=22,
        )
        self.dark_mode_switch.grid(row=0, column=1, sticky="e")

        self.language_label = ttk.Label(
            self.theme_overlay,
            text="Linguagem",
        )
```

MBSniffer v3.2/mb_gui.py — lines 1081-1120

```
        self.language_label.grid(
            row=0, column=2, sticky="e", padx=(18, 6)
        )
        self.language_var = tk.StringVar(value=self.current_language)
        self.language_combo = ttk.Combobox(
            self.theme_overlay,
            textvariable=self.language_var,
            values=LANGUAGE_CHOICES,
            state="readonly",
            width=10,
        )
        self.language_combo.grid(row=0, column=3, sticky="e")
        self.language_combo.bind(
            "<<ComboboxSelected>>", self.on_language_changed
        )
        self.theme_overlay.lift()

        # ====================================================================
        # Compact top workspace: configuration on the left, controls/stats
        # on the right.  This reclaims vertical space for the traffic view.
        # ====================================================================
        top_area = ttk.Frame(self.sniffer_page)
        top_area.grid(row=0, column=0, sticky="ew", padx=10, pady=(10, 5))
        # Keep the 3:2 workspace split independent of translated text widths.
        # Without a shared uniform group Tk first honours each child widget's
        # requested width, so changing language moves the two main rectangles.
        top_area.columnconfigure(0, weight=3, uniform="top_workspace")
        top_area.columnconfigure(1, weight=2, uniform="top_workspace")
        # The left Configuração card defines the natural top-workspace height.
        # The right stack stretches to exactly the same top/bottom limits.
        top_area.rowconfigure(0, weight=1)

        cfg = ttk.LabelFrame(top_area, text="Configuração")
        cfg.grid(row=0, column=0, sticky="nsew", padx=(0, 6))
        cfg.columnconfigure(0, weight=1)

        # Four compact equal columns.  Widgets keep the same fixed widths as
        # v2.6; only the card itself is narrower, eliminating wasted whitespace.
        config_grid = ttk.Frame(cfg)
        config_grid.grid(row=0, column=0, sticky="ew", padx=10, pady=(8, 4))
```

MBSniffer v3.2/mb_gui.py — lines 1121-1160

```
        for col in range(4):
            config_grid.columnconfigure(col, weight=1, uniform="sniffer_cfg")

        def config_cell(row, column):
            cell = ttk.Frame(config_grid)
            cell.grid(
                row=row, column=column, sticky="w",
                padx=(0 if column == 0 else 8, 8 if column < 3 else 0),
                pady=(0, 7),
            )
            return cell

        # Row 1 — physical mode / COM A / COM B / capture status
        cell = config_cell(0, 0)
        ttk.Label(cell, text="Modo físico").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.mode_var = tk.StringVar(value=sniffer_saved.get("physical_mode", "RS485 2-wire"))
        self.mode_combo = ttk.Combobox(
            cell, textvariable=self.mode_var, width=13, state="readonly",
            values=["RS485 2-wire", "RS232 single RX", "RS232 dual RX"],
        )
        self.mode_combo.grid(row=1, column=0, sticky="w")
        self.mode_combo.bind("<<ComboboxSelected>>", lambda _e: self.update_mode_ui())

        cell = config_cell(0, 1)
        ttk.Label(cell, text="COM A").grid(row=0, column=0, sticky="w", pady=(0, 2))
        com_a_controls = ttk.Frame(cell)
        com_a_controls.grid(row=1, column=0, sticky="w")
        self.port_a_var = tk.StringVar()
        self.port_a_combo = ttk.Combobox(
            com_a_controls, textvariable=self.port_a_var, width=13, state="readonly"
        )
        self.port_a_combo.grid(row=0, column=0, sticky="w")
        self.port_a_combo.bind("<<ComboboxSelected>>", lambda _e: self.update_com_status())
        self.refresh_a_btn = RefreshButton(com_a_controls, command=self.refresh_ports, size=24)
        self.refresh_a_btn.grid(row=0, column=1, sticky="w", padx=(4, 0))

        def _sync_sniffer_refresh_a_square(_event=None):
            height = self.port_a_combo.winfo_height()
            if height > 1:
                self.refresh_a_btn.configure(width=height, height=height)
```

MBSniffer v3.2/mb_gui.py — lines 1161-1200

```
1161
1162        self.port_a_combo.bind("<Configure>", _sync_sniffer_refresh_a_square, add="+")
1163
1164        cell = config_cell(0, 2)
1165        self.port_b_label = ttk.Label(cell, text="COM B")
1166        self.port_b_label.grid(row=0, column=0, sticky="w", pady=(0, 2))
1167        com_b_controls = ttk.Frame(cell)
1168        com_b_controls.grid(row=1, column=0, sticky="w")
1169        self.port_b_var = tk.StringVar()
1170        self.port_b_combo = ttk.Combobox(
1171            com_b_controls, textvariable=self.port_b_var, width=13, state="readonly"
1172        )
1173        self.port_b_combo.grid(row=0, column=0, sticky="w")
1174        self.port_b_combo.bind("<<ComboboxSelected>>", lambda _e: self.update_com_status())
1175        self.refresh_b_btn = RefreshButton(com_b_controls, command=self.refresh_ports, size=24)
1176        self.refresh_b_btn.grid(row=0, column=1, sticky="w", padx=(4, 0))
1177
1178        def _sync_sniffer_refresh_b_square(_event=None):
1179            height = self.port_b_combo.winfo_height()
1180            if height > 1:
1181                self.refresh_b_btn.configure(width=height, height=height)
1182
1183        self.port_b_combo.bind("<Configure>", _sync_sniffer_refresh_b_square, add="+")
1184
1185        cell = config_cell(0, 3)
1186        ttk.Label(cell, text="Estado").grid(row=0, column=0, sticky="w", pady=(0, 2))
1187        self.status_var = tk.StringVar(value="● Parado")
1188        ttk.Label(
1189            cell,
1190            textvariable=self.status_var,
1191            style="MBS.Status.TLabel",
1192            width=22,
1193            anchor="w",
1194        ).grid(
1195            row=1, column=0, sticky="w"
1196        )
1197
1198        # Row 2 — serial format
1199        cell = config_cell(1, 0)
1200        ttk.Label(cell, text="Baud rate").grid(row=0, column=0, sticky="w", pady=(0, 2))
```

MBSniffer v3.2/mb_gui.py — lines 1201-1240

```
        self.baud_var = tk.StringVar(value=sniffer_saved.get("baud", "9600"))
        ttk.Combobox(
            cell, textvariable=self.baud_var, width=13,
            values=["1200", "2400", "4800", "9600", "19200", "38400", "57600", "115200"],
        ).grid(row=1, column=0, sticky="w")

        cell = config_cell(1, 1)
        ttk.Label(cell, text="Data bits").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.data_var = tk.StringVar(value=sniffer_saved.get("data_bits", "8"))
        ttk.Combobox(
            cell, textvariable=self.data_var, width=13, state="readonly",
            values=["5", "6", "7", "8"],
        ).grid(row=1, column=0, sticky="w")

        cell = config_cell(1, 2)
        ttk.Label(cell, text="Parity").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.parity_var = tk.StringVar(value=sniffer_saved.get("parity", "None"))
        ttk.Combobox(
            cell, textvariable=self.parity_var, width=13, state="readonly",
            values=["None", "Even", "Odd", "Mark", "Space"],
        ).grid(row=1, column=0, sticky="w")

        cell = config_cell(1, 3)
        ttk.Label(cell, text="Stop bits").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.stop_var = tk.StringVar(value=sniffer_saved.get("stop_bits", "1"))
        ttk.Combobox(
            cell, textvariable=self.stop_var, width=13, state="readonly",
            values=["1", "1.5", "2"],
        ).grid(row=1, column=0, sticky="w")

        # Row 3 — timing and display options
        cell = config_cell(2, 0)
        ttk.Label(cell, text="Frame gap").grid(row=0, column=0, sticky="w", pady=(0, 2))
        self.gap_mode_var = tk.StringVar(value=sniffer_saved.get("frame_gap_mode", "Auto"))
        self.gap_combo = ttk.Combobox(
            cell, textvariable=self.gap_mode_var, width=13, state="readonly",
            values=["Auto", "Manual"],
        )
        self.gap_combo.grid(row=1, column=0, sticky="w")
        self.gap_combo.bind("<<ComboboxSelected>>", lambda _e: self.update_gap_state())
```

MBSniffer v3.2/mb_gui.py — lines 1241-1280

```
1241
1242        cell = config_cell(2, 1)
1243        ttk.Label(cell, text="Gap manual (ms)").grid(row=0, column=0, sticky="w", pady=(0, 2))
1244        self.gap_ms_var = tk.StringVar(value=sniffer_saved.get("frame_gap_ms", ""))
1245        self.gap_entry = ttk.Entry(cell, textvariable=self.gap_ms_var, width=15, state="disabled")
1246        self.gap_entry.grid(row=1, column=0, sticky="w")
1247
1248        cell = config_cell(2, 2)
1249        ttk.Label(cell, text="Visualização").grid(row=0, column=0, sticky="w", pady=(0, 2))
1250        self.auto_scroll_var = tk.BooleanVar(value=sniffer_saved.get("auto_scroll", True))
1251        ttk.Checkbutton(cell, text="Auto-scroll", variable=self.auto_scroll_var).grid(
1252            row=1, column=0, sticky="w"
1253        )
1254
1255        cell = config_cell(2, 3)
1256        ttk.Label(cell, text="Pending timeout (s)").grid(row=0, column=0, sticky="w", pady=(0, 2))
1257        self.pending_timeout_var = tk.StringVar(value=sniffer_saved.get("pending_timeout_s", f"{DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS:g}"))
1258        # Direct numeric entry is cleaner here than tiny spin arrows. Range
1259        # validation still happens in get_pending_timeout_seconds().
1260        self.pending_timeout_entry = ttk.Entry(
1261            cell,
1262            textvariable=self.pending_timeout_var,
1263            width=13,
1264        )
1265        self.pending_timeout_entry.grid(row=1, column=0, sticky="w")
1266        ttk.Label(
1267            cell,
1268            text=(
1269                f"min {MIN_PENDING_REQUEST_TIMEOUT_SECONDS:g} / "
1270                f"default {DEFAULT_PENDING_REQUEST_TIMEOUT_SECONDS:g} / "
1271                f"max {MAX_PENDING_REQUEST_TIMEOUT_SECONDS:g}"
1272            ),
1273        ).grid(row=2, column=0, sticky="w", pady=(2, 0))
1274
1275        # COM status + mode help: compact footer inside the configuration card.
1276        com_status = ttk.Frame(cfg)
1277        com_status.grid(row=1, column=0, sticky="w", padx=10, pady=(0, 3))
1278        ttk.Label(com_status, text="Estado COM").grid(
1279            row=0, column=0, rowspan=2, sticky="nw", padx=(0, 14)
1280        )
```

MBSniffer v3.2/mb_gui.py — lines 1281-1320

```
1281        self.port_a_status_var = tk.StringVar(value="COM A - —")
1282        self.port_b_status_var = tk.StringVar(value="COM B - —")
1283        ttk.Label(com_status, textvariable=self.port_a_status_var).grid(row=0, column=1, sticky="w")
1284        ttk.Label(com_status, textvariable=self.port_b_status_var).grid(
1285            row=1, column=1, sticky="w", pady=(1, 0)
1286        )
1287
1288        self.mode_help_var = tk.StringVar()
1289        ttk.Label(cfg, textvariable=self.mode_help_var).grid(
1290            row=2, column=0, sticky="w", padx=10, pady=(2, 7)
1291        )
1292
1293        # ------------------------------------------------------------------
1294        # Right-hand command/status panel.
1295        # ------------------------------------------------------------------
1296        side_panel = ttk.Frame(top_area)
1297        side_panel.grid(row=0, column=1, sticky="nsew", padx=(6, 0))
1298        side_panel.columnconfigure(0, weight=1)
1299        side_panel.rowconfigure(0, weight=0)
1300        side_panel.rowconfigure(1, weight=1)
1301
1302        actions = ttk.LabelFrame(side_panel, text="Controlos")
1303        # Same top coordinate as Configuração; only the gap below Controlos
1304        # separates it from Estatísticas.
1305        actions.grid(row=0, column=0, sticky="ew", pady=(0, 6))
1306        actions.columnconfigure(0, weight=1, uniform="action_columns")
1307        actions.columnconfigure(1, weight=1, uniform="action_columns")
1308
1309        self.start_btn = ttk.Button(
1310            actions, text="Iniciar Captura", command=self.start_capture,
1311            style="MBS.Primary.TButton"
1312        )
1313        self.start_btn.grid(row=0, column=0, sticky="ew", padx=(8, 4), pady=(7, 3))
1314
1315        self.stop_btn = ttk.Button(
1316            actions, text="Parar", command=self.stop_capture, state="disabled",
1317            style="MBS.Toolbar.TButton"
1318        )
1319        self.stop_btn.grid(row=0, column=1, sticky="ew", padx=(4, 8), pady=(7, 3))
1320
```

MBSniffer v3.2/mb_gui.py — lines 1321-1360

```
1321        self.restart_btn = ttk.Button(
1322            actions, text="Reiniciar COM", command=self.restart_com_port,
1323            style="MBS.Toolbar.TButton"
1324        )
1325        self.restart_btn.grid(row=1, column=0, sticky="ew", padx=(8, 4), pady=3)
1326
1327        ttk.Button(
1328            actions, text="Limpar", command=self.clear_view,
1329            style="MBS.Toolbar.TButton"
1330        ).grid(row=1, column=1, sticky="ew", padx=(4, 8), pady=3)
1331
1332        ttk.Button(
1333            actions, text="Abrir pasta de logs", command=self.open_log_folder,
1334            style="MBS.Toolbar.TButton"
1335        ).grid(row=2, column=0, columnspan=2, sticky="ew", padx=8, pady=3)
1336
1337        self.sim_btn = None
1338        if debug_sim == 1:
1339            self.sim_btn = ttk.Button(
1340                actions, text="Simulação", command=self.run_simulation,
1341                style="MBS.Toolbar.TButton"
1342            )
1343            self.sim_btn.grid(row=3, column=0, columnspan=2, sticky="ew", padx=8, pady=3)
1344            options_row = 4
1345        else:
1346            options_row = 3
1347
1348        options_bar = ttk.Frame(actions)
1349        options_bar.grid(
1350            row=options_row,
1351            column=0,
1352            columnspan=2,
1353            sticky="ew",
1354            padx=8,
1355            pady=(3, 7),
1356        )
1357        options_bar.columnconfigure(2, weight=1)
1358
1359        self.separate_transactions_var = tk.BooleanVar(
1360            value=sniffer_saved.get("separate_transactions", True)
```

MBSniffer v3.2/mb_gui.py — lines 1361-1400

```
        )
        ttk.Checkbutton(
            options_bar,
            text="Separar transações",
            variable=self.separate_transactions_var,
            command=self.render_filtered_view,
        ).grid(row=0, column=0, sticky="w", padx=(0, 12))

        self.highlight_anomalies_var = tk.BooleanVar(
            value=sniffer_saved.get("highlight_results", False)
        )
        ttk.Checkbutton(
            options_bar,
            text="Realçar resultados",
            variable=self.highlight_anomalies_var,
            command=self.on_highlight_anomalies_changed,
        ).grid(row=0, column=1, sticky="w")

        self.bus_activity_var = tk.StringVar(value="BUS ○")
        ttk.Label(
            options_bar,
            textvariable=self.bus_activity_var,
            style="MBS.Status.TLabel",
            anchor="e",
            width=9,
        ).grid(row=0, column=2, sticky="e")

        stats = ttk.LabelFrame(side_panel, text="Estatísticas da sessão")
        # Expand vertically into all remaining space so the lower border always
        # lines up with the lower border of Configuração, even if Configuração
        # changes height in the future.
        stats.grid(row=1, column=0, sticky="nsew")
        for col in range(4):
            stats.columnconfigure(col, weight=1, uniform="stats")
        # Keep the eight counters centred in a fixed 4 x 2 matrix.  The
        # labels must not hug the LabelFrame border or shift with translation.
        for row in range(2):
            stats.rowconfigure(row, weight=1, uniform="stats_rows")

        self.stat_requests_var = tk.StringVar(value="Pedidos: 0")
```

MBSniffer v3.2/mb_gui.py — lines 1401-1440

```
1401        self.stat_responses_var = tk.StringVar(value="Respostas: 0")
1402        self.stat_pending_var = tk.StringVar(value="Pendentes: 0")
1403        self.stat_timeouts_var = tk.StringVar(value="Timeouts: 0")
1404        self.stat_crc_var = tk.StringVar(value="Erros de CRC: 0")
1405        self.stat_exceptions_var = tk.StringVar(value="Exceções: 0")
1406        self.stat_raw_var = tk.StringVar(value="RAW: 0")
1407        self.stat_slaves_var = tk.StringVar(value="Slaves: 0")
1408
1409        stat_vars = [
1410            self.stat_requests_var,
1411            self.stat_responses_var,
1412            self.stat_pending_var,
1413            self.stat_timeouts_var,
1414            self.stat_crc_var,
1415            self.stat_exceptions_var,
1416            self.stat_raw_var,
1417            self.stat_slaves_var,
1418        ]
1419        for idx, var in enumerate(stat_vars):
1420            ttk.Label(
1421                stats,
1422                textvariable=var,
1423                style="MBS.Stat.TLabel",
1424                anchor="center",
1425            ).grid(
1426                row=idx // 4,
1427                column=idx % 4,
1428                sticky="",
1429                padx=6,
1430                pady=3,
1431            )
1432
1433        # Notebook
1434        notebook = ttk.Notebook(self.sniffer_page)
1435        notebook.grid(row=1, column=0, sticky="nsew", padx=10, pady=(5, 8))
1436
1437        parsed = ttk.Frame(notebook)
1438        parsed.columnconfigure(0, weight=1)
1439        parsed.rowconfigure(1, weight=1)
1440        notebook.add(parsed, text="Tráfego")
```

MBSniffer v3.2/mb_gui.py — lines 1441-1480

```
1441
1442        filter_bar = ttk.Frame(parsed)
1443        filter_bar.grid(
1444            row=0,
1445            column=0,
1446            columnspan=2,
1447            sticky="ew",
1448            padx=4,
1449            pady=(4, 3),
1450        )
1451        filter_bar.columnconfigure(8, weight=1)
1452        # "Pesquisa" is wider than "Search". Keep its canonical PT cell width
1453        # reserved so the search Entry never moves when language changes.
1454        filter_bar.columnconfigure(7, minsize=65)
1455
1456        ttk.Label(filter_bar, text="Filtros").grid(
1457            row=0, column=0, sticky="w", padx=(2, 6)
1458        )
1459
1460        self.advanced_filter_type_var = tk.StringVar(value="Todos")
1461        type_filter = ttk.Combobox(
1462            filter_bar,
1463            textvariable=self.advanced_filter_type_var,
1464            state="readonly",
1465            width=11,
1466            values=(
1467                "Todos",
1468                "Requests",
1469                "Responses",
1470                "Exceptions",
1471                "CRC errors",
1472                "Timeouts",
1473                "RAW",
1474            ),
1475        )
1476        type_filter.grid(row=0, column=1, sticky="w", padx=(0, 8))
1477        self.advanced_filter_type_combo = type_filter
1478        type_filter.bind(
1479            "<<ComboboxSelected>>",
1480            lambda _e: self.render_filtered_view(),
```

MBSniffer v3.2/mb_gui.py — lines 1481-1520

```
        )

        ttk.Label(filter_bar, text="FC (hex)").grid(
            row=0, column=2, sticky="w", padx=(0, 4)
        )
        self.advanced_filter_fc_var = tk.StringVar(value="")
        fc_filter = ttk.Entry(
            filter_bar,
            textvariable=self.advanced_filter_fc_var,
            width=7,
        )
        fc_filter.grid(row=0, column=3, sticky="w", padx=(0, 8))

        ttk.Label(filter_bar, text="Resp. >").grid(
            row=0, column=4, sticky="w", padx=(0, 4)
        )
        self.advanced_filter_response_ms_var = tk.StringVar(value="")
        response_filter = ttk.Entry(
            filter_bar,
            textvariable=self.advanced_filter_response_ms_var,
            width=7,
        )
        response_filter.grid(row=0, column=5, sticky="w")
        ttk.Label(filter_bar, text="ms").grid(
            row=0, column=6, sticky="w", padx=(3, 8)
        )

        ttk.Label(filter_bar, text="Pesquisa").grid(
            row=0, column=7, sticky="w", padx=(0, 4)
        )
        self.advanced_filter_text_var = tk.StringVar(value="")
        search_filter = ttk.Entry(
            filter_bar,
            textvariable=self.advanced_filter_text_var,
            width=22,
        )
        search_filter.grid(row=0, column=8, sticky="ew")

        filter_actions = ttk.Frame(filter_bar)
        filter_actions.grid(row=0, column=9, sticky="e")
```

MBSniffer v3.2/mb_gui.py — lines 1521-1560

```
        ttk.Button(
            filter_actions,
            text="Limpar filtros",
            command=self.clear_advanced_filters,
            style="MBS.Compact.TButton",
        ).grid(row=0, column=0, padx=(6, 4))
        ttk.Button(
            filter_actions,
            text="Exportar CSV",
            command=self.export_session_csv,
            style="MBS.Compact.TButton",
        ).grid(row=0, column=1)

        for variable in (
            self.advanced_filter_fc_var,
            self.advanced_filter_response_ms_var,
            self.advanced_filter_text_var,
        ):
            variable.trace_add(
                "write",
                self.schedule_advanced_filter_refresh,
            )

        cols = ("time", "delta", "response", "channel", "type", "slave", "fc", "details", "crc")
        # Dedicated style so wrapped multi-line rows do not affect unrelated
        # widgets. Treeview has one row height for the whole table, therefore
        # the row height is adjusted to the largest wrapped Details cell that
        # is currently visible.
        self.tree_style_name = "MBSniffer.Treeview"
        self.tree_style = ttk.Style(self)
        self.tree = ttk.Treeview(
            parsed,
            columns=cols,
            show="headings",
            selectmode="browse",
            style=self.tree_style_name
        )

        self._tree_reflow_after_id = None
        self.tree_detail_min_width_px = 220
```

MBSniffer v3.2/mb_gui.py — lines 1561-1600

```
        self.tree_detail_border_reserve_px = 18

        # Internal sort state. The user changes it only by clicking Hora.
        self.time_order_var = tk.StringVar(
            value="Crescente — antigo → recente"
        )

        self.tree_headings = {
            "time": "Hora ↑",
            "delta": "Δt (ms)",
            "response": "Resp. (ms)",
            "channel": "COM",
            "type": "Tipo",
            "slave": "Slave ▾",
            "fc": "FC",
            "details": "Detalhes",
            "crc": "CRC",
        }

        # Keep columns as compact as possible while leaving breathing room
        # between the longest text and the column borders.
        self.tree_column_padding_px = 28  # total horizontal padding
        self.tree_cell_font = tkfont.nametofont("TkDefaultFont")
        try:
            heading_font_name = ttk.Style(self).lookup("Treeview.Heading", "font")
            self.tree_heading_font = (
                tkfont.nametofont(heading_font_name)
                if heading_font_name
                else self.tree_cell_font
            )
        except Exception:
            self.tree_heading_font = self.tree_cell_font

        self.tree_font_linespace = max(
            1,
            int(self.tree_cell_font.metrics("linespace"))
        )
        self.tree_single_line_rowheight = max(
            20,
            self.tree_font_linespace + 5
```

MBSniffer v3.2/mb_gui.py — lines 1601-1640

```
        )
        self.tree_style.configure(
            self.tree_style_name,
            rowheight=self.tree_single_line_rowheight
        )

        for col in cols:
            align = "w" if col == "details" else "center"
            self.tree.heading(
                col,
                text=self.tree_headings[col],
                anchor=align
            )
            initial_width = self.measure_tree_column_width(
                col,
                [],
                heading_text=self.tree_headings[col]
            )
            self.tree.column(
                col,
                width=initial_width,
                minwidth=initial_width,
                anchor=align,
                stretch=(col == "details")
            )

        self.tree.heading(
            "time",
            text=self.tree_headings["time"],
            anchor="center",
            command=self.toggle_time_order
        )
        self.tree.heading(
            "slave",
            text=self.tree_headings["slave"],
            anchor="center",
            command=self.show_slave_filter_menu
        )

        # Popup menu opened directly from the Slave column heading.
```

MBSniffer v3.2/mb_gui.py — lines 1641-1680

```
1641        self.slave_filter_menu = tk.Menu(self, tearoff=False)
1642
1643        # Right-click actions for the selected traffic frame.
1644        self.traffic_context_menu = tk.Menu(self, tearoff=False)
1645        self.traffic_context_menu.add_command(
1646            label="Copiar Raw Hex",
1647            command=self.copy_selected_raw,
1648        )
1649        self.traffic_context_menu.add_command(
1650            label="Copiar frame descodificado",
1651            command=self.copy_selected_decoded,
1652        )
1653        self.traffic_context_menu.add_separator()
1654        self.traffic_context_menu.add_command(
1655            label="Filtrar por este Slave",
1656            command=self.context_filter_selected_slave,
1657        )
1658        self.traffic_context_menu.add_command(
1659            label="Filtrar por este FC",
1660            command=self.context_filter_selected_fc,
1661        )
1662        self.traffic_context_menu.add_command(
1663            label="Mostrar apenas esta transação",
1664            command=self.context_show_selected_transaction,
1665        )
1666        self.traffic_context_menu.add_command(
1667            label="Limpar filtro de transação",
1668            command=self.clear_transaction_filter,
1669        )
1670
1671        ys = ttk.Scrollbar(parsed, orient="vertical", command=self.tree.yview)
1672        self.tree.configure(yscrollcommand=ys.set)
1673        self.tree.grid(row=1, column=0, sticky="nsew")
1674        ys.grid(row=1, column=1, sticky="ns")
1675        self.tree.bind("<<TreeviewSelect>>", self.on_row_select)
1676        self.tree.bind("<Button-3>", self.show_traffic_context_menu, add="+")
1677        self.tree.bind("<Configure>", self.schedule_tree_reflow)
1678        self.configure_anomaly_tags(self.initial_dark_mode)
1679
1680        raw = ttk.Frame(notebook)
```

MBSniffer v3.2/mb_gui.py — lines 1681-1720

```
        raw.columnconfigure(0, weight=1)
        raw.rowconfigure(0, weight=1)
        notebook.add(raw, text="Raw Hex / Log")

        self.raw_text = tk.Text(
            raw,
            wrap="word",
            font=("Consolas", 10)
        )
        self.raw_text_font = tkfont.Font(font=self.raw_text.cget("font"))
        self.raw_text.bind("<Configure>", self.schedule_raw_separator_reflow)

        raw_y = ttk.Scrollbar(raw, orient="vertical", command=self.raw_text.yview)
        self.raw_text.configure(yscrollcommand=raw_y.set)
        self.raw_text.grid(row=0, column=0, sticky="nsew")
        raw_y.grid(row=0, column=1, sticky="ns")

        # Bus Health — session-level diagnostic metrics.
        self.health_frame = ttk.Frame(notebook)
        health_frame = self.health_frame
        health_frame.columnconfigure(0, weight=1, uniform="health")
        health_frame.columnconfigure(1, weight=1, uniform="health")
        health_frame.columnconfigure(2, weight=1, uniform="health")
        health_frame.rowconfigure(0, weight=1)
        notebook.add(health_frame, text="Bus Health")

        traffic_health = ttk.LabelFrame(
            health_frame,
            text="Tráfego",
        )
        traffic_health.grid(
            row=0, column=0, sticky="nsew", padx=(6, 3), pady=6
        )
        traffic_health.columnconfigure(1, weight=1)

        response_health = ttk.LabelFrame(
            health_frame,
            text="Tempos de resposta",
        )
        response_health.grid(
```

MBSniffer v3.2/mb_gui.py — lines 1721-1760

```
            row=0, column=1, sticky="nsew", padx=3, pady=6
        )
        response_health.columnconfigure(1, weight=1)

        quality_health = ttk.LabelFrame(
            health_frame,
            text="Qualidade",
        )
        quality_health.grid(
            row=0, column=2, sticky="nsew", padx=(3, 6), pady=6
        )
        quality_health.columnconfigure(1, weight=1)

        self.health_requests_var = tk.StringVar(value="0")
        self.health_responses_var = tk.StringVar(value="0")
        self.health_crc_count_var = tk.StringVar(value="0")
        self.health_timeout_count_var = tk.StringVar(value="0")
        self.health_exception_count_var = tk.StringVar(value="0")
        self.health_req_rate_var = tk.StringVar(value="0.00 req/s")
        self.health_resp_avg_var = tk.StringVar(value="—")
        self.health_resp_min_var = tk.StringVar(value="—")
        self.health_resp_max_var = tk.StringVar(value="—")
        self.health_p95_var = tk.StringVar(value="—")
        self.health_nonvalidated_rate_var = tk.StringVar(value="0.00 %")
        self.health_timeout_rate_var = tk.StringVar(value="0.00 %")
        self.health_bus_load_var = tk.StringVar(value="0.00 %")
        self.health_slaves_var = tk.StringVar(value="0")
        self.health_total_bytes_var = tk.StringVar(value="0")
        self.health_slowest_var = tk.StringVar(value="—")
        self.health_exception_slaves_var = tk.StringVar(value="—")

        self._health_metric_groups = []

        def health_item(parent, label, variable, *, wraplength=0):
            return (
                ttk.Label(parent, text=label),
                ttk.Label(
                    parent,
                    textvariable=variable,
                    anchor="e",
```

MBSniffer v3.2/mb_gui.py — lines 1761-1800

```
                    wraplength=wraplength,
                    justify="right",
                ),
            )

        traffic_pairs = [
            health_item(traffic_health, "Pedidos", self.health_requests_var),
            health_item(traffic_health, "Respostas", self.health_responses_var),
            health_item(traffic_health, "Taxa de pedidos", self.health_req_rate_var),
            health_item(traffic_health, "Slaves ativos", self.health_slaves_var),
            health_item(traffic_health, "Bytes observados", self.health_total_bytes_var),
            health_item(traffic_health, "Utilização do bus ~", self.health_bus_load_var),
        ]
        response_pairs = [
            health_item(response_health, "Média", self.health_resp_avg_var),
            health_item(response_health, "Mínimo", self.health_resp_min_var),
            health_item(response_health, "Máximo", self.health_resp_max_var),
            health_item(response_health, "P95", self.health_p95_var),
            health_item(response_health, "Slave mais lento", self.health_slowest_var),
        ]
        quality_pairs = [
            health_item(quality_health, "Erros de CRC", self.health_crc_count_var),
            health_item(quality_health, "Bytes não validados", self.health_nonvalidated_rate_var),
            health_item(quality_health, "Timeouts", self.health_timeout_count_var),
            health_item(quality_health, "Taxa de timeout", self.health_timeout_rate_var),
            health_item(quality_health, "Exceções", self.health_exception_count_var),
            health_item(
                quality_health,
                "Exceções / Slave",
                self.health_exception_slaves_var,
                wraplength=190,
            ),
        ]

        self._health_metric_groups = [
            (traffic_health, traffic_pairs),
            (response_health, response_pairs),
            (quality_health, quality_pairs),
        ]
        self._health_cards = (
```

MBSniffer v3.2/mb_gui.py — lines 1801-1840

```
            traffic_health,
            response_health,
            quality_health,
        )
        self._health_layout_compact = False
        self._health_full_required_card_height = None
        self._bus_health_layout_after_id = None

        # Start with the normal layout, then adapt only if the real available
        # height is insufficient.
        self._apply_bus_health_layout(False)

        health_frame.bind(
            "<Configure>",
            self._schedule_bus_health_layout_sync,
            add="+",
        )
        notebook.bind(
            "<<NotebookTabChanged>>",
            self._schedule_bus_health_layout_sync,
            add="+",
        )
        self.after_idle(self._initialize_bus_health_layout)

        # Help tab
        help_frame = ttk.Frame(notebook)
        help_frame.columnconfigure(0, weight=1)
        help_frame.rowconfigure(0, weight=1)
        notebook.add(help_frame, text="Ajuda / Ligações")

        self.help_text = tk.Text(
            help_frame,
            wrap="word",
            font=("Segoe UI", 10),
            padx=14,
            pady=12,
            spacing1=2,
            spacing3=5
        )
        help_scroll = ttk.Scrollbar(help_frame, orient="vertical", command=self.help_text.yview)
```

MBSniffer v3.2/mb_gui.py — lines 1841-1880

```
1841        self.help_text.configure(yscrollcommand=help_scroll.set)
1842        self.help_text.grid(row=0, column=0, sticky="nsew")
1843        help_scroll.grid(row=0, column=1, sticky="ns")
1844
1845        self.help_text.tag_configure("title", font=("Segoe UI", 14, "bold"), spacing1=8, spacing3=8)
1846        self.help_text.tag_configure("heading", font=("Segoe UI", 11, "bold"), spacing1=10, spacing3=4)
1847        self.help_text.tag_configure("subheading", font=("Segoe UI", 10, "bold"), spacing1=6, spacing3=2)
1848        self.help_text.tag_configure("mono", font=("Consolas", 10), lmargin1=18, lmargin2=18)
1849        self.help_text.tag_configure("warning", font=("Segoe UI", 10, "bold"), lmargin1=12, lmargin2=12)
1850        self.help_text.tag_configure("bullet", lmargin1=12, lmargin2=28)
1851
1852        def h(text_value, tag=None):
1853            self.help_text.insert("end", text_value, tag)
1854
1855        max_ui_frames_text = f"{MAX_UI_FRAMES:_}".replace("_", " ")
1856        max_raw_lines_text = f"{MAX_RAW_TEXT_LINES:_}".replace("_", " ")
1857
1858        h("MBSniffer — Ajuda e ligações\n", "title")
1859        h(
1860            "O MBSniffer foi pensado para diagnóstico de comunicações Modbus RTU. "
1861            "O separador Sniffer é passivo: abre a(s) porta(s) COM para receção e "
1862            "não transmite pedidos. O separador Bus Slave Finder é uma ferramenta "
1863            "ativa de pesquisa e deve ser utilizado apenas quando não existe outro "
1864            "master ativo no mesmo barramento.\n"
1865        )
1866
1867        h("Início rápido — Sniffer\n", "heading")
1868        h("1. Escolhe o modo físico correto: RS485 2-wire, RS232 single RX ou RS232 dual RX.\n", "bullet")
1869        h("2. Seleciona a(s) porta(s) COM e os parâmetros série da comunicação observada.\n", "bullet")
1870        h("3. Mantém Frame gap em Auto, salvo quando tens uma razão concreta para o ajustar.\n", "bullet")
1871        h("4. Clica em “Iniciar Captura” e confirma atividade através de BUS ● RX.\n", "bullet")
1872        h("5. Usa Tráfego para diagnóstico rápido e Raw Hex / Log para confirmar os bytes reais.\n", "bullet")
1873        h("6. Seleciona um frame para obter a análise detalhada no Frame Inspector.\n", "bullet")
1874
1875        h("1. RS485 2-wire\n", "heading")
1876        h(
1877            "Neste modo uma única COM observa pedidos e respostas no mesmo par diferencial. "
1878            "O adaptador do sniffer deve ser ligado em paralelo e não deve atuar como master.\n"
1879        )
1880        h("Ligação típica:\n", "subheading")
```

MBSniffer v3.2/mb_gui.py — lines 1881-1920

```
        h(
            "        Equipamento A          Equipamento B\n"
            "             D+ --------------- D+\n"
            "             D- --------------- D-\n"
            "               \\               /\n"
            "                \\             /\n"
            "                 D+          D-\n"
            "                  USB-RS485 sniffer\n\n",
            "mono"
        )
        h("• Liga D+ do sniffer em paralelo com D+ do barramento.\n", "bullet")
        h("• Liga D− do sniffer em paralelo com D− do barramento.\n", "bullet")
        h(
            "• Se existir C/COM/0 V de referência, liga-o apenas quando souberes que "
            "corresponde à referência correta da interface RS485.\n",
            "bullet"
        )
        h(
            "• Não acrescentes uma terceira resistência de 120 Ω se o barramento já "
            "estiver terminado nas duas extremidades.\n",
            "warning"
        )
        h(
            "• Se o adaptador USB-RS485 tiver terminação ou bias selecionáveis, deixa-os "
            "desativados durante uma captura passiva, salvo se forem intencionalmente "
            "necessários na instalação.\n",
            "bullet"
        )
        h(
            "• A/B não é uma nomenclatura universal: fabricantes diferentes podem usar "
            "polaridades opostas. Dá prioridade a D+/D− e à documentação do equipamento.\n",
            "bullet"
        )

        h("2. RS232 single RX\n", "heading")
        h(
            "Observa apenas uma direção RS232. O RS232 é full-duplex: cada direção utiliza "
            "uma linha TX independente, pelo que uma única entrada RX não consegue observar "
            "simultaneamente os dois sentidos.\n"
        )
```

MBSniffer v3.2/mb_gui.py — lines 1921-1960

```
1921        h(
1922            "        Equipamento A                    Equipamento B\n"
1923            "             TX ----------------------------> RX\n"
1924            "              \\\n"
1925            "               \\----> RX do USB-RS232 sniffer (COM A)\n"
1926            "\n"
1927            "             GND --------------------------- GND\n"
1928            "               \\--------------------------> GND sniffer\n\n"
1929            "        TX do sniffer: NÃO LIGAR\n\n",
1930            "mono"
1931        )
1932        h("• Liga RX do sniffer ao TX da direção que pretendes observar.\n", "bullet")
1933        h("• Liga também o GND comum.\n", "bullet")
1934        h("• Deixa fisicamente desligado o TX do adaptador usado como sniffer.\n", "warning")
1935
1936        h("3. RS232 dual RX\n", "heading")
1937        h(
1938            "Permite observar as duas direções RS232 ao mesmo tempo. São necessários "
1939            "dois canais RX independentes, normalmente dois adaptadores USB-RS232.\n"
1940        )
1941        h(
1942            "        Equipamento A                    Equipamento B\n"
1943            "             TX ----------------------------> RX\n"
1944            "              \\----> RX sniffer COM A   (A → B)\n"
1945            "\n"
1946            "             RX <---------------------------- TX\n"
1947            "                                    \\----> RX sniffer COM B   (B → A)\n"
1948            "\n"
1949            "             GND --------------------------- GND\n"
1950            "               \\--------------------------- GND COM A\n"
1951            "                \\--------------------------- GND COM B\n\n"
1952            "        TX de ambos os sniffers: NÃO LIGAR\n\n",
1953            "mono"
1954        )
1955        h("• COM A é apresentada como A→B e COM B como B→A.\n", "bullet")
1956        h("• As duas COM devem usar os mesmos parâmetros série da ligação observada.\n", "bullet")
1957
1958        h("RS232 não é TTL\n", "heading")
1959        h(
1960            "Não ligues sinais RS232 diretamente a interfaces USB-TTL/UART de 3,3 V ou "
```

MBSniffer v3.2/mb_gui.py — lines 1961-2000

```
            "5 V. Usa um adaptador USB-RS232 com transceiver apropriado aos níveis RS232.\n",
            "warning"
        )

        h("Parâmetros da comunicação série\n", "heading")
        h(
            "Baud, Data bits, Parity e Stop bits têm de coincidir com a comunicação real. "
            "Parâmetros errados podem produzir bytes aparentemente aleatórios, frames com "
            "CRC inválido ou ausência total de frames reconhecíveis.\n"
        )
        h("• Baud rate — velocidade em bit/s, por exemplo 9600 ou 19200.\n", "bullet")
        h("• Data bits — normalmente 8 em Modbus RTU.\n", "bullet")
        h("• Parity — None, Even, Odd, Mark ou Space, conforme a instalação.\n", "bullet")
        h("• Stop bits — 1, 1.5 ou 2, conforme a configuração observada.\n", "bullet")

        h("Frame gap\n", "heading")
        h(
            "Em Auto, o programa usa 3,5 tempos de carácter até 19200 bit/s, inclusive. "
            "Acima de 19200 bit/s usa o t3.5 fixo de 1,750 ms recomendado para Modbus RTU. "
            "O modo Manual permite definir outro valor em milissegundos para diagnóstico "
            "de equipamentos ou drivers com comportamento particular.\n"
        )
        h(
            "Os adaptadores USB e o Windows podem agrupar bytes e introduzir latência. "
            "Por isso, os timestamps do MBSniffer são úteis para diagnóstico de sequência "
            "e tempos relativos, mas não substituem um analisador lógico quando são "
            "necessárias medições temporais de elevada precisão.\n"
        )
        h(
            "Quando um burst termina com o início de um frame conhecido ainda incompleto, "
            f"o MBSniffer espera até {SNIFFER_SPLIT_FRAME_HOLD_MS:g} ms adicionais antes de o "
            "fechar. Assim, um frame partido em dois pelo adaptador USB não aparece como RAW. "
            "Frames juntos no mesmo burst mantêm a hora de chegada do seu próprio bloco de bytes.\n"
        )

        h("Pending timeout\n", "heading")
        h(
            "Define durante quanto tempo um REQUEST CRC-válido pode permanecer à espera de "
            "uma RESPONSE/EXCEPTION emparelhável. Ao expirar, o REQUEST original é marcado "
            "como timeout e aumenta os contadores de Timeout/Timeout rate.\n"
```

MBSniffer v3.2/mb_gui.py — lines 2001-2040

```
        )
        h(
            f"Intervalo permitido: {MIN_PENDING_REQUEST_TIMEOUT_SECONDS:g} a "
            f"{MAX_PENDING_REQUEST_TIMEOUT_SECONDS:g} s. O valor deve ser superior ao "
            "tempo de resposta máximo normal da instalação para evitar falsos timeouts.\n"
        )
        h(
            "Pedidos broadcast (Slave 0) não têm resposta por definição: não ficam pendentes, "
            "nunca contam como timeout e não entram no cálculo da Timeout rate.\n"
        )

        h("Como interpretar a tabela Tráfego\n", "heading")
        h("• REQUEST — pedido Modbus RTU reconhecido pelo parser.\n", "bullet")
        h("• RESPONSE — resposta Modbus RTU reconhecida que não é uma EXCEPTION.\n", "bullet")
        h(
            "• FC05/FC06/FC08/FC22 — o pedido e a resposta têm o mesmo formato. A trama é "
            "RESPONSE quando existe um pedido pendente do mesmo Slave/FC (em FC05/06/22 os "
            "bytes têm de ser iguais; em RS232 dual RX tem de vir do canal oposto). Caso "
            "contrário é REQUEST.\n",
            "bullet"
        )
        h(
            "• Respostas FC01/FC02 mostram os bits por ordem de endereço, agrupados por byte. "
            "No Frame Inspector, as respostas de leitura emparelhadas (FC01–FC04) mostram o "
            "PDU Address de cada valor.\n",
            "bullet"
        )
        h(
            "• EXCEPTION — resposta Modbus com Function Code de exceção (bit 7 ativo). "
            "Detalhes mostra o código e a descrição normalizada.\n",
            "bullet"
        )
        h("• CRC OK — CRC recebido coincide com o CRC calculado.\n", "bullet")
        h(
            "• CRC ERROR — a estrutura parece um frame suportado, mas o CRC recebido não "
            "coincide com o CRC calculado.\n",
            "bullet"
        )
        h(
            "• RAW/UNSYNC — bytes descartados durante uma recuperação de sincronismo antes "
```

MBSniffer v3.2/mb_gui.py — lines 2041-2080

```
2041            "de ser encontrado, mais à frente no mesmo burst, um frame de estrutura "
2042            "reconhecida com CRC válido.\n",
2043            "bullet"
2044        )
2045        h(
2046            "• RAW/UNPARSED — restante conjunto de bytes que não pôde ser interpretado nem "
2047            "ressincronizado como um frame Modbus válido. É este estado que aparece, por "
2048            "exemplo, num bloco de ruído sem um frame válido posterior no mesmo burst.\n",
2049            "bullet"
2050        )
2051        h(
2052            "RAW/UNSYNC e RAW/UNPARSED são portanto situações diferentes e ambas podem "
2053            "aparecer na coluna Tipo. Nenhuma delas representa uma resposta Modbus válida.\n"
2054        )
2055
2056        h("Colunas da tabela Tráfego\n", "heading")
2057        h("• Hora — timestamp do início do burst/frame processado.\n", "bullet")
2058        h("• Δt — intervalo desde o frame anterior observada.\n", "bullet")
2059        h(
2060            "• Resp. — tempo entre um REQUEST e a RESPONSE/EXCEPTION emparelhada. "
2061            "Fica vazio quando não existe um par válido.\n",
2062            "bullet"
2063        )
2064        h("• COM — identifica o canal de captura: BUS em RS485, A→B/B→A em RS232 dual RX e SIM no modo de desenvolvimento. Não representa necessariamente o número COM do Windows
    .\n", "bullet")
2065        h("• Tipo — REQUEST, RESPONSE, EXCEPTION ou um estado RAW.\n", "bullet")
2066        h("• Slave — Unit/Slave ID Modbus quando identificável.\n", "bullet")
2067        h("• FC — Function Code hexadecimal observado no frame.\n", "bullet")
2068        h("• Detalhes — endereço, quantidade, dados ou descrição da operação.\n", "bullet")
2069        h("• CRC — OK, ERROR ou ? quando não existe um frame Modbus completo.\n", "bullet")
2070
2071        h("Emparelhamento REQUEST / RESPONSE\n", "heading")
2072        h(
2073            "Modbus RTU não possui Transaction ID. O MBSniffer emparelha frames por "
2074            "(Slave ID, Function Code) em ordem FIFO. Isto é apropriado para tráfego RTU "
2075            "normal, mas pode tornar-se ambíguo se existirem pedidos sobrepostos com o mesmo "
2076            "Slave e FC antes das respostas correspondentes.\n"
2077        )
2078        h(
2079            "Uma EXCEPTION usa internamente o Function Code base para emparelhamento. "
2080            "Exemplo: uma resposta 0x83 é associada a um REQUEST FC03 quando existe um "
```

MBSniffer v3.2/mb_gui.py — lines 2081-2120

```
            "pedido pendente compatível.\n"
        )

        h("PDU Address, 1-based e Qty\n", "heading")
        h(
            "PDU Address é o endereço codificado no pedido Modbus. 1-based é simplesmente "
            "PDU Address + 1 e serve para comparação com manuais que numeram o primeiro "
            "registo/coil como 1. Qty é a quantidade pedida.\n"
        )
        h(
            "Estes campos pertencem ao REQUEST. Uma RESPONSE não repete automaticamente "
            "Address ou Qty no protocolo, por isso o MBSniffer não inventa esses campos na "
            "linha da resposta.\n"
        )

        h("Códigos de exceção Modbus\n", "heading")
        h("• 0x01 — Illegal Function: função não suportada ou não permitida.\n", "bullet")
        h("• 0x02 — Illegal Data Address: endereço/gama de endereços inválida.\n", "bullet")
        h("• 0x03 — Illegal Data Value: valor, quantidade ou campo inválido.\n", "bullet")
        h("• 0x04 — Server Device Failure: falha interna do dispositivo.\n", "bullet")
        h("• 0x05 — Acknowledge: pedido aceite; processamento ainda em curso.\n", "bullet")
        h("• 0x06 — Server Device Busy: dispositivo temporariamente ocupado.\n", "bullet")
        h("• 0x08 — Memory Parity Error.\n", "bullet")
        h("• 0x0A — Gateway Path Unavailable.\n", "bullet")
        h("• 0x0B — Gateway Target Device Failed to Respond.\n", "bullet")

        h("Filtros avançados\n", "heading")
        h(
            "A barra no topo de Tráfego permite combinar vários filtros sem alterar os "
            "dados originais da sessão:\n"
        )
        h("• Tipo — Todos, Requests, Responses, Exceptions, CRC errors, Timeouts ou RAW.\n", "bullet")
        h("• FC (hex) — por exemplo 03, 04, 10 ou 0x04.\n", "bullet")
        h("• Resp. > — mostra apenas respostas acima do limite indicado em ms.\n", "bullet")
        h(
            "• Pesquisa — procura texto em Detalhes, Raw Hex, Slave, FC, Tipo e COM/canal.\n",
            "bullet"
        )
        h(
            "O filtro Slave continua disponível no cabeçalho da coluna e combina-se com "
```

MBSniffer v3.2/mb_gui.py — lines 2121-2160

```
            "os filtros avançados. “Limpar filtros” repõe todos os filtros de visualização. "
            "Os filtros não alteram as estatísticas nem o conteúdo do log de captura.\n"
        )

        h("Realçar resultados\n", "heading")
        h(
            "Esta opção vem desligada por defeito para manter a tabela neutra. Quando está "
            "ativa, os resultados passam a ser diferenciados por cor: uma transação normal "
            "concluída com RESPONSE CRC-válida fica verde; CRC error, timeout, Exception, "
            "RAW e resposta lenta mantêm as respetivas cores de diagnóstico.\n"
        )
        h(
            "Numa transação normal concluída com sucesso, tanto o REQUEST como a RESPONSE "
            "correspondente ficam verdes. Um REQUEST ainda pendente permanece neutro. "
            "Uma resposta lenta continua com a cor de resposta lenta, mesmo sendo válida, "
            "para não esconder o resultado de diagnóstico.\n"
        )
        h(
            f"Uma resposta é considerada lenta a partir de "
            f"{SLOW_RESPONSE_THRESHOLD_MS:g} ms. Este limite é uma referência visual do "
            "MBSniffer; deve ser interpretado no contexto do equipamento e do ciclo de poll.\n"
        )

        h("Código de cores dos resultados\n", "heading")
        h("• Verde — Sucesso: REQUEST/RESPONSE normal, emparelhada e com CRC válido.\n", "bullet")
        h(
            f"• Amarelo — Resposta lenta: tempo de resposta >= "
            f"{SLOW_RESPONSE_THRESHOLD_MS:g} ms.\n",
            "bullet"
        )
        h("• Vermelho — CRC Error ou resposta Modbus Exception.\n", "bullet")
        h("• Laranja — Timeout: REQUEST expirou sem resposta emparelhada.\n", "bullet")
        h("• Roxo — RAW/UNSYNC ou RAW/UNPARSED: bytes sem frame Modbus válido.\n", "bullet")
        h(
            "Com “Realçar resultados” desligado, a tabela permanece neutra. O Frame "
            "Inspector continua a indicar a classificação do resultado em texto; quando "
            "o destaque está ligado, o campo Resultado também usa a respetiva cor.\n"
        )

        h("Frame Inspector\n", "heading")
```

MBSniffer v3.2/mb_gui.py — lines 2161-2200

```
        h(
            "Seleciona uma linha de Tráfego para analisar o frame sem ter de interpretar "
            "manualmente todos os bytes. O Inspector mostra Slave, Tipo, Function Code + "
            "nome, tempo de resposta, Resultado, PDU Address, 1-based, "
            "Qty, ByteCount, CRC recebido, CRC calculado, dados/registos/Exception e Raw Hex.\n"
        )
        h(
            "“Recolher” esconde o conteúdo do Inspector e devolve espaço vertical à tabela. "
            "“Expandir” volta a apresentar a análise.\n"
        )

        h("Menu de contexto do Tráfego\n", "heading")
        h("Clica com o botão direito numa frame real para:\n")
        h("• Copiar Raw Hex.\n", "bullet")
        h("• Copiar o frame descodificado.\n", "bullet")
        h("• Filtrar pelo Slave dessa frame.\n", "bullet")
        h("• Filtrar pelo Function Code dessa frame.\n", "bullet")
        h("• Mostrar apenas a transação REQUEST/RESPONSE correspondente.\n", "bullet")
        h("• Limpar o filtro temporário de transação.\n", "bullet")

        h("Bus Health\n", "heading")
        h(
            "O separador Bus Health resume a qualidade e o desempenho da sessão sem "
            "substituir a tabela de frames.\n"
        )
        h("• Pedidos / Respostas — contadores da sessão.\n", "bullet")
        h("• Taxa de pedidos — número médio de pedidos observados por segundo.\n", "bullet")
        h("• Slaves ativos — IDs Modbus válidos observados na sessão.\n", "bullet")
        h("• Bytes observados — soma dos bytes entregues ao parser.\n", "bullet")
        h(
            "• Utilização bus ~ — estimativa com base em bytes observados, baud, data bits, "
            "parity e stop bits. É aproximada e depende do ponto de captura.\n",
            "bullet"
        )
        h("• Média / Mínimo / Máximo — tempos de resposta emparelhados.\n", "bullet")
        h("• P95 — 95% das respostas medidas ficaram abaixo ou iguais a este valor.\n", "bullet")
        h(
            "• Slave mais lento — Slave com maior tempo médio de resposta entre as amostras "
            "observadas.\n",
            "bullet"
```

MBSniffer v3.2/mb_gui.py — lines 2201-2240

```
2201        )
2202        h("• Bytes não validados — percentagem dos bytes observados que pertencem a frames com CRC inválido ou a blocos RAW. Não prova, por si só, a existência de ruído elétrico
    ; parâmetros série errados, captura incompleta ou funções não suportadas pelo parser também podem produzir RAW.\n", "bullet")
2203        h("• Taxa de timeout — percentagem de REQUEST que expiraram sem resposta emparelhada.\n", "bullet")
2204        h("• Exceções / Slave — distribuição das respostas Exception por Slave ID.\n", "bullet")
2205
2206        h("Indicador BUS RX\n", "heading")
2207        h(
2208            "BUS ● RX pulsa brevemente quando chega um lote de frames à interface. Em repouso "
2209            "é apresentado BUS ○. Serve apenas como confirmação visual de atividade recebida; "
2210            "não representa qualidade elétrica nem garante que os frames sejam válidos.\n"
2211        )
2212
2213        h("Raw Hex / Log\n", "heading")
2214        h(
2215            "Mostra os bytes capturados em hexadecimal com informação de contexto. As linhas "
2216            "longas fazem wrap à largura disponível. A vista acompanha os mesmos filtros "
2217            "visuais usados em Tráfego; o ficheiro TXT em disco, quando existe, mantém a "
2218            "captura completa e não é truncado pelos filtros.\n"
2219        )
2220
2221        h("Exportar CSV\n", "heading")
2222        h(
2223            "Exportar CSV grava os frames atualmente retidos no histórico da interface em "
2224            "UTF-8 compatível com Excel. Inclui Time, Δt, Response time, Channel, Type, "
2225            "Slave, FC, Details, CRC, Raw, PDU Address, 1-based, Qty, ByteCount, Timeout "
2226            "e classe de anomalia.\n"
2227        )
2228        h(
2229            f"A interface mantém aproximadamente os últimos {max_ui_frames_text} frames. "
2230            "Por isso, em sessões maiores o CSV representa o histórico ainda retido pela GUI, "
2231            "enquanto o TXT de uma captura real continua a ser o registo completo em disco.\n"
2232        )
2233
2234        h("Logs de captura\n", "heading")
2235        h(
2236            "O log TXT só é criado depois de chegar tráfego real. Iniciar e parar sem receber "
2237            "frames não cria um ficheiro vazio.\n"
2238        )
2239        h("Pasta dos logs:\n")
2240        h("        <pasta da aplicação>\\MBSniffer Logs\n", "mono")
```

MBSniffer v3.2/mb_gui.py — lines 2241-2280

```
        h(
            "Os filtros, ordenação, Realçar resultados e o Frame Inspector são apenas funções "
            "de visualização e não alteram os frames gravados no log.\n"
        )

        h("Limites do histórico visível\n", "heading")
        h(
            f"Para manter a aplicação responsiva, a interface conserva aproximadamente "
            f"{max_ui_frames_text} frames e {max_raw_lines_text} linhas de Raw Hex. "
            "A remoção é feita por blocos. O log TXT de captura, quando ativo, é independente "
            "deste limite.\n"
        )

        h("Estado e portas COM\n", "heading")
        h(
            "Estado COM distingue uma porta aberta pelo MBSniffer, uma porta apenas detetada "
            "pelo Windows e uma porta que deixou de ser detetada. O estado é atualizado "
            "periodicamente.\n"
        )
        h(
            "Reiniciar COM fecha e volta a abrir a(s) porta(s) usada(s) pelo Sniffer. "
            "Isto reinicia o handle da porta série dentro do MBSniffer; não reinicia o "
            "driver USB/COM do Windows.\n"
        )
        h(
            "Os botões ↻ ao lado das COM atualizam a enumeração de portas disponível no "
            "Sniffer e no Bus Slave Finder.\n"
        )

        h("Bus Slave Finder — pesquisa ativa\n", "heading")
        h(
            "O Bus Slave Finder transmite pedidos Modbus RTU, por isso não é uma função "
            "passiva. Antes de iniciar, confirma que não existe outro master ativo no mesmo "
            "barramento.\n"
        )
        h(
            "Seleciona COM, baud rates, paridades, stop bits e intervalo de Slave IDs. "
            "Data bits é fixo em 8. A pesquisa tenta FC03 Address 0 Qty 1; uma resposta "
            "normal ou uma Exception Modbus com CRC válido confirma que o Slave existe.\n"
        )
```

MBSniffer v3.2/mb_gui.py — lines 2281-2320

```
2281        h(
2282            "FC04 fallback é opcional e aumenta o número máximo de tentativas. O timeout "
2283            "mínimo é adaptado ao baud rate. Um write timeout significa falha ao transmitir "
2284            "pela porta/driver e não deve ser interpretado simplesmente como “slave sem resposta”.\n"
2285        )
2286        h(
2287            "Tentativas (1 a 3) repete cada pedido sem resposta antes de passar ao Slave "
2288            "seguinte. Antes da primeira transmissão, a pesquisa escuta o barramento durante "
2289            f"{DISCOVERY_IDLE_CHECK_SECONDS:g} s; se receber qualquer byte, não transmite e "
2290            "indica que pode existir outro master ativo.\n"
2291        )
2292        h(
2293            "Device Identification (FC43/14) é opcional. Quando ativo, só é enviado depois "
2294            "de um Slave ter sido encontrado e tenta ler Basic Device Identification: "
2295            "VendorName, ProductCode e MajorMinorRevision. Uma Exception ou ausência de "
2296            "resposta a FC43/14 não invalida a descoberta do Slave.\n"
2297        )
2298        h(
2299            "O Finder e o Sniffer partilham o bloqueio global de atividade para impedir que "
2300            "a mesma aplicação faça pesquisa ativa e captura ao mesmo tempo.\n"
2301        )
2302
2303        h("Master — leitura e escrita como master Modbus RTU\n", "heading")
2304        h(
2305            "O separador Master transmite pedidos: só deve ser usado quando não existe "
2306            "outro master ativo no barramento. Ao ligar, a aplicação escuta o barramento "
2307            f"durante {DISCOVERY_IDLE_CHECK_SECONDS:g} s e não transmite se detetar tráfego. "
2308            "Enquanto o Master está ligado, o Sniffer e o Bus Slave Finder ficam bloqueados.\n"
2309        )
2310        h(
2311            "Cada janela lê uma zona de memória de um Slave: Slave ID, função (FC01 Coils, "
2312            "FC02 Discrete Inputs, FC03 Holding Registers ou FC04 Input Registers), endereço, "
2313            "quantidade e scan rate. Base 0 mostra o PDU Address; Base 1 numera a partir de 1, "
2314            "como em muitos manuais. As alterações só valem depois de “Aplicar”.\n"
2315        )
2316        h(
2317            "Todas as janelas partilham a mesma porta COM: o Modbus RTU só admite um pedido "
2318            "de cada vez, por isso as janelas são servidas à vez conforme o scan rate. "
2319            "Escritas e “Ler 1x” passam à frente das leituras cíclicas. Entre dois pedidos "
2320            "fica sempre o “Atraso entre pedidos” (no mínimo t3.5).\n"
```

MBSniffer v3.2/mb_gui.py — lines 2321-2360

```
        )
        h(
            "Formatos: 16 bits com e sem sinal, hex e binário; 32 bits com e sem sinal e "
            "float, nas ordens de bytes ABCD (big-endian), CDAB (palavras trocadas), BADC "
            "(bytes trocados) e DCBA (little-endian). Os valores de 32 bits ocupam dois "
            "registos e aparecem na linha do primeiro.\n"
        )
        h(
            "Escrita: duplo clique num valor de uma janela FC01 ou FC03. Coils com FC05 ou "
            "FC15, registos com FC06 ou FC16; vários valores separados por ";" são escritos "
            "em endereços consecutivos com FC15/FC16. Slave 0 envia em broadcast, sem "
            "resposta. Depois de uma escrita aceite, a janela é lida de novo.\n"
        )
        h(
            "Cada janela mostra Tx, erros, timeouts, o último tempo de resposta e o último "
            "erro (Timeout, Exception ou resposta inválida). "Destacar" abre a janela à parte "
            "do separador; "Ancorar" ou fechar essa janela devolve-a ao separador. "
            ""Guardar workspace" grava a ligação, todas as janelas, formatos e nomes num "
            "ficheiro .mbsw.\n"
        )

        h("Preferências guardadas\n", "heading")
        h(
            "As preferências do utilizador são guardadas em "
            "%APPDATA%\\MBSniffer\\settings.json. Incluem Light/Dark, último separador, "
            "parâmetros série, Frame gap, Pending timeout, Auto-scroll, Separar transações, "
            "Realçar resultados, opções do Bus Slave Finder e parâmetros de ligação do "
            "Master. As janelas do Master guardam-se num workspace (.mbsw).\n"
        )
        h(
            "As portas COM não são guardadas porque são específicas de cada computador. "
            "A confirmação de segurança do Bus Slave Finder também não é persistida.\n"
        )

        h("Boas práticas de diagnóstico\n", "heading")
        h("• Confirma primeiro cablagem, polaridade e parâmetros série.\n", "bullet")
        h("• Em RS485, mantém a derivação do sniffer curta e evita alterar a terminação existente.\n", "bullet")
        h("• Garante que nenhum outro programa tem a mesma COM aberta.\n", "bullet")
        h("• Usa CRC ERROR e RAW como indícios de framing, ruído, parâmetros errados ou captura incompleta.\n", "bullet")
        h("• Usa Exceptions como respostas válidas do protocolo: não são equivalentes a CRC error.\n", "bullet")
```

MBSniffer v3.2/mb_gui.py — lines 2361-2400

```
        h("• Compara tempos de resposta com o comportamento normal do equipamento, não apenas com um limite genérico.\n", "bullet")
        h("• Guarda o TXT quando precisares de análise posterior da captura completa.\n", "bullet")

        if debug_sim == 1:
            h("Modo Simulação\n", "heading")
            h(
                "Este modo de desenvolvimento aparece apenas quando debug_sim = 1 em "
                "MBSniffer.py. Não abre portas série e nunca cria logs TXT.\n"
            )
            h(
                "Cada execução usa sempre o mesmo tipo de cenário de diagnóstico: uma "
                "resposta normal, uma resposta lenta, uma Modbus Exception, um CRC error, "
                "um timeout e um bloco RAW/UNPARSED. Assim é possível comparar diretamente "
                "a mesma informação com e sem destaque visual.\n"
            )
            h(
                "“Realçar resultados” não altera os dados gerados pela Simulação. Desligado, "
                "todas as linhas usam as cores normais da tabela; ligado, a transação normal "
                "concluída fica verde e as linhas de Exception, CRC error, timeout, resposta "
                "lenta e RAW recebem as respetivas cores de diagnóstico.\n"
            )
            h(
                "As frames deste cenário usam o canal SIM para não serem confundidas com uma "
                "captura física real.\n"
            )

        h("Limitação importante\n", "heading")
        h(
            "O MBSniffer interpreta bytes, CRC, Function Code, endereços, quantidades, "
            "emparelhamento e tempos observados pelo sistema operativo. Não mede diretamente "
            "níveis elétricos, ringing, reflexões, bias, common-mode, tempos de subida/descida "
            "ou integridade analógica do sinal. Para esses problemas continua a ser necessário "
            "um osciloscópio ou analisador adequado à interface física.\n"
        )

        self.help_text.configure(state="disabled")

        detail = ttk.LabelFrame(self.sniffer_page, text="Frame Inspector")
        detail.grid(row=2, column=0, sticky="ew", padx=10, pady=(0, 10))
        detail.columnconfigure(0, weight=1)
```

MBSniffer v3.2/mb_gui.py — lines 2401-2440

```
2401
2402        inspector_header = ttk.Frame(detail)
2403        inspector_header.grid(
2404            row=0, column=0, sticky="ew", padx=8, pady=(4, 2)
2405        )
2406        inspector_header.columnconfigure(0, weight=1)
2407        ttk.Label(
2408            inspector_header,
2409            text=(
2410                "Seleciona uma linha no Tráfego para ver a descodificação "
2411                "estruturada do frame."
2412            ),
2413        ).grid(row=0, column=0, sticky="w")
2414
2415        self.inspector_collapsed = False
2416        self.inspector_toggle_btn = ttk.Button(
2417            inspector_header,
2418            text="Recolher",
2419            command=self.toggle_frame_inspector,
2420            style="MBS.Compact.TButton",
2421        )
2422        self.inspector_toggle_btn.grid(row=0, column=1, sticky="e")
2423
2424        self.inspector_body = ttk.Frame(detail)
2425        self.inspector_body.grid(
2426            row=1, column=0, sticky="ew", padx=8, pady=(2, 7)
2427        )
2428        for col in range(4):
2429            self.inspector_body.columnconfigure(
2430                col,
2431                weight=1,
2432                uniform="inspector",
2433            )
2434
2435        self.inspector_vars = {
2436            "slave": tk.StringVar(value="Slave: —"),
2437            "type": tk.StringVar(value="Tipo: —"),
2438            "function": tk.StringVar(value="Function: —"),
2439            "response": tk.StringVar(value="Resp.: —"),
2440            "result": tk.StringVar(value="Resultado: —"),
```

MBSniffer v3.2/mb_gui.py — lines 2441-2480

```
            "pdu": tk.StringVar(value="PDU Address: —"),
            "one_based": tk.StringVar(value="1-based: —"),
            "qty": tk.StringVar(value="Qty: —"),
            "byte_count": tk.StringVar(value="ByteCount: —"),
            "crc": tk.StringVar(value="CRC: —"),
            "data": tk.StringVar(value="Dados: —"),
        }

        inspector_grid = (
            ("slave", 0, 0),
            ("type", 0, 1),
            ("function", 0, 2),
            ("response", 0, 3),
            ("pdu", 1, 0),
            ("one_based", 1, 1),
            ("qty", 1, 2),
            ("byte_count", 1, 3),
        )
        for key, row, col in inspector_grid:
            ttk.Label(
                self.inspector_body,
                textvariable=self.inspector_vars[key],
                anchor="w",
            ).grid(
                row=row,
                column=col,
                sticky="ew",
                padx=(0 if col == 0 else 8, 8 if col < 3 else 0),
                pady=2,
            )

        self.inspector_result_label = ttk.Label(
            self.inspector_body,
            textvariable=self.inspector_vars["result"],
            anchor="w",
            style="MBS.Result.neutral.TLabel",
        )
        self.inspector_result_label.grid(
            row=2,
            column=0,
```

MBSniffer v3.2/mb_gui.py — lines 2481-2520

```
            columnspan=4,
            sticky="ew",
            pady=(3, 2),
        )

        ttk.Label(
            self.inspector_body,
            textvariable=self.inspector_vars["crc"],
            anchor="w",
        ).grid(
            row=3, column=0, columnspan=2, sticky="ew", pady=2
        )
        # Data/Register payloads can be very long. A readonly Entry keeps the
        # inspector compact instead of letting content change requested width.
        ttk.Entry(
            self.inspector_body,
            textvariable=self.inspector_vars["data"],
            state="readonly",
        ).grid(
            row=3,
            column=2,
            columnspan=2,
            sticky="ew",
            padx=(8, 0),
            pady=2,
        )

        self.selected_raw_var = tk.StringVar()
        ttk.Entry(
            self.inspector_body,
            textvariable=self.selected_raw_var,
            state="readonly",
        ).grid(
            row=4,
            column=0,
            columnspan=4,
            sticky="ew",
            pady=(4, 0),
        )

```

MBSniffer v3.2/mb_gui.py — lines 2521-2529

```
        self.build_discovery_tab(self.discovery_page)
        self.build_master_tab(self.master_page)
        self.update_stats_labels()
        self.update_frame_inspector(None)

        # Classic Tk/Canvas widgets are outside ttk.Style; apply the remembered
        # startup palette after every such widget has been created.
        self._apply_classic_widget_palette(self.initial_dark_mode)

```

MBSniffer v3.2/tests/test_regression.py

787 source lines.

MBSniffer v3.2/tests/test_regression.py — lines 1-40

```
import json
import queue
import random
import tempfile
import threading
import time
import unittest
from collections import deque
from pathlib import Path

import mb_config
import mb_slave_finder
from mb_capture import CaptureMixin
from mb_protocol import (
    SessionMetrics,
    append_crc,
    burst_may_continue,
    crc16_modbus,
    crc_ok,
    modbus_rtu_interframe_gap_seconds,
    parse_burst,
    read_bits_details,
    split_capture_buffer,
)
from mb_slave_finder import SlaveFinderMixin
from mb_view import ViewMixin
from mb_diagnostics import inspector_values
from mb_i18n import (
    LANGUAGE_ENGLISH,
    LANGUAGE_PORTUGUESE,
    english_help_blocks,
    normalize_language,
    translate_details,
    translate_runtime_text,
    translate_text,
)

from mb_slave_finder import (
    build_device_identification_probe,
    build_discovery_probe,
```

MBSniffer v3.2/tests/test_regression.py — lines 41-80

```
41      device_identification_timeout_seconds,
42      find_device_identification_response,
43      find_discovery_response,
44      format_device_identification,
45      read_basic_device_identification,
46  )
47
48
49  def with_bad_crc(frame: bytes) -> bytes:
50      damaged = bytearray(frame)
51      damaged[-1] ^= 0xFF
52      return bytes(damaged)
53
54
55  def device_id_response(
56      slave=1,
57      objects=None,
58      *,
59      more_follows=False,
60      next_object_id=0,
61      conformity=0x01,
62  ):
63      if objects is None:
64          objects = {
65              0x00: b"Schneider Electric",
66              0x01: b"TEST-PRODUCT",
67              0x02: b"1.2.3",
68          }
69      payload = bytearray((
70          slave,
71          0x2B,
72          0x0E,
73          0x01,
74          conformity,
75          0xFF if more_follows else 0x00,
76          next_object_id,
77          len(objects),
78      ))
79      for object_id, value in objects.items():
80          value = bytes(value)
```

MBSniffer v3.2/tests/test_regression.py — lines 81-120

```
81          payload.extend((int(object_id) & 0xFF, len(value)))
82          payload.extend(value)
83      return append_crc(bytes(payload))
84
85
86  class FakeSerial:
87      """Minimal non-blocking serial fake for active Device ID regression tests."""
88
89      def __init__(self, response_by_object):
90          self.response_by_object = dict(response_by_object)
91          self.rx = bytearray()
92          self.writes = []
93
94      @property
95      def in_waiting(self):
96          return len(self.rx)
97
98      def reset_input_buffer(self):
99          self.rx.clear()
100
101     def reset_output_buffer(self):
102         pass
103
104     def write(self, data):
105         data = bytes(data)
106         self.writes.append(data)
107         object_id = data[4] if len(data) >= 5 else 0
108         response = self.response_by_object.get(object_id, b"")
109         self.rx.extend(response)
110         return len(data)
111
112     def read(self, size):
113         size = min(int(size), len(self.rx))
114         chunk = bytes(self.rx[:size])
115         del self.rx[:size]
116         return chunk
117
118
119 class FakeSerialModule:
120     class SerialTimeoutException(TimeoutError):
```

MBSniffer v3.2/tests/test_regression.py — lines 121-160

```
121         pass
122
123     class SerialException(OSError):
124         pass
125
126
127 class StaticVar:
128     def __init__(self, value):
129         self.value = value
130
131     def get(self):
132         return self.value
133
134
135 class DummyCapture:
136     calculated_gap_ms = CaptureMixin.calculated_gap_ms
137
138     def __init__(self, *, mode="Auto", baud="9600", data="8", parity="None", stop="1", manual="2.0"):
139         self.gap_mode_var = StaticVar(mode)
140         self.baud_var = StaticVar(baud)
141         self.data_var = StaticVar(data)
142         self.parity_var = StaticVar(parity)
143         self.stop_var = StaticVar(stop)
144         self.gap_ms_var = StaticVar(manual)
145
146
147 class TimingRegressionTests(unittest.TestCase):
148     def test_t35_uses_character_time_at_or_below_19200(self):
149         self.assertAlmostEqual(
150             modbus_rtu_interframe_gap_seconds(9600, 11),
151             3.5 * 11 / 9600,
152             places=9,
153         )
154         self.assertAlmostEqual(
155             modbus_rtu_interframe_gap_seconds(19200, 11),
156             3.5 * 11 / 19200,
157             places=9,
158         )
159
160     def test_t35_is_fixed_above_19200(self):
```

MBSniffer v3.2/tests/test_regression.py — lines 161-200

```
161         for baud in (38400, 57600, 115200):
162             with self.subTest(baud=baud):
163                 self.assertAlmostEqual(
164                     modbus_rtu_interframe_gap_seconds(baud, 10),
165                     0.001750,
166                     places=9,
167                 )
168 
169     def test_sniffer_auto_gap_uses_fixed_1750_us_above_19200(self):
170         capture = DummyCapture(baud="115200", parity="None", stop="1")
171         self.assertAlmostEqual(capture.calculated_gap_ms(), 1.750, places=6)
172 
173     def test_sniffer_manual_gap_is_unchanged(self):
174         capture = DummyCapture(mode="Manual", manual="2,75")
175         self.assertAlmostEqual(capture.calculated_gap_ms(), 2.75, places=6)
176 
177     def test_device_id_timeout_covers_a_full_rtu_frame(self):
178         timeout = device_identification_timeout_seconds(9600, "None", "1", 40)
179         self.assertGreaterEqual(timeout, 256 * 10 / 9600 + 0.001750)
180 
181 
182 class ParserRegressionTests(unittest.TestCase):
183     def test_crc_known_request(self):
184         frame = build_discovery_probe(1, 3)
185         self.assertTrue(crc_ok(frame))
186         self.assertEqual(frame[:6], bytes.fromhex("01 03 00 00 00 01"))
187 
188     def test_invalid_frame_before_valid_frame_stays_crc_error(self):
189         first = with_bad_crc(build_discovery_probe(1, 3))
190         second = build_discovery_probe(2, 4)
191         parts = split_capture_buffer(first + second)
192         self.assertEqual(parts[0], (first, "REQUEST", "ERROR"))
193         self.assertEqual(parts[1], (second, "REQUEST", "OK"))
194 
195     def test_long_unsynchronised_noise_recovers_next_valid_frame(self):
196         noise = bytes([0xFF]) * 4000
197         frame = build_discovery_probe(7, 3)
198         parts = split_capture_buffer(noise + frame)
199         self.assertEqual(parts[-1], (frame, "REQUEST", "OK"))
200         self.assertTrue(parts[0][1].startswith("RAW/"))
```

MBSniffer v3.2/tests/test_regression.py — lines 201-240

```
201         self.assertEqual(sum(len(part[0]) for part in parts), len(noise) + len(frame))
202
203     def test_supported_function_code_shapes(self):
204         frames = {
205             "fc01_req": append_crc(bytes.fromhex("01 01 00 00 00 08")),
206             "fc01_resp": append_crc(bytes.fromhex("01 01 01 5A")),
207             "fc02_req": append_crc(bytes.fromhex("01 02 00 00 00 08")),
208             "fc02_resp": append_crc(bytes.fromhex("01 02 01 A5")),
209             "fc03_req": append_crc(bytes.fromhex("01 03 00 00 00 01")),
210             "fc03_resp": append_crc(bytes.fromhex("01 03 02 00 2A")),
211             "fc04_req": append_crc(bytes.fromhex("01 04 00 00 00 01")),
212             "fc04_resp": append_crc(bytes.fromhex("01 04 02 00 2A")),
213             "fc05": append_crc(bytes.fromhex("01 05 00 01 FF 00")),
214             "fc06": append_crc(bytes.fromhex("01 06 00 01 00 2A")),
215             "fc07_req": append_crc(bytes.fromhex("01 07")),
216             "fc07_resp": append_crc(bytes.fromhex("01 07 AA")),
217             "fc08": append_crc(bytes.fromhex("01 08 00 00 12 34")),
218             "fc15_req": append_crc(bytes.fromhex("01 0F 00 00 00 08 01 5A")),
219             "fc15_resp": append_crc(bytes.fromhex("01 0F 00 00 00 08")),
220             "fc16_req": append_crc(bytes.fromhex("01 10 00 00 00 01 02 00 2A")),
221             "fc16_resp": append_crc(bytes.fromhex("01 10 00 00 00 01")),
222             "fc22": append_crc(bytes.fromhex("01 16 00 01 FF 00 00 2A")),
223             "fc23_req": append_crc(bytes.fromhex(
224                 "01 17 00 00 00 01 00 10 00 01 02 00 2A"
225             )),
226             "fc23_resp": append_crc(bytes.fromhex("01 17 02 00 2A")),
227             "exception": append_crc(bytes.fromhex("01 83 02")),
228         }
229         for name, frame in frames.items():
230             with self.subTest(name=name):
231                 parts = split_capture_buffer(frame)
232                 self.assertEqual(len(parts), 1)
233                 self.assertEqual(parts[0][0], frame)
234                 self.assertEqual(parts[0][2], "OK")
235                 self.assertFalse(parts[0][1].startswith("RAW"))
236
237     def test_random_noise_never_crashes_and_preserves_all_bytes(self):
238         rng = random.Random(20260910)
239         for _ in range(250):
240             length = rng.randint(1, 512)
```

MBSniffer v3.2/tests/test_regression.py — lines 241-280

```
241             data = bytes(rng.randrange(256) for _ in range(length))
242             parts = split_capture_buffer(data)
243             self.assertEqual(sum(len(part[0]) for part in parts), len(data))
244             self.assertTrue(all(len(part[0]) > 0 for part in parts))
245
246     def test_request_response_pairing_and_response_time(self):
247         req = parse_burst(
248             append_crc(bytes.fromhex("01 03 00 00 00 01")),
249             "12:00:00.000",
250             "BUS",
251             100.000,
252         )[0]
253         resp = parse_burst(
254             append_crc(bytes.fromhex("01 03 02 00 2A")),
255             "12:00:00.125",
256             "BUS",
257             100.125,
258         )[0]
259         req["_view_seq"] = 10
260         resp["_view_seq"] = 11
261
262         metrics = SessionMetrics(pending_timeout_seconds=1.0)
263         first = metrics.process(req)
264         second = metrics.process(resp)
265
266         self.assertEqual(first["pending"], 1)
267         self.assertEqual(second["pending"], 0)
268         self.assertEqual(second["matched_request"]["seq"], 10)
269         self.assertAlmostEqual(second["response_ms"], 125.0, places=6)
270
271     def test_timeout_regression(self):
272         req = parse_burst(
273             append_crc(bytes.fromhex("05 04 00 00 00 01")),
274             "12:00:00.000",
275             "BUS",
276             10.0,
277         )[0]
278         req["_view_seq"] = 77
279         metrics = SessionMetrics(pending_timeout_seconds=1.0)
280         metrics.process(req)
```

MBSniffer v3.2/tests/test_regression.py — lines 281-320

```
281         self.assertEqual(metrics.expire_pending(11.1), 1)
282         expired = metrics.pop_expired_requests()
283         self.assertEqual(expired[0]["seq"], 77)
284         self.assertEqual(metrics.timeouts, 1)
285
286
287 class SlaveFinderRegressionTests(unittest.TestCase):
288     def test_normal_discovery_response(self):
289         response = append_crc(bytes.fromhex("04 03 02 12 34"))
290         found = find_discovery_response(response, 4, 3)
291         self.assertIsNotNone(found)
292         self.assertEqual(found["kind"], "RESPONSE")
293
294     def test_discovery_exception_counts_as_response(self):
295         response = append_crc(bytes.fromhex("04 83 02"))
296         found = find_discovery_response(response, 4, 3)
297         self.assertIsNotNone(found)
298         self.assertEqual(found["kind"], "EXCEPTION")
299         self.assertEqual(found["exception_code"], 2)
300
301     def test_device_identification_request(self):
302         request = build_device_identification_probe(4)
303         self.assertEqual(request[:5], bytes.fromhex("04 2B 0E 01 00"))
304         self.assertEqual(len(request), 7)
305         self.assertTrue(crc_ok(request))
306
307     def test_device_identification_response_parser(self):
308         response = device_id_response(slave=4)
309         found = find_device_identification_response(response, 4)
310         self.assertIsNotNone(found)
311         self.assertEqual(found["kind"], "DEVICE_ID")
312         self.assertEqual(found["objects"][0x00], "Schneider Electric")
313         self.assertEqual(found["objects"][0x01], "TEST-PRODUCT")
314         self.assertEqual(found["objects"][0x02], "1.2.3")
315         self.assertEqual(
316             format_device_identification(found),
317             "Schneider Electric | TEST-PRODUCT | 1.2.3",
318         )
319
320     def test_device_identification_parser_ignores_request_echo(self):
```

MBSniffer v3.2/tests/test_regression.py — lines 321-360

```
        request = build_device_identification_probe(4)
        response = device_id_response(slave=4)
        found = find_device_identification_response(request + response, 4)
        self.assertIsNotNone(found)
        self.assertEqual(found["objects"][0x01], "TEST-PRODUCT")

    def test_device_identification_exception(self):
        response = append_crc(bytes.fromhex("04 AB 01"))
        found = find_device_identification_response(response, 4)
        self.assertIsNotNone(found)
        self.assertEqual(found["kind"], "EXCEPTION")
        self.assertEqual(found["exception_code"], 1)
        self.assertIn("Illegal Function", format_device_identification(found))

    def test_device_identification_rejects_bad_crc(self):
        response = with_bad_crc(device_id_response(slave=4))
        self.assertIsNone(find_device_identification_response(response, 4))

    def test_device_identification_follows_more_follows(self):
        page0 = device_id_response(
            slave=4,
            objects={0x00: b"Vendor"},
            more_follows=True,
            next_object_id=0x01,
        )
        page1 = device_id_response(
            slave=4,
            objects={0x01: b"Product", 0x02: b"2.0"},
            more_follows=False,
            next_object_id=0x00,
        )
        serial = FakeSerial({0x00: page0, 0x01: page1})
        result = read_basic_device_identification(
            serial,
            4,
            115200,
            "None",
            "1",
            10,
            threading.Event(),
```

MBSniffer v3.2/tests/test_regression.py — lines 361-400

```
361             FakeSerialModule,
362         )
363         self.assertIsNotNone(result)
364         self.assertEqual(result["objects"][0x00], "Vendor")
365         self.assertEqual(result["objects"][0x01], "Product")
366         self.assertEqual(result["objects"][0x02], "2.0")
367         self.assertEqual(len(serial.writes), 2)
368         self.assertEqual(serial.writes[0][4], 0x00)
369         self.assertEqual(serial.writes[1][4], 0x01)
370
371
372 class InternationalizationRegressionTests(unittest.TestCase):
373     def test_language_normalization_defaults_to_portuguese(self):
374         self.assertEqual(normalize_language("invalid"), LANGUAGE_PORTUGUESE)
375         self.assertEqual(normalize_language(LANGUAGE_ENGLISH), LANGUAGE_ENGLISH)
376
377     def test_static_ui_translation_is_reversible(self):
378         self.assertEqual(translate_text("Configuração", LANGUAGE_ENGLISH), "Configuration")
379         self.assertEqual(translate_text("Configuration", LANGUAGE_PORTUGUESE), "Configuração")
380         self.assertEqual(translate_text("Frame gap", LANGUAGE_ENGLISH), "Frame gap")
381         # "None" is a technical parity value and must remain English in the PT UI.
382         self.assertEqual(translate_text("None", LANGUAGE_PORTUGUESE), "None")
383         self.assertEqual(translate_text("Nenhum", LANGUAGE_ENGLISH), "None")
384
385     def test_dynamic_status_translation_is_reversible(self):
386         pt = "Pesquisa concluída — 7 encontrados — 00:12"
387         en = "Scan complete — 7 found — 00:12"
388         self.assertEqual(translate_runtime_text(pt, LANGUAGE_ENGLISH), en)
389         self.assertEqual(translate_runtime_text(en, LANGUAGE_PORTUGUESE), pt)
390
391     def test_register_details_follow_language(self):
392         self.assertEqual(
393             translate_details("ByteCount=2  Registos=[42]", LANGUAGE_ENGLISH),
394             "ByteCount=2  Registers=[42]",
395         )
396         self.assertEqual(
397             translate_details("ByteCount=2  Registers=[42]", LANGUAGE_PORTUGUESE),
398             "ByteCount=2  Registos=[42]",
399         )
400
```

MBSniffer v3.2/tests/test_regression.py — lines 401-440

```
401     def test_inspector_english_labels_and_crc(self):
402         frame = append_crc(bytes.fromhex("01 03 02 00 2A"))
403         row = parse_burst(frame, "12:00:00.000", "BUS", 1.0)[0]
404         row.update({"response_text": "12.3", "matched": True})
405         values = inspector_values(row, language=LANGUAGE_ENGLISH)
406         self.assertTrue(values["type"].startswith("Type:"))
407         self.assertTrue(values["result"].startswith("Result:"))
408         self.assertTrue(values["data"].startswith("Data:"))
409         self.assertIn("Registers=[42]", values["data"])
410         self.assertIn("received", values["crc"])
411         self.assertIn("calculated", values["crc"])
412 
413     def test_english_help_contains_no_portuguese_diacritics(self):
414         text = "".join(
415             block for _tag, block in english_help_blocks(
416                 "20 000", "20 000", 1.0, 300.0, debug_sim=True
417             )
418         )
419         for character in "ãõçáéíóúâê":
420             self.assertNotIn(character, text.casefold())
421         self.assertIn("Bus Slave Finder — active scan", text)
422         self.assertIn("Device Identification (FC43/14)", text)
423 
424 
425 def bitwise_crc16(data: bytes) -> int:
426     """Reference CRC used up to v3.0, kept to verify the table-driven CRC."""
427     crc = 0xFFFF
428     for b in data:
429         crc ^= b
430         for _ in range(8):
431             crc = (crc >> 1) ^ 0xA001 if crc & 1 else crc >> 1
432     return crc
433 
434 
435 def process_frames(metrics, frames):
436     """Run (bytes, channel, epoch) frames through SessionMetrics."""
437     out = []
438     for index, (frame, channel, epoch) in enumerate(frames):
439         for row in parse_burst(frame, "12:00:00.000", channel, epoch):
440             row["_view_seq"] = index
```

MBSniffer v3.2/tests/test_regression.py — lines 441-480

```
441             out.append((row, metrics.process(row)))
442     return out
443
444
445 class ProtocolV31RegressionTests(unittest.TestCase):
446     def test_crc_table_matches_bitwise_reference(self):
447         rng = random.Random(31)
448         for length in (0, 1, 2, 7, 64, 256):
449             data = bytes(rng.randrange(256) for _ in range(length))
450             self.assertEqual(crc16_modbus(data), bitwise_crc16(data))
451
452     def test_broadcast_request_never_times_out(self):
453         request = append_crc(bytes.fromhex("00 10 00 00 00 01 02 00 05"))
454         metrics = SessionMetrics(pending_timeout_seconds=1.0)
455         process_frames(metrics, [(request, "BUS", 10.0)])
456         self.assertEqual(metrics.requests, 1)
457         self.assertEqual(metrics.broadcast_requests, 1)
458         self.assertEqual(metrics.pending_count(), 0)
459         metrics.expire_pending(20.0)
460         self.assertEqual(metrics.timeouts, 0)
461         self.assertEqual(metrics.health_summary(now=20.0)["timeout_rate_pct"], 0.0)
462
463     def test_fc06_echo_pair_is_request_then_response(self):
464         write = append_crc(bytes.fromhex("01 06 00 01 00 09"))
465         metrics = SessionMetrics(pending_timeout_seconds=1.0)
466         results = process_frames(
467             metrics, [(write, "BUS", 100.0), (write, "BUS", 100.040)]
468         )
469         self.assertEqual(
470             [row["type"] for row, _timing in results], ["REQUEST", "RESPONSE"]
471         )
472         self.assertEqual((metrics.requests, metrics.responses), (1, 1))
473         self.assertAlmostEqual(results[1][1]["response_ms"], 40.0, places=6)
474
475     def test_unanswered_fc05_write_times_out(self):
476         write = append_crc(bytes.fromhex("02 05 00 10 FF 00"))
477         metrics = SessionMetrics(pending_timeout_seconds=1.0)
478         results = process_frames(metrics, [(write, "BUS", 5.0)])
479         self.assertEqual(results[0][0]["type"], "REQUEST")
480         self.assertEqual(metrics.expire_pending(6.5), 1)
```

MBSniffer v3.2/tests/test_regression.py — lines 481-520

```
481
482     def test_different_fc06_writes_are_both_requests(self):
483         first = append_crc(bytes.fromhex("01 06 00 01 00 09"))
484         second = append_crc(bytes.fromhex("01 06 00 02 00 0A"))
485         metrics = SessionMetrics()
486         results = process_frames(
487             metrics, [(first, "BUS", 1.0), (second, "BUS", 1.1)]
488         )
489         self.assertEqual(
490             [row["type"] for row, _timing in results], ["REQUEST", "REQUEST"]
491         )
492
493     def test_dual_rx_echo_must_come_from_opposite_channel(self):
494         write = append_crc(bytes.fromhex("01 06 00 01 00 09"))
495         metrics = SessionMetrics()
496         results = process_frames(metrics, [
497             (write, "A→B", 1.0),
498             (write, "A→B", 1.5),
499             (write, "B→A", 1.52),
500         ])
501         self.assertEqual(
502             [row["type"] for row, _timing in results],
503             ["REQUEST", "REQUEST", "RESPONSE"],
504         )
505
506     def test_split_frame_is_held_until_complete(self):
507         response = append_crc(bytes.fromhex("01 03 08") + bytes(8))
508         self.assertTrue(burst_may_continue(response[:7]))
509         self.assertTrue(burst_may_continue(response[:1]))
510         self.assertFalse(burst_may_continue(response))
511         self.assertFalse(burst_may_continue(bytes((0xFF,) * 6)))
512
513     def test_merged_frames_keep_their_chunk_arrival_times(self):
514         request = append_crc(bytes.fromhex("01 03 00 00 00 01"))
515         response = append_crc(bytes.fromhex("01 03 02 00 2A"))
516         rows = parse_burst(
517             request + response,
518             "12:00:00.000",
519             "BUS",
520             100.0,
```

MBSniffer v3.2/tests/test_regression.py — lines 521-560

```
            chunk_times=[(0, 100.0), (len(request), 100.030)],
        )
        self.assertEqual([row["type"] for row in rows], ["REQUEST", "RESPONSE"])
        self.assertAlmostEqual(rows[1]["_epoch"], 100.030, places=6)
        metrics = SessionMetrics()
        metrics.process(rows[0])
        self.assertAlmostEqual(metrics.process(rows[1])["response_ms"], 30.0, places=6)

    def test_read_bits_details_use_request_qty(self):
        response = append_crc(bytes.fromhex("01 01 02 CD 01"))
        self.assertEqual(
            read_bits_details(response),
            "ByteCount=2  Bits=10110011 10000000",
        )
        self.assertEqual(
            read_bits_details(response, qty=10),
            "ByteCount=2  Bits=10110011 10",
        )


class _ChunkedSerial:
    """Delivers byte chunks at fixed delays, like a USB latency timer."""

    def __init__(self, chunks, stop_event):
        self.started = time.perf_counter()
        self.chunks = list(chunks)  # [(delay_s, bytes)]
        self.stop_event = stop_event

    def _due(self):
        elapsed = time.perf_counter() - self.started
        if self.chunks and elapsed >= self.chunks[0][0]:
            return self.chunks[0][1]
        if not self.chunks and elapsed > 0.3:
            self.stop_event.set()
        return b""

    @property
    def in_waiting(self):
        return len(self._due())

```

MBSniffer v3.2/tests/test_regression.py — lines 561-600

```
561     def read(self, size):
562         data = self._due()
563         if data:
564             self.chunks.pop(0)
565         return data
566 
567     def close(self):
568         pass
569 
570 
571 class _ReaderStub(CaptureMixin):
572     def __init__(self):
573         self.stop_event = threading.Event()
574         self.event_queue = queue.Queue()
575 
576 
577 class ReaderV31RegressionTests(unittest.TestCase):
578     def test_frame_split_by_usb_latency_is_reassembled(self):
579         response = append_crc(bytes.fromhex("01 03 08") + bytes(8))
580         reader = _ReaderStub()
581         serial = _ChunkedSerial(
582             [(0.0, response[:7]), (0.016, response[7:])], reader.stop_event
583         )
584         # 115200 bit/s: Auto t3.5 is 1.75 ms, far shorter than the 16 ms split.
585         reader.reader_loop(serial, 1.75, "BUS", 1)
586         rows = []
587         while not reader.event_queue.empty():
588             name, payload = reader.event_queue.get()
589             if name == "frame":
590                 rows.append(payload)
591         self.assertEqual([(row["type"], row["crc"]) for row in rows], [("RESPONSE", "OK")])
592 
593 
594 class _FakeVar:
595     def __init__(self, value):
596         self.value = value
597 
598     def get(self):
599         return self.value
600 
```

MBSniffer v3.2/tests/test_regression.py — lines 601-640

```
601
602 class _FakeTree:
603     def __init__(self):
604         self.tags = {}
605
606     def item(self, iid, tags=()):
607         self.tags[iid] = tags
608
609
610 class _TimeoutView(ViewMixin):
611     def __init__(self, type_filter):
612         self.tree = _FakeTree()
613         self.advanced_filter_type_var = _FakeVar(type_filter)
614         self.highlight_anomalies_var = _FakeVar(True)
615         self._successful_request_seqs = set()
616         self._frame_history = deque([{
617             "seq": 5, "type": "REQUEST", "crc": "OK", "timed_out": False,
618             "response_text": "",
619         }])
620         self._tree_iids_by_seq = {5: ["I001", "I002"]}
621         self.full_renders = 0
622
623     def render_filtered_view(self):
624         self.full_renders += 1
625
626     def selected_record(self):
627         return None
628
629
630 class ViewV31RegressionTests(unittest.TestCase):
631     def test_timeout_retags_rows_without_full_rebuild(self):
632         view = _TimeoutView("Todos")
633         self.assertEqual(view.mark_timed_out_records([{"seq": 5}]), 1)
634         self.assertEqual(view.full_renders, 0)
635         self.assertEqual(view.tree.tags["I001"], ("anomaly_timeout",))
636         self.assertEqual(view.tree.tags["I002"], ("anomaly_timeout",))
637
638     def test_timeouts_filter_still_rebuilds_view(self):
639         view = _TimeoutView("Timeouts")
640         view.mark_timed_out_records([{"seq": 5}])
```

MBSniffer v3.2/tests/test_regression.py — lines 641-680

```
641         self.assertEqual(view.full_renders, 1)
642
643     def test_inspector_shows_address_of_each_register(self):
644         frame = append_crc(bytes.fromhex("01 03 04 00 2A 00 2B"))
645         row = parse_burst(frame, "12:00:00.000", "BUS", 1.0)[0]
646         row["request_pdu_address"] = 100
647         pt = inspector_values(row, language=LANGUAGE_PORTUGUESE)
648         en = inspector_values(row, language=LANGUAGE_ENGLISH)
649         self.assertIn("Registos (PDU Address=valor): 100=42, 101=43", pt["data"])
650         self.assertIn("Registers (PDU Address=value): 100=42, 101=43", en["data"])
651
652     def test_inspector_shows_address_of_each_bit(self):
653         frame = append_crc(bytes.fromhex("01 02 01 05"))
654         row = parse_burst(frame, "12:00:00.000", "BUS", 1.0)[0]
655         row.update({"request_pdu_address": 8, "request_qty": 3})
656         values = inspector_values(row, language=LANGUAGE_PORTUGUESE)
657         self.assertIn("Bits (PDU Address=valor): 8=1, 9=0, 10=1", values["data"])
658
659
660 class _FakeFinderSerial:
661     def __init__(self, incoming=b""):
662         self.incoming = bytearray(incoming)
663         self.writes = []
664
665     @property
666     def in_waiting(self):
667         return len(self.incoming)
668
669     def read(self, size):
670         data = bytes(self.incoming[:size])
671         del self.incoming[:size]
672         return data
673
674     def write(self, data):
675         self.writes.append(bytes(data))
676         return len(data)
677
678     def reset_input_buffer(self):
679         pass
680
```

MBSniffer v3.2/tests/test_regression.py — lines 681-720

```
681     def reset_output_buffer(self):
682         pass
683 
684     def close(self):
685         pass
686 
687 
688 class _FinderStub(SlaveFinderMixin):
689     current_language = LANGUAGE_PORTUGUESE
690 
691     def __init__(self, fake_serial):
692         self.fake_serial = fake_serial
693         self.event_queue = queue.Queue()
694         self.discovery_stop_event = threading.Event()
695         self.discovery_serial = None
696 
697     def tr(self, text):
698         return translate_text(text, self.current_language)
699 
700     def trf(self, text, **values):
701         return self.tr(text).format(**values)
702 
703     def open_discovery_serial_port(self, serial, port, baud, parity, stopbits):
704         return self.fake_serial
705 
706     def run_worker(self, slaves, attempts):
707         self.discovery_worker(
708             None, "COM9", [(115200, "None", "1")], slaves, 10, False, False,
709             attempts,
710         )
711         events = []
712         while not self.event_queue.empty():
713             events.append(self.event_queue.get())
714         return events
715 
716 
717 class FinderV31RegressionTests(unittest.TestCase):
718     def setUp(self):
719         self._idle_check = mb_slave_finder.DISCOVERY_IDLE_CHECK_SECONDS
720         mb_slave_finder.DISCOVERY_IDLE_CHECK_SECONDS = 0.05
```

MBSniffer v3.2/tests/test_regression.py — lines 721-760

```
721
722     def tearDown(self):
723         mb_slave_finder.DISCOVERY_IDLE_CHECK_SECONDS = self._idle_check
724
725     def test_each_silent_slave_gets_all_attempts(self):
726         stub = _FinderStub(_FakeFinderSerial())
727         events = stub.run_worker([1, 2], attempts=3)
728         done = [payload for name, payload in events if name == "discovery_done"][0]
729         self.assertIsNone(done["error"])
730         self.assertEqual(len(stub.fake_serial.writes), 6)
731
732     def test_traffic_before_scan_blocks_transmission(self):
733         stub = _FinderStub(_FakeFinderSerial(b"\x01\x03\x00"))
734         events = stub.run_worker([1], attempts=1)
735         done = [payload for name, payload in events if name == "discovery_done"][0]
736         self.assertIn("Foi detetado tráfego em COM9", done["error"])
737         self.assertEqual(stub.fake_serial.writes, [])
738
739     def test_legacy_default_timeout_migrates(self):
740         with tempfile.TemporaryDirectory() as folder:
741             path = Path(folder) / "settings.json"
742             original = mb_config.get_settings_file
743             mb_config.get_settings_file = lambda: path
744             try:
745                 path.write_text(
746                     json.dumps({"bus_slave_finder": {"min_timeout_ms": 40}}),
747                     encoding="utf-8",
748                 )
749                 finder = mb_config.load_ui_settings()["bus_slave_finder"]
750                 self.assertEqual(finder["min_timeout_ms"], "100")
751                 self.assertEqual(finder["attempts"], "1")
752
753                 path.write_text(
754                     json.dumps({"bus_slave_finder": {
755                         "min_timeout_ms": 40, "attempts": 2,
756                     }}),
757                     encoding="utf-8",
758                 )
759                 finder = mb_config.load_ui_settings()["bus_slave_finder"]
760                 self.assertEqual(finder["min_timeout_ms"], "40")
```

MBSniffer v3.2/tests/test_regression.py — lines 761-787

```
                self.assertEqual(finder["attempts"], "2")
            finally:
                mb_config.get_settings_file = original


class SourceHygieneV31Tests(unittest.TestCase):
    def test_no_inline_language_branches(self):
        package = Path(__file__).resolve().parents[1]
        for source in package.glob("*.py"):
            text = source.read_text(encoding="utf-8")
            self.assertNotIn(
                'current_language == "English"', text, source.name
            )

    def test_message_templates_translate(self):
        self.assertEqual(
            translate_text("Encontrados: {count}", LANGUAGE_ENGLISH).format(count=3),
            "Found: 3",
        )
        self.assertEqual(
            translate_text("A verificar se o barramento está livre…", LANGUAGE_ENGLISH),
            "Checking that the bus is idle…",
        )


if __name__ == "__main__":
    unittest.main(verbosity=2)
```

MBSniffer v3.2/tests/test_master.py

300 source lines.

MBSniffer v3.2/tests/test_master.py — lines 1-40

```python
import queue
import struct
import time
import unittest

from mb_master import (
    BYTE_ORDERS,
    MasterEngine,
    PollDefinition,
    build_read_request,
    build_write_request,
    encode_coil_value,
    encode_register_value,
    find_master_response,
    format_register_values,
)
from mb_master_ui import MasterMixin, MasterWindow
from mb_protocol import append_crc, crc_ok


class FakeSlaveSerial:
    """In-memory Modbus RTU slave behind a fake serial port.

    Answers FC01/03 reads and FC05/06/15/16 writes for ``slave_id`` from its
    coil/register banks, can echo every request like some RS485 adapters, and
    can hold bytes that are already on the bus before the master transmits.
    """

    def __init__(self, slave_id=1, *, echo=False, bus_traffic=b"", exception_for=None):
        self.slave_id = slave_id
        self.echo = echo
        self.exception_for = exception_for or {}
        self.registers = {i: i * 10 for i in range(200)}
        self.coils = {i: i % 2 for i in range(200)}
        self.rx = bytearray(bus_traffic)
        self.requests = []

    @property
    def in_waiting(self):
        return len(self.rx)
```

MBSniffer v3.2/tests/test_master.py — lines 41-80

```
41
42      def read(self, size):
43          data = bytes(self.rx[:size])
44          del self.rx[:size]
45          return data
46
47      def reset_input_buffer(self):
48          self.rx.clear()
49
50      def reset_output_buffer(self):
51          pass
52
53      def cancel_read(self):
54          pass
55
56      def cancel_write(self):
57          pass
58
59      def close(self):
60          pass
61
62      def write(self, data):
63          data = bytes(data)
64          self.requests.append(data)
65          if self.echo:
66              self.rx.extend(data)
67          if crc_ok(data) and data[0] == self.slave_id:
68              self.rx.extend(self._answer(data))
69          elif crc_ok(data) and data[0] == 0:
70              self._answer(data)  # broadcast: execute, never answer
71          return len(data)
72
73      def _answer(self, req):
74          fc = req[1]
75          if fc in self.exception_for:
76              return append_crc(bytes((req[0], fc | 0x80, self.exception_for[fc])))
77          address = (req[2] << 8) | req[3]
78          qty = (req[4] << 8) | req[5]
79          if fc == 3:
80              payload = b"".join(struct.pack(">H", self.registers.get(address + i, 0))
```

MBSniffer v3.2/tests/test_master.py — lines 81-120

```
                               for i in range(qty))
            return append_crc(bytes((req[0], 3, len(payload))) + payload)
        if fc == 1:
            data = bytearray((qty + 7) // 8)
            for i in range(qty):
                if self.coils.get(address + i):
                    data[i // 8] |= 1 << (i % 8)
            return append_crc(bytes((req[0], 1, len(data))) + bytes(data))
        if fc == 6:
            self.registers[address] = qty
            return req
        if fc == 5:
            self.coils[address] = 1 if qty == 0xFF00 else 0
            return req
        if fc == 16:
            for i in range(qty):
                self.registers[address + i] = (req[7 + 2 * i] << 8) | req[8 + 2 * i]
            return append_crc(req[:6])
        if fc == 15:
            for i in range(qty):
                self.coils[address + i] = (req[7 + i // 8] >> (i % 8)) & 1
            return append_crc(req[:6])
        return append_crc(bytes((req[0], fc | 0x80, 1)))


def drain(event_queue):
    events = []
    while not event_queue.empty():
        events.append(event_queue.get())
    return events


def make_engine(serial, **overrides):
    options = dict(baud=115200, parity="None", stopbits="1",
                   response_timeout_ms=30, delay_ms=0)
    options.update(overrides)
    return MasterEngine(serial, queue.Queue(), **options)


class MasterFormatTests(unittest.TestCase):
```

MBSniffer v3.2/tests/test_master.py — lines 121-160

```
121     def test_32_bit_formats_round_trip_in_every_byte_order(self):
122         for order in BYTE_ORDERS:
123             for kind, text in (("int32", "-123456"), ("uint32", "4000000000"),
124                                ("float32", "123.456")):
125                 fmt = f"{kind}_{order}"
126                 registers = encode_register_value(text, fmt)
127                 self.assertEqual(len(registers), 2)
128                 self.assertEqual(format_register_values(registers, fmt)[0], text)
129 
130     def test_float_byte_orders_match_wire_layout(self):
131         big_endian = struct.pack(">f", 1.5)  # 3F C0 00 00 = A B C D
132         self.assertEqual(encode_register_value("1.5", "float32_ABCD"), [0x3FC0, 0x0000])
133         self.assertEqual(encode_register_value("1.5", "float32_CDAB"), [0x0000, 0x3FC0])
134         self.assertEqual(encode_register_value("1.5", "float32_BADC"), [0xC03F, 0x0000])
135         self.assertEqual(encode_register_value("1.5", "float32_DCBA"), [0x0000, 0xC03F])
136         self.assertEqual(big_endian, bytes.fromhex("3FC00000"))
137 
138     def test_16_bit_formats(self):
139         self.assertEqual(format_register_values([0xFFFE], "int16"), ["-2"])
140         self.assertEqual(format_register_values([0xFFFE], "uint16"), ["65534"])
141         self.assertEqual(format_register_values([0x00AB], "hex16"), ["0x00AB"])
142         self.assertEqual(format_register_values([0x0105], "bin16"), ["00000001 00000101"])
143         self.assertEqual(encode_register_value("-2", "int16"), [0xFFFE])
144         self.assertEqual(encode_register_value("0x1A2B", "hex16"), [0x1A2B])
145         self.assertEqual(encode_register_value("0000 0001 0000 0101", "bin16"), [0x0105])
146 
147     def test_values_outside_the_format_are_rejected(self):
148         for text, fmt in (("40000", "int16"), ("-1", "uint16"), ("abc", "float32_ABCD"),
149                           ("1e40", "float32_ABCD"), ("", "int16")):
150             with self.assertRaises(ValueError):
151                 encode_register_value(text, fmt)
152         with self.assertRaises(ValueError):
153             encode_coil_value("2")
154 
155     def test_unpaired_register_in_32_bit_format(self):
156         self.assertEqual(format_register_values([1, 2, 3], "uint32_ABCD"), ["65538", "", "—"])
157 
158 
159 class MasterFrameTests(unittest.TestCase):
160     def test_write_frames_match_modbus_specification_examples(self):
```

MBSniffer v3.2/tests/test_master.py — lines 161-200

```
161         # Modbus Application Protocol V1.1b3 examples (CRC appended for RTU).
162         self.assertEqual(build_write_request(0x11, 16, 1, [0x000A, 0x0102])[:-2],
163                          bytes.fromhex("11 10 0001 0002 04 000A 0102"))
164         coils = [1, 0, 1, 1, 0, 0, 1, 1, 1, 0]
165         self.assertEqual(build_write_request(0x11, 15, 0x13, coils)[:-2],
166                          bytes.fromhex("11 0F 0013 000A 02 CD 01"))
167         self.assertEqual(build_write_request(0x11, 5, 0xAC, [1])[:-2],
168                          bytes.fromhex("11 05 00AC FF00"))
169         self.assertEqual(build_write_request(0x11, 6, 1, [3])[:-2],
170                          bytes.fromhex("11 06 0001 0003"))
171 
172     def test_invalid_definitions_are_rejected(self):
173         for args in ((0, 3, 0, 1), (1, 7, 0, 1), (1, 3, 0, 126), (1, 1, 0, 2001),
174                      (1, 3, 65535, 2)):
175             with self.assertRaises(ValueError):
176                 build_read_request(*args)
177         with self.assertRaises(ValueError):
178             build_write_request(1, 6, 0, [1, 2])
179 
180     def test_read_response_found_after_adapter_echo(self):
181         request = build_read_request(1, 3, 0, 2)
182         response = append_crc(bytes.fromhex("01 03 04 0001 0002"))
183         found = find_master_response(request + response, request)
184         self.assertEqual(found["kind"], "RESPONSE")
185         self.assertEqual(found["values"], [1, 2])
186 
187     def test_coil_response_is_trimmed_to_quantity(self):
188         request = build_read_request(1, 1, 0, 3)
189         found = find_master_response(append_crc(bytes.fromhex("01 01 01 FD")), request)
190         self.assertEqual(found["values"], [1, 0, 1])
191 
192     def test_exception_and_bad_crc(self):
193         request = build_read_request(1, 3, 0, 1)
194         exception = append_crc(bytes.fromhex("01 83 02"))
195         self.assertEqual(find_master_response(exception, request)["exception_code"], 2)
196         damaged = bytearray(append_crc(bytes.fromhex("01 03 02 0001")))
197         damaged[-1] ^= 0xFF
198         self.assertIsNone(find_master_response(bytes(damaged), request))
199 
200     def test_write_response_must_match_request(self):
```

MBSniffer v3.2/tests/test_master.py — lines 201-240

```
        request = build_write_request(1, 16, 5, [1, 2])
        self.assertIsNotNone(find_master_response(append_crc(request[:6]), request))
        other = append_crc(bytes.fromhex("01 10 0006 0002"))
        self.assertIsNone(find_master_response(other, request))


class MasterEngineTests(unittest.TestCase):
    def run_engine(self, engine, seconds):
        engine.start()
        time.sleep(seconds)
        engine.stop()
        return drain(engine.event_queue)

    def test_scheduler_serves_writes_then_read_once_then_due_polls(self):
        engine = make_engine(FakeSlaveSerial())
        engine.set_poll(PollDefinition(1, 1, 3, 0, 1, 1000))
        engine.set_poll(PollDefinition(2, 1, 3, 0, 1, 1000))
        engine.read_once(2)
        engine.queue_write(1, build_write_request(1, 6, 0, [5]))
        self.assertEqual(engine.next_job(now=10.0)[0], "write")
        self.assertEqual(engine.next_job(now=10.0)[1:], (engine._polls[2][0], True))
        first = engine.next_job(now=10.0)
        second = engine.next_job(now=10.0)
        self.assertEqual({first[1].window_id, second[1].window_id}, {1, 2})
        self.assertEqual(engine.next_job(now=10.5)[0], "wait")
        self.assertEqual(engine.next_job(now=11.01)[0], "read")

    def test_fast_and_slow_windows_share_one_bus(self):
        serial = FakeSlaveSerial()
        engine = make_engine(serial)
        engine.set_poll(PollDefinition(1, 1, 3, 0, 4, 0))
        engine.set_poll(PollDefinition(2, 1, 1, 0, 8, 10000))
        events = self.run_engine(engine, 0.4)
        results = [p for name, p in events if name == "master_result"]
        fast = [r for r in results if r["window_id"] == 1]
        slow = [r for r in results if r["window_id"] == 2]
        self.assertGreater(len(fast), 3)
        self.assertEqual(len(slow), 1)
        self.assertEqual(fast[0]["values"], [0, 10, 20, 30])
        self.assertEqual(slow[0]["values"], [0, 1, 0, 1, 0, 1, 0, 1])
```

MBSniffer v3.2/tests/test_master.py — lines 241-280

```
241         self.assertEqual(events[0][0], "master_ready")
242         self.assertEqual(events[-1], ("master_stopped", {"error": None}))
243 
244     def test_write_is_executed_and_reported(self):
245         serial = FakeSlaveSerial(echo=True)
246         engine = make_engine(serial)
247         engine.queue_write(1, build_write_request(1, 16, 10, [0xABCD, 0x1234]), "FC16")
248         results = [p for n, p in self.run_engine(engine, 0.15) if n == "master_result"]
249         self.assertEqual(results[0]["status"], "ok")
250         self.assertEqual((serial.registers[10], serial.registers[11]), (0xABCD, 0x1234))
251 
252     def test_broadcast_write_does_not_wait_for_a_response(self):
253         serial = FakeSlaveSerial()
254         engine = make_engine(serial, response_timeout_ms=5000)
255         engine.queue_write(1, build_write_request(0, 6, 3, [77]))
256         started = time.perf_counter()
257         result = engine.transact(engine._writes.popleft()[1])
258         self.assertLess(time.perf_counter() - started, 1.0)
259         self.assertEqual(result["status"], "broadcast")
260         self.assertEqual(serial.registers[3], 77)
261 
262     def test_timeout_exception_and_invalid_results(self):
263         engine = make_engine(FakeSlaveSerial(slave_id=9))
264         self.assertEqual(engine.transact(build_read_request(1, 3, 0, 1))["status"], "timeout")
265         engine = make_engine(FakeSlaveSerial(exception_for={3: 2}))
266         result = engine.transact(build_read_request(1, 3, 0, 1))
267         self.assertEqual((result["status"], result["exception_code"]), ("exception", 2))
268 
269     def test_traffic_on_the_bus_blocks_the_master(self):
270         serial = FakeSlaveSerial(bus_traffic=b"\x01\x03\x00")
271         engine = make_engine(serial, idle_check_seconds=0.05)
272         engine.set_poll(PollDefinition(1, 1, 3, 0, 1, 0))
273         events = self.run_engine(engine, 0.2)
274         self.assertEqual(serial.requests, [])
275         self.assertEqual(events[-1][0], "master_stopped")
276         self.assertEqual(events[-1][1]["error"][0], "traffic")
277 
278 
279 class MasterWindowTests(unittest.TestCase):
280     def test_workspace_entry_round_trip_and_validation(self):
```

MBSniffer v3.2/tests/test_master.py — lines 281-300

```
281         window = MasterWindow(1, 1, slave=5, fc=3, address=100, qty=4, scan_ms=250, base=1,
282                               fmt="float32_CDAB", names={100: "Temperatura"})
283         data = MasterWindow.validated_kwargs(window.to_json())
284         self.assertEqual(data["names"], {100: "Temperatura"})
285         self.assertEqual((data["slave"], data["fmt"], data["base"]), (5, "float32_CDAB", 1))
286         for bad in ({"slave": 0}, {"fc": 6}, {"qty": 200}, {"scan_ms": -1}, "x"):
287             with self.assertRaises((ValueError, TypeError)):
288                 MasterWindow.validated_kwargs(bad)
289 
290     def test_write_dialog_builds_fc16_for_float_values(self):
291         window = MasterWindow(1, 1, slave=2, fc=3, address=0, qty=10, base=1,
292                               fmt="float32_ABCD")
293         request, _ = MasterMixin.master_build_write(None, window, "2", "11", "1.5;2", 16)
294         self.assertEqual(request[:-2], bytes.fromhex("02 10 000A 0004 08 3FC0 0000 4000 0000"))
295         with self.assertRaises(ValueError):
296             MasterMixin.master_build_write(None, window, "2", "11", "1.5", 6)
297 
298 
299 if __name__ == "__main__":
300     unittest.main(verbosity=2)
```

MBSniffer.bat

210 source lines.

MBSniffer.bat — lines 1-40

```
@echo off
setlocal EnableExtensions
title MBSniffer 3.2
set "ROOT=%~dp0"
set "APPDIR=%~dp0MBSniffer v3.2"
set "SCRIPT=%APPDIR%\MBSniffer.py"
cd /d "%APPDIR%"
set "PYRUN="
set "PYEXE="
set "PYWEXE="

if not exist "%SCRIPT%" (
    echo.
    echo ERRO: os ficheiros do MBSniffer nao foram encontrados em:
    echo   %SCRIPT%
    echo.
    pause
    exit /b 1
)

rem ============================================================
rem 1. Procurar uma instalacao funcional de Python 3
rem ============================================================
call :DETECT_PYTHON

if defined PYRUN goto PYTHON_OK

echo.
echo Python 3 nao foi encontrado.
echo A tentar instalar automaticamente Python 3.13 com WinGet...
echo.

where winget >nul 2>&1
if errorlevel 1 goto NO_WINGET

winget install --id Python.Python.3.13 --exact --scope user --silent ^
    --accept-package-agreements --accept-source-agreements

if errorlevel 1 (
    echo.
```

MBSniffer.bat — lines 41-80

```
41      echo ERRO: A instalacao automatica do Python falhou.
42      echo O WinGet pode estar bloqueado pela politica da empresa ou sem acesso a Internet.
43      echo.
44      pause
45      exit /b 1
46  )
47
48  rem O instalador pode alterar o PATH sem atualizar esta janela CMD.
49  rem Por isso voltamos a procurar tambem nos caminhos standard.
50  call :DETECT_PYTHON
51
52  if not defined PYRUN (
53      echo.
54      echo ERRO: Python foi instalado, mas nao consegui localizar python.exe.
55      echo Fecha esta janela e volta a executar MBSniffer.bat.
56      echo.
57      pause
58      exit /b 1
59  )
60
61  :PYTHON_OK
62  rem ============================================================
63  rem 2. Descobrir o python.exe real e o respetivo pythonw.exe
64  rem ============================================================
65  for /f "usebackq delims=" %%P in (`%PYRUN% -c "import sys; print(sys.executable)" 2^>nul`) do (
66      set "PYEXE=%%P"
67  )
68
69  if not defined PYEXE (
70      echo.
71      echo ERRO: Nao foi possivel determinar a instalacao do Python.
72      echo.
73      pause
74      exit /b 1
75  )
76
77  for %%P in ("%PYEXE%") do set "PYWEXE=%%~dpPpythonw.exe"
78
79  rem ============================================================
80  rem 3. Verificar tkinter
```

MBSniffer.bat — lines 81-120

```
rem ============================================================
%PYRUN% -c "import tkinter" >nul 2>&1
if not errorlevel 1 goto TK_OK

echo.
echo tkinter nao esta disponivel nesta instalacao do Python.
echo A tentar instalar/reparar Python 3.13 com WinGet...
echo.

where winget >nul 2>&1
if errorlevel 1 goto TK_ERROR

winget install --id Python.Python.3.13 --exact --scope user --silent --force ^
    --accept-package-agreements --accept-source-agreements

call :DETECT_PYTHON
if not defined PYRUN goto TK_ERROR

%PYRUN% -c "import tkinter" >nul 2>&1
if errorlevel 1 goto TK_ERROR

for /f "usebackq delims=" %%P in (`%PYRUN% -c "import sys; print(sys.executable)" 2^>nul`) do (
    set "PYEXE=%%P"
)
for %%P in ("%PYEXE%") do set "PYWEXE=%%~dpPpythonw.exe"

:TK_OK
rem ============================================================
rem 4. Verificar pyserial (obrigatorio)
rem ============================================================
%PYRUN% -c "import serial" >nul 2>&1
if not errorlevel 1 goto PYSERIAL_OK

echo.
echo pyserial nao esta instalado.
echo A instalar pyserial...
echo.

%PYRUN% -m pip --version >nul 2>&1
if errorlevel 1 (
```

MBSniffer.bat — lines 121-160

```
121     %PYRUN% -m ensurepip --upgrade
122     if errorlevel 1 goto DEPENDENCIES_ERROR
123 )
124
125 %PYRUN% -m pip install --user pyserial
126 if errorlevel 1 goto DEPENDENCIES_ERROR
127
128 %PYRUN% -c "import serial" >nul 2>&1
129 if errorlevel 1 goto DEPENDENCIES_ERROR
130
131 :PYSERIAL_OK
132 rem ============================================================
133 rem 5. Dependencias verificadas
134 rem ============================================================
135 :DEPENDENCIES_OK
136 rem ============================================================
137 rem 6. Executar o MBSniffer
138 rem ============================================================
139 if exist "%PYWEXE%" (
140     start "" "%PYWEXE%" "%SCRIPT%"
141     exit /b 0
142 )
143
144 rem Fallback caso pythonw.exe nao exista.
145 start "" %PYRUN% "%SCRIPT%"
146 exit /b 0
147
148
149 :DETECT_PYTHON
150 set "PYRUN="
151
152 rem Python Launcher for Windows
153 py -3 -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
154 if not errorlevel 1 (
155     set "PYRUN=py -3"
156     exit /b 0
157 )
158
159 rem Python no PATH
160 python -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
```

MBSniffer.bat — lines 161-200

```
161 if not errorlevel 1 (
162     set "PYRUN=python"
163     exit /b 0
164 )
165
166 rem Caminhos standard para Python 3.13 instalado por utilizador/WinGet.
167 if exist "%LocalAppData%\Programs\Python\Python313\python.exe" (
168     set PYRUN="%LocalAppData%\Programs\Python\Python313\python.exe"
169     exit /b 0
170 )
171
172 if exist "%ProgramFiles%\Python313\python.exe" (
173     set PYRUN="%ProgramFiles%\Python313\python.exe"
174     exit /b 0
175 )
176
177 if exist "%ProgramFiles(x86)%\Python313\python.exe" (
178     set PYRUN="%ProgramFiles(x86)%\Python313\python.exe"
179     exit /b 0
180 )
181
182 exit /b 0
183
184
185 :NO_WINGET
186 echo.
187 echo ERRO: Python 3 nao esta instalado e o WinGet nao esta disponivel.
188 echo Nao e possivel fazer a instalacao automatica neste PC.
189 echo Instala Python 3 manualmente e volta a executar MBSniffer.bat.
190 echo.
191 pause
192 exit /b 1
193
194
195 :TK_ERROR
196 echo.
197 echo ERRO: tkinter continua indisponivel.
198 echo E necessario usar uma instalacao normal do Python para Windows com Tcl/Tk.
199 echo.
200 pause
```

MBSniffer.bat — lines 201-210

```
exit /b 1


:DEPENDENCIES_ERROR
echo.
echo ERRO: Nao foi possivel instalar pyserial.
echo Verifica acesso a Internet, proxy/firewall e permissoes do Python.
echo.
pause
exit /b 1
```

MBSniffer v3.2/build_exe.bat

151 source lines.

MBSniffer v3.2/build_exe.bat — lines 1-40

```
@echo off
setlocal EnableExtensions
title Build MBSniffer 3.2

rem Normalizar caminhos sem barra final. Isto evita problemas de parsing
rem do Windows/Python quando um argumento quoted termina em "\".
for %%I in ("%~dp0.") do set "APPDIR=%%~fI"
for %%I in ("%APPDIR%\..") do set "DISTDIR=%%~fI"

set "SCRIPT=%APPDIR%\MBSniffer.py"
set "ICON=%APPDIR%\MBSniffer.ico"
set "WORKDIR=%APPDIR%\build"
set "SPECFILE=%APPDIR%\MBSniffer.spec"
set "PYRUN="

cd /d "%APPDIR%"

if not exist "%SCRIPT%" (
    echo.
    echo ERRO: MBSniffer.py nao foi encontrado.
    echo.
    pause
    exit /b 1
)

rem ============================================================
rem 1. Procurar Python 3
rem ============================================================
py -3 -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
if not errorlevel 1 (
    set "PYRUN=py -3"
    goto PYTHON_OK
)

python -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
if not errorlevel 1 (
    set "PYRUN=python"
    goto PYTHON_OK
)

```

MBSniffer v3.2/build_exe.bat — lines 41-80

```
41 if exist "%LocalAppData%\Programs\Python\Python313\python.exe" (
42     set PYRUN="%LocalAppData%\Programs\Python\Python313\python.exe"
43     goto PYTHON_OK
44 )
45
46 echo.
47 echo ERRO: Python 3 nao foi encontrado.
48 echo Executa primeiro MBSniffer.bat para instalar/verificar as dependencias.
49 echo.
50 pause
51 exit /b 1
52
53
54 :PYTHON_OK
55 rem ============================================================
56 rem 2. Instalar/atualizar dependencias de build
57 rem ============================================================
58 echo.
59 echo [1/4] A verificar pyserial e PyInstaller...
60 %PYRUN% -m pip install --upgrade pyserial pyinstaller
61 if errorlevel 1 (
62     echo.
63     echo ERRO ao instalar pyserial/PyInstaller.
64     echo.
65     pause
66     exit /b 1
67 )
68
69 rem ============================================================
70 rem 3. Executar testes de regressao antes do build
71 rem ============================================================
72 echo.
73 echo [2/4] A executar testes de regressao...
74 %PYRUN% -m unittest discover -s tests -p "test_*.py" -v
75 if errorlevel 1 (
76     echo.
77     echo ERRO: os testes de regressao falharam. Build cancelado.
78     echo.
79     pause
80     exit /b 1
```

MBSniffer v3.2/build_exe.bat — lines 81-120

```
)

rem ============================================================
rem 4. Limpar builds anteriores
rem ============================================================
echo.
echo [3/4] A limpar build anterior...
if exist "%WORKDIR%" rmdir /s /q "%WORKDIR%"
if exist "%SPECFILE%" del /q "%SPECFILE%"
if exist "%DISTDIR%\MBSniffer.exe" del /q "%DISTDIR%\MBSniffer.exe"

rem ============================================================
rem 5. Criar EXE
rem ============================================================
echo.
echo [4/4] A criar MBSniffer.exe...

if exist "%ICON%" (
    %PYRUN% -m PyInstaller ^
      --noconfirm ^
      --clean ^
      --onefile ^
      --windowed ^
      --name MBSniffer ^
      --distpath "%DISTDIR%" ^
      --workpath "%WORKDIR%" ^
      --icon "%ICON%" ^
      --add-data "%ICON%;." ^
      "%SCRIPT%"
) else (
    echo AVISO: MBSniffer.ico nao foi encontrado.
    %PYRUN% -m PyInstaller ^
      --noconfirm ^
      --clean ^
      --onefile ^
      --windowed ^
      --name MBSniffer ^
      --distpath "%DISTDIR%" ^
      --workpath "%WORKDIR%" ^
      "%SCRIPT%"
```

MBSniffer v3.2/build_exe.bat — lines 121-151

```
)

if errorlevel 1 (
    echo.
    echo ERRO: o build falhou.
    echo.
    pause
    exit /b 1
)

if not exist "%DISTDIR%\MBSniffer.exe" (
    echo.
    echo ERRO: o PyInstaller terminou sem criar:
    echo   %DISTDIR%\MBSniffer.exe
    echo.
    pause
    exit /b 1
)

echo.
echo ============================================================
echo BUILD CONCLUIDO
echo ============================================================
echo.
echo Executavel criado em:
echo   %DISTDIR%\MBSniffer.exe
echo.
echo O icone MBSniffer.ico foi aplicado ao EXE e incluido
echo como recurso para a janela/barra de tarefas.
echo.
pause
```

MBSniffer v3.2/run_tests.bat

45 source lines.

MBSniffer v3.2/run_tests.bat — lines 1-40

```
@echo off
setlocal EnableExtensions
title MBSniffer regression tests

for %%I in ("%~dp0.") do set "APPDIR=%%~fI"
cd /d "%APPDIR%"
set "PYRUN="

py -3 -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
if not errorlevel 1 set "PYRUN=py -3"

if not defined PYRUN (
    python -c "import sys; assert sys.version_info.major == 3" >nul 2>&1
    if not errorlevel 1 set "PYRUN=python"
)

if not defined PYRUN (
    echo.
    echo ERRO: Python 3 nao foi encontrado.
    echo.
    pause
    exit /b 1
)

echo.
echo A executar testes de regressao do MBSniffer...
echo.
%PYRUN% -m unittest discover -s tests -p "test_*.py" -v
if errorlevel 1 (
    echo.
    echo ============================================================
    echo TESTES FALHARAM - build/release nao deve prosseguir.
    echo ============================================================
    echo.
    pause
    exit /b 1
)

echo.
echo ============================================================
```

MBSniffer v3.2/run_tests.bat — lines 41-45

```
41 echo TODOS OS TESTES PASSARAM
42 echo ============================================================
43 echo.
44 pause
45 exit /b 0
```